Libertinus and diacritics

To do

Long term: add mark-to-mark positioning (stacking) for the common above-marks in all four styles: ring, dieresis, macron, acute, grave, circumflex, caron, tilde, breve. The designer's stacked precomposed glyphs (è, ȍ, ã, …) could supply the mark-to-mark anchors, as the precomposed capitals do for stage 4. Only glyphs drawn as a true stack are evidence: others among the short list of precomposed letters with two above-marks are composed more compactly in some other configuration (Ǻ, Ű, Ő), so each needs to be verified by eye before it is used.

Write general instructions for using CGJ U+034F to block unwanted normalization. A below-mark typed after a precomposed letter that already has an above-mark is reordered and recomposed by normalization: ì + tilde below becomes ḭ + grave, which brings back the dot of i, and Ǻ + dot below becomes Ạ + ring + acute (130 cases among the precomposed letters; see "Precomposed characters"). That is a decomposition problem, to be avoided with a CGJ (or a dotless base) whether or not the font has mark-to-mark positioning; the sections "Lowercase i with tilde below and grave above, in italic and semibold italic" and following explore it, and "Floating tones" uses CGJ for a similar reason. The instructions should say when CGJ is needed, where it goes, and its cost (a precomposed glyph before CGJ has no below-anchor, so the below-mark falls back to a default position).

Audit the above anchors of the vowels against their precomposed glyphs, which are many and good evidence (survey of 2026-11-07). Where the patched anchor differs from the median mark centre of the precomposed glyphs by more than 15: italic o $+20$, O $+22$, E -20, Y -17; semibold y -19, Æ -16; semibold italic y $+29$ (regular's anchor, copied), Y -58, a $+18$, A -15, Æ -19. Regular O is within 12, but its marks land 16–19 left of the centre line of O. Two cautions: italic and semibold italic Y and U were just curated so that shaped marks land on their axes, and their precomposed glyphs disagree by 14–58; and the semibold italic precomposed glyphs run 30–40 right of the anchor model for g and E too. The vowel base anchors have not changed since; with the symmetric marks on their axes (2026-11-09), the median offset of composed marks from the precomposed glyphs per letter confirms the lowercase figures (italic o $+19$, semibold y -18, semibold italic y $+30$, a $+22$), and adds i: marks on ɪ land 12–18 right of ì í î ï in every style (regular $+16$, italic $+18$, semibold $+18$, semibold italic $+12$). The capitals were not remeasured. O and o lean to the left, unlike most of the other humanist letters, so their optical centre at the top may lie right of the bbox midpoint: a reason to move their above-anchors right, to weigh in the audit.

Anchor model

Libertinus has an anchor model where the base anchors are set to locations where the marks would be placed relative to the base. See the blue dots in the left panel of Figure 1; from top to bottom, the above-anchor is the blue dot above the base glyph, the overlay-anchor is blue dot inside the base glyph outline, and the below-anchor is the blue dot below the base glyph. The mark anchor is generally near the optical center of the mark glyph outline. See the red square in the middle panel of Figure 1; the mark's below-anchor is at the optical center of the mark. The base and mark are combined in the right panel of Figure 1 by superimposing their below-anchors.

Every font has five vertical reference lines (with respective Y coordinate values) – ascender line, capline, x-height or meanline, baseline, descender line – within which each glyph lives. These vertical reference lines are the gray vertical lines in Figures 1, 2, 3 and 4. Each glyph has a (tight) bounding box for real minimum and maximum X and Y coordinate values, as well as a so-called ideal bounding box where the glyph seems to live, but accounting for overshoots of bowls and serifs. The entire design goal is that the top and bottom of the ideal bounding box are two vertical reference lines, except for glyphs that are non-full ascenders or non-full descenders.

By clearance, I mean the (vertical) distance between a base and a mark, measured between the ideal bounding boxes of the base and the mark. For some designed fonts, the clearance is consistent for each vertical reference line, so that all glyphs that live at that reference line are separated from all marks with the same clearance; e.g. dots should not dance below the baseline for different letters that sit on the baseline, and acutes and graves sit at about the same height for different letters with similar top vertical reference line.

In roman styles of Libertinus, the clearance is encoded in the base anchor. The Y coordinate of an above-anchor is the top of the base's ideal bounding box plus the clearance of the nearest vertical reference line, and the Y coordinate of a below-anchor is the bottom of the ideal bounding box minus that clearance. For most glyphs that edge of the ideal bounding box is the reference line itself. For a non-full ascender or non-full descender it is not, but the clearance is the same; see the section on vertical reference lines below. The X coordinate of the anchor is at some horizontal optical center of the base glyph, but the meaning of the optical center depends on the glyph letterform, or whatever the designer thinks looks best. For a mostly symmetric letterform in the roman styles, a candidate for this X coordinate is the midpoint of the bounding box of the glyph.

The anchor of a mark is at the horizontal center of the mark, if the mark is symmetric. An asymmetric mark, such as the acute or grave, has its anchor a few units to one side of the center of its outline. Vertically, the anchor of an above-mark is about 140 units above the bottom of the mark outline. That is near the top of a tall mark, such as the circumflex, and above the top

of a short mark, such as the macron or dot. Because the above-anchor of a base at the x-height is 216 units above the x-height, the bottom of a tall mark is about 75 units above the base, and a short mark sits a little higher.

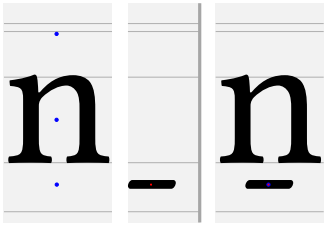


Figure 1: Libertinus n and macron with anchors.

Other fonts use a different geometric model for anchors. In Noto, the base's below-anchor is at the baseline, the overlay-anchor is inside the glyph, and the above-anchor is at the x-height. See the blue dots in the left panel of Figure 2; from the bottom to the top, the mark's below-anchor is the blue dot on the baseline, the overlay-anchor is blue dot inside the base glyph outline, and the above-anchor is the blue dot at the x-height. In Noto, the mark's below-anchor is at the origin $(0, 0)$, and the mark glyph outline is drawn below the origin. In the middle panel of Figure 2, the mark's below-anchor is the red square at the origin. The base and mark are combined in the right panel of Figure 2 by superimposing their below-anchors. So in Noto the clearance is encoded in the mark anchor, and in Libertinus it is encoded in the base anchor.

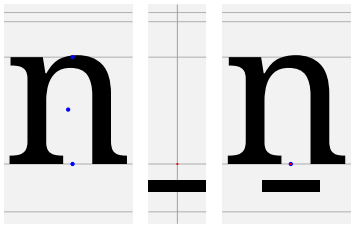


Figure 2: Noto n and macron with anchors.

Italic anchor model

In italic, the Libertinus anchor model is similar to the roman styles, but it is applied along the italic axis. The italic lane of a base glyph is its advance-width box, slanted by the italic angle θ (12°). The midline of the italic lane is the line $X = W/2 + Y \tan\theta$, where W is the advance width of the glyph. See the gray band and the dashed blue line in the left panel of Figure 3.

The Y coordinate of an above- or below-anchor is set exactly as in the roman styles: the top or bottom of the glyph's ideal bounding box plus or minus the clearance of the nearest vertical reference line, measured vertically. So marks sit at the same height in roman and italic. The X coordinate is the point of the midline at that height: $ax = W/2 + ay \tan\theta$ and $bx = W/2 + by \tan\theta$.

The height in this formula is the height of the anchor itself, that is, the edge of the ideal bounding box plus or minus the clearance, not the height of the reference line or of the ideal bounding box. So the Y coordinate of the anchor must be determined before the X coordinate can be calculated. In italic, the above-anchor is at $W/2 + 137$ for a letter at the x-height ($ay = 645$), at $W/2 + 181$ for a capital ($ay = 850$), and at $W/2 + 189$ for an ascender ($ay = 890$). The below-anchor is at $W/2 - 23$ for a letter on the baseline ($by = -110$) and at $W/2 - 68$ for a descender ($by = -319$). An error of 10 units in the Y coordinate moves the X coordinate by about 2 units.

The anchor of a mark lies on the mark's own slanted axis, through the center of its outline if the mark is symmetric. See the dashed red lines in the middle panel of Figure 3. That is, the mark anchor is at $X = C + Y \tan\theta$, where Y is the height of the anchor, as in the roman styles, and C is where the axis of the mark meets the baseline: the horizontal center of the mark

outline after the slant is removed. An asymmetric mark keeps the same offset from its axis as in the roman styles. Superimposing the anchors lays the axis of the mark on the midline of the base, as in the right panel. The mark is displaced from the base along the italic axis, not vertically above or below it.

The midline is a geometric center, not an optical center, but it is a good candidate for the anchor X coordinate of a mostly symmetric letterform. Asymmetric letterforms, and letters that overhang their advance width such as f and j, need curated anchors.

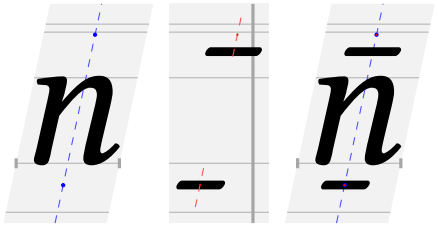


Figure 3: Libertinus italic n, macron and macron below, with anchors. The gray band is the italic lane and the dashed blue line is its midline.

Semibold italic anchor model

Semibold italic has its own anchor model, which is not the italic anchor model, and the patch keeps it. The outlines are slanted (the italic angle θ is $11.5°$), but the anchors are, almost all, the anchors of regular, copied unchanged: 42 of the 44 above- and below-mark anchors, and 319 of the 384 above- and below-anchors of bases, have exactly the coordinates of regular.

So the anchors do not follow the slant. A slanted outline moves further right the higher it goes, but an anchor copied from regular stays where it was. An above-anchor therefore sits left of where the italic anchor model would put it, by roughly the amount the slant has moved the outline at that height. This is true of the base and of the mark alike: the base's above-anchor is left of the midline of its italic lane, and the mark's anchor is left of the mark's own slanted axis, by about the same amount. When the two anchors are superimposed, the two errors cancel, and the axis of the mark lands near the midline of the base, as it would in italic. See Figure 4, where the filled symbols are the designer's anchors and the hollow symbols are the positions given by the italic anchor model: the above-anchor of n and the anchor of the macron are both far left of the hollow symbols, but by nearly the same distance.

Each anchor in the font has its own shift k: how far it lies left of the position the italic anchor model gives it. For a base anchor, $k = W/2 + Y \tan\theta - X$. For a mark anchor, $k = C + Y \tan\theta - (X - d)$, where C is where the mark's axis meets the baseline and d is the mark's offset from its axis in regular ($d = 0$ for a symmetric mark; for the acute it is -32). In Figure 4, the above-anchor of n has $k = 170$ and the macron has $k = 141$. A mark lands $k_{\text{mark}} - k_{\text{base}}$ units right of the base's midline (left, if negative), so the macron lands $141 - 170 = -29$, that is, 29 units left of the midline of n.

The shifts differ from anchor to anchor, because the designer copied regular's X coordinates, which carry no slant. So an anchor's shift is about $Y \tan\theta$ at its own height, plus a small amount that depends on how the slanted outline was drawn. A higher anchor has a larger shift: comma above U+0313, with its anchor at $Y = 605$, has $k = 125$; hook above U+0309, at $Y = 765$, has $k = 168$. Only the difference between a mark's shift and a base's shift moves the mark off the midline, so any single value works, provided every base anchor and every mark anchor uses it.

The semibold italic anchor model therefore fixes one value of K for each anchor class and gives every anchor of that class the same shift: the base anchor is at $X = W/2 + Y \tan\theta - K$ and the mark anchor is at $X = C + Y \tan\theta + d - K$, with $\tan\theta = 0.2035$ and the same Y coordinates as in the other styles.

- Above-anchors: $K = 150$, for every above-mark and every base's above-anchor. Above-anchors are high, where the slant has moved the outline far to the right, so their shift is large. 150 is the median shift of the designer's above-marks; most are between 132 and 168.
- Below-anchors: $K = -12$, for every below-mark and every base's below-anchor. Below-anchors are just under the baseline, where the slant has moved the outline a little to the left, so their shift is small and negative. -12 is the median shift of the designer's below-marks; most are between -23 and -4.

The values are taken from the marks rather than the bases so that most of the designer's mark anchors already conform. The designer's base anchors have median shifts of 139 above and -1 below on lowercase Latin and IPA letters, and 173 above and -5 below on capitals. An anchor conforms if its own shift k is close to K. The italic anchor model is the case $K = 0$.

The candidates for a base are $ax = W/2 + ay \tan\theta - 150$ and $bx = W/2 + by \tan\theta + 12$. So the above-anchor is at $W/2 - 19$ for a letter at the x-height ($ay = 645$), at $W/2 + 23$ for a capital ($ay = 850$), and at $W/2 + 31$ for an ascender ($ay = 890$). The below-anchor is at $W/2 - 10$ for a letter on the baseline ($by = -110$) and at $W/2 - 53$ for a descender ($by = -319$). For semibold italic n, $W = 606$, so $ax = 284$ and $bx = 293$; the font has $ax = 264$ and $bx = 265$, the values of regular n, and the macron lands 29 units left of the midline.

The cancellation is only approximate, because the designer's anchors do not follow the slant at all, whereas the model's anchors do, apart from the constant K. A mark's shift is fixed by the height of its own anchor: for a mark, $k = Y \tan\theta$ plus a small amount, where Y is the height of the mark anchor. But the designer's base anchors are near $W/2 - 9$ at every height, so they do not move right for a taller base, such as a capital or an ascender. The error of a mark on a base therefore grows with the difference in height of the two anchors, by 2 units for every 10. The median distance of a designer's base anchor from the candidate is about 20 units for a lowercase letter and 30 units for a capital.

The patch does not move this style to the italic anchor model. It keeps the designer's anchors where they conform to the semibold italic anchor model, and corrects only the anchors that do not, so that they conform to it: a base anchor is moved to the candidate above, and a mark anchor is moved to $X = C + Y \tan\theta - 150$ for an above-mark or $X = C + Y \tan\theta + 12$ for a below-mark. An anchor set by the italic anchor model would be wrong here: a base anchor without the shift would put every above-mark about 150 units too far right, because the marks carry the shift.

The following do not conform to the semibold italic anchor model and probably need to be patched, some of them manually.

- Capitals. The designer's above-anchors are about 25 units left of the candidate, since they are 205 units higher than the anchors of lowercase letters but no further right. The worst are B, D, I, N, O, R, W (40 to 60 units left) and L (159 units left; its anchor is over the stem, as in regular).
- The capital forms of the grave, acute, circumflex and breve (`grave.cap`, `acute.cap`, `circumflex.cap`, `breve.cap`), which the font substitutes over capitals. Their mark anchors were set with the slant (k between -13 and 29), but the above-anchors of capitals were not, so these four marks land 160 to 190 units left of the middle of a capital. The other above-marks have no capital form in use and land 20 to 50 units left. The capital forms need mark anchors at $X = C + Y \tan\theta - 150$; `caron.cap` and `dieresis.cap` are not substituted.
- Above-anchors that were set with the slant, as in italic, so that an above-mark lands 130 to 190 units too far right: a, s, h and J.
- Other base anchors more than 40 units from the candidate. Above: c, g, Æ, Ɖ, ƕ, Ƣ, Ƥ, ɳ, ɗ, ƒ, ʄ, Ħ, ɵ, ɦ. Below: A, C, D, J, a, d, f, j, q, r, t, Æ, ĸ, ƕ, Ƣ, Ƥ, ɳ, ɾ, ɑ, ɖ, ɗ, ɤ, ɥ, ɯ, ɻ, Ħ, ɵ. Several of these are copies of the anchors of another glyph. The anchors of ɥ and ɻ are transposed, as in the other styles.
- Marks whose own shift k is far from K (150 above, -12 below), so that no base anchor can be right for them and for the other marks; each lands $k - K$ units right of where a conforming mark lands: dieresis U+0308 ($k = 111$, so it lands about 40 units left of the other above-marks), comma above U+0313 (125), inverted breve U+0311 (132), hook above U+0309 and double grave U+030F (168, about 18 units right), koronis U+0343 (371), right tack below U+0319 (-30) and tilde below U+0330 ($+10$). The double acute U+030B (156) conforms.
- Tilde U+0303, one of the two mark anchors that is not a copy of regular; its shift is 139. (These are the designer's values. In the patched font the dieresis and tilde are on the model by the mark-axis rule, and the hook above, inverted breve, U+0314 and tilde below by curated anchors; see "Mark anchor model".)
- The Hebrew ligatures U+05F0, U+05F1, U+05F2, which have above- and below-anchors in this lookup table in semibold italic only.
- Letters that overhang their advance width, such as f and j, which need curated anchors in every italic style.

The anchors of precomposed glyphs were not compared with these candidates.

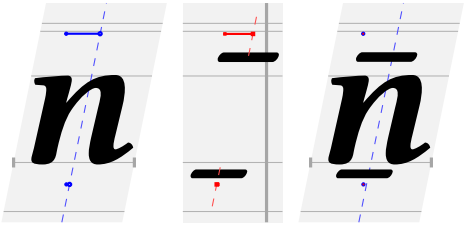


Figure 4: Libertinus semibold italic n, macron and macron below, with the designer's anchors (filled) and the positions given by the italic anchor model (hollow). The above-anchor of n is 170 units left of the midline ($k = 170$) and the anchor of the macron is 141 units left of its axis ($k = 141$), so the macron lands 29 units left of the midline of n. The model's own positions, with $K = 150$, would be 20 units right of the filled circle and 9 units left of the filled square.

Mark anchor model

The anchor of a mark follows the same model in every style. Horizontally, it lies on the mark's own axis, $X = C + Y \tan\theta - K$ ($\theta = 0$ and $K = 0$ in the roman styles, $K = 0$ in italic, $K = 150$ above and -12 below in semibold italic). An asymmetric mark, such as the grave, acute, double acute or double grave, keeps the offset from its axis that it has in regular. Vertically, the anchor is at a fixed distance from the mark's outline: the anchor of an above-mark is a certain height above the bottom of the outline, and the anchor of a below-mark is a certain depth below the top of the outline. These distances differ from mark to mark (a short mark such as the macron sits a little higher than a tall one), but since the base anchors have the same Y coordinates in every style, a mark sits at the same height in every style only if its distance is the same as in regular.

So each mark can be checked in two ways: its X offset from its axis, compared with regular, and the distance between its anchor and its outline, compared with regular. A third check is the designer's precomposed glyphs. They are not consistent between styles (the macron of ō was 18 units lower than the combining macron in regular, but 32 units higher in italic), and horizontally they also carry the offsets of the base anchors, so for X they can confirm a finding but not settle one: the X of a symmetric mark is set by the model (the mark-axis rule). Vertically they are better evidence: every x-height base has its above-anchor at 645, so where the offset of a composed mark from the precomposed glyphs is the same on every base, it belongs to the mark's anchor, and the anchor is moved by it ("Vertical moves from the precomposed glyphs"). In what follows, "lower" and "higher" and "left" and "right" describe where the mark lands on a base whose anchor conforms. A mark lands lower when its anchor is further above its outline, and left when its anchor is further right.

Status. There is no audit of mark anchors. One stage moves designer mark anchors to the model, and only their X: the mark-axis rule for symmetric above marks (stage 4, 2026-11-09; see the end of this section). A wholesale move of every mark anchor to the model was rejected (also at the end of this section). Otherwise mark anchors are changed one at a time, as curated anchors, where the precomposed glyphs support the change (including the vertical moves from Δy), and checked by eye against the mark zone figures. A curated mark anchor is the final anchor and obeys the mark-axis rule itself. The patch changes mark anchors in five places, and one change of base anchors (in stage 1) serves the marks.

- Stage 1, semibold only: the designer's second above-anchor class `top_punkt` (class 1) is merged into the above class. Its 12 marks (dot above, diaeresis, hook above, ring above, U+0312, U+0313, U+0314, U+031A, U+0351, U+0357, U+0363 and `hookabovecomb.cap`) move to class 0, and their anchors are raised by 96 units, the difference between the two classes' base anchors. This does not move these marks on a lowercase letter, but it fixes their coverage: before, they attached only to the 24 bases with a `top_punkt` anchor; now they attach to every base with an above-anchor. It does not change how high these marks sit compared with regular (see below).
- Stage 3, semibold and semibold italic: the combining marks the style lacks are added, and semibold's U+0358–U+0363, which were copies of the capital accents, are replaced. Their anchors are regular's. In semibold italic, regular's anchor is used with italic's outline moved right by 12 units. There, the added above-marks landed 7 to 15 units left of the model ($K = 150$), inside the spread of the designer's own above-marks (k from 111 to 168); the symmetric ones 10 or more off are now moved onto their axes by the mark-axis rule (stage 4).
- Stage 4, all styles: the capital forms (`grave.cap`, `acute.cap`, …) get anchors computed from the designer's precomposed capitals, on the axis for a symmetric mark, at the precomposed offset for an asymmetric one, and with $K = 150$ in semibold italic. This fixes semibold italic, whose capital forms had anchors without the shift K and landed 120 to 160 units left,

and italic, where only `grave.cap` had an anchor (in the wrong class). In italic the `.cap` anchors are curated (2026-11-07), fitted to the precomposed capitals: each X is moved so that the accent lands, as a median over the capitals, where the designer drew it in the precomposed glyph, as in regular, where every capital form is within a few units of them. Stage 4 had left the symmetric marks about 14 units left of the precomposed accents, and the earlier curated caron (placed at its vertex, ignoring the slant) about 15 right. The fit was made after italic O, U and Y took the median accent centre of their own precomposed glyphs as their above-anchor X (493, 508, 441; the designer's O was 515, 22 right, which Ø inherited).

- Stage 4, all styles, the mark-axis rule (2026-11-09): every symmetric above mark whose k is 10 or more from K gets its anchor X on its own axis; 37 anchors move, Y unchanged. Marks with a curated anchor are skipped. See "Mark-axis rule for symmetric above marks" at the end of this section.
- Stage 7, curated mark anchors in `data/fontanchors_human/`:
  - Vertical moves from the precomposed glyphs (2026-11-09): dot above in all four styles, macron in all but italic, semibold tilde, semibold italic ring, and the double acute; each anchor is moved by the median Δy (with X along the slant in the italic styles). See "Vertical moves from the precomposed glyphs" at the end of this section.

  Earlier curated mark anchors:

  - Regular, left and right tack below (U+0318, U+0319): the designer's anchors were 25 units right and 36 units left of the centre of the outline; the curated anchors are at the centre. The other styles keep the designer's X, so the tacks now differ between regular and the other styles.
  - Semibold, half rings above U+0351 and U+0357 (2026-11-08): their anchors were the designer's class 1 (`top_punkt`, Y 547, 549), raised 96 by stage 1, so they sat 156–158 above the mark's bottom against regular's 126–136, and the marks landed 20–32 units low (40–53 above x-height letters, against 67–88 in regular). They now have regular's anchor-to-outline offsets, measured from the outline's centre and bottom: $(90, 623)$ and $(82, 613)$, so they land as in regular. (The X of $+100$ was not itself wrong: semibold draws these two marks right of the origin.) Regular's left half ring lands 16 units right of centre, as it shares the right half ring's anchor; Gentium Plus puts it about 11 left. Left as it is.
  - All styles, seven tall below-marks (2026-11-08): left and right tack U+0318, U+0319, left and right half ring U+031C, U+0339, plus sign U+031F, ring U+0325 and up arrow U+034E. They rose 82 to 94 units above their own anchor, against 33 to 77 for the other below-marks, so at the descender's below-anchor ($Y = -319$, 87 below the descender) each overlapped 1 to 13 of the 26 to 31 Latin and IPA descenders (p, q, g, þ, ȝ, …), and under baseline letters their median gap was 7 to 26 units, against about 40 for marks such as the vertical line below. Each anchor is moved up, X unchanged, so that the mark rises 62 above it, as U+0329 and U+0348 do: the marks sit 20 to 32 units lower under every base. No below-mark now overlaps a descender, and their median gap under baseline letters is 38 to 47. The precomposed Ạ̊ and ạ̊ confirm the ring: their ring is within 2 units of where the combining ring now lands (regular, italic, semibold italic; semibold's precomposed ring is a different outline). The alternative, lowering the descender reference to -342 (the baseline's clearance of 110), also removed the overlaps but left the crowding under baseline letters and moved every mark under descenders. The up tack U+031D (rising 65 to 77) clears every descender, by 4 to 7 units at the closest.
  - Semibold, grave and acute: 30 units lower (713 to 683), which makes them sit as in regular. The grave keeps the designer's $X = -127$; an earlier curated -156 had no recorded reason and put the grave 37 units (median) from the precomposed glyphs, against 10. The grave and acute tone marks U+0340 and U+0341, which stage 3 adds, still have 713, so they sit 30 units lower than the grave and acute.
  - Mark-axis positions, $X = C + Y \tan\theta + d - K$, where the precomposed glyphs confirm the move (median distance of the mark from the precomposed glyphs, before and after; added 2026-11-06): semibold inverted breve U+0311 -217 (11 to 4) and tilde below U+0330 -200 (13 to 4); semibold italic hook above U+0309 -142 (26 to 8), inverted breve U+0311 -223 (19 to 3) and U+0314 -91 (19 to 4, on one glyph). The semibold italic capital forms `acute.cap`, `grave.cap`, `hungarumlaut.cap` and `uni030F.cap` were also curated here (2026-11-06) and removed on 2026-11-08 in favour of stage 4; see the capital alternates. The inverted breves are symmetric marks and obey the mark-axis rule.
  - Italic, `acute.cap`: the curated $(-65, 724)$ had no recorded reason and has been removed, so stage 4's $(-37, 724)$ applies.
  - Italic and semibold italic, tilde below: moved to the centre of the outline. In semibold italic this corrects an offset 22 units different from regular's; in italic it moves the anchor by 8 units, harmlessly.
  - Italic, double acute: the designer's anchor $(-74, 714)$ had regular's offset; the curated $(-57, 714)$ lands the mark 17 units left. Since 2026-11-09 it is $(-55, 725)$, after the vertical move.

– Semibold italic, grave $(0, 683)$ and double acute $(-28, 719)$: these were wrong, and have been removed. The designer's anchors, $(-142, 683)$ and $(-219, 714)$, conform to the model. The curated ones lacked the shift K, so on every conforming base the grave landed 142 units left and the double acute 190 units left (ò: 126 units left of the precomposed grave; ő: 150 units left). They had been adjusted on a, whose above-anchor carried the slant and landed every above-mark about 150 units too far right (see the list above Figure 4). The fix: stage 1 now deletes the above-anchors of a, s, h and J (as `"delete"` entries in `data/fontanchors_audit/overrides.py`), stage 6 replaces them with the candidates, and the designer's grave and double acute are back. The designer's a, s, h and J were 137 to 179 units right of the candidate, and h's anchor was at the x-height instead of the ascender. On a, the above-marks now land 7 to 33 units from where the precomposed glyphs put them (before: 113 to 161 units right); on s within 11; the circumflex of ĥ, which landed 219 units too low, now sits 26 units higher than in the precomposed glyph, and that of ĵ 34 units right and 34 higher.

Remaining departures from the model. Compared with regular, after the patch:

- Regular. The marks conform. The dot above U+0307 sat 45 units lower than the dot of the precomposed ȯ; since 2026-11-09 it sits as in the precomposed glyphs, and so does the macron (18 units lower than before). The dot below U+0323 and diaeresis below U+0324 are drawn 43 units lower, relative to their anchors, than in the other three styles. Regular's precomposed ạ agrees with the combining dot below, but in ụ the diaeresis is 35 units higher. Whether regular or the other styles should change is a design decision.
- Italic. The marks conform. The tilde was 11 units off its axis; the mark-axis rule puts it on its axis. The dot above sat 10 units higher than in regular, still 11 units lower than in the precomposed ȯ; since 2026-11-09 it sits as in the precomposed glyphs. The macron is not moved: its precomposed glyphs disagree (-1 on ā ē ḡ, -32 on ī ō ū ȳ).
- Semibold. Horizontally, the marks are within about 12 units of regular's offsets. Vertically, about a quarter of the above-marks sat lower than in regular: U+0357 32 and U+0351 20 (fixed, see stage 7), double acute 19, diaeresis 14, dot above 13 (ȯ confirms, 21), hook above 13, macron 13 (ō confirms, 12), circumflex 10. The tilde sits 12 units higher (õ confirms, 14), and so does the breve below. Semibold's x-height is 5 units higher than regular's, so differences under 10 do not matter. Since 2026-11-09 the macron, dot above and tilde sit as in the precomposed glyphs (vertical moves of -12, -21, $+14$), and the dieresis is on its axis (the mark-axis rule, 12 units left).
- Semibold italic. Horizontally, the marks whose k is far from 150 are listed above Figure 4. Of these, the hook above, inverted breve and U+0314 have curated anchors on the model, and the dieresis (38 units left) and the tilde (11 right) are moved onto it by the mark-axis rule; the comma above, double grave, koronis and right tack below remain, the precomposed glyphs not supporting a move of the first two. Vertically, the designer drew some outlines at a different height from regular's but kept regular's anchors: the macron sits 32 units lower than in regular (ō confirms, 32), the macron below 52 units higher (n̲ confirms, 26), the dot above 21 units higher (which agrees with the precomposed ȯ), the up tack below U+031D 12 units lower and the down tack below U+031E 12 units higher. Since 2026-11-09 the macron sits as in the precomposed glyphs (32 units higher than before), the dot above 11 units lower (it was 11 above the precomposed ċ ė ġ …), the ring 23 lower and the double acute 11 lower.

Changes needed.

- Semibold italic: move the anchors of the marks whose k is far from 150 to the model (done for the hook above, inverted breve and U+0314, and by the mark-axis rule for the dieresis and tilde; the precomposed glyphs do not support moving U+0313 or U+030F); move the macron's anchor 32 units down (done, see below). The macron below's anchor was moved 52 units up, to $(-233, -58)$, 23 units below its outline as in the other styles (curated, 2026-11-06): it sat 25 units under n, against 86 to 96 in the other styles.
- Semibold: lower the anchors of the above-marks that sit low by the amounts listed, so that they sit as in regular (done for U+0351 and U+0357, and against the precomposed glyphs for the macron and dot above, and for the tilde, which sat high; see below). The double acute, dieresis, hook above and circumflex are within 10 of their precomposed glyphs vertically, which do not support moving them; give U+0340 and U+0341 the curated height of the grave and acute (683).
- Regular: lower the anchor of the dot above by about 45 units (and by about 11 in italic and 21 in semibold) to match the precomposed glyphs (done, see below); decide between regular and the other styles for the dot below and diaeresis below, and for the tacks below.
- Italic: the curated double acute keeps its X; its Y is now set from the precomposed glyphs (below).

Vertical moves from the precomposed glyphs (2026-11-09). For each above mark, Δy is the median vertical offset of the composed mark (base + CGJ + mark) from the designer's precomposed glyphs, over the lowercase letters with an x-height base. Every such base has its above-anchor at 645, and Δy is the same on every base to within a few units, so it belongs to the mark's anchor. Where $|\Delta y| \geq 10$ the mark anchor is curated (stage 7) Δy higher, so that the mark sits Δy lower; in the italic styles its X moves with the slant, by $\Delta y \tan\theta$, so that the mark stays on its axis and keeps its k. Regular: macron $+18$, dot above -45, double acute $+11$. Italic: dot above -11, double acute $+11$. Semibold: tilde $+14$, macron -12, dot above -21. Semibold italic: macron -32, dot above $+11$, ring $+23$, double acute $+11$. Not moved: the italic macron, whose precomposed glyphs disagree (-1 on ā ē ḡ, -32 on ī ō ū ȳ). A curated mark anchor is the final anchor: it obeys the mark-axis rule itself, and stage 4 skips it (and reports it if it does not).

Rejected: moving every mark anchor to the model (2026-11-08). A stage that moved every mark anchor onto the model, $X = C + Y\tan\theta + d - K$, was tested and rejected: most moves were under 20 units, and as many marks got further from the precomposed glyphs as got closer. Since the precomposed glyphs are used wherever they exist, a composed combination should match them; mark anchors are therefore changed one at a time, as curated anchors, not by an audit of mark anchors.

Mark-axis rule for symmetric above marks (2026-11-09). The exception is one rule, applied in stage 4 (`font_patching_markaxis.py`): a symmetric above mark has its anchor X on its own axis, $k = K$ (0; 150 in semibold italic), wherever $|k - K| \geq 10$, with k measured on the mark's ink. The symmetric above marks are the mirror-symmetric ones (circumflex, tilde, macron, breve, dot above, dieresis, ring, caron, vertical line and double vertical line above, candrabindu, inverted breve, x above, vertical tilde, double overline, perispomeni, bridge above, not tilde, homothetic, almost equal to, fermata, zigzag) and the combining Latin letters U+0363–U+036F, centred on their ink as base letters are. Smaller deviations are within the designer's spread and are left. Matching each mark to the precomposed glyphs ($\Delta x = 0$ per mark) would be the wrong target: what a composed mark differs from the precomposed glyphs by is partly the base anchor's offset, common to every mark on that base, and fitting it into each mark would carry that offset to every base without a precomposed form (most IPA letters). Under the rule what remains is common to the marks of a style (semibold italic about $+12$ to $+16$), a question for the base anchors. The rule moves 37 mark anchors. Among the common marks: italic tilde (k 11, 11 units left), semibold dieresis (12, 12 left), semibold italic tilde (139, 11 right) and dieresis (112, 38 right); among the others, the vertical tilde (about 18, in regular, italic and semibold), the perispomeni (18, in regular and italic) and most of semibold italic's combining letters (k 130–140, 10–20 right). It does not repeat the rejected move above: it moves only symmetric marks, only their X, and only where they are 10 or more off. The semibold italic dieresis, which landed 26 units left of its precomposed glyphs where every other mark landed 5–16 right, now lands 12 right, like the others; the semibold italic tilde moves from 12 to 23 right of its precomposed glyphs, the rule's one cost. Asymmetric marks keep their designed k: grave, acute and their tone-mark forms, double acute and double grave (a symmetric pair placed off-centre), hook above, the comma-shaped marks, the half rings and the right arrowhead. The rule does not touch Y. What the threshold leaves is under 10 units (the ring: $+2$, $+3$, 0, $+7$); the threshold is likely to be lowered. That first needs a more precise mark axis than the centre of the ink bounds, which does not always find the midline, determined per glyph and checked by eye: for instance the line about which the outline best matches its mirror image (circumflex, ring, dieresis), or its rotation by 180° for marks that are symmetric that way (tilde, zigzag, vertical tilde), whose terminals can push the bounds off centre.

Mark outline zones

Figures 5, 6, 7 and 8 overlay the outlines of all the above and below marks of the patched fonts on a solid o, at three times the scale of the anchor model figures. Each mark is placed so that its anchor is on the anchor of the o for its class (the blue dots), which is where the shaper puts it. The common marks are black outlines. The other marks are gray with a light fill, so the shading builds up where many marks overlap. The grid is in font units of the o, with lines every 100 units. In semibold italic the above marks lie well to the right of the blue dot, because both the base and the mark anchor carry the shift K.

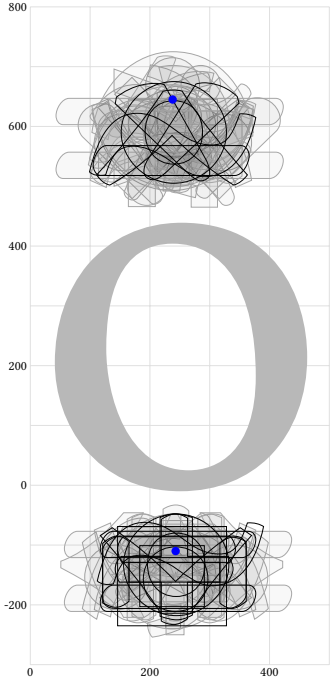


Figure 5: Mark zones, regular.

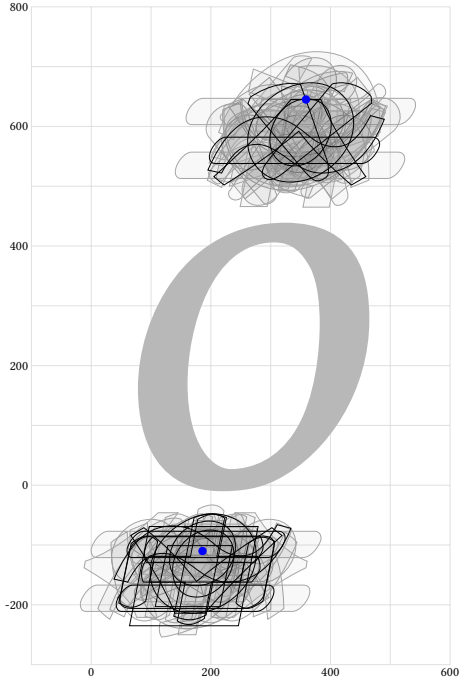


Figure 6: Mark zones, italic.

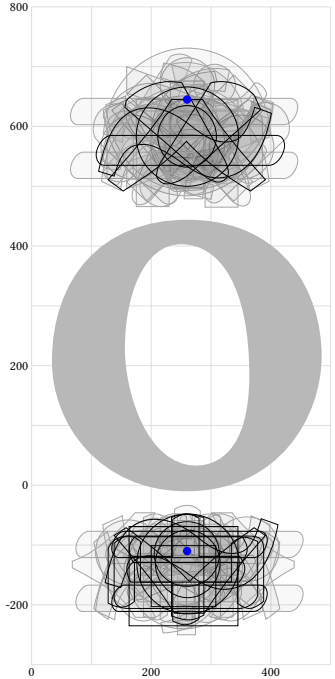


Figure 7: Mark zones, semibold.

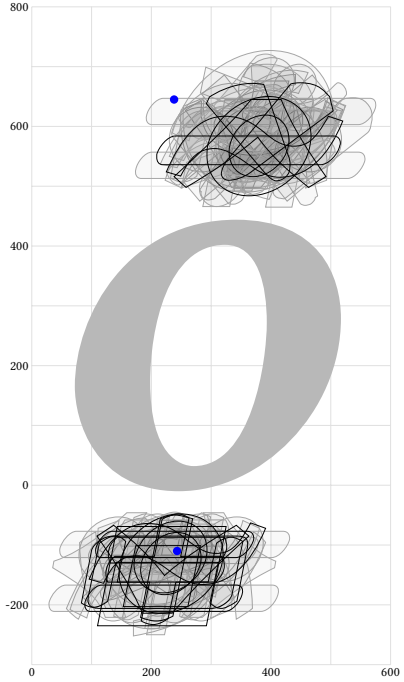


Figure 8: Mark zones, semibold italic.

Font metrics per glyph

See above- and below-anchor coordinates for A–Z, a–z, and IPA letters for Libertinus serif fonts (regular, italic, semibold, semibold italic) in the following tables (Tables 1–11), and the mark anchors of above- and below-marks (Tables 12–15). Recall that Language Science Press uses semibold Libertinus fonts, e.g. `LibertinusSerif-Semibold.otf`, for boldface styles.

Table row headings

hex: Unicode character codepoint in 4-hex.

cat: Unicode character category. L = letter. Lu = uppercase (capital) letter, Ll = lowercase (small) letter, Lo = other letter. Used to determine expected ideal bounding box. M = mark. Mn = nonspacing mark.

reg, it, sb, si: regular, italic, semibold, semibold italic. In that row, the character is printed alone, with combining dot above, and with combining macron below, to show anchor positioning (via combining grapheme joiner) when the base has an anchor; otherwise HarfBuzz default positioning applies.

patch: Use the patched Libertinus font to print the character alone, with combining dot above, and with combining macron below to show anchor positioning of the patched font (via combining grapheme joiner). Where the original font has no above- or below-anchor, the patched font has the candidate (a?, b?), unless a human-curated anchor overrules it; existing anchors are not moved.

ideal: top and bottom ideal bounding box reference lines, with offset from $Y_{\max}$ or $Y_{\min}$ for non-full descender or non-full ascender. c = capline, x = x-height, a = ascender, b = baseline, d = descender.

a: X, Y coordinates of the above-anchor.

a?: X, Y coordinates of the above-anchor candidate.

aδ: X, Y coordinate differences between above-anchor and above-anchor candidate

b: X, Y coordinates of the below-anchor.

b?: X, Y coordinates of the below-anchor candidate.

bδ: X, Y coordinate differences between below-anchor and below-anchor candidate

In the tables of marks, each column is a mark and the rows are as follows.

reg, it, sb, si, patch: the mark on n and on H (via combining grapheme joiner), in the original and patched fonts.

m: X, Y coordinates of the mark anchor. A class in parentheses is an anchor class other than above (0) or below (2): semibold puts the dot above, diaeresis, ring, hook above and comma above in class 1, whose base anchors are lower (n has 550 in class 1 and 646 in class 0). This is the designer's anchor class `top_punkt`; it places the marks at the same height as class 0, but only 24 bases have it. The patched semibold merges it into class 0.

m?: X, Y coordinates of the mark-anchor candidate. Y is the bottom of the mark outline plus the style's median distance of an above-mark anchor above it (regular 143, italic 142, semibold 157 in class 0 and 60 in class 1, semibold italic 147), or the top of the outline minus the median distance of a below-mark anchor below it (regular 46, italic 48, semibold 54, semibold italic 54). X is on the mark's own axis at that height, $C + Y \tan\theta - K$.

mδ: X, Y coordinate differences between mark anchor and mark-anchor candidate. An asymmetric mark, such as the acute or grave, has an X difference by design.

Table 1: Font metrics for A–Z 1/2

hex	0041	0042	0043	0044	0045	0046	0047	0048	0049	004A	004B	004C	004D
cat	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu
reg	AȦẠ	BḂḄ	CĊÇ	DḊḌ	EĖẸ	FḞF	GĠĢ	HḢḤ	IİỊ	JJ̇J	KḴḲ	LĿḶ	MṀṂ
patch	AȦẠ	BḂḄ	CĊÇ	DḊḌ	EĖẸ	FḞF	GĠĢ	HḢḤ	IİỊ	JJ̇J	KḴḲ	LĿḶ	MṀṂ
ideal	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+60	c, b	c, b	c, b
a	354, 850	267, 850	340, 850	306, 850	281, 850	284, 850	365, 850	361, 850	150, 850	182, 850	336, 850	151, 850	431, 850
a?	346, 850	280, 850	324, 850	336, 850	271, 850	244, 850	352, 850	362, 850	148, 850	120, 850	323, 850	260, 850	418, 850
aδ	8, 0	-14, 0	16, 0	-30, 0	10, 0	40, 0	14, 0	-1, 0	2, 0	62, 0	13, 0	-110, 0	13, 0
b	308, -110	268, -110	363, -110	309, -110	287, -110	209, -110	367, -110	362, -110	147, -110	109, -319	342, -110	262, -110	393, -110
b?	346, -110	280, -110	324, -110	336, -110	271, -110	244, -110	352, -110	362, -110	148, -110	120, -257	323, -110	260, -110	418, -110
bδ	-38, 0	-12, 0	39, 0	-27, 0	16, 0	-36, 0	16, 0	0, 0	-1, 0	-11, -62	19, 0	2, 0	-25, 0
it	*AȦẠ*	*BḂḄ*	*CĊÇ*	*DḊḌ*	*EĖẸ*	*FḞF*	*GĠĢ*	*HḢḤ*	*IİỊ*	*JJ̇J*	*KḴḲ*	*LĿḶ*	*MṀṂ*
patch	*AȦẠ*	*BḂḄ*	*CĊÇ*	*DḊḌ*	*EĖẸ*	*FḞF*	*GĠĢ*	*HḢḤ*	*IİỊ*	*JJ̇J*	*KḴḲ*	*LĿḶ*	*MṀṂ*
ideal	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c+32, d+32	c, b	c, b	c, b
a	505, 850	447, 850	500, 850	396, 850	413, 850	406, 850	513, 850	481, 850	316, 850	392, 850	466, 850	389, 850	557, 850
a?	514, 850	459, 850	489, 850	514, 850	444, 850	409, 850	513, 850	517, 850	321, 850	388, 850	499, 850	440, 850	583, 850
aδ	-9, 0	-12, 0	11, 0	-118, 0	-31, 0	-3, 0	0, 0	-36, 0	-5, 0	4, 0	-33, 0	-51, 0	-26, 0
b	308, -108	238, -108	333, -110	279, -107	269, -107	131, -111	345, -110	320, -107	127, -109	109, -319	292, -110	252, -110	363, -110
b?	310, -110	255, -110	285, -110	310, -110	240, -110	205, -110	309, -110	313, -110	117, -110	146, -285	295, -110	236, -110	379, -110
bδ	-2, 2	-17, 2	48, 0	-31, 3	29, 3	-74, -1	36, 0	7, 3	10, 1	-37, -34	-3, 0	16, 0	-16, 0
sb	**AȦẠ**	**BḂḄ**	**CĊÇ**	**DḊḌ**	**EĖẸ**	**FḞF**	**GĠĢ**	**HḢḤ**	**IİỊ**	**JJ̇J**	**KḴḲ**	**LĿḶ**	**MṀṂ**
patch	**AȦẠ**	**BḂḄ**	**CĊÇ**	**DḊḌ**	**EĖẸ**	**FḞF**	**GĠĢ**	**HḢḤ**	**IİỊ**	**JJ̇J**	**KḴḲ**	**LĿḶ**	**MṀṂ**
ideal	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+50	c, b	c, b	c, b
a										203, 806			
a?	350, 850	300, 850	332, 850	352, 850	283, 850	268, 850	368, 850	362, 850	160, 850	131, 850	340, 850	267, 850	428, 850
aδ										72, -44			
b								362, -110		109, -319			
b?	350, -110	300, -110	332, -110	352, -110	283, -110	268, -110	368, -110	362, -110	160, -110	131, -267	340, -110	267, -110	428, -110
bδ								0, 0		-22, -52			
si	***AȦẠ***	***BḂḄ***	***CĊÇ***	***DḊḌ***	***EĖẸ***	***FḞF***	***GĠĢ***	***HḢḤ***	***IİỊ***	***JJ̇J***	***KḴḲ***	***LĿḶ***	***MṀṂ***
patch	***AȦẠ***	***BḂḄ***	***CĊÇ***	***DḊḌ***	***EĖẸ***	***FḞF***	***GĠĢ***	***HḢḤ***	***IİỊ***	***JJ̇J***	***KḴḲ***	***LĿḶ***	***MṀṂ***
ideal	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c+34, d+68	c, b	c, b	c, b
a	354, 850	267, 850	340, 850	306, 850	273, 850	276, 850	365, 850	361, 850	150, 850	432, 850	336, 850	149, 850	431, 850
a?	382, 850	319, 850	341, 850	367, 850	292, 850	269, 850	371, 850	396, 850	194, 850	249, 850	369, 850	310, 850	441, 850
aδ	-28, 0	-52, 0	-1, 0	-61, 0	-19, 0	7, 0	-6, 0	-35, 0	-44, 0	183, 0	-33, 0	-161, 0	-10, 0
b	308, -110	268, -110	363, -110	279, -110	279, -110	201, -110	367, -110	365, -110	147, -110	109, -319	342, -110	260, -110	393, -110
b?	349, -110	286, -110	308, -110	334, -110	259, -110	236, -110	338, -110	363, -110	161, -110	187, -249	336, -110	277, -110	408, -110
bδ	-41, 0	-18, 0	55, 0	-55, 0	20, 0	-35, 0	29, 0	2, 0	-14, 0	-78, -70	6, 0	-17, 0	-15, 0

Table 2: Font metrics for A–Z 2/2

hex	004E	004F	0050	0051	0052	0053	0054	0055	0056	0057	0058	0059	005A
cat	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu
reg	NN̈N̲	OÖO̲	PP̈P̲	QQ̈Q̲	RR̈R̲	SS̈S̲	TT̈T̲	UU̇Ụ	VV̇V̲	WẄW̲	XẌX̲	YŸY̲	ZŻZ̲
patch	NN̈N̲	OÖO̲	PP̈P̲	QQ̈Q̲	RR̈R̲	SS̈S̲	TT̈T̲	UU̇Ụ	VV̇V̲	WẄW̲	XẌX̲	YŸY̲	ZŻZ̲
ideal	c, b	c, b	c, b	c, d+23	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b
a	346, 850	335, 850	277, 850	348, 850	269, 850	232, 850	307, 850	346, 850	336, 850	450, 850	332, 850	295, 850	320, 850
a?	353, 850	352, 850	260, 850	359, 850	301, 850	236, 850	297, 850	331, 850	326, 850	474, 850	323, 850	288, 850	310, 850
aδ	-7, 0	-16, 0	18, 0	-11, 0	-32, 0	-4, 0	10, 0	15, 0	10, 0	-24, 0	9, 0	8, 0	10, 0
b	348, -110	350, -110	240, -110		258, -110	234, -110	300, -110	328, -110	318, -110	479, -110	307, -110	295, -110	287, -110
b?	353, -110	352, -110	260, -110	359, -294	301, -110	236, -110	297, -110	331, -110	326, -110	474, -110	323, -110	288, -110	310, -110
bδ	-5, 0	-2, 0	-20, 0		-43, 0	-2, 0	3, 0	-3, 0	-8, 0	5, 0	-16, 0	8, 0	-22, 0
it	*NN̈N̲*	*OÖO̲*	*PP̈P̲*	*QQ̈Q̲*	*RR̈R̲*	*SS̈S̲*	*TT̈T̲*	*UU̇Ụ*	*VV̇V̲*	*WẄW̲*	*XẌX̲*	*YŸY̲*	*ZŻZ̲*
patch	*NN̈N̲*	*OÖO̲*	*PP̈P̲*	*QQ̈Q̲*	*RR̈R̲*	*SS̈S̲*	*TT̈T̲*	*UU̇Ụ*	*VV̇V̲*	*WẄW̲*	*XẌX̲*	*YŸY̲*	*ZŻZ̲*
ideal	c, b	c, b	c, b	c, d+23	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b
a	476, 850	515, 850	426, 850	498, 850	419, 850	402, 850	457, 850	486, 850	481, 850	610, 850	482, 850	435, 850	450, 850
a?	514, 850	515, 850	430, 850	515, 850	458, 850	408, 850	453, 850	498, 850	479, 850	635, 850	495, 850	457, 850	470, 850
aδ	-38, 0	0, 0	-4, 0	-17, 0	-39, 0	-6, 0	4, 0	-12, 0	2, 0	-25, 0	-13, 0	-22, 0	-20, 0
b	328, -109	310, -113	140, -110		248, -111	204, -111	277, -109	308, -109	263, -110	439, -107	287, -107	270, -109	267, -107
b?	310, -110	311, -110	226, -110	272, -294	254, -110	204, -110	249, -110	294, -110	275, -110	431, -110	291, -110	253, -110	266, -110
bδ	18, 1	-1, -3	-86, 0		-6, -1	0, -1	28, 1	14, 1	-12, 0	8, 3	-4, 3	17, 1	1, 3
sb	**NNN̲**	**OOO̲**	**PPP̲**	**QQ̈Q̲**	**RR̈R̲**	**SS̈S̲**	**TT̈T̲**	**UUU̲**	**VV̇V̲**	**WẄW̲**	**XẌX̲**	**YŸY̲**	**ZŻZ̲**
patch	**NN̈N̲**	**OÖO̲**	**PP̈P̲**	**QQ̈Q̲**	**RR̈R̲**	**SS̈S̲**	**TT̈T̲**	**UU̇Ụ**	**VV̇V̲**	**WẄW̲**	**XẌX̲**	**YŸY̲**	**ZŻZ̲**
ideal	c, b	c, b	c, b	c, d	c, b	c, b	c+28, b	c, b	c, b	c, b	c, b	c, b	c, b
a		363, 802		363, 802	304, 808	252, 802	298, 802		331, 805	493, 805	364, 804	316, 805	321, 803
a?	354, 850	366, 850	278, 850	366, 850	320, 850	238, 850	299, 850	339, 850	318, 850	489, 850	354, 850	308, 850	309, 850
aδ		-2, -48		-2, -48	-16, -42	14, -48	-1, -48		12, -45	4, -45	10, -46	8, -45	12, -47
b					292, -111	234, -111	301, -107		303, -108	470, -107	318, -107	316, -109	308, -107
b?	354, -110	366, -110	278, -110	366, -319	320, -110	238, -110	299, -110	339, -110	318, -110	489, -110	354, -110	308, -110	309, -110
bδ					-28, -1	-4, -1	2, 3		-16, 2	-19, 3	-36, 3	8, 1	-1, 3
si	***NN̈N̲***	***OÖO̲***	***PP̈P̲***	***QQ̈Q̲***	***RR̈R̲***	***SS̈S̲***	***TT̈T̲***	***UU̇Ụ***	***VV̇V̲***	***WẄW̲***	***XẌX̲***	***YŸY̲***	***ZŻZ̲***
patch	***NN̈N̲***	***OÖO̲***	***PP̈P̲***	***QQ̈Q̲***	***RR̈R̲***	***SS̈S̲***	***TT̈T̲***	***UU̇Ụ***	***VV̇V̲***	***WẄW̲***	***XẌX̲***	***YŸY̲***	***ZŻZ̲***
ideal	c, b	c, b	c, b	c, d	c, b	c, b	c+28, b	c, b	c, b	c, b	c, b	c, b	c, b
a	346, 850	335, 850	316, 850	348, 850	269, 850	232, 850	307, 850	346, 850	336, 850	450, 850	332, 850	295, 850	320, 850
a?	394, 850	390, 850	284, 850	390, 850	315, 850	250, 850	309, 850	364, 850	328, 850	501, 850	362, 850	319, 850	322, 850
aδ	-48, 0	-55, 0	32, 0	-42, 0	-46, 0	-18, 0	-2, 0	-18, 0	8, 0	-51, 0	-30, 0	-24, 0	-2, 0
b	348, -110	350, -110	240, -110		258, -110	234, -110	300, -110	328, -110	318, -110	479, -110	307, -110	295, -110	287, -110
b?	361, -110	357, -110	251, -110	314, -319	282, -110	217, -110	276, -110	331, -110	295, -110	468, -110	329, -110	286, -110	289, -110
bδ	-13, 0	-7, 0	-11, 0		-24, 0	17, 0	24, 0	-3, 0	23, 0	11, 0	-22, 0	9, 0	-2, 0

Table 3: Font metrics for a–z 1/2

hex	0061	0062	0063	0064	0065	0066	0067	0068	0069	006A	006B	006C	006D
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	aȧạ	bḃḅ	cċc̣	dḋḍ	eėẹ	fḟf̣	gġg̣	hḣḥ	iiį	jjj̣	kk̇ḳ	l̇l̇ḷ	ṁṁṃ
patch	aȧạ	bḃḅ	cċc̣	dḋḍ	eėẹ	fḟf̣	gġg̣	hḣḥ	iiį	jjj̣	kk̇ḳ	l̇l̇ḷ	ṁṁṃ
ideal	x, b	a, b	x, b	a, b	x, b	a, b	x+33, d	a, b	a-51, b	a-51, d+9	a, b	a, b	x, b
a	209, 645		242, 645	218, 645	243, 645		231, 645	299, 645	137, 795		306, 784	135, 885	392, 645
a?	246, 645	232, 885	218, 645	270, 885	222, 645	208, 885	256, 676	270, 885	142, 834	76, 834	261, 885	134, 885	402, 645
aδ	-36, 0		24, 0	-52, -240	21, 0		-26, -31	28, -240	-5, -39		45, -101	1, 0	-10, 0
b	196, -110	234, -110	236, -110	221, -110	234, -110	132, -110	229, -319	272, -110	136, -110	109, -319	290, -214	135, -110	408, -110
b?	246, -110	232, -110	218, -110	270, -110	222, -110	208, -110	256, -319	270, -110	142, -110	76, -308	261, -110	134, -110	402, -110
bδ	-50, 0	2, 0	18, 0	-50, 0	12, 0	-76, 0	-28, 0	2, 0	-6, 0	34, -11	29, -104	1, 0	6, 0
it	*aȧạ*	*bḃḅ*	*cċc̣*	*dḋḍ*	*eėẹ*	*fḟf̣*	*gġg̣*	*hḣḥ*	*iiį*	*jjj̣*	*kk̇ḳ*	*l̇l̇ḷ*	*ṁṁṃ*
patch	*aȧạ*	*bḃḅ*	*cċc̣*	*dḋḍ*	*eėẹ*	*fḟf̣*	*gġg̣*	*hḣḥ*	*iiį*	*jjj̣*	*kk̇ḳ*	*l̇l̇ḷ*	*ṁṁṃ*
ideal	x, b	a, b	x, b	a, b	x, b	a, d	x, d	a, b	a-80, b	a-80, d+36	a, b	a, b	x, b
a	341, 647		343, 645	218, 645	332, 645		360, 645	409, 645			405, 646	292, 890	522, 645
a?	380, 645	428, 890	332, 645	434, 890	338, 645	346, 890	376, 645	449, 890	310, 810	302, 810	432, 890	322, 890	529, 645
aδ	-39, 2		11, 0	-216, -245	-6, 0		-16, 0	-40, -245			-27, -244	-30, 0	-7, 0
b	174, -108	214, -108	163, -113	221, -107	158, -108	222, -111	197, -319	229, -111	136, -107	0, -319	223, -107	115, -107	382, -105
b?	220, -110	216, -110	171, -110	221, -110	177, -110	89, -319	171, -319	236, -110	115, -110	70, -281	220, -110	110, -110	368, -110
bδ	-46, 2	-2, 2	-8, -3	0, 3	-19, 2	133, 208	26, 0	-7, -1	21, 3	-70, -38	3, 3	5, 3	14, 5
sb	**aȧạ**	**bḃḅ**	**cċc̣**	**dḋḍ**	**eėẹ**	**fḟf̣**	**gġg̣**	**hḣḥ**	**iiį**	**jjj̣**	**kk̇ḳ**	**l̇l̇ḷ**	**ṁṁṃ**
patch	**aȧạ**	**bḃḅ**	**cċc̣**	**dḋḍ**	**eėẹ**	**fḟf̣**	**gġg̣**	**hḣḥ**	**iiį**	**jjj̣**	**kk̇ḳ**	**l̇l̇ḷ**	**ṁṁṃ**
ideal	x, b	a, b-13	x, b	a, b	x, b	a, b	x+33, d	a, b	a-48, b	a-48, d	a, b	a, b	x, b
a							231, 645		137, 795			142, 885	396, 645
a?	255, 645	236, 885	224, 645	274, 885	232, 645	226, 885	264, 676	275, 885	156, 837	78, 837	262, 885	148, 885	408, 645
aδ							-32, -31		-18, -42			-6, 0	-12, 0
b							229, -319	277, -110	136, -107	109, -319		135, -110	396, -105
b?	255, -110	236, -121	224, -110	274, -110	232, -110	226, -110	264, -319	275, -110	156, -110	78, -319	262, -110	148, -110	408, -110
bδ							-34, 0	2, 0	-20, 3	31, 0		-14, 0	-12, 5
si	***aȧạ***	***bḃḅ***	***cċc̣***	***dḋḍ***	***eėẹ***	***fḟf̣***	***gġg̣***	***hḣḥ***	***iiį***	***jjj̣***	***kk̇ḳ***	***l̇l̇ḷ***	***ṁṁṃ***
patch	***aȧạ***	***bḃḅ***	***cċc̣***	***dḋḍ***	***eėẹ***	***fḟf̣***	***gġg̣***	***hḣḥ***	***iiį***	***jjj̣***	***kk̇ḳ***	***l̇l̇ḷ***	***ṁṁṃ***
ideal	x, b	a, b	x, b	a, b	x, b	a, d	x, d	a, b	a-67, b	a-67, d+25	a, b	a, b	x, b
a	379, 645		242, 645	218, 645	243, 645		291, 645	469, 645	137, 795		306, 784	174, 890	392, 645
a?	251, 645	290, 890	203, 645	309, 890	218, 645	216, 890	220, 645	331, 890	180, 823	192, 823	316, 890	194, 890	407, 645
aδ	128, 0		39, 0	-91, -245	25, 0		71, 0	138, -245	-43, -28		-10, -106	-20, 0	-15, 0
b	196, -110	234, -110	236, -110	221, -110	234, -110	132, -110	199, -319	272, -110	136, -110	67, -319	290, -214	135, -110	400, -110
b?	260, -110	248, -110	212, -110	268, -110	227, -110	132, -319	186, -319	290, -110	153, -110	127, -292	275, -110	153, -110	415, -110
bδ	-64, 0	-14, 0	24, 0	-47, 0	7, 0	0, 209	13, 0	-18, 0	-17, 0	-60, -27	15, -104	-18, 0	-15, 0

Table 4: Font metrics for a–z 2/2

hex	006E	006F	0070	0071	0072	0073	0074	0075	0076	0077	0078	0079	007A
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	nn̰	oo̰	pp̰	qq̰	rr̰	ss̰	tt̰	uṵ	vv̰	ww̰	xx̰	yy̰	zz̰
patch	nn̰	oo̰	pp̰	qq̰	rr̰	ss̰	tt̰	uṵ	vv̰	ww̰	xx̰	yy̰	zz̰
ideal	x, b	x, b	x, d	x, d	x, b	x, b	a-116, b	x, b	x, b	x, b	x, b	x, d	x, b
a	264, 646	238, 645	271, 646	238, 645	191, 646	206, 645		255, 645	256, 646	345, 646	256, 646	262, 645	220, 646
a?	276, 645	252, 645	242, 645	268, 645	190, 645	199, 645	163, 769	271, 645	249, 645	373, 645	245, 645	260, 645	220, 645
aδ	-12, 1	-14, 0	28, 1	-30, 0	1, 1	7, 0		-16, 0	7, 1	-28, 1	11, 1	2, 0	0, 1
b	265, -110	243, -110	179, -319	348, -319	135, -110	196, -110	163, -110	239, -110	241, -110	352, -110	224, -110	242, -319	207, -110
b?	276, -110	252, -110	242, -319	268, -319	190, -110	199, -110	163, -110	271, -110	249, -110	373, -110	245, -110	260, -319	220, -110
bδ	-10, 0	-9, 0	-64, 0	80, 0	-55, 0	-3, 0	0, 0	-32, 0	-8, 0	-21, 0	-21, 0	-18, 0	-14, 0
it	*nn̰*	*oo̰*	*pp̰*	*qq̰*	*rr̰*	*ss̰*	*tt̰*	*uṵ*	*vv̰*	*ww̰*	*xx̰*	*yy̰*	*zz̰*
patch	*nn̰*	*oo̰*	*pp̰*	*qq̰*	*rr̰*	*ss̰*	*tt̰*	*uṵ*	*vv̰*	*ww̰*	*xx̰*	*yy̰*	*zz̰*
ideal	x, b	x, b	x+62, d	x, d	x, b	x, b	x+132, b	x, b	x, b	x, b	x, b	x, d	x, b
a	394, 646	359, 645	411, 646	369, 645	312, 646	327, 645		386, 646		348, 646	366, 646	367, 644	360, 646
a?	396, 645	361, 645	394, 705	383, 645	316, 645	314, 645	318, 775	398, 645	373, 645	481, 645	375, 645	389, 645	355, 645
aδ	-2, 1	-2, 0	17, -59	-14, 0	-4, 1	13, 0		-12, 1		-133, 1	-9, 1	-22, -1	5, 1
b	235, -111	186, -107	248, -110	169, -110	113, -110	169, -111	133, -107	226, -111		355, -110	224, -110	368, -107	187, -111
b?	236, -110	200, -110	177, -319	178, -319	155, -110	153, -110	130, -110	237, -110	213, -110	321, -110	214, -110	184, -319	195, -110
bδ	-1, -1	-14, 3	71, 209	-9, 209	-42, 0	16, -1	3, 3	-11, -1		34, 0	10, 0	184, 212	-8, -1
sb	**nn̰**	**oo̰**	**pp̰**	**qq̰**	**rr̰**	**ss̰**	**tt̰**	**uṵ**	**vv̰**	**ww̰**	**xx̰**	**yy̰**	**zz̰**
patch	**nn̰**	**oo̰**	**pp̰**	**qq̰**	**rr̰**	**ss̰**	**tt̰**	**uṵ**	**vv̰**	**ww̰**	**xx̰**	**yy̰**	**zz̰**
ideal	x, b	x, b	x, d	x, d	x, b	x, b	a-103, b	x, b	x, b	x, b	x, b	x, d	x, b
a	264, 646							264, 646					
a?	290, 645	260, 645	254, 645	273, 645	212, 645	201, 645	168, 782	276, 645	262, 645	384, 645	260, 645	264, 645	222, 645
aδ	-26, 1							-12, 1					
b	265, -111						163, -107	248, -111					
b?	290, -110	260, -110	254, -319	273, -319	212, -110	201, -110	168, -110	276, -110	262, -110	384, -110	260, -110	264, -319	222, -110
bδ	-24, -1						-5, 3	-28, -1					
si	***nn̰***	***oo̰***	***pp̰***	***qq̰***	***rr̰***	***ss̰***	***tt̰***	***uṵ***	***vv̰***	***ww̰***	***xx̰***	***yy̰***	***zz̰***
patch	***nn̰***	***oo̰***	***pp̰***	***qq̰***	***rr̰***	***ss̰***	***tt̰***	***uṵ***	***vv̰***	***ww̰***	***xx̰***	***yy̰***	***zz̰***
ideal	x, b	x, b	x+36, d	x, d	x, b	x, b	a-125, b	x, b	x, b	x, b	x, b	x, d	x, b
a	264, 646	238, 645	271, 646	238, 645	191, 646	326, 645		255, 645	256, 646	345, 646		262, 645	220, 646
a?	284, 645	235, 645	257, 679	267, 645	203, 645	190, 645	189, 765	262, 645	220, 645	332, 645	240, 645	236, 645	210, 645
aδ	-20, 1	3, 0	14, -33	-29, 0	-12, 1	136, 0		-7, 0	36, 1	13, 1		26, 0	10, 1
b	265, -110	243, -110	179, -319	348, -319	135, -110	196, -110	133, -107	239, -110	241, -110	352, -110		242, -319	207, -110
b?	293, -110	243, -110	216, -319	233, -319	212, -110	198, -110	173, -110	270, -110	228, -110	341, -110	248, -110	202, -319	218, -110
bδ	-28, 0	0, 0	-37, 0	115, 0	-77, 0	-2, 0	-40, 3	-31, 0	13, 0	11, 0		40, 0	-11, 0

Table 5: Font metrics for Latin and IPA 1/7

hex	00C6	00D0	00D8	00DE	00DF	00E6	00F0	00F8	00FE	0110	0111	0126	0127	0131	0141	0142	014B	0152
cat	Lu	Lu	Lu	Lu	Ll	Ll	Ll	Ll	Ll	Lu	Ll	Lu	Ll	Ll	Lu	Ll	Ll	Lu
reg	Æ Ȧ Ӕ	Ð Ḋ Ḏ	Ø Ȯ Ǿ	Þ Ṗ Ꝑ	ß ß̊ ß	æ ȧ ӕ	ð ð̊ ð	ø ȯ ø	þ ṗ ꝑ	Đ Ḋ Ḏ	đ ḋ ḍ	Ħ Ḣ Ḥ	ħ ḣ ḥ	ı i̊ ı	Ł Ł̇ Ł	ł l̊ ł	ŋ ŋ̇ ŋ	Œ Ė ɶ
patch	Æ Ȧ Ӕ	Ð Ḋ Ḏ	Ø Ȯ Ǿ	Þ Ṗ Ꝑ	ß ß̊ ß	æ ȧ ӕ	ð ð̊ ð	ø ȯ ø	þ ṗ ꝑ	Đ Ḋ Ḏ	đ ḋ ḍ	Ħ Ḣ Ḥ	ħ ḣ ḥ	ı i̊ ı	Ł Ł̇ Ł	ł l̊ ł	ŋ ŋ̇ ŋ	Œ Ė ɶ
ideal	c, b	c, b	c, b-13	c, b	a, b	x, b	a, b	x+25, b-28	a, d	c, b	a, b	c, b	a, b	x, b	c, b	a, b	x, d+9	c, b
a	480, 850	341, 850	351, 850	281, 850		350, 645			278, 646					132, 645			264, 646	460, 850
a?	426, 850	333, 850	352, 850	250, 850	259, 885	344, 645	240, 885	250, 668	236, 885	336, 850	270, 885	362, 850	270, 885	142, 645	256, 850	133, 885	240, 645	436, 850
aδ	54, 0	8, 0	0, 0	32, 0		6, 0			42, -239					-10, 0			24, 1	24, 0
b	549, -107		350, -110			327, -109		243, -110						139, -108			265, -111	
b?	426, -110	333, -110	352, -121	250, -110	259, -110	344, -110	240, -110	250, -136	236, -319	336, -110	270, -110	362, -110	270, -110	142, -110	256, -110	133, -110	240, -308	436, -110
bδ	122, 3		-2, 11			-16, 1		-8, 26						-3, 2			25, 197	
it	*Æ Ȧ Ӕ*	*Ð Ḋ Ḏ*	*Ø Ȯ Ǿ*	*Þ Ṗ Ꝑ*	*ß ß̊ ß*	*æ ȧ ӕ*	*ð ð̊ ð*	*ø ȯ ø*	*þ ṗ ꝑ*	*Đ Ḋ Ḏ*	*đ ḋ ḍ*	*Ħ Ḣ Ḥ*	*ħ ḣ ḥ*	*ı i̊ ı*	*Ł Ł̇ Ł*	*ł l̊ ł*	*ŋ ŋ̇ ŋ*	*Œ Ė ɶ*
patch	*Æ Ȧ Ӕ*	*Ð Ḋ Ḏ*	*Ø Ȯ Ǿ*	*Þ Ṗ Ꝑ*	*ß ß̊ ß*	*æ ȧ ӕ*	*ð ð̊ ð*	*ø ȯ ø*	*þ ṗ ꝑ*	*Đ Ḋ Ḏ*	*đ ḋ ḍ*	*Ħ Ḣ Ḥ*	*ħ ḣ ḥ*	*ı i̊ ı*	*Ł Ł̇ Ł*	*ł l̊ ł*	*ŋ ŋ̇ ŋ*	*Œ Ė ɶ*
ideal	c, b	c, b	c+25, b-43	c, b	a, d	x, b	a+6, b	x+48, b-48	a, d-7	c, b	a, b	c, b	a, b	x, b	c, b	a, b	x, d	c, b
a	615, 850					480, 645			278, 646					267, 645			264, 646	575, 850
a?	595, 850	514, 850	515, 850	431, 850	461, 890	481, 645	429, 896	370, 691	439, 890	514, 850	434, 890	517, 850	449, 890	275, 645	440, 850	322, 890	382, 645	587, 850
aδ	20, 0					-1, 0			-161, -244					-8, 0			-118, 1	-12, 0
b	549, -107		310, -113			327, -109		186, -107						139, -108			265, -111	
b?	391, -110	310, -110	302, -151	227, -110	204, -319	320, -110	216, -110	190, -156	181, -324	310, -110	221, -110	313, -110	236, -110	115, -110	236, -110	110, -110	177, -319	383, -110
bδ	158, 3		8, 38			7, 1		-4, 49						24, 2			88, 208	
sb	**Æ Ȧ Ӕ**	**Ð Ḋ Ḏ**	**Ø Ȯ Ǿ**	**Þ Ṗ Ꝑ**	**ß ß̊ ß**	**æ ȧ ӕ**	**ð ð̊ ð**	**ø ȯ ø**	**þ ṗ ꝑ**	**Đ Ḋ Ḏ**	**đ ḋ ḍ**	**Ħ Ḣ Ḥ**	**ħ ḣ ḥ**	**ı i̊ ı**	**Ł Ł̇ Ł**	**ł l̊ ł**	**ŋ ŋ̇ ŋ**	**Œ Ė ɶ**
patch	**Æ Ȧ Ӕ**	**Ð Ḋ Ḏ**	**Ø Ȯ Ǿ**	**Þ Ṗ Ꝑ**	**ß ß̊ ß**	**æ ȧ ӕ**	**ð ð̊ ð**	**ø ȯ ø**	**þ ṗ ꝑ**	**Đ Ḋ Ḏ**	**đ ḋ ḍ**	**Ħ Ḣ Ḥ**	**ħ ḣ ḥ**	**ı i̊ ı**	**Ł Ł̇ Ł**	**ł l̊ ł**	**ŋ ŋ̇ ŋ**	**Œ Ė ɶ**
ideal	c, b	c, b	c+25, b-45	c, b	a, b	x, b	a+19, b	x+41, b-34	a, d	c, b	a, b	c, b	a, b	x, b	c, b	a, b	x, d+36	c, b
a	510, 850					355, 645								151, 645				450, 850
a?	446, 850	349, 850	366, 850	287, 850	284, 885	362, 645	256, 904	260, 684	260, 885	349, 850	274, 885	362, 850	275, 885	156, 645	258, 850	142, 885	248, 645	432, 850
aδ	64, 0					-8, 0								-4, 0				18, 0
b														135, -108				
b?	446, -110	349, -110	366, -153	287, -110	284, -110	362, -110	256, -110	260, -142	260, -319	349, -110	274, -110	362, -110	275, -110	156, -110	258, -110	142, -110	248, -281	432, -110
bδ														-20, 2				
si	***Æ Ȧ Ӕ***	***Ð Ḋ Ḏ***	***Ø Ȯ Ǿ***	***Þ Ṗ Ꝑ***	***ß ß̊ ß***	***æ ȧ ӕ***	***ð ð̊ ð***	***ø ȯ ø***	***þ ṗ ꝑ***	***Đ Ḋ Ḏ***	***đ ḋ ḍ***	***Ħ Ḣ Ḥ***	***ħ ḣ ḥ***	***ı i̊ ı***	***Ł Ł̇ Ł***	***ł l̊ ł***	***ŋ ŋ̇ ŋ***	***Œ Ė ɶ***
patch	***Æ Ȧ Ӕ***	***Ð Ḋ Ḏ***	***Ø Ȯ Ǿ***	***Þ Ṗ Ꝑ***	***ß ß̊ ß***	***æ ȧ ӕ***	***ð ð̊ ð***	***ø ȯ ø***	***þ ṗ ꝑ***	***Đ Ḋ Ḏ***	***đ ḋ ḍ***	***Ħ Ḣ Ḥ***	***ħ ḣ ḥ***	***ı i̊ ı***	***Ł Ł̇ Ł***	***ł l̊ ł***	***ŋ ŋ̇ ŋ***	***Œ Ė ɶ***
ideal	c, b	c, b	c+25, b-40	c, b	a, d	x, b	a+19, b	x, b-30	a, d	c, b	a, b	c, b	a, b	x, b	c, b	a, b	x, d+16	c, b
a	500, 850	341, 850	351, 850	281, 850		350, 645			278, 646					132, 645			264, 646	480, 850
a?	459, 850	367, 850	390, 850	303, 850	325, 890	339, 645	291, 909	235, 645	300, 890	367, 850	309, 890	396, 850	331, 890	144, 645	300, 850	190, 890	253, 645	450, 850
aδ	41, 0	-26, 0	-39, 0	-22, 0		11, 0			-22, -244					-12, 0			11, 1	30, 0
b	549, -107		350, -110			327, -109		243, -110						139, -108			265, -111	
b?	426, -110	334, -110	349, -148	270, -110	241, -319	347, -110	246, -110	237, -138	216, -319	334, -110	268, -110	363, -110	290, -110	153, -110	267, -110	148, -110	222, -301	417, -110
bδ	123, 3		1, 38			-20, 1		6, 28						-14, 2			43, 190	

Table 6: Font metrics for Latin and IPA 2/7

hex	0153	017F	0181	0182	0186	0187	0189	018A	018B	018E	018F	0190	0191	0192	0193	0194	0196	0197
cat	Ll	Ll	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Ll	Lu	Lu	Lu	Lu
reg	œɶɶ̧	ſſ̱	Ɓ̇Ɓ̱	ƁƁ̱	ƆƆ̱	ƇƇ̱	ƉƉ̱	ƊƊ̇Ɗ̱	ƋƋ̇	ƎƎ̱	ƏƏ̱	ƐƐ̱	ƑƑ̱	ƒƒ̱	ƓƓ̱	ƔƔ̱	ƖƖ̱	ƗƗ̱
patch	œɶɶ̧	ſẛ̱	Ɓ̇Ɓ̱	ƁƁ̱	ƆƆ̱	ƇƇ̱	ƉƉ̱	ƊƊ̇Ɗ̱	ƋƋ̇	ƎƎ̇	ƏƏ̱	ƐƐ̱	ƑƑ̱	ƒƒ̱	ƓƓ̱	ƔƔ̱	ƖƖ̱	ƗƗ̱
ideal	x, b	a, b	c, b	c, b	c, b	c+98, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+60	a, d	c+98, b	c, d	a, b	c, b
a	405, 645		327, 805					306, 805								331, 805		
a?	388, 645	190, 885	303, 850	274, 850	322, 850	369, 850	333, 850	368, 850	292, 850	280, 850	326, 850	256, 850	183, 850	201, 885	372, 850	308, 850	152, 885	148, 850
aδ	18, 0		24, -45					-62, -45								22, -45		
b			328, -108					369, -107										
b?	388, -110	190, -110	303, -110	274, -110	322, -110	369, -110	333, -110	368, -110	292, -110	280, -110	326, -110	256, -110	183, -257	201, -319	372, -110	308, -319	152, -110	148, -110
bδ			25, 2					0, 3										
it	*œɶɶ̧*	*ſſ̱*	*Ɓ̇Ɓ̱*	*ƁƁ̱*	*ƆƆ̱*	*ƇƇ̱*	*ƉƉ̱*	*ƊƊ̇Ɗ̱*	*ƋƋ̇*	*ƎƎ̱*	*ƏƏ̱*	*ƐƐ̱*	*ƑƑ̱*	*ƒƒ̱*	*ƓƓ̱*	*ƔƔ̱*	*ƖƖ̱*	*ƗƗ̱*
patch	*œɶɶ̧*	*ſẛ̱*	*Ɓ̇Ɓ̱*	*ƁƁ̱*	*ƆƆ̱*	*ƇƇ̱*	*ƉƉ̱*	*ƊƊ̇Ɗ̱*	*ƋƋ̇*	*ƎƎ̇*	*ƏƏ̱*	*ƐƐ̱*	*ƑƑ̱*	*ƒƒ̱̇*	*ƓƓ̱*	*ƔƔ̱*	*ƖƖ̱*	*ƗƗ̱*
ideal	x, b	a, d	c, b	c, b	c+14, b	c+98, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+60	a, d+60	c+98, b	c, d+20	a, b	c, b
a	480, 645		327, 805					306, 805								331, 805		
a?	490, 645	339, 890	488, 850	465, 850	503, 850	503, 850	534, 850	553, 850	450, 850	465, 850	515, 850	439, 850	419, 850	414, 890	541, 850	462, 850	322, 890	321, 850
aδ	-10, 0		-161, -45					-247, -45								-131, -45		
b			328, -108					369, -107										
b?	329, -110	82, -319	284, -110	261, -110	299, -110	299, -110	330, -110	349, -110	246, -110	261, -110	311, -110	235, -110	184, -257	170, -257	337, -110	218, -297	110, -110	117, -110
bδ			44, 2					20, 3										
sb	**œɶɶ̧**	**ſſ̱**	**Ɓ̇Ɓ̱**	**ƁƁ̱**	**ƆƆ̱**	**ƇƇ̱**	**ƉƉ̱**	**ƊƊ̇Ɗ̱**	**ƋƋ̇**	**ƎƎ̱**	**ƏƏ̱**	**ƐƐ̱**	**ƑƑ̱**	**ƒƒ̱**	**ƓƓ̱**	**ƔƔ̱**	**ƖƖ̱**	**ƗƗ̱**
patch	**œɶɶ̧**	**ſẛ̱**	**Ɓ̇Ɓ̱**	**ƁƁ̱**	**ƆƆ̱**	**ƇƇ̱**	**ƉƉ̱**	**ƊƊ̇Ɗ̱**	**ƋƋ̇**	**ƎƎ̇**	**ƏƏ̱**	**ƐƐ̱**	**ƑƑ̱**	**ƒƒ̱̇**	**ƓƓ̱**	**ƔƔ̱**	**ƖƖ̱**	**ƗƗ̱**
ideal	x, b	a, b	c, b	c, b	c, b	c+98, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+36	a, d+62	c+98, b	c, d+22	a, b	c, b
a	430, 645				275, 833													
a?	402, 645	217, 885	316, 850	268, 850	314, 850	368, 850	349, 850	381, 850	292, 850	302, 850	353, 850	269, 850	236, 850	221, 885	374, 850	294, 850	170, 885	158, 850
aδ	28, 0				-38, -17													
b																		
b?	402, -110	217, -110	316, -110	268, -110	314, -110	368, -110	349, -110	381, -110	292, -110	302, -110	353, -110	269, -110	236, -281	221, -255	374, -110	294, -295	170, -110	158, -110
bδ																		
si	***œɶɶ̧***	***ſſ̱***	***Ɓ̇Ɓ̱***	***ƁƁ̱***	***ƆƆ̱***	***ƇƇ̱***	***ƉƉ̱***	***ƊƊ̇Ɗ̱***	***ƋƋ̇***	***ƎƎ̱***	***ƏƏ̱***	***ƐƐ̱***	***ƑƑ̱***	***ƒƒ̱***	***ƓƓ̱***	***ƔƔ̱***	***ƖƖ̱***	***ƗƗ̱***
patch	***œɶɶ̧***	***ſẛ̱***	***Ɓ̇Ɓ̱***	***ƁƁ̱***	***ƆƆ̱***	***ƇƇ̱***	***ƉƉ̱***	***ƊƊ̇Ɗ̱***	***ƋƋ̇***	***ƎƎ̇***	***ƏƏ̱***	***ƐƐ̱***	***ƑƑ̱***	***ƒƒ̱̇***	***ƓƓ̱***	***ƔƔ̱***	***ƖƖ̱***	***ƗƗ̱***
ideal	x, b	a+13, d-9	c, b	c, b	c, b	c+98, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+36	a, d+62	c+98, b	c, d+22	a, b	c, b
a	415, 645		327, 805					306, 805								331, 805		
a?	384, 645	209, 903	359, 850	303, 850	357, 850	359, 850	388, 850	424, 850	304, 850	320, 850	369, 850	293, 850	288, 850	269, 890	395, 850	316, 850	194, 890	181, 850
aδ	31, 0		-32, -45					-118, -45								15, -45		
b			328, -108					369, -107										
b?	393, -110	121, -326	326, -110	270, -110	324, -110	326, -110	355, -110	391, -110	271, -110	287, -110	336, -110	260, -110	220, -281	198, -255	362, -110	245, -295	153, -110	148, -110
bδ			2, 2					-22, 3										

Table 7: Font metrics for Latin and IPA 3/7

hex	0198	019C	019D	019F	01B7	0237	0250	0251	0252	0253	0254	0255	0256	0257	0258	0259	025A	025B
cat	Lu	Lu	Lu	Lu	Lu	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	KKK̦	ШШШ̦	ƝƝƝ̦	ƟƟƟ̦	ȝȝȝ	jjj̦	ɐɐɐ̦	ɑɑɑ̦	ɒɒɒ̦	ɓɓɓ̦	ɔɔɔ̦	ɕɕɕ̦	ɖɖɖ̦	ɗɗɗ̦	ɘɘɘ̦	əəə̦	ɚɚɚ̦	ɛɛɛ̦
patch	KḰK̦	ШШ́Ш̦	ƝƝ́Ɲ̦	ƟƟ́Ɵ̦	ȝȝ́ȝ̦	jjj̦	ɐɐɐ̦	ɑɑɑ̦	ɒɒɒ̦	ɓɓɓ̦	ɔɔɔ̦	ɕɕɕ̦	ɖɖɖ̦	ɗɗɗ̦	ɘɘɘ̦	əəə̦	ɚɚɚ̦	ɛɛɛ̦
ideal	c, b	c, b	c, d+45	c, b	c+22, d+21	x, d+9	x, b	x, b	x, b	a, b	x, b	x, b-106	a, d+44	a, b	x, b	x, b	x+45, b	x, b
a		431, 801				140, 645	228, 645	228, 645	264, 645	480, 630	216, 645	209, 645	218, 645	218, 645	204, 645	195, 645	229, 645	209, 645
a?	339, 850	434, 850	324, 850	352, 850	303, 850	74, 645	216, 645	272, 645	247, 645	265, 885	216, 645	222, 645	314, 885	322, 885	220, 645	232, 645	306, 688	202, 645
aδ		-2, -49				66, 0	12, 0	-44, 0	17, 0	215, -255	0, 0	-13, 0	-96, -240	-104, -240	-16, 0	-37, 0	-78, -43	8, 0
b		450, -108		350, -110	272, -319		229, -114	212, -114	259, -114	212, -114	177, -114	212, -114	212, -114	212, -114	212, -114	186, -114	212, -114	212, -114
b?	339, -110	434, -110	324, -272	352, -110	303, -296	74, -308	216, -110	272, -110	247, -110	265, -110	216, -110	222, -214	314, -273	322, -110	220, -110	232, -110	306, -110	202, -110
bδ		16, 2		-2, 0	-31, -23		12, -4	-60, -4	12, -4	-53, -4	-38, -4	-10, 100	-102, 159	-110, -4	-8, -4	-46, -4	-94, -4	10, -4
it	*KKK̦*	*ШШШ̦*	*ƝƝƝ̦*	*ƟƟƟ̦*	*ȝȝȝ*	*jjj̦*	*ɐɐɐ̦*	*ɑɑɑ̦*	*ɒɒɒ̦*	*ɓɓɓ̦*	*ɔɔɔ̦*	*ɕɕɕ̦*	*ɖɖɖ̦*	*ɗɗɗ̦*	*ɘɘɘ̦*	*əəə̦*	*ɚɚɚ̦*	*ɛɛɛ̦*
patch	*KḰK̦*	*ШШ́Ш̦*	*ƝƝ́Ɲ̦*	*ƟƟ́Ɵ̦*	*ȝȝ́ȝ*	*jjj̦*	*ɐɐɐ̦*	*ɑɑɑ̦*	*ɒɒɒ̦*	*ɓɓɓ̦*	*ɔɔɔ̦*	*ɕɕɕ̦*	*ɖɖɖ̦*	*ɗɗɗ̦*	*ɘɘɘ̦*	*əəə̦*	*ɚɚɚ̦*	*ɛɛɛ̦*
ideal	c, b	c, b	c, d+45	c, b	c+22, d+21	x, d+36	x, b	x, b	x, b	a+6, b	x, b	x, b-100	a+9, d+44	a, b	x, b	x, b	x+44, b	x, b
a		431, 801				140, 645					322, 654		218, 645	218, 645	204, 645			322, 654
a?	499, 850	609, 850	528, 850	515, 850	485, 850	267, 645	365, 645	380, 645	391, 645	431, 896	332, 645	352, 645	465, 899	464, 890	356, 645	356, 645	418, 687	338, 645
aδ		-178, -49				-127, 0					-10, 9		-247, -254	-246, -245	-152, 0			-16, 9
b		450, -108					229, -114	212, -114	243, -114	212, -114	172, -114	212, -114	212, -114	212, -114	212, -114	212, -114	212, -114	212, -114
b?	295, -110	405, -110	290, -272	311, -110	241, -296	70, -281	205, -110	220, -110	231, -110	218, -110	171, -110	171, -208	215, -273	251, -110	195, -110	195, -110	248, -110	178, -110
bδ		45, 2					24, -4	-8, -4	12, -4	-6, -4	1, -4	41, 94	-3, 159	-39, -4	17, -4	17, -4	-36, -4	34, -4
sb	**KKK̦**	**ШШШ̦**	**ƝƝƝ̦**	**ƟƟƟ̦**	**ȝȝȝ**	**jjj̦**	**ɐɐɐ̦**	**ɑɑɑ̦**	**ɒɒɒ̦**	**ɓɓɓ̦**	**ɔɔɔ̦**	**ɕɕɕ̦**	**ɖɖɖ̦**	**ɗɗɗ̦**	**ɘɘɘ̦**	**əəə̦**	**ɚɚɚ̦**	**ɛɛɛ̦**
patch	**KḰK̦**	**ШШ́Ш̦**	**ƝƝ́Ɲ̦**	**ƟƟ́Ɵ̦**	**ȝȝ́ȝ**	**jjj̦**	**ɐɐɐ̦**	**ɑɑɑ̦**	**ɒɒɒ̦**	**ɓɓɓ̦**	**ɔɔɔ̦**	**ɕɕɕ̦**	**ɖɖɖ̦**	**ɗɗɗ̦**	**ɘɘɘ̦**	**əəə̦**	**ɚɚɚ̦**	**ɛɛɛ̦**
ideal	c, b	c, b	c, d+47	c, b	c+29, d+24	x, d	x, b-14	x, b	x, b	a, b	x, b-14	x, b-101	a, d+46	a, b	x, b	x, b	x+50, b	x, b
a						163, 645					199, 645							229, 645
a?	338, 850	444, 850	336, 850	366, 850	316, 850	78, 645	207, 645	276, 645	251, 645	268, 885	199, 645	232, 645	322, 885	324, 885	230, 645	228, 645	319, 693	212, 645
aδ						85, 0					0, 0							16, 0
b					272, -319		228, -114				184, -114					212, -114		
b?	338, -110	444, -110	336, -270	366, -110	316, -293	78, -319	207, -122	276, -110	251, -110	268, -110	199, -122	232, -209	322, -271	324, -110	230, -110	228, -110	319, -110	212, -110
bδ					-44, -26		21, 8				-15, 8					-16, -4		
si	***KKK̦***	***ШШШ̦***	***ƝƝƝ̦***	***ƟƟƟ̦***	***ȝȝȝ***	***jjj̦***	***ɐɐɐ̦***	***ɑɑɑ̦***	***ɒɒɒ̦***	***ɓɓɓ̦***	***ɔɔɔ̦***	***ɕɕɕ̦***	***ɖɖɖ̦***	***ɗɗɗ̦***	***ɘɘɘ̦***	***əəə̦***	***ɚɚɚ̦***	***ɛɛɛ̦***
patch	***KḰK̦***	***ШШ́Ш̦***	***ƝƝ́Ɲ̦***	***ƟƟ́Ɵ̦***	***ȝȝ́ȝ***	***jjj̦***	***ɐɐɐ̦***	***ɑɑɑ̦***	***ɒɒɒ̦***	***ɓɓɓ̦***	***ɔɔɔ̦***	***ɕɕɕ̦***	***ɖɖɖ̦***	***ɗɗɗ̦***	***ɘɘɘ̦***	***əəə̦***	***ɚɚɚ̦***	***ɛɛɛ̦***
ideal	c, b	c, b	c, d+47	c, b	c+28, d+24	x, d+25	x, b	x, b	x, b	a, b	x, b	x, b-101	a, d+46	a, b	x, b	x, b	x+50, b	x, b
a		431, 801				140, 645		228, 645	264, 645		160, 645		218, 645	218, 645	204, 645			209, 645
a?	353, 850	463, 850	383, 850	387, 850	329, 850	121, 645	221, 645	247, 645	247, 645	284, 890	199, 645	204, 645	302, 890	318, 890	207, 645	218, 645	299, 693	194, 645
aδ		-32, -49				19, 0		-19, 0	17, 0		-39, 0		-84, -245	-100, -245	-3, 0			15, 0
b		450, -108			236, -319		229, -114	212, -114	259, -114	212, -114	177, -114	212, -114	212, -114	212, -114	212, -114		212, -114	212, -114
b?	320, -110	430, -110	317, -270	354, -110	258, -293	93, -292	230, -110	256, -110	256, -110	243, -110	207, -110	192, -209	227, -271	276, -110	216, -110	227, -110	298, -110	203, -110
bδ		20, 2			-22, -26		-1, -4	-44, -4	3, -4	-31, -4	-30, -4	20, 95	-15, 157	-64, -4	-4, -4		-86, -4	9, -4

Table 8: Font metrics for Latin and IPA 4/7

hex	025C	025D	025E	025F	0260	0261	0262	0263	0264	0265	0266	0267	0268	0269	026A	026B	026C	026D
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	ɜ̡	ɝ̡	ɞ̡	ɟ̡	ɠ̡	g̡	ɢ̡	ɣ̡	ɤ̡	ɥ̡	ɦ̡	ɧ̡	ɨ̡	ɩ̡	ɪ̡	ᵻ̡	ᶖ̡	ᶗ̡
patch	ɜ̡	ɝ̡	ɞ̡	ɟ̡	ɠ̡	g̡	ɢ̡	ɣ̡	ɤ̡	ɥ̡	ɦ̡	ɧ̡	ɨ̡	ɩ̡	ɪ̡	ᵻ̡	ᶖ̡	ᶗ̡
ideal	x, b	a-90, b	x, b	x, d+26	a, d	x, d	x, b	x, d+30	x, b	x, d	a, b	a, d+11	a-51, b	x, b	x, b	a, b	a, b	a, d+44
a	194, 645	209, 645	239, 645	169, 645	209, 645	239, 645	265, 645	259, 645	248, 645	268, 645	299, 565	299, 565	155, 795	123, 645	133, 645	209, 645	209, 645	209, 645
a?	196, 645	234, 795	242, 645	124, 645	320, 885	228, 645	259, 645	258, 645	221, 645	272, 645	262, 885	228, 885	150, 834	170, 645	136, 645	144, 885	182, 885	182, 885
aδ	-2, 0	-26, -150	-2, 0	44, 0	-112, -240	11, 0	6, 0	1, 0	27, 0	-4, 0	36, -320	72, -320	6, -39	-47, 0	-3, 0	65, -240	27, -240	27, -240
b	184, -114	212, -114	256, -114	196, -110	236, -110	226, -320	248, -110	251, -277	216, -114	219, -129	240, -114	240, -114	134, -114	153, -114	139, -114	146, -114	218, -120	196, -110
b?	196, -110	234, -110	242, -110	124, -291	320, -319	228, -319	259, -110	258, -287	221, -110	272, -319	262, -110	228, -306	150, -110	170, -110	136, -110	144, -110	182, -110	182, -273
bδ	-12, -4	-22, -4	14, -4	72, 181	-84, 209	-2, -1	-11, 0	-7, 10	-5, -4	-52, 190	-22, -4	12, 192	-16, -4	-17, -4	3, -4	2, -4	36, -10	14, 163
it	*ɜ̡*	*ɝ̡*	*ɞ̡*	*ɟ̡*	*ɠ̡*	*g̡*	*ɢ̡*	*ɣ̡*	*ɤ̡*	*ɥ̡*	*ɦ̡*	*ɧ̡*	*ɨ̡*	*ɩ̡*	*ɪ̡*	*ᵻ̡*	*ᶖ̡*	*ᶗ̡*
patch	*ɜ̡*	*ɝ̡*	*ɞ̡*	*ɟ̡*	*ɠ̡*	*g̡*	*ɢ̡*	*ɣ̡*	*ɤ̡*	*ɥ̡*	*ɦ̡*	*ɧ̡*	*ɨ̡*	*ɩ̡*	*ɪ̡*	*ᵻ̡*	*ᶖ̡*	*ᶗ̡*
ideal	x, b	x+132, b	x, b-25	x, d	a+6, d+31	x, d+31	x, b	x, d+45	x, b+2	x, d	a, b	a, d	a-80, b	x, b	x, b	a, b	a, b	a, d+30
a							265, 645			258, 645	299, 565			123, 645	128, 645			
a?	331, 645	413, 775	375, 645	305, 645	431, 896	378, 645	388, 645	387, 645	338, 645	396, 645	449, 890	449, 890	316, 810	266, 645	281, 645	333, 890	368, 890	341, 890
aδ							-123, 0			-138, 0	-150, -325			-143, 0	-153, 0			
b	184, -114	212, -114	256, -114				248, -114		177, -114	219, -129	240, -114		134, -114	153, -114	139, -114	146, -114	218, -120	
b?	170, -110	225, -110	209, -133	100, -319	180, -286	180, -286	227, -110	192, -272	178, -106	191, -319	236, -110	192, -319	121, -110	106, -110	120, -110	120, -110	156, -110	91, -287
bδ	14, -4	-13, -4	47, 19				21, -4		-1, -8	28, 190	4, -4		13, -4	47, -4	19, -4	26, -4	62, -10	
sb	**ɜ̡**	**ɝ̡**	**ɞ̡**	**ɟ̡**	**ɠ̡**	**g̡**	**ɢ̡**	**ɣ̡**	**ɤ̡**	**ɥ̡**	**ɦ̡**	**ɧ̡**	**ɨ̡**	**ɩ̡**	**ɪ̡**	**ᵻ̡**	**ᶖ̡**	**ᶗ̡**
patch	**ɜ̡**	**ɝ̡**	**ɞ̡**	**ɟ̡**	**ɠ̡**	**g̡**	**ɢ̡**	**ɣ̡**	**ɤ̡**	**ɥ̡**	**ɦ̡**	**ɧ̡**	**ɨ̡**	**ɩ̡**	**ɪ̡**	**ᵻ̡**	**ᶖ̡**	**ᶗ̡**
ideal	x, b	a-128, b	x, b	x, d	a+6, d+33	x, d	x, b	x, d+32	x, b	x, d	a, b	a, d	a-67, b	x, b	x, b	a, b	a, b	a, d
a															146, 645			
a?	216, 645	276, 757	249, 645	144, 645	336, 891	235, 645	266, 645	272, 645	202, 645	278, 645	274, 885	243, 885	152, 818	164, 645	152, 645	156, 885	190, 885	194, 885
aδ															-6, 0			
b													134, -114		156, -114	146, -114	218, -120	
b?	216, -110	276, -110	249, -110	144, -319	336, -284	235, -319	266, -110	272, -285	202, -110	278, -319	274, -110	243, -319	152, -110	164, -110	152, -110	156, -110	190, -110	194, -319
bδ													-18, -4		4, -4	-10, -4	28, -10	
si	***ɜ̡***	***ɝ̡***	***ɞ̡***	***ɟ̡***	***ɠ̡***	***g̡***	***ɢ̡***	***ɣ̡***	***ɤ̡***	***ɥ̡***	***ɦ̡***	***ɧ̡***	***ɨ̡***	***ɩ̡***	***ɪ̡***	***ᵻ̡***	***ᶖ̡***	***ᶗ̡***
patch	***ɜ̡***	***ɝ̡***	***ɞ̡***	***ɟ̡***	***ɠ̡***	***g̡***	***ɢ̡***	***ɣ̡***	***ɤ̡***	***ɥ̡***	***ɦ̡***	***ɧ̡***	***ɨ̡***	***ɩ̡***	***ɪ̡***	***ᵻ̡***	***ᶖ̡***	***ᶗ̡***
ideal	x, b	a-128, b	x, b	x, d	a, d+33	x, d	x, b	x, d+32	x, b	x, d	a, b	a, d	a-67, b	x, b	x, b	a, b	a, b	a, d
a	194, 645						265, 645			258, 645	299, 565	299, 565	137, 795	123, 645	128, 645			
a?	194, 645	261, 762	232, 645	141, 645	309, 890	238, 645	244, 645	255, 645	185, 645	258, 645	302, 890	301, 890	180, 823	138, 645	137, 645	187, 890	223, 890	187, 890
aδ	0, 0						21, 0			0, 0	-3, -325	-2, -325	-43, -28	-15, 0	-9, 0			
b	184, -114	212, -114	256, -114				248, -110		177, -114	219, -129	240, -114	240, -114	136, -110	153, -114	139, -114	146, -114	218, -120	
b?	203, -110	246, -110	241, -110	107, -319	232, -284	204, -319	252, -110	228, -285	194, -110	224, -319	260, -110	217, -319	153, -110	147, -110	145, -110	145, -110	181, -110	103, -319
bδ	-19, -4	-34, -4	15, -4				-4, 0		-17, -4	-5, 190	-20, -4	23, 205	-17, 0	6, -4	-6, -4	1, -4	37, -10	

Table 9: Font metrics for Latin and IPA 5/7

hex	026E	026F	0270	0271	0272	0273	0274	0275	0276	0277	0278	0279	027A	027B	027C	027D	027E	027F
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	ɮɮ	ɯɯ	ɰɰ	ɱɱ	ɲɲ	ɳɳ	ɴɴ	ɵɵ	ɶɶ	ɷɷ	ɸɸ	ɹɹ	ɺɺ	ɻɻ	ɼɼ	ɽɽ	ɾɾ	ɿɿ
patch	ɮ̇ɮ	ɯ̇ɯ	ɰ̇ɰ	ɱ̇ɱ	ɲ̇ɲ	ɳ̇ɳ	ɴ̇ɴ	ɵ̇ɵ	ɶ̇ɶ	ɷ̇ɷ	ɸ̇ɸ	ɹ̇ɹ	ɺ̇ɺ	ɻ̇ɻ	ɼ̇ɼ	ɽ̇ɽ	ɾ̇ɾ	ɿ̇ɿ
ideal	a, d	x, b-13	x, d	x, d+11	x, d+11	x, d+11	x, b	x, b	x, b	x, b	a, d	x, b-13	a-51, b-13	x, d+58	x, d	x, d	x, b	x, b
a	209, 645	407, 645	209, 645	383, 645	264, 646	263, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645		222, 703	209, 645	209, 645	209, 645
a?	250, 885	390, 645	379, 645	366, 645	234, 645	296, 645	278, 645	252, 645	340, 645	357, 645	286, 885	178, 645	180, 834	226, 645	186, 645	186, 645	174, 645	170, 645
aδ	-42, -240	16, 0	-170, 0	18, 0	30, 1	-32, 0	-70, 0	-43, 0	-132, 0	-148, 0	-76, -240	32, 0	30, -189		36, 58	22, 0	34, 0	38, 0
b	196, -110	410, -114	294, -114	383, -105	265, -111	262, -112	247, -114	196, -110	196, -110	341, -114	196, -110	190, -114	203, -114	196, -110	127, -305	196, -110	196, -110	196, -110
b?	250, -319	390, -121	379, -319	366, -306	234, -306	296, -306	278, -110	252, -110	340, -110	357, -110	286, -319	178, -121	180, -121	226, -259	186, -319	186, -319	174, -110	170, -110
bδ	-54, 209	20, 7	-85, 205	18, 201	30, 195	-34, 194	-32, -4	-56, 0	-144, 0	-16, -4	-90, 209	12, 7	24, 7	-30, 149	-60, 14	10, 209	22, 0	26, 0
it	*ɮɮ*	*ɯɯ*	*ɰɰ*	*ɱɱ*	*ɲɲ*	*ɳɳ*	*ɴɴ*	*ɵɵ*	*ɶɶ*	*ɷɷ*	*ɸɸ*	*ɹɹ*	*ɺɺ*	*ɻɻ*	*ɼɼ*	*ɽɽ*	*ɾɾ*	*ɿɿ*
patch	*ɮ̇ɮ*	*ɯ̇ɯ*	*ɰ̇ɰ*	*ɱ̇ɱ*	*ɲ̇ɲ*	*ɳ̇ɳ*	*ɴ̇ɴ*	*ɵ̇ɵ*	*ɶ̇ɶ*	*ɷ̇ɷ*	*ɸ̇ɸ*	*ɹ̇ɹ*	*ɺ̇ɺ*	*ɻ̇ɻ*	*ɼ̇ɼ*	*ɽ̇ɽ*	*ɾ̇ɾ*	*ɿ̇ɿ*
ideal	a, d+8	x, b-15	x, d	x, d	x, d	x, d	x, b	x, b	x, b	x, b-18	a, d+32	x, b	a-71, b-19	x, d+58	x, d+34	x, d+38	x, b	x, b
a		407, 645		383, 645	264, 646	263, 645									222, 703			
a?	447, 890	517, 645	515, 645	515, 645	396, 645	367, 645	418, 645	379, 645	457, 645	494, 645	481, 890	315, 645	362, 819	352, 645	325, 645	320, 645	311, 645	307, 645
aδ		-110, 0		-132, 0	-132, 1	-104, 0									-103, 58			
b		410, -114	294, -114	383, -105	265, -111	262, -112	247, -114			341, -114		190, -114	203, -114		127, -305			
b?	192, -309	354, -123	310, -319	310, -319	191, -319	162, -319	258, -110	218, -110	296, -110	330, -126	231, -285	155, -110	161, -127	160, -259	128, -283	124, -279	151, -110	147, -110
bδ		56, 9	-16, 205	73, 214	74, 208	100, 207	-11, -4			11, 12		35, -4	42, 13		-1, -22			
sb	**ɮɮ**	**ɯɯ**	**ɰɰ**	**ɱɱ**	**ɲɲ**	**ɳɳ**	**ɴɴ**	**ɵɵ**	**ɶɶ**	**ɷɷ**	**ɸɸ**	**ɹɹ**	**ɺɺ**	**ɻɻ**	**ɼɼ**	**ɽɽ**	**ɾɾ**	**ɿɿ**
patch	**ɮ̇ɮ**	**ɯ̇ɯ**	**ɰ̇ɰ**	**ɱ̇ɱ**	**ɲ̇ɲ**	**ɳ̇ɳ**	**ɴ̇ɴ**	**ɵ̇ɵ**	**ɶ̇ɶ**	**ɷ̇ɷ**	**ɸ̇ɸ**	**ɹ̇ɹ**	**ɺ̇ɺ**	**ɻ̇ɻ**	**ɼ̇ɼ**	**ɽ̇ɽ**	**ɾ̇ɾ**	**ɿ̇ɿ**
ideal	a, d	x-2, b-17	x, d	x, d	x, d	x, d	x, b	x, b	x, b	x, b	a, d+34	x-2, b-17	a-69, b	x, d+60	x, d+36	x, d	x, b	x, b
a		377, 645																
a?	274, 885	354, 641	388, 645	363, 645	246, 645	306, 645	290, 645	260, 645	334, 645	369, 645	297, 885	144, 641	205, 816	238, 645	206, 645	198, 645	186, 645	182, 645
aδ		23, 4																
b		380, -114						229, -114				190, -114						
b?	274, -319	354, -125	388, -319	363, -319	246, -319	306, -319	290, -110	260, -110	334, -110	369, -110	297, -283	144, -125	205, -110	238, -257	206, -281	198, -319	186, -110	182, -110
bδ		26, 11						-30, -4				46, 11						
si	***ɮɮ***	***ɯɯ***	***ɰɰ***	***ɱɱ***	***ɲɲ***	***ɳɳ***	***ɴɴ***	***ɵɵ***	***ɶɶ***	***ɷɷ***	***ɸɸ***	***ɹɹ***	***ɺɺ***	***ɻɻ***	***ɼɼ***	***ɽɽ***	***ɾɾ***	***ɿɿ***
patch	***ɮ̇ɮ***	***ɯ̇ɯ***	***ɰ̇ɰ***	***ɱ̇ɱ***	***ɲ̇ɲ***	***ɳ̇ɳ***	***ɴ̇ɴ***	***ɵ̇ɵ***	***ɶ̇ɶ***	***ɷ̇ɷ***	***ɸ̇ɸ***	***ɹ̇ɹ***	***ɺ̇ɺ***	***ɻ̇ɻ***	***ɼ̇ɼ***	***ɽ̇ɽ***	***ɾ̇ɾ***	***ɿ̇ɿ***
ideal	a, d	x-3, b	x, d	x, d	x, d	x, d	x, b	x, b	x-1, b	x, b	a, d+34	x-3, b	a-69, b	x, d+60	x, d+36	x, d	x, b	x, b
a		407, 645		383, 645	264, 646	263, 645									222, 703			
a?	301, 890	372, 640	373, 645	373, 645	254, 645	254, 645	275, 645	229, 645	313, 642	350, 645	328, 890	171, 640	217, 821	194, 645	181, 645	177, 645	167, 645	167, 645
aδ		35, 5		10, 0	10, 1	9, 0									41, 58			
b		410, -114	294, -114	383, -105	265, -111	262, -112	247, -114			341, -114		190, -114	203, -114		127, -305			
b?	217, -319	382, -110	339, -319	339, -319	220, -319	220, -319	283, -110	238, -110	322, -110	359, -110	251, -283	180, -110	190, -110	172, -257	155, -281	143, -319	176, -110	176, -110
bδ		28, -4	-45, 205	44, 214	45, 208	42, 207	-36, -4			-18, -4		10, -4	13, -4		-28, -24			

Table 10: Font metrics for Latin and IPA 6/7

hex	0280	0281	0282	0283	0284	0285	0286	0287	0288	0289	028A	028B	028C	028D	028E	028F	0290	0291
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	ʀ̀ʀ̄	ʁ̀ʁ̄	ʂ̀ʂ̄	ʃ̀ʃ̄	ʄ̀ʄ̄	ɱ̀ɱ̄	ʃ̀ʃ̄	ɨ̀ɨ̄	t̀t̄	ʉ̀ʉ̄	ʊ̀ʊ̄	ʊ̀ʊ̄	ʌ̀ʌ̄	ʍ̀ʍ̄	ʎ̀ʎ̄	ʏ̀ʏ̄	ʐ̀ʐ̄	ʑ̀ʑ̄
patch	ʀ̀ʀ̄	ʁ̀ʁ̄	ʂ̀ʂ̄	ʃ̀ʃ̄	ʄ̀ʄ̄	ɱ̀ɱ̄	ʃ̀ʃ̄	ɨ̀ɨ̄	t̀t̄	ʉ̀ʉ̄	ʊ̀ʊ̄	ʊ̀ʊ̄	ʌ̀ʌ̄	ʍ̀ʍ̄	ʎ̀ʎ̄	ʏ̀ʏ̄	ʐ̀ʐ̄	ʑ̀ʑ̄
ideal	x, b	x, b	x, d+56	a, d+18	a, d+18	x+121, d+18	a, d+7	x, d+79	a-114, d+11	x, b	x, b	x, b	x, b	x, b	a-37, b	x, b	x, d+47	x+19, d+110
a	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	259, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645
a?	238, 645	238, 645	199, 645	140, 885	140, 885	170, 764	150, 885	150, 645	158, 771	271, 645	254, 645	235, 645	244, 645	367, 645	248, 848	236, 645	262, 645	218, 662
aδ	-30, 0	-30, 0	10, 0	68, -240	68, -240	38, -119	60, -240	59, 0	52, -126	-62, 0	-44, 0	24, 0	-35, 0	-158, 0	-39, -203	-27, 0	-54, 0	-8, -17
b	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	258, -114	251, -114	196, -110	196, -110	196, -110	239, -107	196, -110	196, -110
b?	238, -110	238, -110	199, -261	140, -299	140, -299	170, -299	150, -310	150, -238	158, -306	271, -110	254, -110	235, -110	244, -110	367, -110	248, -110	236, -110	262, -270	218, -207
bδ	-42, 0	-42, 0	-3, 151	56, 189	56, 189	26, 189	46, 200	46, 128	38, 196	-75, 0	4, -4	16, -4	-48, 0	-171, 0	-52, 0	3, 3	-66, 160	-22, 97
it	*ʀ̀ʀ̄*	*ʁ̀ʁ̄*	*ʂ̀ʂ̄*	*ʃ̀ʃ̄*	*ʄ̀ʄ̄*	*ɱ̀ɱ̄*	*ʃ̀ʃ̄*	*ɨ̀ɨ̄*	*t̀t̄*	*ʉ̀ʉ̄*	*ʊ̀ʊ̄*	*ʊ̀ʊ̄*	*ʌ̀ʌ̄*	*ʍ̀ʍ̄*	*ʎ̀ʎ̄*	*ʏ̀ʏ̄*	*ʐ̀ʐ̄*	*ʑ̀ʑ̄*
patch	*ʀ̀ʀ̄*	*ʁ̀ʁ̄*	*ʂ̀ʂ̄*	*ʃ̀ʃ̄*	*ʄ̀ʄ̄*	*ɱ̀ɱ̄*	*ʃ̀ʃ̄*	*ɨ̀ɨ̄*	*t̀t̄*	*ʉ̀ʉ̄*	*ʊ̀ʊ̄*	*ʊ̀ʊ̄*	*ʌ̀ʌ̄*	*ʍ̀ʍ̄*	*ʎ̀ʎ̄*	*ʏ̀ʏ̄*	*ʐ̀ʐ̄*	*ʑ̀ʑ̄*
ideal	x, b	x, b	x, d+52	a+10, d+18	a+10, d+18	a+10, d+18	a+10, d+34	x, d+78	a-114, d+6	x, b	x, b-13	x, b-16	x, b	x, b	a-52, b	x, b	x, d+41	x, d+110
a												259, 645						
a?	358, 645	349, 645	314, 645	343, 900	343, 900	343, 900	343, 900	288, 645	314, 776	398, 645	392, 645	382, 645	383, 645	506, 645	433, 838	372, 645	343, 645	352, 645
aδ												-123, 0						
b											258, -114	251, -114				239, -107		
b?	197, -110	189, -110	121, -265	88, -299	88, -299	88, -299	92, -283	100, -239	83, -311	238, -110	229, -121	218, -124	222, -110	346, -110	231, -110	212, -110	147, -276	171, -207
bδ											29, 7	33, 10				27, 3		
sb	**ʀ̀ʀ̄**	**ʁ̀ʁ̄**	**ʂ̀ʂ̄**	**ʃ̀ʃ̄**	**ʄ̀ʄ̄**	**ɱ̀ɱ̄**	**ʃ̀ʃ̄**	**ɨ̀ɨ̄**	**t̀t̄**	**ʉ̀ʉ̄**	**ʊ̀ʊ̄**	**ʊ̀ʊ̄**	**ʌ̀ʌ̄**	**ʍ̀ʍ̄**	**ʎ̀ʎ̄**	**ʏ̀ʏ̄**	**ʐ̀ʐ̄**	**ʑ̀ʑ̄**
patch	**ʀ̀ʀ̄**	**ʁ̀ʁ̄**	**ʂ̀ʂ̄**	**ʃ̀ʃ̄**	**ʄ̀ʄ̄**	**ɱ̀ɱ̄**	**ʃ̀ʃ̄**	**ɨ̀ɨ̄**	**t̀t̄**	**ʉ̀ʉ̄**	**ʊ̀ʊ̄**	**ʊ̀ʊ̄**	**ʌ̀ʌ̄**	**ʍ̀ʍ̄**	**ʎ̀ʎ̄**	**ʏ̀ʏ̄**	**ʐ̀ʐ̄**	**ʑ̀ʑ̄**
ideal	x, b	x, b	x, d+53	a, d+21	a, d+21	a, d+21	a, d	x, d+67	a-103, d	x, b	x, b	x, b	x, b	x, b	a-36, b	x, b	x, d+41	x, d+107
a											266, 645							
a?	236, 645	220, 645	201, 645	152, 885	152, 885	152, 885	162, 885	157, 645	182, 782	274, 645	267, 645	256, 645	231, 645	356, 645	243, 849	244, 645	254, 645	230, 645
aδ											-1, 0							
b																		
b?	236, -110	220, -110	201, -264	152, -296	152, -296	152, -296	162, -319	157, -250	182, -319	274, -110	267, -110	256, -110	231, -110	356, -110	243, -110	244, -110	254, -276	230, -210
bδ																		
si	***ʀ̀ʀ̄***	***ʁ̀ʁ̄***	***ʂ̀ʂ̄***	***ʃ̀ʃ̄***	***ʄ̀ʄ̄***	***ɱ̀ɱ̄***	***ʃ̀ʃ̄***	***ɨ̀ɨ̄***	***t̀t̄***	***ʉ̀ʉ̄***	***ʊ̀ʊ̄***	***ʊ̀ʊ̄***	***ʌ̀ʌ̄***	***ʍ̀ʍ̄***	***ʎ̀ʎ̄***	***ʏ̀ʏ̄***	***ʐ̀ʐ̄***	***ʑ̀ʑ̄***
patch	***ʀ̀ʀ̄***	***ʁ̀ʁ̄***	***ʂ̀ʂ̄***	***ʃ̀ʃ̄***	***ʄ̀ʄ̄***	***ɱ̀ɱ̄***	***ʃ̀ʃ̄***	***ɨ̀ɨ̄***	***t̀t̄***	***ʉ̀ʉ̄***	***ʊ̀ʊ̄***	***ʊ̀ʊ̄***	***ʌ̀ʌ̄***	***ʍ̀ʍ̄***	***ʎ̀ʎ̄***	***ʏ̀ʏ̄***	***ʐ̀ʐ̄***	***ʑ̀ʑ̄***
ideal	x-1, b	x, b	x, d+52	a, d+21	a, d+21	a, d+21	a, d	x, d+67	a-103, d	x, b	x-1, b	x, b	x, b	x, b	a-56, b	x, b	x, d+38	x, d+107
a												259, 645						
a?	214, 642	214, 645	178, 645	195, 890	195, 890	195, 890	189, 890	149, 645	172, 787	255, 645	248, 642	241, 645	239, 645	363, 645	286, 834	228, 645	202, 645	208, 645
aδ												18, 0						
b											258, -114	251, -114				239, -107		
b?	223, -110	223, -110	155, -265	116, -296	116, -296	116, -296	105, -319	129, -250	109, -319	264, -110	257, -110	249, -110	247, -110	371, -110	256, -110	237, -110	176, -279	196, -210
bδ											1, -4	2, -4				2, 3		

Table 11: Font metrics for Latin and IPA 7/7

hex	0292	0293	0294	0295	0296	0297	0299	029B	029C	029D	029F	03B2
cat	Ll	Ll	Lu	Lu	Lu	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg												
patch												
ideal	x, d	x, d	c, b	c, b	c, b	x, d+11	x, b	a-121, b	x, b	a-51, d+55	x, b	a, d
a	218, 646	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	
a?	216, 645	220, 645	211, 850	221, 850	208, 850	221, 645	232, 645	294, 764	293, 645	126, 834	207, 645	262, 885
aδ	2, 1	-11, 0	-2, -205	-12, -205	1, -205	-12, 0	-24, 0	-85, -119	-84, 0	83, -189	2, 0	
b	185, -319	196, -110	202, -107	230, -107	196, -110	196, -110	196, -110	196, -110	305, -110	196, -110	196, -110	275, -114
b?	216, -319	220, -319	211, -110	221, -110	208, -110	221, -306	232, -110	294, -110	293, -110	126, -262	207, -110	262, -319
bδ	-32, 0	-24, 209	-9, 3	9, 3	-12, 0	-25, 196	-36, 0	-98, 0	12, 0	70, 152	-11, 0	14, 205
it												
patch												
ideal	x, d	x, d-80	c+27, b	c+38, b	c, b-23	x, d	x, b	a-94, b	x, b	a-80, d+63	x, b	a, d
a	328, 646											
a?	350, 645	355, 645	373, 850	373, 850	373, 850	411, 645	352, 645	410, 796	416, 645	303, 810	355, 645	425, 890
aδ	-22, 1											
b	130, -319		202, -107	230, -107					273, -110			275, -114
b?	145, -319	133, -397	169, -110	169, -110	164, -131	206, -319	192, -110	217, -110	256, -110	77, -254	194, -110	168, -319
bδ	-15, 0		33, 3	61, 3					17, 0			107, 205
sb												
patch												
ideal	x, d	x, d-96	c, b	c, b	c, b	x, d	x, b	a-121, b	x, b	a-64, d+65	x, b	a, d
a	234, 645											
a?	227, 645	236, 645	206, 850	206, 850	206, 850	273, 645	246, 645	290, 764	316, 645	132, 821	219, 645	290, 885
aδ	7, 0											
b	193, -319								351, -110			
b?	227, -319	236, -413	206, -110	206, -110	206, -110	273, -319	246, -110	290, -110	316, -110	132, -252	219, -110	290, -319
bδ	-34, 0								35, 0			
si												
patch												
ideal	x, d	x, d-96	c, b	c, b	c, b	x, d+15	x, b	a-94, b	x+28, b	a-78, d+65	x, b	a, d
a	218, 646											
a?	206, 645	211, 645	227, 850	227, 850	227, 850	268, 645	213, 645	278, 796	342, 671	158, 812	211, 645	302, 890
aδ	12, 1											
b	141, -319		202, -107	230, -107					327, -110			275, -114
b?	172, -319	157, -413	194, -110	194, -110	194, -110	237, -302	221, -110	256, -110	345, -110	103, -252	219, -110	218, -319
bδ	-31, 0		8, 3	36, 3					-18, 0			57, 205

Table 12: Font metrics for above marks 1/2

hex	0300	0301	0302	0303	0304	0306	0307	0308	0309	030A	030B	030C	030D	030F	0310
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	ǹH̀	ńH́	n̂H̀	ñH̃	n̄H̄	n̆H̆	ṅḢ	n̈Ḧ	n̉H̉	n̊H̊	ő H̋	n̆H̆	ṅḢ	n̏H̏	n̑H̑
patch	ǹH̀	ńH́	n̂H̀	ñH̃	n̄H̄	n̆H̆	ṅḢ	n̈Ḧ	n̉H̉	n̊H̊	ő H̋	n̆H̆	ṅḢ	n̏H̏	n̑H̑
m	-142, 683	-191, 682	-139, 697	-149, 697	-209, 644	-161, 681	-180, 667	-210, 645	-160, 765	-184, 708	-219, 714	-205, 676	-155, 688	-140, 758	-251, 675
m?	-159, 690	-160, 691	-140, 697	-149, 711	-212, 678	-161, 672	-180, 667	-210, 669	-161, 755	-182, 711	-170, 700	-204, 676	-154, 691	-168, 724	-251, 672
mδ	17, -7	-31, -9	1, 0	0, -14	3, -34	0, 9	0, 0	0, -24	1, 10	-2, -3	-49, 14	-1, 0	-1, -3	28, 34	0, 3
it	*ǹH*	*ńH*	*n̂H*	*ñH̃*	*n̄H̄*	*n̆H*	*ṅḢ*	*n̈Ḧ*	*n̉H̉*	*n̊H̊*	*ő H̋*	*n̆H̆*	*ṅḢ*	*n̏H̏*	*n̑H̑*
patch	*ǹH̀*	*ńH́*	*n̂H̀*	*ñH̃*	*n̄H̄*	*n̆H̆*	*ṅḢ*	*n̈Ḧ*	*n̉H̉*	*n̊H̊*	*ő H̋*	*n̆H̆*	*ṅḢ*	*n̏H̏*	*n̑H̑*
m	-3, 683	-52, 682	3, 697	-17, 697	-78, 645	-22, 681	-40, 689	-79, 645	-4, 765	-40, 708	-74, 714	-67, 676	-15, 688	14, 758	-114, 675
m?	-18, 689	-18, 690	3, 696	-3, 716	-71, 680	-24, 671	-35, 698	-74, 669	-8, 754	-37, 712	-26, 699	-66, 675	-13, 690	-19, 723	-114, 671
mδ	15, -6	-34, -8	0, 1	-14, -19	-7, -35	2, 10	-5, -9	-5, -24	4, 11	-3, -4	-48, 15	-1, 1	-2, -2	33, 35	0, 4
sb	**ǹH**	**ńH**	**n̂H**	**ñH**	**n̄H**	**n̆H**	**ṅḢ**	**n̈Ḧ**	**n̉H̉**	**n̊H̊**	**ő H̋**	**n̆H̆**	**ṅḢ**	**n̏H̏**	**n̑H̑**
patch	**ǹH̀**	**ńH́**	**n̂H̀**	**ñH̃**	**n̄H̄**	**n̆H̆**	**ṅḢ**	**n̈Ḧ**	**n̉H̉**	**n̊H̊**	**ő H̋**	**n̆H̆**	**ṅḢ**	**n̏H̏**	**n̑H̑**
m	-127, 713	-180, 713	-127, 700	-133, 679	-205, 625	-149, 684	-169, 603 (1)	-198, 557 (1)	-162, 682 (1)	-170, 646 (1)	-197, 733	-193, 676	-137, 719	-127, 758	-239, 675
m?	-144, 702	-144, 699	-128, 704	-136, 719	-200, 660	-149, 680	-168, 603	-186, 580	-148, 672	-170, 630	-158, 714	-192, 686	-137, 696	-152, 737	-235, 680
mδ	17, 11	-36, 14	1, -4	3, -40	-5, -35	0, 4	-1, 0	-12, -23	-14, 10	0, 16	-39, 19	-1, -10	0, 23	25, 21	-4, -5
si	***ǹH̀***	***ńH́***	***n̂H̀***	***ñH̃***	***n̄H̄***	***n̆H̆***	***ṅḢ***	***n̈Ḧ***	***n̉H̉***	***n̊H̊***	***ő H̋***	***n̆H̆***	***ṅḢ***	***n̏H̏***	***n̑H̑***
patch	***ǹH̀***	***ńH́***	***n̂H̀***	***ñH̃***	***n̄H̄***	***n̆H̆***	***ṅḢ***	***n̈Ḧ***	***n̉H̉***	***n̊H̊***	***ő H̋***	***n̆H̆***	***ṅḢ***	***n̏H̏***	***n̑H̑***
m	-142, 683	-191, 682	-139, 697	-134, 700	-209, 645	-161, 681	-180, 667	-210, 645	-160, 765	-184, 708	-219, 714	-205, 676	-155, 688	-140, 758	-251, 675
m?	-154, 694	-152, 694	-136, 697	-141, 717	-217, 651	-162, 673	-177, 692	-243, 671	-145, 759	-174, 717	-166, 704	-205, 676	-147, 686	-156, 728	-248, 673
mδ	12, -11	-39, -12	-3, 0	7, -17	8, -6	1, 8	-3, -25	33, -26	-15, 6	-10, -9	-53, 10	0, 0	-8, 2	16, 30	-3, 2

Table 13: Font metrics for above marks 2/2

hex	0311	0312	033D	033E	033F	030E	0346	034A	034B	034C	0350	0351	0352	0357	035B
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	n̂H̄	ṅḢ	n̆H̆	ṅḢ	n̄H̄	n̋H̋	n̂H̄	n̆H̆	n̊H̊	n̄H̄	n̆H̆	ṅḢ	n̂Ĥ	ṅḢ	n̆H̆
patch	n̂H̄	ṅḢ	n̆H̆	ṅḢ	n̄H̄	n̋H̋	n̂H̄	n̆H̆	n̊H̊	n̄H̄	n̆H̆	ṅḢ	n̂Ĥ	ṅḢ	n̆H̆
m	-205, 645	-110, 605	-265, 605	-265, 605	-235, 611	-245, 719	-251, 621	-215, 645	-205, 645	-235, 645	-201, 645	-171, 645	-251, 645	-171, 645	-164, 645
m?	-208, 641	-102, 611	-260, 582	-247, 584	-240, 622	-244, 682	-255, 624	-217, 637	-217, 611	-238, 647	-183, 643	-156, 652	-251, 652	-141, 662	-174, 632
mδ	3, 4	-8, -6	-5, 23	-18, 21	5, -11	-1, 37	4, -3	2, 8	12, 34	3, -2	-18, 2	-15, -7	0, -7	-30, -17	10, 13
it	*n̂H̄*	*ṅḢ*	*n̆H̆*	*ṅḢ*	*n̄H̄*	*n̋H̋*	*n̂H̄*	*n̆H̆*	*n̊H̊*	*n̄H̄*	*n̆H̆*	*ṅḢ*	*n̂Ĥ*	*ṅḢ*	*n̆H̆*
patch	*n̂H̄*	*ṅḢ*	*n̆H̆*	*ṅḢ*	*n̄H̄*	*n̋H̋*	*n̂H̄*	*n̆H̆*	*n̊H̊*	*n̄H̄*	*n̆H̆*	*ṅḢ*	*n̂Ĥ*	*ṅḢ*	*n̆H̆*
m	-74, 645	13, 605	-142, 605	-142, 605	-111, 611	-99, 719	-125, 621	-84, 645	-74, 645	-104, 645	-70, 645	-40, 645	-120, 645	-40, 645	-33, 645
m?	-76, 640	24, 610	-142, 581	-128, 583	-113, 621	-105, 681	-128, 623	-87, 636	-92, 610	-107, 646	-53, 642	-23, 651	-118, 651	-6, 661	-46, 631
mδ	2, 5	-11, -5	0, 24	-14, 22	2, -10	6, 38	3, -2	3, 9	18, 35	3, -1	-17, 3	-17, -6	-2, -6	-34, -16	13, 14
sb	**n̂H̄**	**ṅḢ**	**n□H□**	**n□H□**	**n□H□**	**n̋H̋**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**ṅH‘**	**n□H□**	**ṅH’**	**n̆H̆**
patch	**n̂H̄**	**ṅḢ**	**n̆H̆**	**ṅḢ**	**n̄H̄**	**n̋H̋**	**n̂H̄**	**n̆H̆**	**n̊H̊**	**n̄H̄**	**n̆H̆**	**ṅḢ**	**n̂Ĥ**	**ṅḢ**	**n̆H̆**
m	-205, 645	-97, 504 (1)				-232, 719						100, 547 (1)		100, 549 (1)	-180, 639
m?	-220, 650	-102, 528				-232, 696						106, 547		112, 547	-184, 681
mδ	15, -5	5, -24				0, 23						-6, 0		-12, 2	4, -42
si	***n̂H̄***	***ṅḢ***	***n□H□***	***n□H□***	***n□H□***	***n̋H̋***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***
patch	***n̂H̄***	***ṅḢ***	***n̆H̆***	***ṅḢ***	***n̄H̄***	***n̋H̋***	***n̂H̄***	***n̆H̆***	***n̊H̊***	***n̄H̄***	***n̆H̆***	***ṅḢ***	***n̂Ĥ***	***ṅḢ***	***n̆H̆***
m	-205, 645	-110, 605				-245, 719									
m?	-226, 644	-116, 614				-242, 686									
mδ	21, 1	6, -9				-3, 33									

Table 14: Font metrics for below marks 1/2

hex	0316	0317	0318	0319	031C	0320	0323	0324	0325	0326	0329	032A	032B	032C	032D	032E
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	n̖H	n̗H	n̘H	n̙H	n̜H	n̝H	n̞H	n̟H	n̠H	n̡H	n̢H	n̤H	n̥H	n̦H	ņH	n̨H
patch	n̖H	n̗H	n̘H	n̙H	n̜H	n̝H	n̞H	n̟H	n̠H	n̡H	n̢H	n̤H	n̥H	n̦H	ņH	n̨H
m	-219, -133	-210, -132	-224, -148	-285, -147	-216, -146	-253, -148	-216, -113	-218, -116	-218, -116	-255, -82	-257, -122	-253, -113	-251, -108	-204, -116	-195, -133	-251, -110
m?	-248, -124	-224, -114	-249, -102	-249, -101	-262, -98	-252, -170	-216, -152	-217, -152	-217, -78	-245, -87	-256, -106	-255, -134	-252, -119	-204, -123	-195, -128	-252, -109
mδ	29, -9	14, -18	25, -46	-36, -46	46, -48	-1, 22	0, 39	-1, 36	-1, -38	-10, 5	-1, -16	2, 21	1, 11	0, 7	0, -5	1, -1
it	*n̖H*	*n̗H*	*n̘H*	*n̙H*	*n̜H*	*n̝H*	*n̞H*	*n̟H*	*n̠H*	*n̡H*	*n̢H*	*n̤H*	*n̥H*	*n̦H*	*ņH*	*n̨H*
patch	*n̖H*	*n̗H*	*n̘H*	*n̙H*	*n̜H*	*n̝H*	*n̞H*	*n̟H*	*n̠H*	*n̡H*	*n̢H*	*n̤H*	*n̥H*	*n̦H*	*ņH*	*n̨H*
m	-246, -133	-237, -132	-254, -148	-315, -147	-246, -146	-283, -148	-239, -113	-242, -116	-242, -116	-272, -82	-282, -122	-276, -113	-273, -108	-228, -116	-222, -133	-275, -110
m?	-274, -126	-248, -116	-270, -104	-270, -103	-281, -100	-287, -172	-238, -111	-239, -111	-233, -82	-263, -89	-278, -108	-283, -136	-277, -121	-229, -125	-221, -130	-269, -111
mδ	28, -7	11, -16	16, -44	-45, -44	35, -46	4, 24	-1, -2	-3, -5	-9, -34	-9, 7	-4, -14	7, 23	4, 13	1, 9	-1, -3	-6, 1
sb	**n̖H**	**n̗H**	**n̘H**	**n̙H**	**n̜H**	**n̝H**	**n̞H**	**n̟H**	**n̠H**	**n̡H**	**n̢H**	**n̤H**	**n̥H**	**n̦H**	**ņH**	**n̨H**
patch	**n̖H**	**n̗H**	**n̘H**	**n̙H**	**n̜H**	**n̝H**	**n̞H**	**n̟H**	**n̠H**	**n̡H**	**n̢H**	**n̤H**	**n̥H**	**n̦H**	**ņH**	**n̨H**
m	-219, -133	-210, -132	-224, -148	-285, -147	-216, -146	-253, -149	-216, -113	-218, -116	-218, -116	-255, -73	-257, -124	-253, -113	-251, -108	-208, -116	-195, -133	-239, -110
m?	-248, -132	-224, -122	-249, -110	-249, -110	-262, -106	-251, -179	-216, -118	-218, -113	-216, -88	-244, -86	-256, -116	-255, -142	-252, -127	-204, -131	-195, -136	-240, -105
mδ	29, -1	14, -10	25, -38	-36, -37	46, -40	-2, 30	0, 5	0, -3	-2, -28	-11, 13	-1, -8	2, 29	1, 19	-4, 15	0, 3	1, -5
si	***n̖H***	***n̗H***	***n̘H***	***n̙H***	***n̜H***	***n̝H***	***n̞H***	***n̟H***	***n̠H***	***n̡H***	***n̢H***	***n̤H***	***n̥H***	***n̦H***	***ņH***	***n̨H***
patch	***n̖H***	***n̗H***	***n̘H***	***n̙H***	***n̜H***	***n̝H***	***n̞H***	***n̟H***	***n̠H***	***n̡H***	***n̢H***	***n̤H***	***n̥H***	***n̦H***	***ņH***	***n̨H***
m	-219, -133	-210, -132	-224, -148	-285, -147	-216, -146	-253, -148	-216, -113	-218, -116	-218, -116	-255, -82	-257, -122	-253, -113	-251, -108	-204, -116	-195, -133	-251, -110
m?	-251, -132	-225, -122	-246, -110	-259, -110	-260, -106	-263, -179	-214, -117	-216, -113	-211, -88	-237, -86	-256, -116	-260, -142	-253, -127	-206, -127	-199, -136	-244, -113
mδ	32, -1	15, -10	22, -38	-26, -37	44, -40	10, 31	-2, 4	-2, -3	-7, -28	-18, 4	-1, -6	7, 29	2, 19	2, 11	4, 3	-7, 3

Table 15: Font metrics for below marks 2/2

hex	032F	031D	031E	031F	0330	0331	0333	0339	033A	033B	033C	0347	0348	0349	034D	034E
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯
patch	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯
m	-250, -116	-249, -143	-247, -106	-249, -145	-213, -70	-244, -110	-244, -110	-216, -146	-253, -113	-246, -113	-251, -108	-244, -110	-197, -110	-192, -110	-222, -110	-227, -110
m?	-251, -125	-249, -112	-249, -112	-252, -99	-205, -68	-240, -133	-240, -123	-178, -98	-255, -134	-252, -118	-252, -119	-240, -123	-195, -94	-189, -111	-228, -92	-228, -74
mδ	1, 9	0, -31	2, 6	3, -46	-8, -2	-4, 23	-4, 13	-38, -48	2, 21	6, 5	1, 11	-4, 13	-2, -16	-3, 1	6, -18	1, -36
it	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*
patch	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*
m	-274, -116	-278, -143	-269, -106	-278, -145	-227, -70	-266, -110	-266, -110	-246, -146	-276, -113	-269, -113	-273, -108	-266, -110	-219, -110	-214, -110	-244, -110	-249, -110
m?	-276, -127	-271, -114	-272, -114	-271, -101	-220, -74	-268, -135	-266, -125	-198, -100	-283, -136	-277, -120	-275, -121	-266, -125	-214, -96	-213, -113	-247, -94	-243, -76
mδ	2, 11	-7, -29	3, 8	-7, -44	-7, 4	2, 25	0, 15	-48, -46	7, 23	8, 7	2, 13	0, 15	-5, -14	-1, 3	3, -16	-6, -34
sb	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**
patch	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**
m	-250, -116	-249, -143	-247, -106	-249, -146	-213, -70	-244, -58										
m?	-263, -127	-249, -121	-249, -120	-251, -108	-192, -70	-240, -89										
mδ	13, 11	0, -22	2, 14	2, -38	-21, 0	-4, 31										
si	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***
patch	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***
m	-250, -116	-249, -143	-247, -106	-249, -145	-213, -70	-244, -110										
m?	-265, -131	-252, -132	-247, -108	-249, -108	-184, -77	-234, -89										
mδ	15, 15	3, -11	0, 2	0, -37	-29, 7	-10, -21										

Vertical references, anchor Y coordinates and reference, and clearance

A font has at least five vertical reference lines where glyph outlines live inside: descender height, baseline, x-height, ascender height, capital height. The values or ranges of these Y coordinates are in the first five rows of Table 16. These vertical aspects are the gray horizontal lines of Figure 1. These vertical aspects are displayed in a FontForge glyph outline window as black horizontal guidelines with red or blue shaded bands for ranges. I denote these as yra, yrc, yrx, $yr0$, yrd because they are Y values, and r indicates a reference value. a, c, x, 0, d indicate the five vertical aspects. These values (and the ranges that set the bands) are in the SFD, in a line that starts with `BlueValues`, e.g. in Libertinus serif regular,

```
BlueValues 47 [-12 0 429 442 460 472 600 610 645 658 688 698]
OtherBlues 11 [-238 -227]
```

means $yra = 698, [688, 698]$; $yrc = 645, [645, 658]$; $yrx = 429, [429, 442]$; $yr0 = 0, [-12, 0]$; and $yrd = -232, [-238, -227]$.

In the anchor model for Libertinus, the Y coordinate of an above-anchor (ay) or a below-anchor (by) is at a stable value depending on where the glyph outline lives relative to a vertical aspect. See the rows a and b in Tables 1–4 and find the common values according to the glyph's vertical aspects. There is no set name in font design for these above- or below-anchor Y coordinate reference values by ascender aspect, so I will denote them $ayra$, $ayrc$, $ayrx$, $byr0$, $byrd$, These are the middle five rows of Table 16. For example, all glyphs that sit on the baseline have below-anchor Y coordinate $by = -110$, and thus $byr0 = -110$. All capitals A–Z have $ay = 850$, and thus $ayrc = 850$. Any glyph that descends to the descender value $yrd = -232$ or descender range $[-238, -227]$ should have $by = byrd = -319$.

The clearance is defined as the difference between the anchor reference value and the vertical aspect Y value. I will denote these clearances as $ay\delta a$, $ay\delta c$, $ay\delta x$, $by\delta 0$, $by\delta d$. So, $ay\delta a = ayra - yra$ etc. For example, capital letters A–Z have a capital height $yrc = 645$, and all capital letters A–Z have an above-anchor Y coordinate at $ayrc = 850$. So the above-anchor clearance for capitals is $ay\delta c = ayrc - yec = 850 - 645 = 205$. Because of the weird anchor model for Libertinus, this clearance value is not the exact vertical distance between a base and a mark, but this clearance value is still useful.

A glyph may not be a full ascender or a full descender, and its bounding box may not live closely within two vertical aspects. In such a case, the anchor Y coordinate is set by looking for the nearest, or most suitable, vertical aspect, and considering the clearance of that vertical aspect relative to the bounding box of the glyph.

Here is the heuristic algorithm for calculating the Y coordinate of the above- or below-anchor (ay or by) of a glyph that is not a full ascender or descender.

Consider a glyph that is not a full ascender. Find the nearest, or most suitable, vertical aspect – ascender a, capital c, or x-height x. Then $ay = Y_{\text{max}} + ay\delta[acx] - \epsilon$, where Y_{max} is the maximum Y coordinate of bounding box of the glyph, and ϵ is the correction for the vertical aspect range, usually for an overshoot.

Capitals are the exception: their top reference line is always the capline, and their above-anchor always has the capital reference Y ($ay = 850$), even where the outline rises above the capline. In humanist letterforms capitals do not have ascenders: an ascender is a stroke of the pen, not merely a high Y_{max}. What rises above the capline in Libertinus is away from the horizontal center of the glyph, where the mark sits: the top-left stroke of J (italic and semibold italic, 34 units above the capline), the serifs of T (semibold and semibold italic, 28), the slash of Ø (25), the top of Ʒ (up to 29), and the hooks of Ƈ and Ɠ (98). A mark at the capital reference Y does not collide with these. The one capital with a genuine ascender is the capital iota ɩ (U+0196), whose stem reaches the ascender line; it takes the ascender reference ($ay = 885$; italic styles 890). The glottal stop ʔ (U+0294) and pharyngeal ʕ (U+0295), counted here as capitals, rise above the capline only in italic (672 and 683); there they have curated anchors at $ay = 850 + (Y_{\text{max}} - 645) - 10$, with the bowl overshoot correction $\epsilon = 10$ instead of the default 2, which gives 867 and 878.

Consider a glyph that is not a full descender. Find the nearest, or most suitable, vertical aspect – baseline 0 or desender d. Then $by = Y_{\text{min}} + by\delta[0d] + \epsilon$, where Y_{min} in the minimum Y coordinate of the bounding box of the glyph, and ϵ is the correction for the vertical aspect range, usually for an overshoot.

The capital anchor Y reference and clearance of semibold are the patched values. The x-height bands marked * are widened above the designer's BlueValues (442, 447, 447) on 2026-11-07: several x-height letters of those styles rise 1–11 units above the band (ɐ ɔ ɡ ɵ ɽ ʂ ʐ ʑ in italic, ɢ ʑ in semibold, ʐ ʑ in semibold italic), which gave their candidates a Y of 657–666, above the base letters s, r, z, o at 645. Snapped to 645, no above-mark of the IPA set overlaps them (the nearest is 19 away); semibold italic ʜ (462) and p (470) stay outside.

The designer set semibold's capitals near 805 (clearance 160) and compensated only in the capital alternates of the marks, so a mark without a capital alternate crashed into the capital: in 225 of the 780 combinations of A–Z with 30 above marks, the bottom of the mark was below the top of the letter. The patch moves the capitals to 850 and adds 45 to the mark anchors of the capital alternates, which stay where they were. That leaves 4 such combinations, all on T, whose serif tips rise to 673 away from the marks; under the marks the gap is 21–26. A few capitals have the same problem in the other styles (Ŋ Ƅ Ƈ Ɋ Ħ Ɪ Ƙ Ƶ at 798–805), so since 2026-11-08 stage 5 moves every capital with an above-anchor in 795–835 to 850 in every style. In italic and semibold italic the designer's X of Ŋ Ƅ Ƈ Ƙ Ƶ was regular's, without the slant (italic Ƈ 355 against the candidate 622), so they are deleted in stage 1 and get the candidate; Ħ, outside the Latin/IPA scope, keeps regular's X there. These capitals are also in `@capitals`, so the twelve marks with a capital alternate take it on them.

Table 16: Vertical aspects, anchor Y references, and clearances of Libertinus

	v. aspect	regular	italic	semibold	semibold italic
yra	ascender	$698, [688, 698]$	$698, [688, 698]$	$698, [690, 698]$	$698, [696, 705]$
yrc	capital	$645, [645, 658]$	$645, [645, 658]$	$645, [645, 662]$	$645, [645, 662]$
yrx	x-height	$429, [429, 442]$	$429, [429, 450]^*$	$434, [433, 457]^*$	$434, [434, 458]^*$
$yr0$	baseline	$0, [-12, 0]$	$0, [-12, 1]$	$0, [-12, 1]$,	$0, [-20, 0]$
yrd	descender	$-232, [-238, -227]$	$-232, [-238, -227]$	$-232, [-238, -212]$	$-232, [-239, -219]$
$ayra$	ascender	885	890	885	890
$ayrc$	capital	850	850	850	850
$ayrx$	x-height	645	645	645	645
$byr0$	baseline	-110	-110	-110	-110
$byrd$	descender	-319	-319	-319	-319
$ay\delta a$	ascender	187	192	187	192
$ay\delta c$	capital	205	205	205	205
$ay\delta x$	x-height	216	216	211	211
$by\delta 0$	baseline	110	110	110	110
$by\delta d$	descender	87	87	87	87

Of the other designer anchors more than 5 from a reference (stage 5 lists them), most are legitimate: stacked marks on precomposed glyphs, superscripts at 805–815, small capitals at their own height, 685. Fixed on 2026-11-08: above and below swapped on ɥ ɥ̨ (regular, italic, semibold italic) and ƚ (regular), now deleted in stage 1 and given the candidates; and į i̇ ɨ, whose above-anchors were 9–59 units below i's, so that a mark keeping the dot crowded it, now curated at i's anchor.

For the vertical aspect, the first single value is the optical value of the vertical aspect. In practice, this is the y-coordinate of the central control point of a (vertical) stem that meets that aspect. In the table, the range for the vertical aspects are the BlueValues set by the designer in the SFD and seen as guidelines in a FontForge glyph outline window view.

I need to keep track of serif and bowl corrections relative to the (single) vertical aspect value. In Libertinus serif regular, a glyph that meets an aspect with a stem needs a 2 unit correction. For example, the letter x meets the baseline and x-height aspects, with stems, but $y_{\text{min}} = 0 - 2 = -2$ and $y_{\text{max}} = 429 + 2 = 431$. For bowls, the corrections tend to be different for each aspect. For Libertinus serif regular, a glyph that is a bowl full-descender (usually) has a 6 unit overshoot, i.e. $y_{\text{min}} = -232 - 6 = -238$. A bowl on the baseline (usually) has a 10 unit overshoot, i.e. $y_{\text{min}} = 0 - 10 = -10$. A bowl at the x-height aspect (usually) has a 10 unit overshoot, i.e. $y_{\text{max}} = 429 + 10 = 439$. A bowl at the capital aspect (usually) has a 13 unit overshoot, i.e. $y_{\text{max}} = 645 + 13 = 658$. For full ascenders, there is no correction in practice, regardless of the shape of the glyph's ascender, because $y_{\text{max}} = 698$ for all such glyphs.

That is, these stem and bowl corrections are 2 and 6, 10, 10, 13.

For the other font styles (italic, semibold, semibold italic), these corrections are remarkably consistent for all aspects, even though e.g. italic small letters do not really have flat terminal stems. But I think that the correction principle is valid for glyphs that are not within the threshold of an expected vertical aspect. For a non-full descender, if y_{min} is closest to the descender aspect, and the glyph looks like a bowl at this lowest point, then the correction is 6; otherwise I assume that the glyph must look like a flat stem and the correction is 2. But for a non-full descender that is closer to the baseline, the bowl correction is 10. Full descender bowls tend to be flatter because the lower threhold of the descender really is a hard lower bound.

Derived letters

Many letters are a base letter plus a bar, stroke, hook, tail, curl or belt: Đ is D with a bar, Ɗ is D with a hook, ɖ is d with a tail. Where the added part stays clear of a mark, the derived letter should attach the mark where its base letter does. The designer often drew the derived letter in a shifted frame: Ɗ is D moved about 80 units right, to make room for the hook. So the derived letter's anchor is the base letter's anchor plus that shift. The heuristic candidate cannot know this: a hook or tail moves the midpoint of the bounding box that the candidate is

centred on.

Semibold ɭ is redrawn by the patch (stage 2, `font_patching_redraw.py`, 2026-11-08). The designer's glyph had a regular-weight stem (75 units; semibold l, ɬ and ꞎ have 110–112), a plain slab head instead of l's serif, and a hook deeper than ɖ's. Regular ɭ is regular l with the foot serif replaced by the retroflex hook of regular ɖ, to about 5 units per control point; so semibold ɭ is now semibold l with its foot serif replaced by the hook of semibold ɖ, moved 236 units left onto l's stem, its first control point on the stem line as in regular ɭ and its return curve 7 units left, because ɖ's stem is 117 wide and l's 110 (advance 275). Its stem now matches l's exactly, so stage 8 inherits l's anchors (stem match 100%): above $(150, 885)$ and below $(150, -319)$, on the stem's midline, below the hook. The curated above-anchor that replaced the failed inheritance is commented out.

Stage 8 of the patch inherits these anchors. `data/derived_letters.py` lists each derived letter with its base letter and the anchor classes it may inherit: both above and below for a bar or stroke through the letter, or a hook at the top away from the centre; above and below for a hook, tail or curl at the bottom (see below); below only for a hook at the top in the way of an above mark (ȝ, ȣ, ʃ). The classes can differ by style. In italic and semibold italic the designer put the above-anchor of d and h at the x-height, over the bowl beside the ascender, where the bar or hook of đ ɗ ħ ɦ is in the mark's way (in semibold italic, 27 above marks overlapped the ascender of d itself). Both are now at the ascender reference (deleted in stage 1, candidate in stage 6), and đ ɗ inherit above in every style; ħ ɦ still inherit only the below-anchor there. The shift is found by matching outlines: at every height on a 10-unit grid, the positions where the base letter's outline crosses that height are compared with the derived letter's, and the shift that pairs the most crossings within 6 units wins. The share of the base letter's crossings so paired measures how much of the base letter the derived letter really contains. Below 70% the derived letter was redrawn rather than built on the base, and nothing is inherited in that style. Stage 8 runs after the curated anchors, so it inherits the base letter's final anchor, and it never overwrites a curated anchor of the derived letter.

In the tables, each column is a derived letter, with its base letter's codepoint below it. For each style: the derived letter with circumflex above and macron below, in the original and patched fonts (`reg`, `patch`), the base letter likewise in the patched font (`base`), and the outline shift and match (in italics if below 70%). Then, for the above anchor: a, the designer's anchor; a?, the heuristic candidate; a<, the base letter's final anchor plus the shift, in bold where the patched font has it; a!, a curated anchor, which stage 8 keeps. The same for the below anchor (b).

Letters with a tail at the bottom. ʃ Ɲ ɕ ɖ ɭ ɱ ɲ ɳ ɻ ɽ ʂ ʃ ʈ ʐ ʑ ʒ (`bottom_tail` in `data/derived_letters.py`) follow one rule, decided 2026-11-06, whether the tail curls toward the centre (ɱ, ɭ, ɽ, ʂ) or away from it (ɖ, ɳ, ɲ, Ɲ). A dot below clears an outward tail at the baseline, but most below marks are much wider: at the designer's baseline anchors, 4 to 29 of the 32 below marks overlapped the tail of ɖ, ɱ, ɲ or ɳ in some style.

- The shift is matched on the stems and bowl sides above the tail (heights from 60 to the x-height less 60), not on the whole outline, since the tail replaces the base letter's foot. Each ink interval of the base is paired with the derived letter's interval of about the same width at the same height, and a line through the paired centres allows for a letter drawn a little narrower or wider (ɱ ɲ ɳ are about 3% narrower than m and n in regular). The whole-outline match was 46–59% for regular ɱ ɲ ɳ ɽ; the stem match is 72–97%. It fails where there are no stems (ʐ ʑ ʒ in the italic styles) and where the italic letter was redrawn (italic ɖ ɲ).
- Below, where the tail has ink within 195 units of the mark's axis (the half-width of the widest below mark, U+0347), Y is the anchor model's candidate, the descender reference with its clearance from the letter's own outline, and X is the base letter's plus the shift, following the slant down. Where the tail is clear of the mark, Y is the base letter's: Ɲ in regular and italic, whose hook is more than 195 units left of the axis. In semibold and semibold italic Ɲ keeps a descender anchor: at the baseline U+0347 and U+034D would overlap its hook.
- Where the stems match less than 70%, the above anchor is not inherited, and a below anchor over a tail gets the candidate, X and Y.
- The curated below anchors of regular ɭ, ɳ and ʂ were removed: the rule gives ɳ and ʂ within 7 units of them, and ɭ X from the stem of l (133 against 196), where one below mark overlaps instead of four.

After the change, no below mark overlaps ɖ ɱ ɲ ɳ in any style. New overlaps, all with the tall half rings and tacks: regular ʐ with U+031C, italic ʈ with U+0339; these marks have since been lowered (see "Mark anchor model", stage 7), and no below mark overlaps a descender.

ɯ has the candidate in every style (its designer anchors are deleted in stage 1): the candidate lies on the central stem, within 2 to 7 units. ɯ̧ is not derived from ɯ: it is narrower and its stems are spaced differently (the outline match is 21% in regular). Its below anchor is curated under the middle stem, at ɯ's candidate Y; only U+0347 overlaps its long leg, in semibold and semibold italic.

Two letters have an anchor set apart from these rules. The curl of ɕ reaches about -100, so at c's $Y = -110$ most below marks overlap it. Its below-anchor keeps the Y of the anchor model, under the curl (-214 regular, -208 italic, -209 semibold and semibold italic), and takes X from c's final below-anchor plus the shift, following the slant down to that height. ɾ

matches r too little to inherit, but its above-anchor takes X from r in the same way, at the x-height reference, 645. Its below-anchor is curated on the axis of its stem, at the descender reference, -319. This is `derived_y` in `data/derived_letters.py`. It does not depend on the outline match, since only X is borrowed.

What the comparison shows:

- In regular and semibold most derived letters are built on the base letter (90% or more of its outline). In italic and semibold italic most were redrawn, and inherit only where the bar or stroke was added to a traced base: Đ Ð đ Ħ ħ Ø ø Ɨ ɨ ɜ ɕ, and some others.
- Several designer anchors are copies: (209, 645) and (196, -110) recur on many IPA letters in regular, including ɖ and ɭ, which have ascenders and now inherit 885 from d and l, and ʈ, which inherits t's anchor (769). ʃ, which descends, had a below-anchor at -110; it now inherits -319 from ɟ.
- The heuristic candidate is pulled sideways by a hook or bar: for Ƒ, 183 against 276 from F; for Ł, 256 against 147 from L (over the stem, as the designer placed it on L).
- Ɓ and Ɗ inherit B's and D's anchors moved about 58 and 82 units right, which the designer's copies lacked.
- The inheritance carries the base letter's own anchor, right or wrong. Where the base letter's anchor is questionable (D in regular is 76 units left of the centre of its body), Đ, Đ, Đ and Ɗ follow it.
- Checked for collisions with a circumflex and a macron: none in what is inherited. ɜ and ɚ inherit only below, because their hook rises into the above mark at the inherited height; italic and semibold italic ħ ɦ inherit only below (see above). ɗ is matched on its stems, like the letters with a tail below: its hook replaces the top of the ascender, and the whole-outline match was 50–53% in semibold and semibold italic. The candidate for ɗ and ɖ is pulled right by the hook or tail (by the bounding box in roman, the advance width in italic): regular ɗ 322 against 233 inherited from d. The curated regular ɗ (218, 885), 17 units left of d's anchor, was removed. Italic ɖ was redrawn wider than d (bowl 16 units left, stem 21 right), so it does not match; its above anchor is curated at d's plus the 4-unit shift of the centre of bowl and stem, (440, 899). ɕ is much better inherited: the designer's anchors put the marks on the hook. Italic ɦ's designer above-anchor (299, 565) put marks inside the hook; it is deleted in stage 1, and stage 6 gives it the ascender candidate.

Table 17: Derived letters and inherited anchors 1/3

hex	026E	00D0	0110	0189	0111	0126	0127	0141	0142	00D8	00F8	019F	0275	0197	0268	0289
base	006C	0044	0044	0044	0064	0048	0068	004C	006C	004F	006F	004F	006F	0049	0069	0075
reg	ɮ̲	Ð̲	Đ̲	Đ̲	đ̲	Ħ̲	ħ̲	Ł̲	ł̲	Ø̲	ø̲	Ɵ̲	ɵ̲	Ɨ̲	ɨ̲	ʉ̲
patch	ɮ̲	Ð̲	Đ̲	Đ̲	đ̲	Ħ̲	ħ̲	Ł̲	ł̲	Ø̲	ø̲	Ɵ̲	ɵ̲	Ɨ̲	ɨ̲	ʉ̲
base	l̲	D̲	D̲	D̲	d̲	H̲	h̲	L̲	l̲	O̲	o̲	O̲	o̲	I̲	i̲	u̲
shift	+2, 95%	+0, 97%	+0, 97%	+0, 97%	+0, 97%	+0, 95%	+0, 97%	-4, 96%	+0, 96%	+0, 95%	-2, 93%	+0, 97%	+0, 95%	+0, 95%	+14, 93%	+0, 91%
a	209, 645	341, 850								351, 850			209, 645		139, 645	209, 645
a?	250, 885	333, 850	336, 850	333, 850	270, 885	362, 850	270, 885	256, 850	133, 885	352, 850	250, 668	352, 850	252, 645	148, 850	150, 834	271, 645
a<	**137, 885**	**306, 850**	**306, 850**	**306, 850**	**235, 885**	**361, 850**	**282, 885**	**147, 850**	**135, 885**	**335, 850**	**236, 645**	**335, 850**	**238, 645**	**150, 850**	156, 645	**255, 645**
a!															142, 834	
b	196, -110									350, -110	243, -110	350, -110	196, -110		134, -114	196, -110
b?	250, -319	333, -110	336, -110	333, -110	270, -110	362, -110	270, -110	256, -110	133, -110	352, -121	250, -136	352, -110	252, -110	148, -110	150, -110	271, -110
b<		**309, -110**	**309, -110**	**309, -110**	**235, -110**	**362, -110**	**272, -110**	**258, -110**	**135, -110**	**350, -110**	**241, -110**	**350, -110**	**243, -110**	**147, -110**	**156, -110**	**239, -110**
b!	275, -319															
it	*ɮ̲*	*Ð̲*	*Đ̲*	*Đ̲*	*đ̲*	*Ħ̲*	*ħ̲*	*Ł̲*	*ł̲*	*Ø̲*	*ø̲*	*Ɵ̲*	*ɵ̲*	*Ɨ̲*	*ɨ̲*	*ʉ̲*
patch	*ɮ̲*	*Ð̲*	*Đ̲*	*Đ̲*	*đ̲*	*Ħ̲*	*ħ̲*	*Ł̲*	*ł̲*	*Ø̲*	*ø̲*	*Ɵ̲*	*ɵ̲*	*Ɨ̲*	*ɨ̲*	*ʉ̲*
base	*l̲*	*D̲*	*D̲*	*D̲*	*d̲*	*H̲*	*h̲*	*L̲*	*l̲*	*O̲*	*o̲*	*O̲*	*o̲*	*I̲*	*i̲*	*u̲*
shift	+16, 77%	+0, 97%	+0, 97%	*+36, 37%*	+0, 97%	+0, 95%	+0, 97%	*+6, 56%*	*+6, 42%*	-4, 96%	+0, 91%	*+2, 42%*	*+6, 38%*	+0, 94%	+14, 93%	*-12, 32%*
a															267, 645	
a?	447, 890	514, 850	514, 850	534, 850	434, 890	517, 850	449, 890	440, 850	322, 890	515, 850	370, 691	515, 850	379, 645	321, 850	316, 810	398, 645
a<	**328, 890**	**471, 850**	**471, 850**	507, 850	**434, 890**	**504, 850**		320, 850	318, 890	**489, 850**	**359, 645**	495, 850	365, 645	**316, 850**	**289, 645**	374, 645
a!								314, 850								
b										310, -113	186, -107				134, -114	
b?	192, -309	310, -110	310, -110	330, -110	221, -110	313, -110	236, -110	236, -110	110, -110	302, -151	190, -156	311, -110	218, -110	117, -110	121, -110	238, -110
b<		**279, -110**	**279, -110**	315, -110	**221, -110**	**320, -110**	**229, -110**	258, -110	105, -110	**306, -110**	**178, -148**	312, -110	192, -110	**127, -110**	**129, -110**	214, -110
b!	207, -319															
sb	**ɮ̲**	**Ð̲**	**Đ̲**	**Đ̲**	**đ̲**	**Ħ̲**	**ħ̲**	**Ł̲**	**ł̲**	**Ø̲**	**ø̲**	**Ɵ̲**	**ɵ̲**	**Ɨ̲**	**ɨ̲**	**ʉ̲**
patch	**ɮ̲**	**Ð̲**	**Đ̲**	**Đ̲**	**đ̲**	**Ħ̲**	**ħ̲**	**Ł̲**	**ł̲**	**Ø̲**	**ø̲**	**Ɵ̲**	**ɵ̲**	**Ɨ̲**	**ɨ̲**	**ʉ̲**
base	**l̲**	**D̲**	**D̲**	**D̲**	**d̲**	**H̲**	**h̲**	**L̲**	**l̲**	**O̲**	**o̲**	**O̲**	**o̲**	**I̲**	**i̲**	**u̲**
shift	+2, 94%	+0, 96%	+0, 96%	+0, 96%	+0, 97%	+0, 94%	+0, 97%	+0, 94%	+0, 94%	+0, 94%	+0, 91%	+0, 96%	+0, 96%	+0, 94%	+0, 77%	+0, 91%
a															151, 645	
a?	274, 885	349, 850	349, 850	349, 850	274, 885	362, 850	275, 885	258, 850	142, 885	366, 850	260, 684	366, 850	260, 645	158, 850	152, 818	274, 645
a<	**152, 885**	**352, 850**	**352, 850**	**352, 850**	**274, 885**	**362, 850**	**275, 885**	**161, 850**	**150, 885**	**363, 850**	**260, 645**	**363, 850**	**260, 645**	**160, 850**	156, 645	**264, 645**
a!															156, 837	
b													229, -114		134, -114	
b?	274, -319	349, -110	349, -110	349, -110	274, -110	362, -110	275, -110	258, -110	142, -110	366, -153	260, -142	366, -110	260, -110	158, -110	152, -110	274, -110
b<		**352, -110**	**352, -110**	**352, -110**	**274, -110**	**362, -110**	**277, -110**	**267, -110**	**150, -110**	**366, -110**	**260, -110**	**366, -110**	**260, -110**	**160, -110**	**156, -110**	**248, -110**
b!	283, -319															
si	***ɮ̲***	***Ð̲***	***Đ̲***	***Đ̲***	***đ̲***	***Ħ̲***	***ħ̲***	***Ł̲***	***ł̲***	***Ø̲***	***ø̲***	***Ɵ̲***	***ɵ̲***	***Ɨ̲***	***ɨ̲***	***ʉ̲***
patch	***ɮ̲***	***Ð̲***	***Đ̲***	***Đ̲***	***đ̲***	***Ħ̲***	***ħ̲***	***Ł̲***	***ł̲***	***Ø̲***	***ø̲***	***Ɵ̲***	***ɵ̲***	***Ɨ̲***	***ɨ̲***	***ʉ̲***
base	***l̲***	***D̲***	***D̲***	***D̲***	***d̲***	***H̲***	***h̲***	***L̲***	***l̲***	***O̲***	***o̲***	***O̲***	***o̲***	***I̲***	***i̲***	***u̲***
shift	+10, 78%	+0, 96%	+0, 97%	*+50, 45%*	+0, 96%	+12, 94%	+0, 97%	+12, 94%	+16, 94%	-2, 94%	+0, 92%	+0, 96%	*+36, 50%*	-14, 73%	+2, 93%	*+16, 51%*
a		341, 850								351, 850					276, 645	
a?	301, 890	367, 850	367, 850	388, 850	309, 890	396, 850	331, 890	300, 850	190, 890	390, 850	235, 645	387, 850	229, 645	181, 850	180, 823	255, 645
a<	**190, 890**	**306, 850**	**306, 850**	356, 850	**309, 890**	**408, 850**		**221, 850**	**196, 890**	**388, 850**	**238, 645**	**390, 850**	274, 645	**180, 850**	144, 645	271, 645
a!															180, 823	
b										350, -110	243, -110				136, -110	
b?	217, -319	334, -110	334, -110	355, -110	268, -110	363, -110	290, -110	267, -110	148, -110	349, -148	237, -138	354, -110	238, -110	148, -110	153, -110	264, -110
b<		**279, -110**	**279, -110**	329, -110	**221, -110**	**377, -110**	**272, -110**	**272, -110**	**154, -110**	**348, -110**	**243, -110**	**350, -110**	279, -110	**155, -110**	**155, -110**	255, -110
b!	231, -319															

Table 18: Derived letters and inherited anchors 2/3

hex	026B	026C	025F	0269	0181	018A	0187	0193	0198	0257	0266	029B	025D	025A	0284	0191
base	006C	006C	0237	0131	0042	0044	0043	0047	004B	0064	0068	0262	025C	0259	0283	0046
reg	ꞎ	ꞎ	ȷ̵	ɩ̵	B̵	D̵	C̵	G̵	K̵	d̵	h̵	ɢ̵	ɜ̵	ə̵	ʃ̵	F̵
patch	ꞎ	ꞎ	ȷ̵	ɩ̵	B̵	D̵	C̵	G̵	K̵	d̵	h̵	ɢ̵	ɜ̵	ə̵	ʃ̵	F̵
base	l̵	l̵	ȷ̵	ɩ̵	B̵	D̵	C̵	G̵	K̵	d̵	h̵	ɢ̵	ɜ̵	ə̵	ʃ̵	F̵
shift	+12, 92%	+88, 94%	*+12, 65%*	-20, 79%	+58, 94%	+82, 95%	+0, 100%	+0, 100%	-4, 86%	-2, 94% s	*-4, 45%*	+0, 100%	+0, 100%	*-12, 61%*	+0, 95%	-8, 99% s
a	209, 645	209, 645	169, 645	123, 645	327, 805	306, 805				218, 645	299, 565	209, 645	209, 645	229, 645	209, 645	
a?	144, 885	182, 885	124, 645	170, 645	303, 850	368, 850	369, 850	372, 850	339, 850	322, 885	262, 885	294, 764	234, 795	306, 688	140, 885	183, 850
a<	**147, 885**	**223, 885**	164, 645	**122, 645**	**325, 850**	**388, 850**	**340, 850**	**365, 850**	**332, 850**	**233, 885**	278, 885	**265, 663**			**140, 885**	**276, 850**
a!																
b	146, -114	218, -120	196, -110	153, -114	328, -108	369, -107				212, -114	240, -114	196, -110	212, -114	212, -114	196, -110	
b?	144, -110	182, -110	124, -291	170, -110	303, -110	368, -110	369, -110	372, -110	339, -110	322, -110	262, -110	294, -110	234, -110	306, -110	140, -299	183, -257
b<	**147, -110**	**223, -110**	121, -319	**122, -119**	**326, -110**	**391, -110**	**363, -110**	**367, -110**	**338, -110**	**233, -110**	268, -110	**248, -110**	**184, -110**	200, -110	**129, -319**	**202, -257**
b!														184, -110		
it	*ꞎ*	*ꞎ*	*ȷ̵*	*ɩ̵*	*B̵*	*D̵*	*C̵*	*G̵*	*K̵*	*d̵*	*h̵*	*ɢ̵*	*ɜ̵*	*ə̵*	*ʃ̵*	*F̵*
patch	*ꞎ*	*ꞎ*	*ȷ̵*	*ɩ̵*	*B̵*	*D̵*	*C̵*	*G̵*	*K̵*	*d̵*	*h̵*	*ɢ̵*	*ɜ̵*	*ə̵*	*ʃ̵*	*F̵*
base	*l̵*	*l̵*	*ȷ̵*	*ɩ̵*	*B̵*	*D̵*	*C̵*	*G̵*	*K̵*	*d̵*	*h̵*	*ɢ̵*	*ɜ̵*	*ə̵*	*ʃ̵*	*F̵*
shift	+12, 94%	+86, 94%	*+42, 60%*	-18, 92%	+58, 99%	+80, 98%	*+18, 34%*	*+0, 54%*	*+0, 48%*	+0, 99% s	+6, 76%	-10, 81%	+0, 99%	*+10, 40%*	+4, 95%	+16, 100% s
a				123, 645	327, 805	306, 805				218, 645	299, 565					
a?	333, 890	368, 890	305, 645	266, 645	488, 850	553, 850	503, 850	541, 850	499, 850	464, 890	449, 890	410, 796	413, 775	418, 687	343, 900	419, 850
a<	**324, 890**	**398, 890**	325, 645	**257, 645**	**505, 850**	**551, 850**	518, 850	513, 850	466, 850	**434, 890**		**378, 645**			**347, 900**	**422, 850**
a!																
b	146, -114	218, -120		153, -114	328, -108	369, -107				212, -114	240, -114		212, -114	212, -114		
b?	120, -110	156, -110	100, -319	106, -110	284, -110	349, -110	299, -110	337, -110	295, -110	251, -110	236, -110	217, -110	225, -110	248, -110	88, -299	184, -257
b<	**111, -110**	**185, -110**	104, -320	**97, -110**	**296, -110**	**359, -110**	351, -110	345, -110	292, -110	**221, -110**	**235, -110**	**238, -110**	**143, -110**	179, -110	**65, -319**	**113, -257**
b!														161, -110		
sb	**ꞎ**	**ꞎ**	**ȷ̵**	**ɩ̵**	**B̵**	**D̵**	**C̵**	**G̵**	**K̵**	**d̵**	**h̵**	**ɢ̵**	**ɜ̵**	**ə̵**	**ʃ̵**	**F̵**
patch	**ꞎ**	**ꞎ**	**ȷ̵**	**ɩ̵**	**B̵**	**D̵**	**C̵**	**G̵**	**K̵**	**d̵**	**h̵**	**ɢ̵**	**ɜ̵**	**ə̵**	**ʃ̵**	**F̵**
base	**l̵**	**l̵**	**ȷ̵**	**ɩ̵**	**B̵**	**D̵**	**C̵**	**G̵**	**K̵**	**d̵**	**h̵**	**ɢ̵**	**ɜ̵**	**ə̵**	**ʃ̵**	**F̵**
shift	+12, 91%	+88, 94%	*+18, 60%*	-16, 83%	+58, 90%	+86, 96%	+0, 99%	+0, 100%	+2, 75%	-4, 92% s	-2, 71%	+0, 100%	+0, 98%	*+6, 58%*	+0, 95%	*+18, 62% s*
a																
a?	156, 885	190, 885	144, 645	164, 645	316, 850	381, 850	368, 850	374, 850	338, 850	324, 885	274, 885	290, 764	276, 757	319, 693	152, 885	236, 850
a<	**162, 885**	**238, 885**	181, 645	**140, 645**	**341, 850**	**438, 850**	**332, 850**	**385, 850**	**342, 850**	**270, 885**	**273, 885**	**266, 645**			**152, 885**	
a!																
b	146, -114	218, -120														
b?	156, -110	190, -110	144, -319	164, -110	316, -110	381, -110	368, -110	374, -110	338, -110	324, -110	274, -110	290, -110	276, -110	319, -110	152, -296	236, -281
b<	**162, -110**	**238, -110**	96, -319	**140, -120**	**358, -110**	**438, -110**	**332, -110**	**368, -110**	**342, -110**	**270, -110**	**275, -110**	**266, -110**	**204, -110**	218, -110	**141, -319**	**236, -281**
b!														199, -110		
si	***ꞎ***	***ꞎ***	***ȷ̵***	***ɩ̵***	***B̵***	***D̵***	***C̵***	***G̵***	***K̵***	***d̵***	***h̵***	***ɢ̵***	***ɜ̵***	***ə̵***	***ʃ̵***	***F̵***
patch	***ꞎ***	***ꞎ***	***ȷ̵***	***ɩ̵***	***B̵***	***D̵***	***C̵***	***G̵***	***K̵***	***d̵***	***h̵***	***ɢ̵***	***ɜ̵***	***ə̵***	***ʃ̵***	***F̵***
base	***l̵***	***l̵***	***ȷ̵***	***ɩ̵***	***B̵***	***D̵***	***C̵***	***G̵***	***K̵***	***d̵***	***h̵***	***ɢ̵***	***ɜ̵***	***ə̵***	***ʃ̵***	***F̵***
shift	+28, 74%	+106, 75%	*-4, 48%*	-12, 77%	*+70, 41%*	*+98, 26%*	*+20, 68%*	*+20, 46%*	*-30, 62%*	-5, 82% s	*-4, 55%*	+12, 100%	+0, 98%	*+12, 29%*	+0, 95%	*+15, 61% s*
a				123, 645	327, 805	306, 805				218, 645	299, 565					
a?	187, 890	223, 890	141, 645	138, 645	359, 850	424, 850	359, 850	395, 850	353, 850	318, 890	302, 890	278, 796	261, 762	299, 693	195, 890	288, 850
a<	**208, 890**	**286, 890**	167, 645	**134, 654**	337, 850	404, 850	360, 850	385, 850	339, 850	**310, 890**		**284, 681**			**195, 890**	
a!																
b	146, -114	218, -120		153, -114	328, -108	369, -107				212, -114	240, -114		212, -114	212, -114		
b?	145, -110	181, -110	107, -319	147, -110	326, -110	391, -110	326, -110	362, -110	320, -110	276, -110	260, -110	256, -110	246, -110	298, -110	116, -296	220, -281
b<	**166, -110**	**244, -110**	85, -314	**141, -110**	338, -110	377, -110	383, -110	387, -110	312, -110	**220, -110**	268, -110	**260, -110**	**193, -110**	198, -110	**88, -319**	**220, -281**
b!														188, -110		

Table 19: Derived letters and inherited anchors 3/3

hex	019D	0255	0256	026D	0271	0272	0273	027B	027D	0282	0286	0288	0290	0291	0293
base	004E	0063	0064	006C	006D	006E	006E	0279	0072	0073	0283	0074	007A	007A	0292
reg	Ɲ̲Ɲ̲	ɕ̲	ɖ̲	ɭ̲	ɱ̲ɱ̲	ɲ̲ɲ̲	ɳ̲ɳ̲	ɻ̲	ɽ̲	ʂ̲	ʃ̲	ʈ̲	ʐ̲	ʑ̲	ʒ̲
patch	Ɲ̲Ɲ̲	ɕ̲	ɖ̲	ɭ̲	ɱ̲ɱ̲	ɲ̲ɲ̲	ɳ̲ɳ̲	ɻ̲	ɽ̲	ʂ̲	ʃ̲	ʈ̲	ʐ̲	ʑ̲	ʒ̲
base	N̲N̲	c̲	d̲	l̲	m̲m̲	n̲n̲	n̲n̲	ɹ̲	r̲	s̲	ʃ̲	t̲	z̲	z̲	ʒ̲
shift	-1, 96% s	+1, 91% s	-3, 95% s	-3, 100% s	-3, 97% s	-11, 96% s	-11, 96% s	+20, 75% s	-16, 72% s	+0, 100% s	+0, 100% s	-3, 86% s	+0, 100% s	-14, 79% s	*+4, 38% s*
a		209, 645	218, 645	209, 645	383, 645	264, 646	263, 645		209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645
a?	324, 850	222, 645	314, 885	182, 885	366, 645	234, 645	296, 645	226, 645	186, 645	199, 645	150, 885	158, 771	262, 645	218, 662	220, 645
a<	**345, 850**	**243, 645**	**232, 885**	**132, 885**	**389, 645**	**253, 645**	**253, 645**	247, 645	**175, 645**	**206, 645**	**140, 885**	**136, 769**	**220, 645**	**206, 645**	
a!								248, 645							222, 645
b		212, -114	212, -114	196, -110	383, -105	265, -111	262, -112	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110
b?	324, -272	222, -214	314, -273	182, -273	366, -306	234, -306	296, -306	226, -259	186, -319	199, -261	150, -310	158, -306	262, -270	218, -207	220, -319
b<	**347, -110**		**232, -273**	**132, -273**	**398, -306**	**254, -306**	**254, -306**	**198, -259**	**124, -319**	**196, -261**	**129, -310**	**160, -306**	**207, -270**	**192, -207**	**220, -319**
b!															
it	*Ɲ̲Ɲ̲*	*ɕ̲*	*ɖ̲*	*ɭ̲*	*ɱ̲ɱ̲*	*ɲ̲ɲ̲*	*ɳ̲ɳ̲*	*ɻ̲*	*ɽ̲*	*ʂ̲*	*ʃ̲*	*ʈ̲*	*ʐ̲*	*ʑ̲*	*ʒ̲*
patch	*Ɲ̲Ɲ̲*	*ɕ̲*	*ɖ̲*	*ɭ̲*	*ɱ̲ɱ̲*	*ɲ̲ɲ̲*	*ɳ̲ɳ̲*	*ɻ̲*	*ɽ̲*	*ʂ̲*	*ʃ̲*	*ʈ̲*	*ʐ̲*	*ʑ̲*	*ʒ̲*
base	*N̲N̲*	*c̲*	*d̲*	*l̲*	*m̲m̲*	*n̲n̲*	*n̲n̲*	*ɹ̲*	*r̲*	*s̲*	*ʃ̲*	*t̲*	*z̲*	*z̲*	*ʒ̲*
shift	+7, 99% s	+7, 83% s	*+10, 4% s*	+0, 91% s	*-13, 70% s*	+18, 79% s	-1, 95% s	*+111, 58% s*	*+4, 61% s*	*+11, 19% s*	+0, 100% s	+2, 84% s	*-5, 22% s*	*-1, 15% s*	*+28, 26% s*
a			218, 645		383, 645	264, 646	263, 645								
a?	528, 850	352, 645	465, 899	341, 890	515, 645	396, 645	367, 645	352, 645	320, 645	314, 645	343, 900	314, 776	343, 645	352, 645	355, 645
a<	**483, 850**	**350, 645**		312, 890		**414, 654**	**393, 645**				**343, 900**	**325, 798**			
a!			440, 899	312, 890				389, 645							
b		212, -114	212, -114		383, -105	265, -111	262, -112								
b?	290, -272	171, -208	215, -273	91, -287	310, -319	191, -319	162, -319	160, -259	124, -279	121, -265	92, -283	83, -311	147, -276	171, -207	133, -397
b<	**337, -110**		**215, -273**	**61, -287**	**310, -319**	**209, -319**	**190, -319**	**160, -259**	**124, -279**	**121, -265**	**69, -283**	**92, -311**	**147, -276**	**171, -207**	133, -397
b!															133, -399
sb	**Ɲ̲Ɲ̲**	**ɕ̲**	**ɖ̲**	**ɭ̲**	**ɱ̲ɱ̲**	**ɲ̲ɲ̲**	**ɳ̲ɳ̲**	**ɻ̲**	**ɽ̲**	**ʂ̲**	**ʃ̲**	**ʈ̲**	**ʐ̲**	**ʑ̲**	**ʒ̲**
patch	**Ɲ̲Ɲ̲**	**ɕ̲**	**ɖ̲**	**ɭ̲**	**ɱ̲ɱ̲**	**ɲ̲ɲ̲**	**ɳ̲ɳ̲**	**ɻ̲**	**ɽ̲**	**ʂ̲**	**ʃ̲**	**ʈ̲**	**ʐ̲**	**ʑ̲**	**ʒ̲**
base	**N̲N̲**	**c̲**	**d̲**	**l̲**	**m̲m̲**	**n̲n̲**	**n̲n̲**	**ɹ̲**	**r̲**	**s̲**	**ʃ̲**	**t̲**	**z̲**	**z̲**	**ʒ̲**
shift	*+8, 37% s*	+2, 75% s	-2, 93% s	*+0, 0% s*	-10, 97% s	-13, 97% s	-15, 89% s	+55, 97% s	-16, 92% s	+0, 100% s	-1, 100% s	-6, 94% s	+0, 100% s	*-2, 65% s*	*+9, 49% s*
a															
a?	336, 850	232, 645	322, 885	194, 885	363, 645	246, 645	306, 645	238, 645	198, 645	201, 645	162, 885	182, 782	254, 645	230, 645	236, 645
a<		**226, 645**	**272, 885**		**386, 645**	**277, 645**	**275, 645**	**260, 645**	**196, 645**	**201, 645**	**151, 885**	**162, 782**	**222, 645**		
a!															231, 645
b															
b?	336, -270	232, -209	322, -271	194, -319	363, -319	246, -319	306, -319	238, -257	198, -319	201, -264	162, -319	182, -319	254, -276	230, -210	236, -413
b<	**336, -270**		**272, -271**	194, -319	**386, -319**	**253, -319**	**250, -319**	**196, -257**	**135, -319**	**201, -264**	**141, -319**	**157, -319**	**222, -276**	**230, -210**	236, -413
b!															236, -415
si	***Ɲ̲Ɲ̲***	***ɕ̲***	***ɖ̲***	***ɭ̲***	***ɱ̲ɱ̲***	***ɲ̲ɲ̲***	***ɳ̲ɳ̲***	***ɻ̲***	***ɽ̲***	***ʂ̲***	***ʃ̲***	***ʈ̲***	***ʐ̲***	***ʑ̲***	***ʒ̲***
patch	***Ɲ̲Ɲ̲***	***ɕ̲***	***ɖ̲***	***ɭ̲***	***ɱ̲ɱ̲***	***ɲ̲ɲ̲***	***ɳ̲ɳ̲***	***ɻ̲***	***ɽ̲***	***ʂ̲***	***ʃ̲***	***ʈ̲***	***ʐ̲***	***ʑ̲***	***ʒ̲***
base	***N̲N̲***	***c̲***	***d̲***	***l̲***	***m̲m̲***	***n̲n̲***	***n̲n̲***	***ɹ̲***	***r̲***	***s̲***	***ʃ̲***	***t̲***	***z̲***	***z̲***	***ʒ̲***
shift	-23, 99% s	*+1, 67% s*	-2, 81% s	+19, 93% s	*-7, 66% s*	*-3, 69% s*	*-5, 67% s*	+35, 86% s	*-10, 50% s*	*+7, 39% s*	+1, 100% s	-9, 84% s	*-4, 11% s*	*+0, 4% s*	*+3, 56% s*
a			218, 645		383, 645	264, 646	263, 645								
a?	383, 850	204, 645	302, 890	187, 890	373, 645	254, 645	254, 645	194, 645	177, 645	178, 645	189, 890	172, 787	202, 645	208, 645	211, 645
a<	**366, 850**		**311, 890**	205, 890				**246, 645**			**195, 890**	**184, 787**			
a!				205, 890											213, 645
b		212, -114	212, -114		383, -105	265, -111	262, -112								
b?	317, -270	192, -209	227, -271	103, -319	339, -319	220, -319	220, -319	172, -257	143, -319	155, -265	105, -319	109, -319	176, -279	196, -210	157, -413
b<	**288, -270**		**189, -271**	**120, -319**	**339, -319**	**220, -319**	**220, -319**	**185, -257**	**143, -319**	**155, -265**	**88, -319**	**81, -319**	**176, -279**	**196, -210**	157, -413
b!															157, -415

Anchor survey, with patches

Libertinus serif regular has good above- and below-anchor base coverage for A–Z, a–z, and IPA letters, in the sense that most anchors are defined. But I will show that some anchors are not set to good values. The base coverage gets worse for italic and semibold.

I will systematically survey above- and below anchors in the four styles, commenting on any features and patching any problems. Refer to Tables 1–4 for anchor values. Missing anchors are readily apparaent as blank entries. Large anchor-midpoint offset values — $a\delta$ and $b\delta$ — indicate possible problems, but these can often be explained by glyph symmetry.

Copied anchors

Many designer anchors are copies: within a style, clusters of letters share one above- or below-anchor exactly (a's $(209, 645)$ and $(196, -110)$ on some 40 IPA letters each. Some copies are harmless coincidences of similar letterforms; most are not.

`run_anchor_copy_report.py` lists the clusters (`tex/input/copied_anchors.tex`, no longer printed here). On 2026-11-07 every member of the regular clusters, and its counterparts in the other styles, was reviewed per character: each anchor is now the candidate, inherited from its base letter, or curated by a stated rule, the same way in all four styles.

A–Z, regular, above- and below-anchor

The above-anchors of A–Z regular are good.

The above-anchor of D is just right of the major stem, not right enough to be centered over the bowl. For D, $ax = 306$. Consider the precomposed characters Ď (U+010E) and Ḋ (U+1E0A). The designer drew their glyphs with the mark centered at $X = 312$ and 296 respectively. So, this is a deliberate design decision to keep the mark closer to the stem, not centered over the bowl.

Similarly, the above anchor of F is just right of the major stem, not right enough to be centered over cross strokes. For F, $ax = 284$. For Ḟ (U+1E1E), the dot is centered at $X = 254$. So, the designer is deliberately choosing to keep the mark cloder to the stem.

The above-anchor of L is centered over the major stem, which is a valid design choice.

The below-anchors of A–Z regular are good. But J's below anchor is too low, and Q is missing a below-anchor. Even though J and Q arguably do not need below-anchors for linguistic notation, I will set this anchor by the heuristic algorithm.

Patch J regular, below-anchor

J's below-anchor is too low. For J, $by = -319$, but J is not a full descender; its bbox $Y_{\text{min}} = -172$, which is well above $yrd = -232$. Yes, the designer set $by = byrd = -319$ for J in all fout font styles, but this is arguably too low. According to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\text{min}} - by\delta d + \epsilon = -172 - 87 + 5 = -254$, where $\epsilon = 5$ accounts for the overshoot of J's tail.

Patch Q regular, below-anchor

Q has bbox $Y_{\text{min}} = -209$. Again, according to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\text{min}} - by\delta d + \epsilon = -209 - 87 + 5 = -291$, where $\epsilon = 5$ accounts for the overshoot of Q's tail. For bx, observe that O and Q have the same counter outline, literally. So, the below-anchor for Q should be at $bx = 350$, the same as that for O.

a–z, regular, above- and below-anchor

b, f, j, and t are missing above-anchors. Some letters (n, p, r, v, w, x, z) should have ay regularized to 645, not 646. For d, h, and k, the above-anchor is centered above the body, not the ascender; this is good design choice, but care must be taken to avoid collision between an above-mark and an ascender. For d and h, ay is at the x-height anchor reference, and therefore

wide marks collide with the ascender. For k, ay is a bit higher, but not at the full ascender anchor references, and therefore the widest marks collide with the ascender.
The below-anchor of k is too low. Otherwise, the below-anchors of a–z are good.

Patch k regular, below-anchor

For k, $by - 214$. But it should be at $byr0 = -110$, like all other letters that sit on the baseline.
Here is this positioning of k in a patched version of the font (with o for comparison only).
Original: k̲ k̲o o̲k̲
Patched: k̲ k̲o o̲k̲

Patch b, d, f, h regular, above-anchor

For b, d, f, and h, set $ay = ayra = 885$.
For ax for b, d, f, and h, there are at least two strategies. First, set ax at the bowl apex. So, for f, set $ax = 279$ at the apex of the bowl. But for b, d, and h, you could set ax centered horizontally over the bowl, in which case you also need to decide which two vertical aspects of the bowl should be used for the centering.

Patch j regular, above-anchor

j needs an above anchor for the rare case of an above-mark that is placed over the dot. The ccmp feature swaps i and j for their dotless forms before every above-mark in its list, so only a mark outside the list attaches to i or j itself; among the IPA marks that is only the double overline U+033F. Such a mark must clear the dot. The designer's $(137, 795)$ on i (and the earlier curated j at $(126, 795)$) put the double overline only 13–16 above the dot in regular and semibold, against 69–74 above o. The patch (2026-11-07) curates i and j in every style on the stem's axis, at the candidate's Y, the dot top plus the ascender clearance (regular 834, italic 810, semibold 837, semibold italic 823), so the double overline sits 55–60 above the dot. The candidate's X is right for i but not for j, whose hook pulls the bounding-box midpoint left of the stem (by 84 in semibold).

Patch n, p, r, v, w, x, z regular, above-anchor

In my patched version of the font, I regularize ay to $ayrx = 645$ for all letters that sit at the x-height, including n, p, r, v, w, x, z.

Above-anchor, a–z regular

b, f, j, and t are missing the above-anchor. d and h have an above-anchor above the meanline but not above the ascender. l has an above-anchor above the ascender. k has an above-anchor, but it is just below the ascender, so any marks other than the do crash into the ascender. These are all legitimate design choices. There are no above-marks, other than those with precomposed characters, the would ever be combined with b, f, or t.

Patch j dot removal for three above-marks

The dot of i and j should be removed before adding any above-mark. But the dot is not removed for three above-marks: x, turned tilde, double macron. So the GSUB table should be patched to fir the dotless j substitution in this context.
To do.

Capital mark alternates

Libertinus has custom glyphs for some marks (acute, grave, cicrumflex, breve, a.o.) that are designed better for capital letters. These alternates are flatter. GSUB lookups 23 and 24 implement the substitution. This substitution is not fired on some capital consonants (T, V, W, X) because these marks are not expected on these bases.

Libertinus has custom glyphs for some marks that are designed better for superscripts. These alternates are smaller. Regrettably?, no GSUB table implements their substitution.

A cirmumflex Â Ã̂ *Â* **Ã̂** ***Â***
O circumflex Ô Õ̂ *Ô* **Õ̂** ***Ô***
AE ligature Æ Æ̀ *Æ* **Æ̀** ***Æ***
Ṡ Ṫ V̇ *Ṡ* *Ṫ* *V̇* **Ṡ** **Ṫ** **V̇** ***Ṡ*** ***Ṫ*** ***V̇***
Ṡ Ṫ V̇ *Ṡ* *Ṫ* *V̇* **Ṡ** **Ṫ** **V̇** ***Ṡ*** ***Ṫ*** ***V̇***

Regular, above-anchor, capitals, X coordinate

Figure 9 (left panel) shows pairs of bases aligned at their above-anchor. If a capital letter has a bad X coordinate for its above-anchor, then it will look always off, right or left, in its black column relative to the gray letters that it overlays. Note that the above-anchor X coordinate is not in the geometric (horizontal) center of the glyph, by design. The above-anchor X coordinate is always off center by a few UPM units, sometimes left and sometimes right. But it very hard to discern a few UPM units. See the right panel of Figure 9 for an illustration of increasingly bad (spoofed) X coordinates of N's above-anchor.

Note that the above-anchor of L is above the major (vertical) stem of the L. For E, F, P and R, the above-anchor is optically centered above the glyph, not above the major stem.

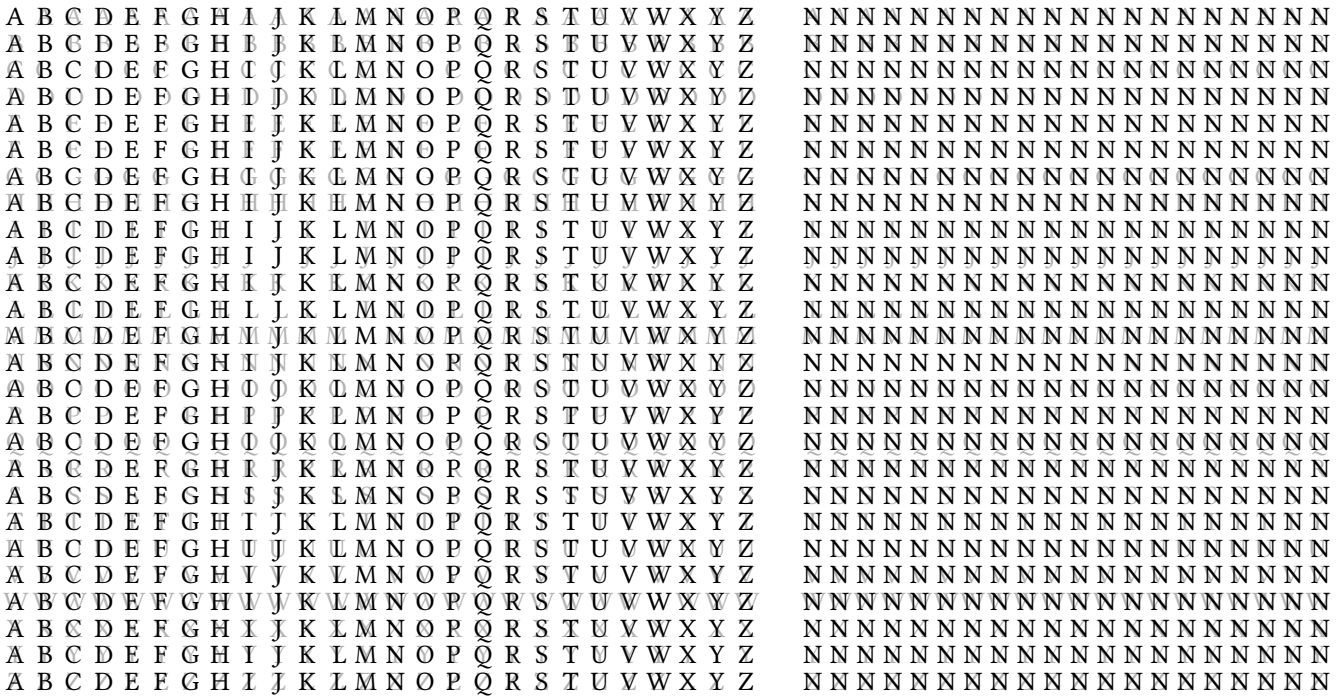


Figure 9: Above-anchor X coordinate alignment. In the left panel, the 26×26 grid shows letters A–Z aligned at the their above-anchor. The right panel shows how hard it is to discern bad X coordinates. The right panel shows N aligned with letters at their above-anchor, starting at the first column with Ns above-anchor as currently set by the font, and then spoofing increasing bad values for the X coordinate of N's above-anchor, by 5 UPM per column. At the rightmost column, it becomes evident that the (spoofed) X coordinate of N's above-anchor is bad, because in that column the black Ns are too far right of the gray letters below them.

Comma-style marks

Libertinus has a some (precomposed) glyphs for Unicode characters (with code points) where the caron or cedilla has a comma-style shape, instead of a wedge or hook as might be naively expected. But all of these are appropriate for Czech/Slovak orthography (caron or *háček*) or Latvian orthography (cedilla, but really the comma-above mark or *komatiņš*, by a known mistake in Unicode).

Czech/Slovak *háček*

In Czech/Slovak orthography, the d, l, L, n and t may take a reduced caron that may look like a comma, not a wedge. In Libertinus, only the d, L, l, and t have the reduced comma-style caron in their precomposed form, and the n has wedge caron precomposed form.

`\char"010F` ď *ď* **ď** ***ď***

`\char"013D` Ľ *Ľ* **Ľ** ***Ľ***

`\char"013E` ľ *ľ* **ľ** ***ľ***

`\char"0148` ň *ň* **ň** ***ň***

`\char"0165` ť *ť* **ť** ***ť***

Libertinus even has a custom glyph for a small capital L with a caron, and it is comma-style.

`\textsc{l\char"030C}` ʟ̬

`\textsc{\char"013E}` ʟ̬

Cedilla and Latvian *komatiņš*

Here are all precomposed characters with a cedilla.

`\char"00C7` Ç *Ç* **Ç** ***Ç***	`\char"00E7` ç *ç* **ç** ***ç***
`\char"1E08` Ḉ *Ḉ* **Ḉ** ***Ḉ***	`\char"1E08` ḉ *ḉ* **ḉ** ***ḉ***
`\char"1E10` Ḑ *Ḑ* **Ḑ** ***Ḑ***	`\char"1E11` ḑ *ḑ* **ḑ** ***ḑ***
`\char"0228` Ȩ *Ȩ* **Ȩ** ***Ȩ***	`\char"0229` ȩ *ȩ* **ȩ** ***ȩ***
`\char"1E1C` Ḝ *Ḝ* **Ḝ** ***Ḝ***	`\char"1E1D` ḝ *ḝ* **ḝ** ***ḝ***
`\char"0122` Ģ *Ģ* **Ģ** ***Ģ***	`\char"0123` ģ *ģ* **ģ** ***ģ***
`\char"1E28` Ḩ *Ḩ* **Ḩ** ***Ḩ***	`\char"1E29` ḩ *ḩ* **ḩ** ***ḩ***
`\char"0136` Ķ *Ķ* **Ķ** ***Ķ***	`\char"0137` ķ *ķ* **ķ** ***ķ***
`\char"013B` Ļ *Ļ* **Ļ** ***Ļ***	`\char"013C` ļ *ļ* **ļ** ***ļ***
`\char"0145` Ņ *Ņ* **Ņ** ***Ņ***	`\char"0146` ņ *ņ* **ņ** ***ņ***
`\char"0156` Ŗ *Ŗ* **Ŗ** ***Ŗ***	`\char"0157` ŗ *ŗ* **ŗ** ***ŗ***
`\char"015E` Ş *Ş* **Ş** ***Ş***	`\char"015F` ş *ş* **ş** ***ş***
`\char"0162` Ţ *Ţ* **Ţ** ***Ţ***	`\char"0163` ţ *ţ* **ţ** ***ţ***

These characters – Ģ ģ Ķ ķ Ļ ļ Ņ ņ Ŗ ŗ – use the Latvian *komatiņš* which is proper for Latvian orthography, but the result of a historical mistake in Unicode naming.

The Unicode characters of D and d with cedilla exist for scholarly and historical reasons, not Latvian orthography. Fonts design these Unicode characters differently. Libertinus designs them inconsistently: comma-style in regular and italic, and hook-style in (semi)bold and (semi)bold italic. I think the cedilla on a D or d should be hook-like, but I do not really know.

The other characters – c, e, h, s, t – with a cedilla should definitely be hook-like, and they are so in Libertinus.

Somewhat mysteriously, if the LaTeX source tries to combine any of these letters – c, d, e, g, h, k, l, n, r, s, t – with the combining cedilla, U+0327, then the precomposed character is produced.

c\char"0327 ç *ç* **ç** ***ç***
d\char"0327 ḑ *ḑ* **ḑ** ***ḑ***
e\char"0327 ȩ *ȩ* **ȩ** ***ȩ***
g\char"0327 ģ *ģ* **ģ** ***ģ***
h\char"0327 ḩ *ḩ* **ḩ** ***ḩ***
k\char"0327 ķ *ķ* **ķ** ***ķ***
l\char"0327 ļ *ļ* **ļ** ***ļ***
n\char"0327 ņ *ņ* **ņ** ***ņ***
r\char"0327 ŗ *ŗ* **ŗ** ***ŗ***
s\char"0327 ş *ş* **ş** ***ş***
t\char"0327 ţ *ţ* **ţ** ***ţ***

This behavior does not make sense for the Latvian *komatiņš* bases – g, k, l, n, r – because any person trying to produce Latvian g with *komatiņš* would not try `g\char"0327` or `\c{g}`.

This behavior is tolerable for the other bases – c, d, e, h, s, t.

Æ: a letter the midpoint cannot centre

Why Æ has curated anchors in every style rather than the anchor model's candidate (see "Where candidates fail" in `ANCHOR-HEURISTICS.md`).

Over Æ, accents do not go at the midpoint of the glyph. The left leg of the A stops short of the top, so the top edge runs from the apex of the A to the end of the arm of the E, and its centre lies well to the right. Other fonts agree: Charis SIL, Gentium Plus, EB Garamond, Junicode, DejaVu Serif and Andika put the above-anchor at the macron of Ǣ, within 5 units, and below they use the common stroke or the same X. In Libertinus the designer's above-anchors missed the accents of Ǣ and Ǽ by 55–88 and 16–68 units, so the curated above-anchors lie between the two (555, 674, 555, 545 for regular, italic, semibold, semibold italic). In the italic styles the macron and the acute cannot both be met, and each lands about 30 units from its precomposed position. Below, semibold's anchor is curated under the common stroke, at 510. The designer's 549 of the other three styles stands: nothing collides there.

IJ

The ij and IJ ligatures are Unicode characters (with code points) and are supported by Libertinus.

IJ `\char"0132` U+0132 capital ligature IJ

ij `\char"0133` U+0133 small ligature ij

Libertinus has custom glyphs for these ligatures with an acute, and the small capital ligature with an acute. The GSUB table that activates these custom glyphs only works when the language is set to Dutch by a font loading command. For example,

```
\newfontfamily\SerifRegularDutch[Language = Dutch]{LibertinusSerif-Regular.otf}
```

ĲĳĲ `{\SerifRegularDutch \char"0132\char"0301}` Dutch capital ligature IJ with acute

ĳ́ `{\SerifRegularDutch \char"0133\char"0301}` Dutch small ligature ij with acute

ĳ́ `{\SerifRegularDutch \textsc{\char"0133\char"0301}}` Dutch small capital ligature IJ with acute

In an English LaTeX document, `ij` and `IJ` are kerned by the GPOS table to produce ij and IJ, so they look close together, but they are not substituted by the single glyph for the ligature.

Lowercase i with tilde below and grave above, in italic and semibold italic

First, start with i with tilde below (precomposed).

ḭ *ḭ* **ḭ** ***ḭ*** `\char"1E2D` i with tilde below (precomposed)

You cannot simply put a grave on it because it collides with the dot.

ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"1E2D\char"0300` i with tilde below (precomposed) with grave

Start over with i with grave (precomposed).

ì *ì* **ì** ***ì*** `\char"00EC` i with grave (precomposed)

Try putting a grave below.

ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"00EC\char"0330` i with grave (precomposed) with tilde below

This fails, because normalization breaks down the input “i with grave (precomposed) with tilde below” into “i with grave with tilde below”, then into canonical order “i with tilde below with grave”, the recomposes it into “i with tilde below (precomposed) with grave”, which reintroduces i’s dot.

Next, try to block normalization with the combining grapheme joiner (CGJ).

ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"00EC\cgj\char"0330` i with grave (precomposed) with combining grapheme joiner with tilde below

This looks bad because i with grave (precomposed) does not have a below-anchor and fallback shaping looks bad.

Finally, try starting with dotless i, which has OK above- and below-anchors.

ı̰ *ı̰* **ı̰** ***ı̰*** `\char"0131\char"0330\char"0300` dotless i with tilde below with grave

Normalization does not affect the dotless i, so it looks the same with the CGJ.

ı̰ *ı̰* **ı̰** ***ı̰*** `\char"0131\cgj\char"0330\char"0300` dotless i with CGJ with tilde below with grave

All together now, for comparison.

ḭ ḭ̀ *ḭ* *ḭ̀* **ḭ** **ḭ̀** ***ḭ*** ***ḭ̀*** `\char"1E2D \char"00EC \char"0131\char"0330\char"0300` i with tilde below (precomposed); i with grave (precomposed); dotless i with tilde below with grave

The same strategy works with other above marks.

ı̰̏ *ı̰̏* **ı̰̏** ***ı̰̏*** `\char"0131\char"0330\char"030F` dotless i with tilde below with double grave

ı̰́ *ı̰́* **ı̰́** ***ı̰́*** `\char"0131\char"0330\char"0301` dotless i with tilde below with acute

ı̰̋ *ı̰̋* **ı̰̋** ***ı̰̋*** `\char"0131\char"0330\char"030B` dotless i with tilde below with double acute

Lowercase a with double grave, followed by floating double grave, in italic

Superseded by “Floating tones” below: write `\floattone{a}{\char"030F}`.

In italic, we want to produce a (precomposed) vowel with a double grave, followed by floating double grave. Ideally, the two double graves should be aligned. We cannot use the combining double grave as the floater, because it has nothing to combine to, and we cannot control its height. So we use a spacing character, specifically the double prime character (‶ `\char"2036`), which looks like a double grave, in italic.

ä‶ `\textit{ä\char"2036}}`

Then we lower it by a few points.

ä‶ `\textit{ä\raisebox{-0.05em}{\char"2036}}`

The distance to move the glyph can be calculated from its SFD splineset, or by trial and error.

Note that only a, e, and i have precomposed glyphs with double acute.

ä‶ `\textit{ä\raisebox{-0.05em}{\char"2036}}`

ë" \textit{è\raisebox{-0.05em}{\char"2036}}

ë" \textit{è\raisebox{-0.045em}{\char"2036}}

ɛ" \textit{\char"025B\raisebox{-0.05em}{\char"2036}}

Tilde below, double acute above, double grave above, on vowels, in italic and semibold italic

Consider the tilde below. There is the combining tilde below, U+0330, encoded in LaTeX as the codepoint \char"0330. Remember to put the base before the combining mark in LaTeX, e.g. a\char"0330 a̰. Three vowels (e, i, u) have precomposed characters – lowercase and capital – encoded in LaTeX as these codepoints:

\char"1E1A \char"1E1B \char"1E2C \char"1E2D \char"1E74 \char"1E75 Ḛ ḛ Ḭ ḭ Ṵ ṵ *Ḛ ḛ Ḭ ḭ Ṵ ṵ* **Ḛ ḛ Ḭ ḭ Ṵ ṵ** ***Ḛ ḛ Ḭ ḭ Ṵ ṵ***

E, e, I, i, U, or u followed by \char"0330 will produce the precomposed character glyph. A, a, O, and o, as well as any other IPA vowel with a below-anchor, followed by \char"0330 will be shaped by attaching the combining below tilde to that vowel at their below-anchor. Note the specific case of i. i\char"0330 = \char"1E2D ḭ = ḭ is a dotted i (a precomposed glyph).

Suppose you want a vowel with a tilde below, possibly also with a mark above, especially acute, grave, caron, double acute, or double grave. In the specific case of i, the above mark will crash into the dotted i unless you take care to encode the triplet starting with a dotless i, U+0131. For example, consider the combining acute U+0301.

i\char"0330\char"0301 = \char"1E2D\char"0301 ḭ́ = ḭ́ *ḭ́ = ḭ́*

\char"0131\char"0330\char"0301 ı̰́ *ı̰́*

Recall that the canonical order for a triplet of a base and two combining marks (one below and one above) is base + combining below mark + combining above mark, not base + above + below.

Consider the wrong order:

i\char"0301\char"0330 ḭ́ *ḭ́* i\char"0301 í *í* i\char"0330 ḭ *ḭ*

\char"00EC\char"0330 ḭ̀ *ḭ̀*

\char"1E2D\char"0301 ḭ́ *ḭ́*

But note that the position of the combining tilde below is bad in italic and semibold italic. This is because the below-anchor for the combining tilde below mark is bad. For tilde below, the precomposed character glyphs with i, e, and u are sometimes good and sometimes bad in italic and semibold italic.

Precomposed characters are blue and therefore cannot be changed unless their glyph outline is changed. Black combinations of vowels and marks are shaped by anchors and therefore can be changed by setting good anchors.

Use the combining grapheme joiner (CGJ) character U+034F, \char"034F, or \cgj if loaded by some package, to force the shaper to use combining marks instead of precomposed glyphs.

A E I O U Ɛ Ɔ a e i o u ɛ ɔ *A E I O U Ɛ Ɔ a e i o u ɛ ɔ* **A E I O U Ɛ Ɔ a e i o u ɛ ɔ** ***A E I O U Ɛ Ɔ a e i o u ɛ ɔ***

1 Á É Í Ó Ú Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́ *Á É Í Ó Ú Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́* **Á É Í Ó Ú Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́** ***Á É Í Ó Ú Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́***

2 À È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀ *À È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀* **À È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀** ***À È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀***

3 Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆* **Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆** ***Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆***

4 Ắ Ĕ́ Ĭ́ Ő Ű Ɛ̆́ Ɔ̆́ ắ ĕ́ ĭ́ ő ű ɛ̆́ ɔ̆́ *Ắ Ĕ́ Ĭ́ Ő Ű Ɛ̆́ Ɔ̆́ ắ ĕ́ ĭ́ ő ű ɛ̆́ ɔ̆́* **Ắ Ĕ́ Ĭ́ Ő Ű Ɛ̆́ Ɔ̆́ ắ ĕ́ ĭ́ ő ű ɛ̆́ ɔ̆́** ***Ắ Ĕ́ Ĭ́ Ő Ű Ɛ̆́ Ɔ̆́ ắ ĕ́ ĭ́ ő ű ɛ̆́ ɔ̆́***

5 Ằ Ĕ̀ Ĭ̀ Ȍ Ű̀ Ɛ̆̀ Ɔ̆̀ ằ ĕ̀ ĭ̀ ȍ ű̀ ɛ̆̀ ɔ̆̀ *Ằ Ĕ̀ Ĭ̀ Ȍ Ű̀ Ɛ̆̀ Ɔ̆̀ ằ ĕ̀ ĭ̀ ȍ ű̀ ɛ̆̀ ɔ̆̀* **Ằ Ĕ̀ Ĭ̀ Ȍ Ű̀ Ɛ̆̀ Ɔ̆̀ ằ ĕ̀ ĭ̀ ȍ ű̀ ɛ̆̀ ɔ̆̀** ***Ằ Ĕ̀ Ĭ̀ Ȍ Ű̀ Ɛ̆̀ Ɔ̆̀ ằ ĕ̀ ĭ̀ ȍ ű̀ ɛ̆̀ ɔ̆̀***

6 Ą Ę Į Ǫ Ų Ɛ̨ Ɔ̨ ą ę į ǫ ų ɛ̨ ɔ̨ *Ą Ę Į Ǫ Ų Ɛ̨ Ɔ̨ ą ę į ǫ ų ɛ̨ ɔ̨* **Ą Ę Į Ǫ Ų Ɛ̨ Ɔ̨ ą ę į ǫ ų ɛ̨ ɔ̨** ***Ą Ę Į Ǫ Ų Ɛ̨ Ɔ̨ ą ę į ǫ ų ɛ̨ ɔ̨***

7 Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́***

7c Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́***

8 Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀***

9 Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆ *Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆* **Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆** ***Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆***

10 Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ̨̋ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ̨̋ ɔ̨̋ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ̨̋ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ̨̋ ɔ̨̋* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ̨̋ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ̨̋ ɔ̨̋** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ̨̋ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ̨̋ ɔ̨̋***

11 A̰ Ḛ Ḭ O̰ Ṵ Ɛ̰ Ɔ̰ a̰ ḛ ḭ o̰ ṵ ɛ̰ ɔ̰ *A̰ Ḛ Ḭ O̰ Ṵ Ɛ̰ Ɔ̰ a̰ ḛ ḭ o̰ ṵ ɛ̰ ɔ̰* **A̰ Ḛ Ḭ O̰ Ṵ Ɛ̰ Ɔ̰ a̰ ḛ ḭ o̰ ṵ ɛ̰ ɔ̰** ***A̰ Ḛ Ḭ O̰ Ṵ Ɛ̰ Ɔ̰ a̰ ḛ ḭ o̰ ṵ ɛ̰ ɔ̰***
Patched
A E I O U Ɛ Ɔ a e i o u ɛ ɔ *A E I O U Ɛ Ɔ a e i o u ɛ ɔ* **A E I O U Ɛ Ɔ a e i o u ɛ ɔ** ***A E I O U Ɛ Ɔ a e i o u ɛ ɔ***
Á É Í Ó Ú Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́ *Á É Í Ó Ú Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́* **Á É Í Ó Ú Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́** ***Á É Í Ó Ú Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́***
À È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀ *À È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀* **À È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀** ***À È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀***
Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆* **Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆** ***Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆***
Ő Ű Ɛ̋ Ɔ̋ ő ű ɛ̋ ɔ̋ *Ő Ű Ɛ̋ Ɔ̋ ő ű ɛ̋ ɔ̋* **Ő Ű Ɛ̋ Ɔ̋ ő ű ɛ̋ ɔ̋** ***Ő Ű Ɛ̋ Ɔ̋ ő ű ɛ̋ ɔ̋***
Ä Ë Ï Ö Ü Ɛ̈ Ɔ̈ ä ë ï ö ü ɛ̈ ɔ̈ *Ä Ë Ï Ö Ü Ɛ̈ Ɔ̈ ä ë ï ö ü ɛ̈ ɔ̈* **Ä Ë Ï Ö Ü Ɛ̈ Ɔ̈ ä ë ï ö ü ɛ̈ ɔ̈** ***Ä Ë Ï Ö Ü Ɛ̈ Ɔ̈ ä ë ï ö ü ɛ̈ ɔ̈***
Ą Ę Į Ǫ Ų Ɛ̨ Ɔ̨ ą ę į ǫ ų ɛ̨ ɔ̨ *Ą Ę Į Ǫ Ų Ɛ̨ Ɔ̨ ą ę į ǫ ų ɛ̨ ɔ̨* **Ą Ę Į Ǫ Ų Ɛ̨ Ɔ̨ ą ę į ǫ ų ɛ̨ ɔ̨** ***Ą Ę Į Ǫ Ų Ɛ̨ Ɔ̨ ą ę į ǫ ų ɛ̨ ɔ̨***
cgj Ą Ę Į Ǫ Ų Ɛ̨ Ɔ̨ ą ę į ǫ ų ɛ̨ ɔ̨ *Ą Ę Į Ǫ Ų Ɛ̨ Ɔ̨ ą ę į ǫ ų ɛ̨ ɔ̨* **Ą Ę Į Ǫ Ų Ɛ̨ Ɔ̨ ą ę į ǫ ų ɛ̨ ɔ̨** ***Ą Ę Į Ǫ Ų Ɛ̨ Ɔ̨ ą ę į ǫ ų ɛ̨ ɔ̨***
Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́***
cgj Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́***
Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ɔ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́***
Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀***
cgj Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ɔ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀***
Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆ *Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆* **Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆** ***Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆***
cgj Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆ *Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆* **Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆** ***Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ɔ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆***
Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ̨̋ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ̨̋ ɔ̨̋ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ̨̋ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ̨̋ ɔ̨̋* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ̨̋ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ̨̋ ɔ̨̋** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ̨̋ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ̨̋ ɔ̨̋***
cgj Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ̨̋ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ̨̋ ɔ̨̋ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ̨̋ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ̨̋ ɔ̨̋* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ̨̋ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ̨̋ ɔ̨̋** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ̨̋ Ɔ̨̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ̨̋ ɔ̨̋***
Ä̰ Ḛ̈ Ḭ̈ Ö̰ Ṵ̈ Ɛ̰̈ Ɔ̰̈ ä̰ ḛ̈ ḭ̈ ö̰ ṵ̈ ɛ̰̈ ɔ̰̈ *Ä̰ Ḛ̈ Ḭ̈ Ö̰ Ṵ̈ Ɛ̰̈ Ɔ̰̈ ä̰ ḛ̈ ḭ̈ ö̰ ṵ̈ ɛ̰̈ ɔ̰̈* **Ä̰ Ḛ̈ Ḭ̈ Ö̰ Ṵ̈ Ɛ̰̈ Ɔ̰̈ ä̰ ḛ̈ ḭ̈ ö̰ ṵ̈ ɛ̰̈ ɔ̰̈** ***Ä̰ Ḛ̈ Ḭ̈ Ö̰ Ṵ̈ Ɛ̰̈ Ɔ̰̈ ä̰ ḛ̈ ḭ̈ ö̰ ṵ̈ ɛ̰̈ ɔ̰̈***
cgj Ä̰ Ḛ̈ Ḭ̈ Ö̰ Ṵ̈ Ɛ̰̈ Ɔ̰̈ ä̰ ḛ̈ ḭ̈ ö̰ ṵ̈ ɛ̰̈ ɔ̰̈ *Ä̰ Ḛ̈ Ḭ̈ Ö̰ Ṵ̈ Ɛ̰̈ Ɔ̰̈ ä̰ ḛ̈ ḭ̈ ö̰ ṵ̈ ɛ̰̈ ɔ̰̈* **Ä̰ Ḛ̈ Ḭ̈ Ö̰ Ṵ̈ Ɛ̰̈ Ɔ̰̈ ä̰ ḛ̈ ḭ̈ ö̰ ṵ̈ ɛ̰̈ ɔ̰̈** ***Ä̰ Ḛ̈ Ḭ̈ Ö̰ Ṵ̈ Ɛ̰̈ Ɔ̰̈ ä̰ ḛ̈ ḭ̈ ö̰ ṵ̈ ɛ̰̈ ɔ̰̈***
Patched not grave
À B̀ È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀ *À B̀ È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀* **À B̀ È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀** ***À B̀ È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀***
Patched grave
À B̀ È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀ *À B̀ È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀* **À B̀ È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀** ***À B̀ È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀***

Floating tones

A floating tone is written as a tone mark with no vowel of its own: after a vowel that bears the same tone (ë̋), or standing alone as a morpheme in a gloss. By Unicode convention, a mark that stands alone is applied to U+00A0 no-break space (the Unicode Standard prefers NBSP over U+0020 space, which allows a line break between the space and its mark). The patched fonts give U+00A0 an above and a below anchor at the lowercase references (645 above, −110 below), with X on each style's anchor model, so a mark on NBSP sits at exactly the height it has on a lowercase vowel, in all four styles.

How to write a floating tone in XeLaTeX

1. Load the patched fonts with `fontspec`, e.g.

   ```
   \setmainfont{LibertinusSerif-Regular-patch.otf}[
   ```

```
Path           = ../fonts/,
ItalicFont     = LibertinusSerif-Italic-patch.otf,
BoldFont       = LibertinusSerif-Semibold-patch.otf,
BoldItalicFont = LibertinusSerif-SemiboldItalic-patch.otf]
```

2. Define three commands in the preamble (this report does):

```
\newcommand{\cgj}{\char"034F }                % combining grapheme joiner
\newcommand{\nbsp}{\char"00A0 }               % no-break space, base for a floating mark
\newcommand{\floattone}[2]{#1\cgj#2\nbsp#2}   % vowel + mark, then the same mark floating
```

3. Write a vowel followed by a floating tone of the same specification as `\floattone{`*vowel*`}{`*mark*`}`, e.g. `\floattone{e}{\char"030F}` gives ȅ̏. The mark may also be typed as the literal combining character between the braces. The vowel may carry other marks (`\floattone{a\char"0330}{\char"0300}` gives à̗), and may be any lowercase tone-bearing unit, such as a syllabic nasal (`\floattone{ŋ}{\char"0300}` gives ŋ̀).

4. Write a floating tone on its own as `\nbsp` followed by the mark: `-\nbsp\char"0300` gives -̀.

5. Punctuation may follow directly: `\floattone{e}{\char"0300}.` gives è̀. NBSP is a quarter em wide (250 units; 274 in semibold italic), so there is that much space before the punctuation.

Two things to avoid. Do not write the base as ~ or `\nobreakspace`: in TeX these are glue, not a character, so HarfBuzz sees no base and the mark is set on its own at its default position. Do not leave out the CGJ when the floating tone repeats the vowel's mark: without it, HarfBuzz substitutes the precomposed glyph where one exists (è), and a precomposed glyph may carry its mark at a different height from the anchor (the dot above U+0307 is 45 units higher in the precomposed glyphs of regular); with it, both marks are placed by anchors at the same height. `\floattone` inserts the CGJ. When typing `\nbsp` by hand before a literal combining mark, write `\nbsp{}`, since XeTeX reads a combining mark as a letter of the command name; `\floattone` has no such problem.

The anchors are lowercase ones, so a floating mark after a capital does not take its `.cap` form or the capital height. In fonts without NBSP anchors (original Libertinus, Noto Serif) the mark still appears, at its default position; Gentium Plus and Charis SIL have NBSP anchors of their own.

Comparison

Patched fonts, with `\floattone`: `-k\floattone{e}{\char"030F}`

-kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Patched fonts, without CGJ: `-ke\char"030F\nbsp\char"030F`

-kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Original Libertinus, with `\floattone`

-kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Gentium Plus, with `\floattone`

-kȅ̏

Note that in Libertinus, the two graves in the U+030F glyph are not translates of an identical outline; this is a deliberate design choice by the designer.

Showcase

Each item is `\floattone{`*vowel*`}{`*mark*`}` for the vowels a e i o u ɛ ɔ ə ɪ ʊ ɨ ʉ ʌ æ and the syllabic nasals m n ŋ, in the patched fonts (regular, italic, semibold, semibold italic), at `\Large` font size. The floating mark matches the height of the vowel's mark throughout, since every x-height letter and NBSP share the same above reference (645) and below reference (−110).

U+0301 acute

á́ é́ í́ ó́ ú́ ɛ́ ɔ́ ə́ ɪ́ ʊ́ ɨ́ ʉ́ ʌ́ ǽ ḿ ń ŋ́

á́ é́ í́ ó́ ú́ ɛ́ ɔ́ ə́ ɪ́ ʊ́ ɨ́ ʉ́ ʌ́ ǽ ḿ ń ŋ́

á́ é́ í́ ó́ ú́ ɛ́ ɔ́ ə́ ɪ́ ʊ́ ɨ́ ʉ́ ʌ́ ǽ ḿ ń ŋ́

á́ é́ í́ ó́ ú́ ɛ́ ɔ́ ə́ ɪ́ ʊ́ ɨ́ ʉ́ ʌ́ ǽ ḿ ń ŋ́

U+0300 grave

à̀ è̀ ì̀ ò̀ ù̀ ɛ̀ ɔ̀ ə̀ ɪ̀ ʊ̀ ɨ̀ ʉ̀ ʌ̀ æ̀ m̀ ǹ ŋ̀

à̀ è̀ ì̀ ò̀ ù̀ ɛ̀ ɔ̀ ə̀ ɪ̀ ʊ̀ ɨ̀ ʉ̀ ʌ̀ æ̀ m̀ ǹ ŋ̀

à̀ è̀ ì̀ ò̀ ù̀ ɛ̀ ɔ̀ ə̀ ɪ̀ ʊ̀ ɨ̀ ʉ̀ ʌ̀ æ̀ m̀ ǹ ŋ̀

à̀ è̀ ì̀ ò̀ ù̀ ɛ̀ ɔ̀ ə̀ ɪ̀ ʊ̀ ɨ̀ ʉ̀ ʌ̀ æ̀ m̀ ǹ ŋ̀

U+030B double acute

a̋ e̋ ı̋ ő ű ɛ̋ ɔ̋ ə̋ ɪ̋ ʊ̋ ɨ̋ ʉ̋ ʌ̋ æ̋ m̋ n̋ ŋ̋

a̋ e̋ ı̋ ő ű ɛ̋ ɔ̋ ə̋ ɪ̋ ʊ̋ ɨ̋ ʉ̋ ʌ̋ æ̋ m̋ n̋ ŋ̋

a̋ e̋ ı̋ ő ű ɛ̋ ɔ̋ ə̋ ɪ̋ ʊ̋ ɨ̋ ʉ̋ ʌ̋ æ̋ m̋ n̋ ŋ̋

a̋ e̋ ı̋ ő ű ɛ̋ ɔ̋ ə̋ ɪ̋ ʊ̋ ɨ̋ ʉ̋ ʌ̋ æ̋ m̋ n̋ ŋ̋

U+030F double grave

ȁ ȅ ı̏ ȍ ù ɛ̏ ɔ̏ ə̏ ɪ̏ ʊ̏ ɨ̏ ʉ̏ ʌ̏ æ̏ m̏ n̏ ŋ̏

ȁ ȅ ı̏ ȍ ù ɛ̏ ɔ̏ ə̏ ɪ̏ ʊ̏ ɨ̏ ʉ̏ ʌ̏ æ̏ m̏ n̏ ŋ̏

ȁ ȅ ı̏ ȍ ù ɛ̏ ɔ̏ ə̏ ɪ̏ ʊ̏ ɨ̏ ʉ̏ ʌ̏ æ̏ m̏ n̏ ŋ̏

ȁ ȅ ı̏ ȍ ù ɛ̏ ɔ̏ ə̏ ɪ̏ ʊ̏ ɨ̏ ʉ̏ ʌ̏ æ̏ m̏ n̏ ŋ̏

U+0304 macron

ā ē ī ō ū ɛ̄ ɔ̄ ə̄ ɪ̄ ʊ̄ ɨ̄ ʉ̄ ʌ̄ ǣ m̄ n̄ ŋ̄

ā ē ī ō ū ɛ̄ ɔ̄ ə̄ ɪ̄ ʊ̄ ɨ̄ ʉ̄ ʌ̄ ǣ m̄ n̄ ŋ̄

ā ē ī ō ū ɛ̄ ɔ̄ ə̄ ɪ̄ ʊ̄ ɨ̄ ʉ̄ ʌ̄ ǣ m̄ n̄ ŋ̄

ā ē ī ō ū ɛ̄ ɔ̄ ə̄ ɪ̄ ʊ̄ ɨ̄ ʉ̄ ʌ̄ ǣ m̄ n̄ ŋ̄

a̅ ***e̅*** ***ɪ̅*** ***o̅*** ***u̅*** ***ɛ̅*** ***ɔ̅*** ***ə̅*** ***ɪ̅*** ***ʊ̅*** ***ɨ̅*** ***ʉ̅*** ***ʌ̅*** ***æ̅*** ***m̅*** ***n̅*** ***ŋ̅***

U+0302 circumflex

â̂ ê̂ î̂ ô̂ û̂ ɛ̂ ɔ̂ ə̂ ɪ̂ ʊ̂ ɨ̂ ʉ̂ ʌ̂ æ̂ m̂ n̂ ŋ̂

â̂ *ê̂* *î̂* *ô̂* *û̂* *ɛ̂* *ɔ̂* *ə̂* *ɪ̂* *ʊ̂* *ɨ̂* *ʉ̂* *ʌ̂* *æ̂* *m̂* *n̂* *ŋ̂*

â̂ **ê̂** **î̂** **ô̂** **û̂** **ɛ̂** **ɔ̂** **ə̂** **ɪ̂** **ʊ̂** **ɨ̂** **ʉ̂** **ʌ̂** **æ̂** **m̂** **n̂** **ŋ̂**

â̂ ***ê̂*** ***î̂*** ***ô̂*** ***û̂*** ***ɛ̂*** ***ɔ̂*** ***ə̂*** ***ɪ̂*** ***ʊ̂*** ***ɨ̂*** ***ʉ̂*** ***ʌ̂*** ***æ̂*** ***m̂*** ***n̂*** ***ŋ̂***

U+030C caron

ǎ̌ ě̌ ǐ̌ ǒ̌ ǔ̌ ɛ̌ ɔ̌ ə̌ ǐ̌ ʊ̌ ɨ̌ ʉ̌ ʌ̌ æ̌ m̌ ň̌ ŋ̌

ǎ̌ *ě̌* *ǐ̌* *ǒ̌* *ǔ̌* *ɛ̌* *ɔ̌* *ə̌* *ǐ̌* *ʊ̌* *ɨ̌* *ʉ̌* *ʌ̌* *æ̌* *m̌* *ň̌* *ŋ̌*

ǎ̌ **ě̌** **ǐ̌** **ǒ̌** **ǔ̌** **ɛ̌** **ɔ̌** **ə̌** **ǐ̌** **ʊ̌** **ɨ̌** **ʉ̌** **ʌ̌** **æ̌** **m̌** **ň̌** **ŋ̌**

ǎ̌ ***ě̌*** ***ǐ̌*** ***ǒ̌*** ***ǔ̌*** ***ɛ̌*** ***ɔ̌*** ***ə̌*** ***ǐ̌*** ***ʊ̌*** ***ɨ̌*** ***ʉ̌*** ***ʌ̌*** ***æ̌*** ***m̌*** ***ň̌*** ***ŋ̌***

Squat reversed esh (Sinological apical vowels)

U+0285 LATIN SMALL LETTER SQUAT REVERSED ESH ʅ is the apical vowel of Sinological transcription after retroflex sibilants (Mandarin *shī* [ʂʅ^{55}], *zhī* [tʂʅ^{55}], *rì* [ʐʅ^{51}]). Its partner ɿ U+027F follows dental sibilants (*sī* [sɿ^{55}], *cí* [tsʰɿ^{35}]). ʅ is ɿ with a retroflex hook, the hook of ɳ ʈ ʂ ʐ ɭ: its top is ɿ's fishhook, at the x-height.

The designer's ʅ was 550 high in regular, 119 above the x-height, and in italic, semibold and semibold italic it is ∫ mirrored, rising to the ascender (semibold exactly so: every point $x \mapsto 305 - x$, hence the same bounding box as ∫). In a survey of 22 fonts (2026-11-09) every font that draws the letter rather than mirroring ∫ tops it at the x-height, level with its own ɿ: Gentium Plus, Charis SIL, Andika, Noto Serif, STIX Two, Junicode, EB Garamond, DejaVu Serif, Tinos, Roboto. The ascender form occurs only as a mirrored ∫ (Gentium 1.x, which Gentium Plus replaced with the squat form; Lato) and in Libertinus and Langscitinus, which has Libertinus's glyph unchanged.

Stage 2 of the patch redraws ʅ in all four styles by control points: ɿ's fishhook and stem, with ɳ's retroflex hook in place of the foot serif. The hook reaches as deep as ɳ's (−221 in regular, −229 in the other styles), short of the descender (−234), like the other retroflex hooks. The left sidebearing is ɿ's, the right ɳ's at the hook. The above-anchor is ɿ's (stage 8, which treats ʅ as ɿ with a bottom tail); the below-anchor is the anchor model's candidate on the new outline (stage 6), at the descender reference. The designer's anchors, placed for the old outline, are deleted in stage 1, including regular's ogonek and cedilla anchors.

Original Libertinus	ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}	aʅa *ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}*	*aʅa* **ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}**	**aʅa *ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}***	***aʅa***
Langscitinus	ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}	aʅa *ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}*	*aʅa* **ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}**	**aʅa *ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}***	***aʅa***
Patched	ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}	aʅa *ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}*	*aʅa* **ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}**	**aʅa *ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}***	***aʅa***
Gentium Plus	ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sʅ^{55}	aʅa *ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sʅ^{55}*	*aʅa* **ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sʅ^{55}**	**aʅa *ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sʅ^{55}***	***aʅa***

With tone and other marks, patched: [ʂ̄ ʂ́ ʂ̌ ʂ̀] [ʂ̲] [ʅ̣] *[ʂ̄ ʂ́ ʂ̌ ʂ̀] [ʂ̲] [ʅ̣]* **[ʂ̄ ʂ́ ʂ̌ ʂ̀] [ʂ̲] [ʅ̣]** ***[ʂ̄ ʂ́ ʂ̌ ʂ̀] [ʂ̲] [ʅ̣]***

Reversed r with fishhook

U+027F LATIN SMALL LETTER REVERSED R WITH FISHHOOK ɿ is the apical vowel after dental sibilants (see the previous section). Fonts draw it in two ways: on the baseline, as ɾ mirrored (Libertinus, Langscitinus, Gentium 1.x, Junicode, STIX Two, DejaVu Serif, EB Garamond), or with a straight descender (Gentium Plus, Charis SIL, Andika, Noto, Tinos, Roboto). SIL changed it from the baseline form in Gentium 1.x to the descending form in Gentium Plus, and the current SIL fonts offer no way back: they have one glyph for U+027F, no character variant or stylistic set for it, and no PUA alternate. Both forms are established, and the patch keeps Libertinus's baseline ɿ as U+027F.

For users who want the descending form, the patched fonts add it at a Private Use Area code point, U+E27F (glyph `uni027F.desc`), in all four styles. It is ɿ's fishhook and stem continued down to q's descender serif, with ɿ's advance width. The above-anchor is ɿ's (stage 8); the below-anchor is curated, under the stem at the descender reference (−319), as q's.

U+027F, patched	sɿ^{55} tsʰɿ^{35} [sɪ̄ sɪ́ sɪ̌ sɪ̀] [sɪ̲] *sɿ^{55} tsʰɿ^{35} [sɪ̄ sɪ́ sɪ̌ sɪ̀] [sɪ̲]* **sɿ^{55} tsʰɿ^{35} [sɪ̄ sɪ́ sɪ̌ sɪ̀] [sɪ̲]** ***sɿ^{55} tsʰɿ^{35} [sɪ̄ sɪ́ sɪ̌ sɪ̀] [sɪ̲]***
U+E27F, patched	sɻ^{55} tsʰɻ^{35} [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̲] *sɻ^{55} tsʰɻ^{35} [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̲]* **sɻ^{55} tsʰɻ^{35} [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̲]** ***sɻ^{55} tsʰɻ^{35} [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̲]***
Gentium Plus	sʅ^{55} tsʰʅ^{35} [sʅ̄ sʅ́ sʅ̌ sʅ̀] [sʅ̲] *sʅ^{55} tsʰʅ^{35} [sʅ̄ sʅ́ sʅ̌ sʅ̀] [sʅ̲]* **sʅ^{55} tsʰʅ^{35} [sʅ̄ sʅ́ sʅ̌ sʅ̀] [sʅ̲]** ***sʅ^{55} tsʰʅ^{35} [sʅ̄ sʅ́ sʅ̌ sʅ̀] [sʅ̲]***

How to use the descending form

- In XeLaTeX or LuaLaTeX with the patched fonts, write `\char"E27F` or define a command, as this report does:

  ```
  \newcommand{\rfishdesc}{\char"E27F }   % reversed r with fishhook and descender
  ```

  `s\rfishdesc` gives sɻ. Before a literal combining mark write `\rfishdesc{}`, since XeTeX reads a combining mark as a letter of the command name (`s\rfishdesc{}` gives sɻ́).

- In other applications, enter the code point E27F with the application's Unicode input (e.g. type `e27f` then Alt+X in Word, or Ctrl+Shift+U e27f in GTK applications).

A Private Use code point has no meaning outside the font: in any other font U+E27F shows as a missing glyph, and searching, sorting, spell-checking and copied text see U+E27F, not U+027F. Use it for display where the descending form is wanted, and keep U+027F wherever the text must be interchanged or searched.

f with dot below

tiɣə́ ɾ̣fa

tiɣə́ ɾ̣fa

tiɣə́ ɾ̣fa

tiɣə́ ɾ̣fa

ɾ̣ = `\char"1E5B` is the precomposed character U+1E5B Latin small letter r with dot below.

f̣ = `f\char"0323` is f with U+0323 combining dot below.

Here are some letters with a dot below.

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̣ g̣ j̣ p̣ q̣ x̣

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̣ g̣ j̣ p̣ q̣ x̣

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̣ g̣ j̣ p̣ q̣ x̣

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̣ g̣ j̣ p̣ q̣ x̣

The left group of letters – a b d e h i k l m n o r s t u v w y z – have a precomposed character. For the right group of letters – c f g j p q x – the dot is placed according to the below-anchor (if the below-anchor is defined for the letter) or by HarfBuzz fallback shaping (placing the dot below the baseline, regardless of the letter's bounding box).

In all styles – regular, italic, semibold, and semibold italic – the precomposed glyph of y with dot below places the dot beside the descender, even though the y has a below-anchor below the descender.

Do the letters c, f, g, j, p, q, and x have good below-anchors? In regular, the below-anchor of all these combining bases is below the glyph outline (good). In italic, the below-anchor of f, p, and q is beside the descender (bad), the below-anchor of g and j is below the descender (good), and the below-anchor of c and x is below the baseline (good). In semibold, the below-anchor of g and j is below the descender (good), but the other bases – c, f, p, q – have no below-anchor, and HarfBuzz fallback positioning places the combining dot below the baseline, not below the bounding box; for c, f and x this is good, but for p and q this is bad, because of the descenders. In semibold italic, the below-anchor of c, g, j, p and q is below the glyph outline (good), the below-anchor of f is inside the descender (very bad), and x has no below-anchor but HarfBuzz fallback positioning places the combining dot below the baseline (good).

Which dots are used in precomposed characters and how big are the dots? In all styles (except semibold italic), the precomposed characters reference the dot of `uni0307`, combining dot above, U+0307, and take care to translate this above-mark to a position below the letter outline, except for m, which references `dotbelowcomb`, combining dot below, U+0323. In semibold italic, `dotaccent` is used instead of `uni0307`. These two dots are four-point Bézier circles (in regular and semibold, but ovals in italic and semibold italic), but of different diameter. In regular, `uni0307` has diameter 100, and `dotbelowcomb` has diameter 108. That is, the precomposed characters (except m) use a smaller dot, and the non-precomposed letters (and m) take a bigger dot. In semibold, the size difference is reversed: `uni0307` has larger diameter 124, and `dotbelowcomb` has small diameter 108. Actually, semibold `dotbelowcomb` is not a Bézier circle – the horizontal span is 109, and the vertical span is 108. In semibold italic, `dotaccent` is a skewed four-point Bézier oval – the horizontal and vertical spans are 120, offset by $(\pm 6, \pm 7.5)$ from the center — and `dotbelowcomb` is slightly larger, slightly flatter skewed six-point Bézier oval – horizontal span is 122 and the vertical span is 120. So, in semibold italic the two dots in precomposed and combining form are closest in size. Recall that f has a bad below-anchor, inside the descender, but g, j, p and q have a good below-anchor, below the glyph outline.

So, in order to solve the problem of the bad dot below f in italic, I need to decide the scope of the badness to correct. The simplest patch is move the below-anchor in italic and semibold italic below the descender. But this does not solve inconsistent below dot size across styles and letters. In italic and semibold italic, g and j have a (good) below-anchor at $Y = -319$, so that is also good Y coordinate for f's below-anchor.

I asked Copilot/GPT-5.1 to calculate a good X coordinate for f's below-anchor in italic and semibold italic, taking into account the U-pocket of f's descender and the below-anchor of `dotbelowcomb`. I decided that this U-pocket strategy was appropriate after testing other strategies that accounted for the slant of onf the major stroke. Here is a reasonbly faithful summary of the calculations and reasoning by Copilot/GPT-5.1.

For italic, the descender's U-pocket is defined by the control points at $(14, -238)$ and $(33, -205)$, giving a U-center around $x \approx 24$. Using the (symmetric) italic `dotbelowcomb` anchor as the dot's optical center, the dot should sit under this U-pocket, not under the ball or the stem. A balanced, U-centered, slightly inner-biased choice is $X \approx 30$.

For semibold italic, the U-pocket of the semibold italic descender is wider and more asymmetric, with a geometric U-center of $X = -33.5$. The semibold italic `dotbelowcomb` itself is also skewed, with its anchor already shifted relative to its mass. In practice, the visually correct position for the dot below f is under the open side of the U-pocket, clearly right of the ball and closer the right of the stem. The anchor candidate is $X \approx 62$.

Here are examples with the patched fonts, with f's below-anchor in italic at $(30, -319)$ and semibold italic at $(62, -319)$. This does not solve the problem of inconsistent dot size below letters, and I did not touch p, q, or y.

Original	Patched italics (f only)
tiɣə ṛfa	tiɣə ṛfa
tiɣə ṛfa	*tiɣə ṛfa*
tiɣə ṛfa	**tiɣə ṛfa**
tiɣə ṛfa	***tiɣə ṛfa***

Original (repeated from above)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱

Patched italics (f only)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱

Ashkun

`a\char"0304\char"0303` ã *ã* **ã** ***ã***

`\char"0101\char"0303` ã *ã* **ã** ***ã***

`u\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\char"0303` ũ *ũ* **ũ** ***ũ***

Patched

`u\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`u\cgj\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\cgj\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\char"0303` ũ *ũ* **ũ** ***ũ***

Move box without patching font

`\char"016B\kern-4.132pt\raisebox{0.88pt}{\char"02DC}` ũ *ũ* **ũ** ***ũ***

Tacks

a̟e̟i̟o̟u̟a̟e̟i̟o̟u̟
a̟e̟i̟o̟u̟a̟e̟i̟o̟u̟

Upsilon and grave

Original, then patched, versions of upsilon and combining grave, after setting upsilon's above-anchor.

ὺ ύ̀ **ὺ** **ύ̀**

ὺ ύ̀ **ὺ** **ύ̀**

New upsilon above-anchor: regular $(253, 645)$; italic $(358, 645)$; semibold $(267, 645)$; semibold italic $(369, 645)$

Revised combining grave above-anchor: semibold italic $(0, 683)$. Moved from $(-142, 683)$, which is far left of the optical center of the mark, which has the effect of attaching the mark too far right.

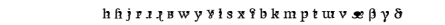
Spoof to move modifier grave (which has the exact same outline as combining grave) to an anchored position, if upsilon had an above-anchor at $(358, 645)$ (keeping combining grave's above-anchor as is). Note upsilon is U+028A and modifier grave is U+02CB.

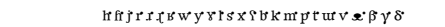
```
\textit{\char"028A\kern-4.66pt\raisebox{-0.38pt}{\char"02CB}}
```

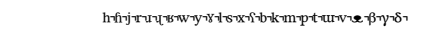
ὺ

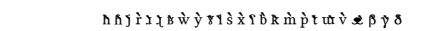
Patch superscript consonant anchors

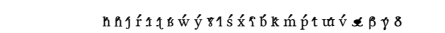
Libertinus serif regular

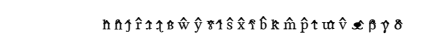




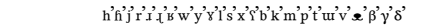


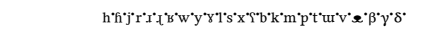


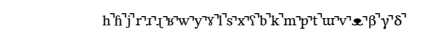


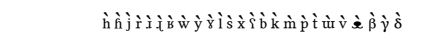


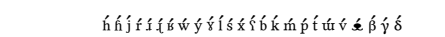
Libertinus serif regular, patched

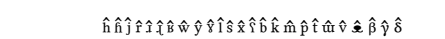




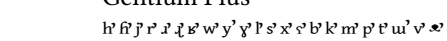


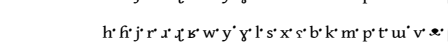


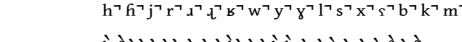


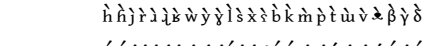


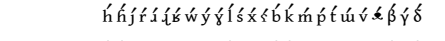
Gentium Plus

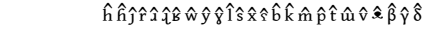


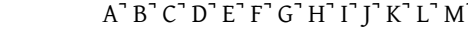


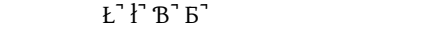


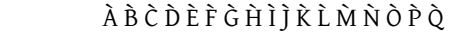




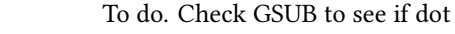




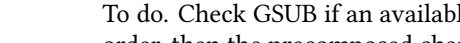












Dot removal

To do. Check GSUB to see if dot removal for i and j is complete.

Precomposed characters

To do. Check GSUB if an available precomposed character is encoded by a doublet of the base and a combining mark, or a triplet of the base and the two combining marks, in either mark order, then the precomposed character should be be produced by the font's GSUB table.

Marks

Here is a catalog of combining marks in Unicode, grouped by the seven GPOS anchor class indices in Libertinus (0–6).

Above marks – GPOS anchor 0

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠
x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠
x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠
x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠

Below marks – GPOS anchor 2

x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠
x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠
x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠
x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠

Left angle – GPOS anchor 3

x̕ *x̕* x⃕ *x⃕*

Below-right marks – GPOS anchor 4

x̙ x̚ x̛
x̙ x̚ x̛
x̙ x̚ x̛
x̙ x̚ x̛

Cedilla – GPOS anchor 5

x̡ *x̡* x̧ *x̧*

Overlay marks – GPOS anchor 6

o̶ o̷ o̸ ø̸
o̶ o̷ o̸ ǿ
o̶ o̷ o̸ o̸ ø
o̶ o̷ o̸ o̸ ǿ

Double marks

x͡x x͡x x͞x x͟x x̃x x̂x x̡x
x͡x x͡x x͞x x͟x x̃x x̂x x̡x
x̋x x̋x x̏x x̑x x̆x x̊x ẍx
x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x

Greek marks

x̀x́x̃x̄x̆ẋẍx̣
x̍x́x̃x̨
x̂x̂x̀x̀x̋x̋x̄ẍx̨
x̀x́x̃x̄x̆ẋx̣
x̍x́x̃x̨
x̂x̂x̀x̀x̋x̋x̋x̋x̨
x̀x́x̃x̄x̆ẋx̣
x̍x́x̃x̨
x̂x̂x̀x̀x̃x̃x̄ẍx̣
x̀x́x̃x̄x̆ẋx̣
x̍x́x̃x̨
x̂x̂x̀x̀x̃x̃x̄ẍx̣

Adding missing marks

Semibold and semibold italic lack many of the combining marks of regular and italic. The patch adds them, with mark anchors, in the range U+0332–U+036F. No mark is drawn from scratch: each is an outline that the designer drew in another style. A semibold mark is the regular outline (an above-mark is moved right by 12 units, with its anchor, as the designer's semibold above-marks are). A semibold italic mark is the italic outline moved right by 12 units, with the anchor of regular, as in the semibold italic anchor model. In semibold, U+0358–U+0363 were copies of the capital accents, and they are replaced. The patch also adds `dotaccent.cap` to italic and semibold italic: the dot of the precomposed İ, at the height it has in the precomposed dotted capitals. It adds `tildecomb.cap` to every style (2026-11-08): the tilde of the precomposed capitals, which is `tildecomb` itself in regular, italic and semibold, and in semibold italic the taller tilde of Ã (also of Ñ Õ Ĩ Ẽ Ỹ Ṽ and ã ñ õ), which differs from `tildecomb` there.

The designer does not thicken below-marks between weights, but thickens above-marks by about 12 percent (mean stroke thickness, measured as twice the area of the outline divided by its perimeter: tilde 40 to 45, caron 36 to 42). So the added above-marks are made bolder. Where a mark is, or contains, another mark, the bold style's own mark is used. The rest of the outline is pushed outward by 7.5 percent of its stroke thickness on each side, about 2 to 3 units. These algorithmically emboldened marks are shown in the table on the next page, on a few bases that have above-anchors in all four patched fonts, with the designer's tilde and caron for comparison of weight. The regular and italic columns show the outlines that were copied.

Also added, but not shown in the table: the low line and double low line (U+0332, U+0333), the tilde, stroke and solidus overlays (U+0334–U+0337), the below-marks U+0339–U+033C, U+0347–U+0349, U+034D, U+034E, U+0353–U+0356, U+0359, U+035A, the double overline (U+033F), the dot above right (U+0358), the double marks (U+035C–U+0362) and the combining letters (U+0363–U+036F). These are copied without emboldening, since the designer's lines, below-marks and modifier letters are not thicker in the bold styles; the dot above right is the bold style's own dot. The double marks have no anchors, as in regular. In semibold, the overlays and the dot above right have no anchors, because the semibold lookup table has no anchor class for them. The half rings above (U+0351, U+0357) already exist in semibold, drawn to the right of the origin, so they sit beside the base; they are left as they are. Not added: the custom-width low lines and overlines of the next section, U+0344, U+0345 and U+0483.

		regular	semibold	italic	semibold italic	
U+0303	tilde (designer's, for comparison)	ãñũÕ	**ãñũÕ**	*ãñũÕ*	***ãñũÕ***	
U+030C	caron (designer's, for comparison)	ǎňǔǑ	**ǎňǔǑ**	*ǎňǔǑ*	***ǎňǔǑ***	
U+033D	x above	x̌x̌ǔx̌	**x̌x̌ǔx̌**	*x̌x̌ǔx̌*	***x̌x̌ǔx̌***	emboldened
U+033E	vertical tilde	a̾n̾u̾O̾	**a̾n̾u̾O̾**	*a̾n̾u̾O̾*	***a̾n̾u̾O̾***	emboldened
U+034A	not tilde above	ãñ̃ũ̃Õ	**ãñ̃ũ̃Õ**	*ãñ̃ũ̃Õ*	***ãñ̃ũ̃Õ***	emboldened
U+034B	homothetic above	ãñ̑ũ̑Õ	**ãñ̑ũ̑Õ**	*ãñ̑ũ̑Õ*	***ãñ̑ũ̑Õ***	own tilde; dots emboldened
U+034C	almost equal to above	ãñ̃ũ̃Õ	**ãñ̃ũ̃Õ**	*ãñ̃ũ̃Õ*	***ãñ̃ũ̃Õ***	emboldened
U+0350	right arrowhead above	ãňǔÒ	**ãňǔÒ**	*ãňǔÒ*	***ãňǔÒ***	emboldened
U+0351	left half ring above	á̜n̜ú̜Ó	**á̜n̜ú̜Ó**	*á̜n̜ú̜Ó*	***á̜n̜ú̜Ó***	emboldened (semibold italic only)
U+0352	fermata	ân̂ûÔ	**ân̂ûÔ**	*ân̂ûÔ*	***ân̂ûÔ***	own inverted breve; dot emboldened
U+0357	right half ring above	a̾n̾u̾O̾	**a̾n̾u̾O̾**	*a̾n̾u̾O̾*	***a̾n̾u̾O̾***	emboldened (semibold italic only)
U+035B	zigzag above	á̴n̴ú̴Ó	**á̴n̴ú̴Ó**	*á̴n̴ú̴Ó*	***á̴n̴ú̴Ó***	emboldened
U+0340	grave tone mark	àǹùÒ	**àǹùÒ**	*àǹùÒ*	***àǹùÒ***	own grave
U+0341	acute tone mark	áńúÓ	**áńúÓ**	*áńúÓ*	***áńúÓ***	own acute
U+0346	bridge above	ãn̄ūŌ	**ãn̄ūŌ**	*ãn̄ūŌ*	***ãn̄ūŌ***	own bridge below, raised

Overline and low line

U+0305 combining overline and U+0332 combining low line are applied by GSUB lookups that substitute a custom-width line mark according to width classes. GSUB lookup 51 effectively classifies the glyphs by width, and GSUB lookups 52–82 provide the appropriate custom-width mark for the base width class. Semibold and semibold italic do not have any combining low lines; the patch adds U+0332 and U+0333 with below-anchors, but not the custom-width variants. Custom-width line marks do not have anchors. They are positioned by HarfBuzz fallback shaping. The overlines are above the ascender line, so they are above all bases. The low lines are below the baseline, but above the descender line, and so they cut though any descenders. If a new glyph is added, and you want to support custom-width line marks for it, then remember to classify the new glyph in GSUB lookup 51.

Above-right marks – GPOS anchor 1

Libertinus has three combining marks with GPOS anchor 1 – U+031B horn, U+0315 comma above right, and U+0358 dot above right.

A few bases in Libertinus have this anchor (anchor 1), but none that possibly need it.

Horn

The Vietnamese alphabet has two horn vowels – Ơơ Ưư– and these vowels may take tonal marks. All these combinations are Unicode characters (with code points).

Ờ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự

Ờ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự

Ờ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự

Ờ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự

There is a Unicode character for the combining horn mark: ̛ `\char"031B` U+031B, but this is not used in practice. The Vietnamese vowels are usually entered in a LaTeX document as Unicode characters, also with `\usepackage[vietnamese]{babel}` for hyphenation and spacing.

Comma above right

There is a Unicode character for the combining comma above right mark: ̕ `\char"0315` U+0315.

U+0315 (comma above right) is used in rare cases when another comma-like mark does not fit.

In IPA, the glottalization diacritic mark uses U+0313 (comma above) as a combining mark. But when a glottalization mark is needed for a superscript consonant (1) or superscript consonant cluster (2), then the combining mark U+0315 (comma above right) is used instead, because its positioning is hopefully better than that of U+0313.

(1) *k̕wkt* [k̕^{wkx}t] ‘to scrape’ (Nuxalk)

(2) *p’ak* [p’^{kx}ak] ‘he/she cut it’ (Upper Chehalis)

In (1), the superscript consonant cluster is ^{wkx} and it is glottalized. In (2), the superscript consonant ^{x} is glottalized.

Because U+0315 is being used in superscript position on a smaller sized glyph, U+0315 should probably be smaller than U+0313, but it is not so in Libertinus.

Dot above right

There is a Unicode character for the combining dot above right mark: ̙ `\char"0358` U+0358 dot above right

U+0358 should be a small, above-right dot, used to mark palatalization or similar secondary articulations on superscript consonants, created for the same typographic reason U+0315 exists – the standard palatalization diacritic (U+0307, dot above) is too big and too high.

In Libertinus, U+0358 is a 104-diameter Bezier circle, and U+0307 is a smaller Bezier oval, inverting the desired size difference.

There are only a few IPA superscript letters that may serve as the possible host of U+0315 or U+0358 on superscript consonants or on superscript consonant clusters. In Libertinus, regrettably none of these characters has the anchor (anchor 1), so the mark (U+0315 and U+0358) is positioned by HarfBuzz fallback shaping, which seems to place the mark at the right boundary of the glyph at a constant vertical aspect (effectively at the superscript x-height).

These IPA marks U+0315 and U+0358 should only be used in regular (not italic and not semibold). Regrettably, in Libertinus serif semibold, U+0358 looks like a grave, not a dot, and there is no U+0358 glyph in Libertinus serif semibold italic.

There are three comma/apostrophe glyphs that could possibly mark glottalization: U+0313 combining comma above, U+0315 combining comma above right, and U+02BC modifier letter apostrophe. In Libertinus, U+0315 has the worst appearance, even though U+0315 is intended for superscript letters. In Gentium Plus (SIL Global), U+0315 is tightly above-right the superscript letter, as expected.

Libertinus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [k̓ʷᵏˣ̓t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k̓ʷᵏˣ̔t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [k̓ʷᵏˣʼt]

Gentium Plus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [k̓ʷᵏˣ̓t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k̓ʷᵏˣ̔t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [k̓ʷᵏˣʼt]

I patched Libertinus to add anchors to those IPA superscript letters to support U+0315.

Patched Libertinus, selected IPA superscript letters with U+0315 and U+0358

ʰ̓ ʱ̓ ʲ̓ ʳ̓ ˠ̓ ᶥ̓ ʷ̓ ʸ̓ ˡ̓ ˢ̓ ˣ̓ ˤ̓ ᵇ̓ ᵏ̓ ᵐ̓ ᵖ̓ ᵗ̓ ᵚ̓ ᵛ̓ ◌͡◌̓ ᵝ̓ ᵞ̓ ᵟ̓

ʰ̔ ʱ̔ ʲ̔ ʳ̔ ˠ̔ ᶥ̔ ʷ̔ ʸ̔ ˡ̔ ˢ̔ ˣ̔ ˤ̔ ᵇ̔ ᵏ̔ ᵐ̔ ᵖ̔ ᵗ̔ ᵚ̔ ᵛ̔ ◌͡◌̔ ᵝ̔ ᵞ̔ ᵟ̔

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k̓ʷᵏˣ̔t]

Gentium Plus, selected IPA superscript letters with U+0315 and U+0358

ʰ̓ ʱ̓ ʲ̓ ʳ̓ ˠ̓ ᶥ̓ ʷ̓ ʸ̓ ˡ̓ ˢ̓ ˣ̓ ˤ̓ ᵇ̓ ᵏ̓ ᵐ̓ ᵖ̓ ᵗ̓ ᵚ̓ ᵛ̓ ◌͡◌̓ ᵝ̓ ᵞ̓ ᵟ̓

ʰ̔ ʱ̔ ʲ̔ ʳ̔ ˠ̔ ᶥ̔ ʷ̔ ʸ̔ ˡ̔ ˢ̔ ˣ̔ ˤ̔ ᵇ̔ ᵏ̔ ᵐ̔ ᵖ̔ ᵗ̔ ᵚ̔ ᵛ̔ ◌͡◌̔ ᵝ̔ ᵞ̔ ᵟ̔

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k̓ʷᵏˣ̔t]

Here are the IPA superscript letters considered and their new anchor (anchor 1).

U+02B0: (347, 648), U+02B1: (339, 648), U+02B2: (193, 648), U+02B3: (284, 648), U+02B4: (274, 648), U+02B5: (319, 648), U+02B6: (308, 648), U+02B7: (497, 648), U+02B8: (368, 648), U+02E0: (341, 648), U+02E1: (193, 648), U+02E2: (270, 648), U+02E3: (344, 648), U+02E4: (282, 648), U+1D47: (319, 648), U+1D4F: (353, 648), U+1D50: (533, 648), U+1D56: (334, 648), U+1D57: (241, 648), U+1D5A: (525, 648), U+1D5B: (354, 648), U+1D5C: (508, 648), U+1D5D: (335, 648), U+1D5E: (372, 648), U+1D5F: (353, 648)

I did not change the anchor-1 of U+315 and U+0358. U+0358 has its anchor at the xMin and yMin of its bounding box. But U+0315 already has some clearance built in to its anchor: its anchor is 62 units below its yMin, but 25 units right of it xMin. So, I decided to provide x-clearance of 12 units and y-clearance of 80 units from the bases. All these IPA superscript letters already have a superscript meanline at $Y = 630$. So their anchor will be at $Y = 630 - 62 + 80 = 648$. The X coordinate depends on their bounding box and its xMax. To provide the desired clearance of 12 units, the anchor is at $X = \text{xMax} + 26 + 12$. These anchor coordinate values were optimized for U+0315 but they work fine for U+0358.

Bases

Here are some subgroups of bases.

Latin

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƃ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƃ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƃ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƃ

IPA

ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞɯɲɵɭʒ2ǁǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɣɤɥɦɧɨɪɿɬɮɭɻɯɰɱɲɳɴɵœɷɸɹɺɽɾʀʁɕʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖǀ
ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞɯɲɵɭʒ2ǁǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɣɤɥɦɧɨɪɿɬɮɭɻɯɰɱɲɳɴɵœɷɸɹɺɽɾʀʁɕʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖǀ
ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞɯɲɵɭʒ2ǁǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɣɤɥɦɧɨɪɿɬɮɭɻɯɰɱɲɳɴɵœɷɸɹɺɽɾʀʁɕʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖǀ
ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞɯɲɵɭʒ2ǁǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɣɤɥɦɧɨɪɿɬɮɭɻɯɰɱɲɳɴɵœɷɸɹɺɽɾʀʁɕʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖǀ

Superscript consonants

ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵡᵛ◌ᵝᵞᵟ
ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵡᵛ◌ᵝᵞᵟ
ʰʱ ʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵡᵛ◌ᵝᵞᵟ
ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵡᵛ◌ᵝᵞᵟ

Greek

ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω

Cyrillic

АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ
АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ
АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ
АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ

Precomposed capital vowels

ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈȊỈỊÒÓÔÕÖŌŎŐƠǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲƯǓǕǗǙǛỤỦỨỪỬỮỰ
ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈȊỈỊÒÓÔÕÖŌŎŐƠǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲƯǓǕǗǙǛỤỦỨỪỬỮỰ
ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈȊỈỊÒÓÔÕÖŌŎŐƠǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲƯǓǕǗǙǛỤỦỨỪỬỮỰ
ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈȊỈỊÒÓÔÕÖŌŎŐƠǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲƯǓǕǗǙǛỤỦỨỪỬỮỰ

Precomposed small vowels

àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻếềểễệìíîïĩīĭįǐȉȋỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűǔųưǖǘǚǜủụứừửữự
àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻếềểễệìíîïĩīĭįǐȉȋỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűǔųưǖǘǚǜủụứừửữự
àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻếềểễệìíîïĩīĭįǐȉȋỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűǔųưǖǘǚǜủụứừửữự
àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻếềểễệìíîïĩīĭįǐȉȋỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűǔųưǖǘǚǜủụứừửữự

Precomposed capital consonants

ḂḄÇḈĈĊČḌĎḐḒḎḐḊĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴĶḲĹḶĽḺḸḼḾṀṂŃṆŇṊṄƝƠṔṖṕŖŘŖŖŚŜṠŠṢṤŠṦṨŞṰŢṪṬṮṰȚƬƮŨṾỤỦẂẀẄẂẊẌÝỲŸỴỶỸȲỾŹŻŽẔẐẒƵ
ḂḄÇḈĈĊČḌĎḐḒḎḐḊĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴĶḲĹḶĽḺḸḼḾṀṂŃṆŇṊṄƝƠṔṖṕŖŘŖŖŚŜṠŠṢṤŠṦṨŞṰŢṪṬṮṰȚƬƮŨṾỤỦẂẀẄẂẊẌÝỲŸỴỶỸȲỾŹŻŽẔẐẒƵ
ḂḄÇḈĈĊČḌĎḐḒḎḐḊĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴĶḲĹḶĽḺḸḼḾṀṂŃṆŇṊṄƝƠṔṖṕŖŘŖŖŚŜṠŠṢṤŠṦṨŞṰŢṪṬṮṰȚƬƮŨṾỤỦẂẀẄẂẊẌÝỲŸỴỶỸȲỾŹŻŽẔẐẒƵ
ḂḄÇḈĈĊČḌĎḐḒḎḐḊĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴĶḲĹḶĽḺḸḼḾṀṂŃṆŇṊṄƝƠṔṖṕŖŘŖŖŚŜṠŠṢṤŠṦṨŞṰŢṪṬṮṰȚƬƮŨṾỤỦẂẀẄẂẊẌÝỲŸỴỶỸȲỾŹŻŽẔẐẒƵ

Precomposed small consonants

ƀƅɓƃbçćĉċčɕƈďđḑḓḋḍḏđƌfĝğġģḡǧǥĥħḫḥḧḣḩȟĵķḱǩḵƙĸĺļľŀłḽḷḻḹŀɬḿṁṃṅṇňŉṋņṉɲŋɵɑɋṗƥƿɍŗřŕȓȑṛṟśŝşšșṡṣßſʂţťŧṭṯṱʈƭʉṽṿẁŵẃẅẋẍýÿŷỳẏỵỹƴźżžẑẓȥʒƨʂpƺ
ƀƅɓƃbçćĉċčɕƈďđḑḓḋḍḏđƌfĝğġģḡǧǥĥħḫḥḧḣḩȟĵķḱǩḵƙĸĺļľŀłḽḷḻḹŀɬḿṁṃṅṇňŉṋņṉɲŋɵɑɋṗƥƿɍŗřŕȓȑṛṟśŝşšșṡṣßſʂţťŧṭṯṱʈƭʉṽṿẁŵẃẅẋẍýÿŷỳẏỵỹƴźżžẑẓȥʒƨʂpƺ
ƀƅɓƃbçćĉċčɕƈďđḑḓḋḍḏđƌfĝğġģḡǧǥĥħḫḥḧḣḩȟĵķḱǩḵƙĸĺļľŀłḽḷḻḹŀɬḿṁṃṅṇňŉṋņṉɲŋɵɑɋṗƥƿɍŗřŕȓȑṛṟśŝşšșṡṣßſʂţťŧṭṯṱʈƭʉṽṿẁŵẃẅẋẍýÿŷỳẏỵỹƴźżžẑẓȥʒƨʂpƺ

ƀƅɓƃbçćĉċčɕƈďđḑḓḋḍḏđƌfĝğġģḡǧǥĥħḫḥḧḣḩȟĵķḱǩḵƙĸĺļľŀłḽḷḻḹŀɬḿṁṃṅṇňŉṋņṉɲŋɵɑɋṗƥƿɍŗřŕȓȑṛṟśŝşšșṡṣßſʂţťŧṭṯṱʈƭʉṽṿẁŵẃẅẋẍýÿŷỳẏỵỹƴźżžẑẓȥʒƨʂpƺ

No marks: rare, digraph etc.

ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj

Small capitals

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ

Precomposed characters of anchor-relevant bases and marks

àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ

Base and mark shaping – Linguistics notation

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic.

The grids use Latin in IPA bases and combining marks that are used in linguistics notation.

A black color means GPOS anchor positioning.

A blue color means a precomposed glyph.

A red color means HarfBuzz fallback shaping (no anchors, no precomposed glyph, no GSUB substitution).

A faded color – gray, light blue, or pink – means that the base and mark combination is not reasonably expected in notation in linguistics.

A gray color can also mean that mark glyph is missing in the font. In this case, just the gray base is printed. This only happens in Libertinus Serif Semibold and Semibold Italic for three below marks: o͇ (U+0333 double macron below), o͚ (U+033A inverted bridge below), and o͟ (U+033B square below). Otherwise, all considered base and mark combinations can be drawn in XeLaTeX documents in Libertinus without tofu for missing glyphs.

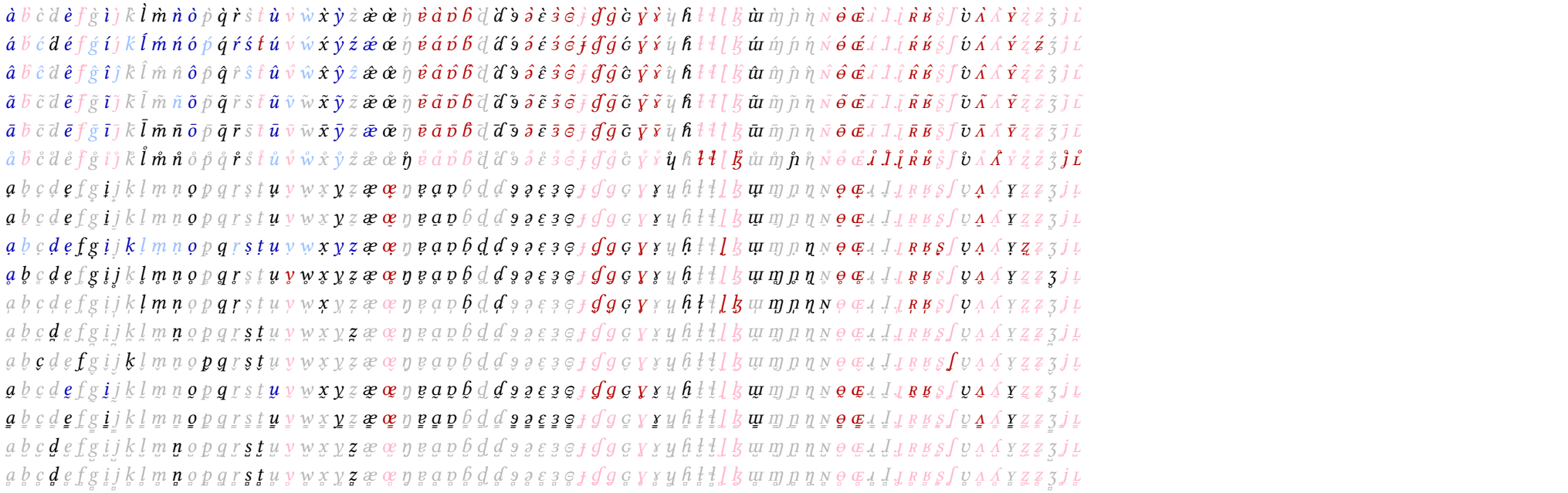
Regular, Bases requiring anchors for common marks, Common combining marks



Regular patched, Bases requiring anchors for common marks, Common combining marks



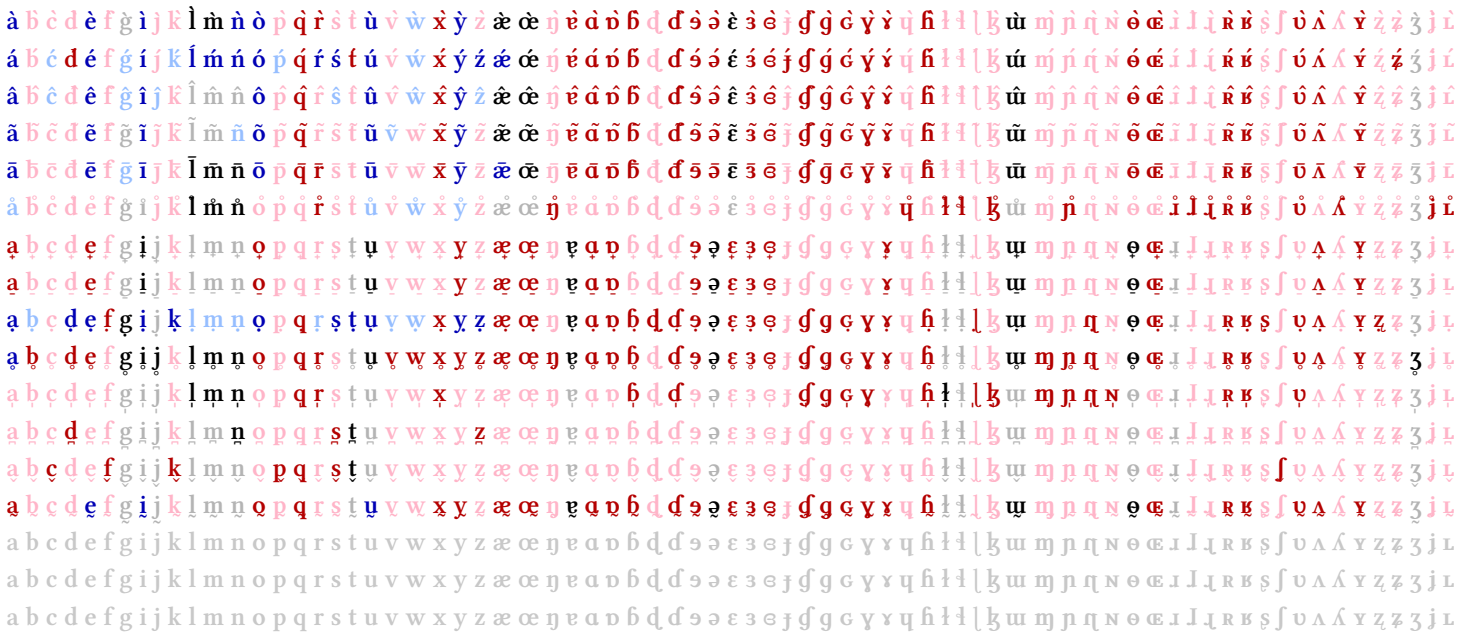
Italic, Bases requiring anchors for common marks, Common combining marks



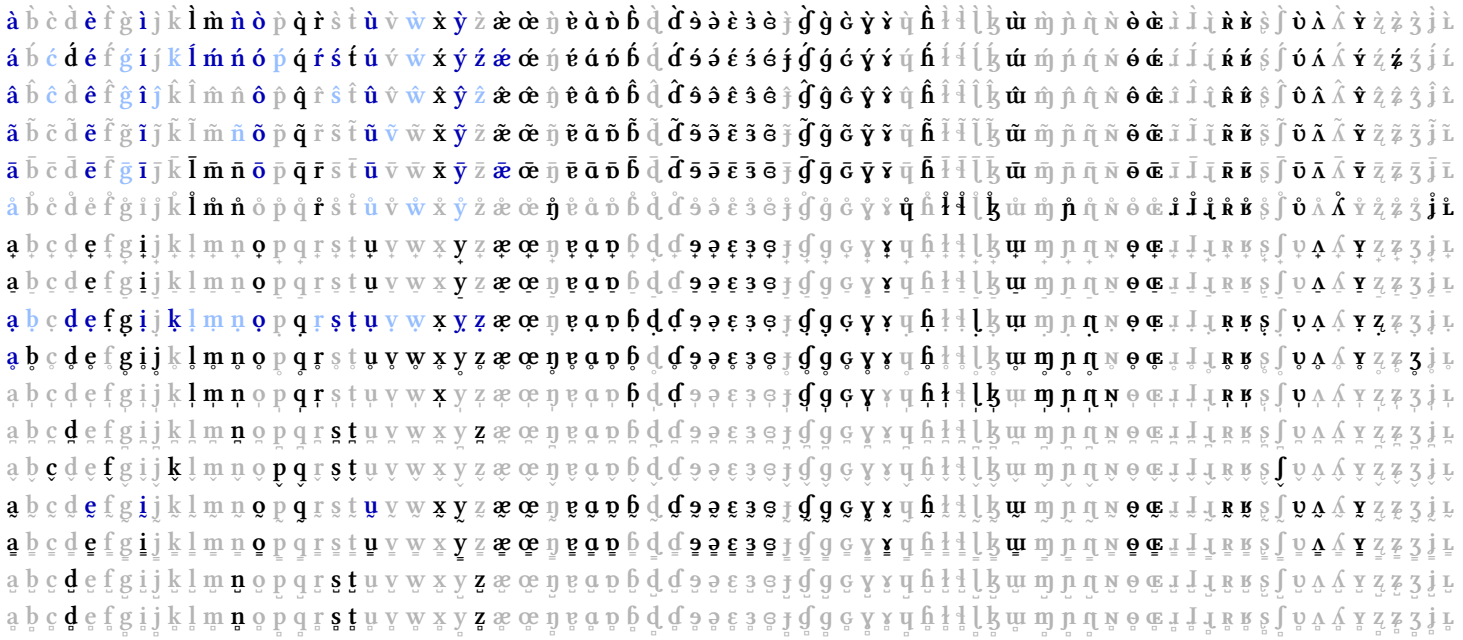
Italic patched, Bases requiring anchors for common marks, Common combining marks



Semibold, Bases requiring anchors for common marks, Common combining marks



Semibold patched, Bases requiring anchors for common marks, Common combining marks



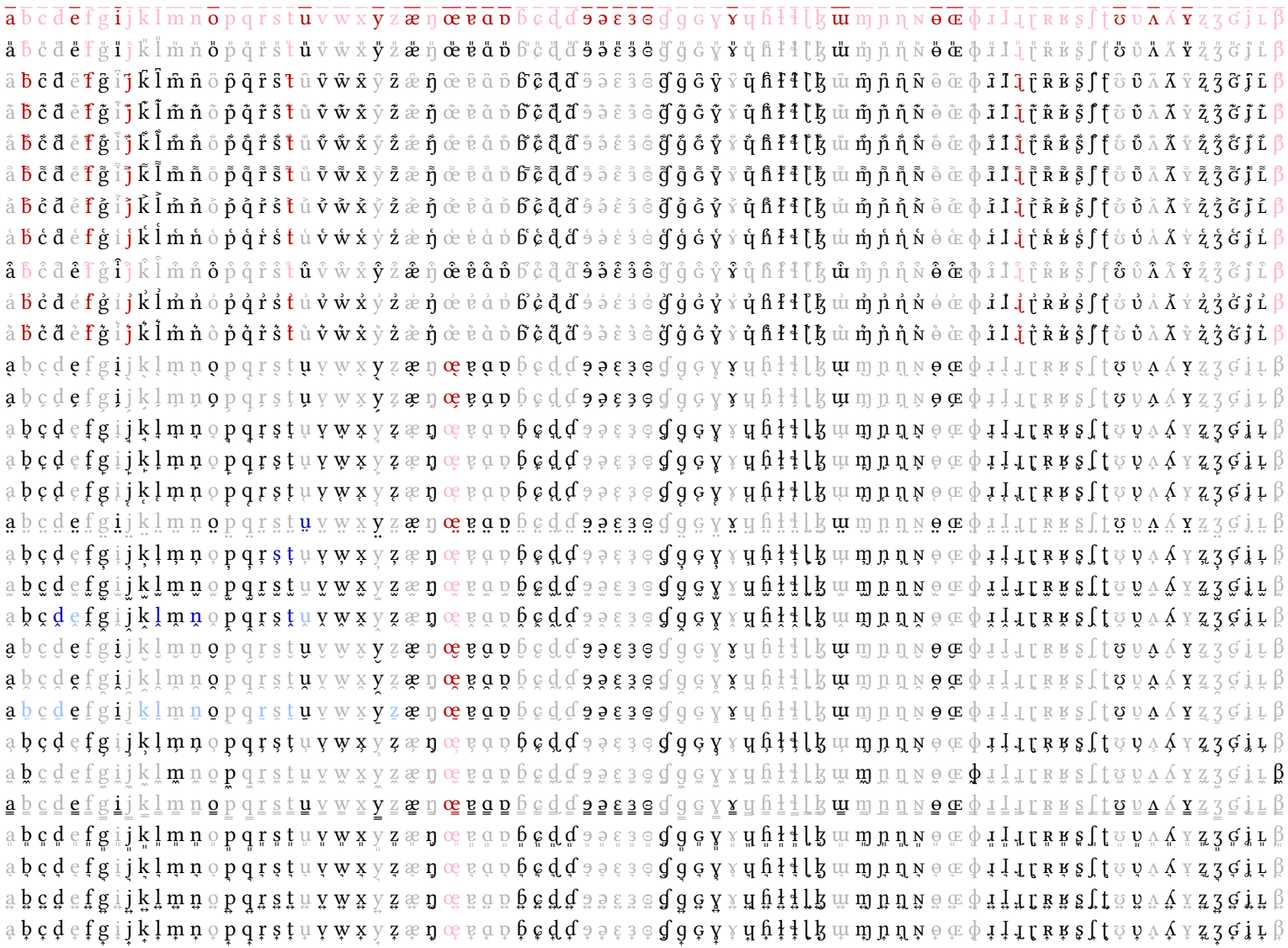
Semibold italic, Bases requiring anchors for common marks, Common combining marks



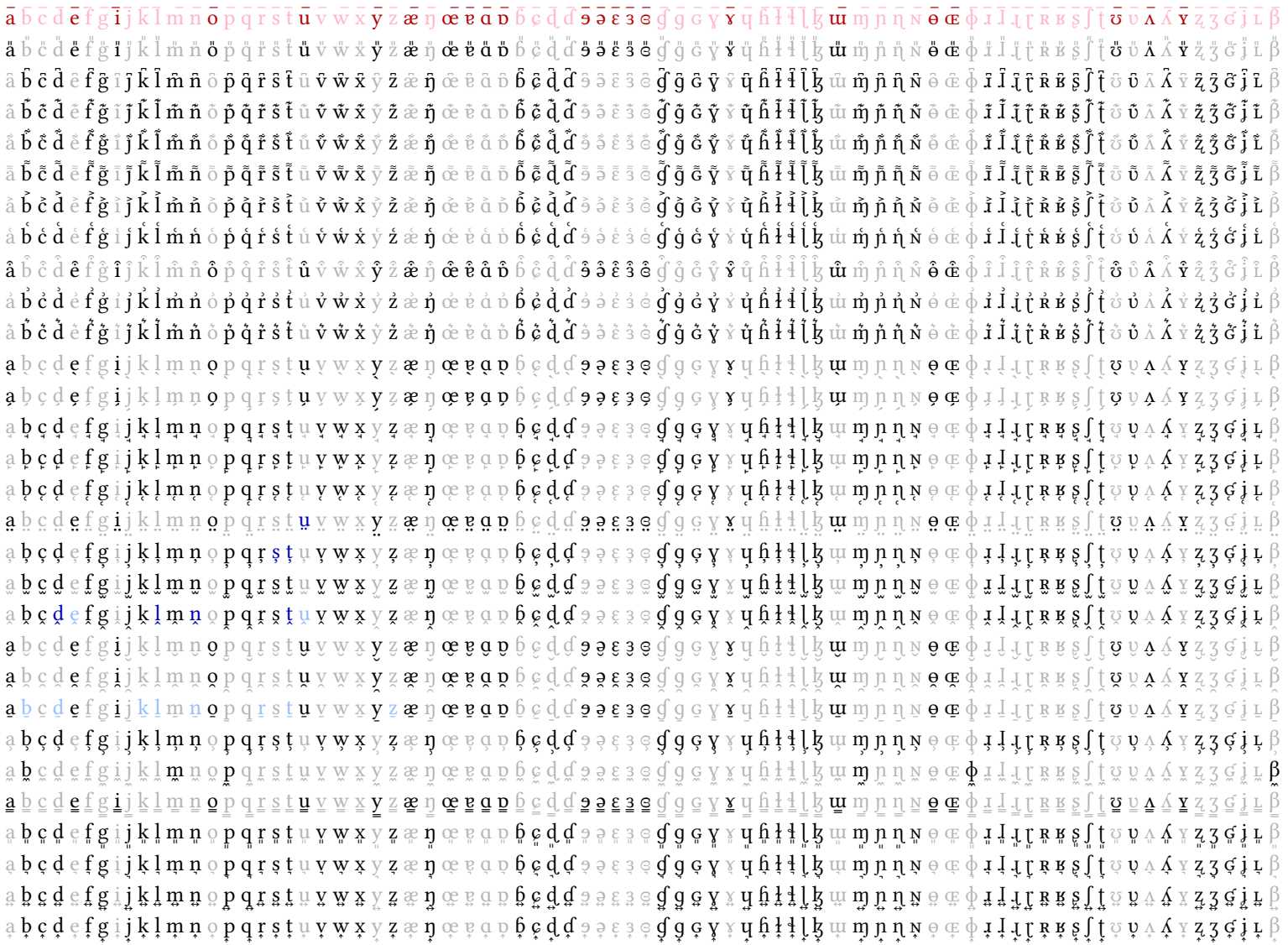
Semibold italic patched, Bases requiring anchors for common marks, Common combining marks



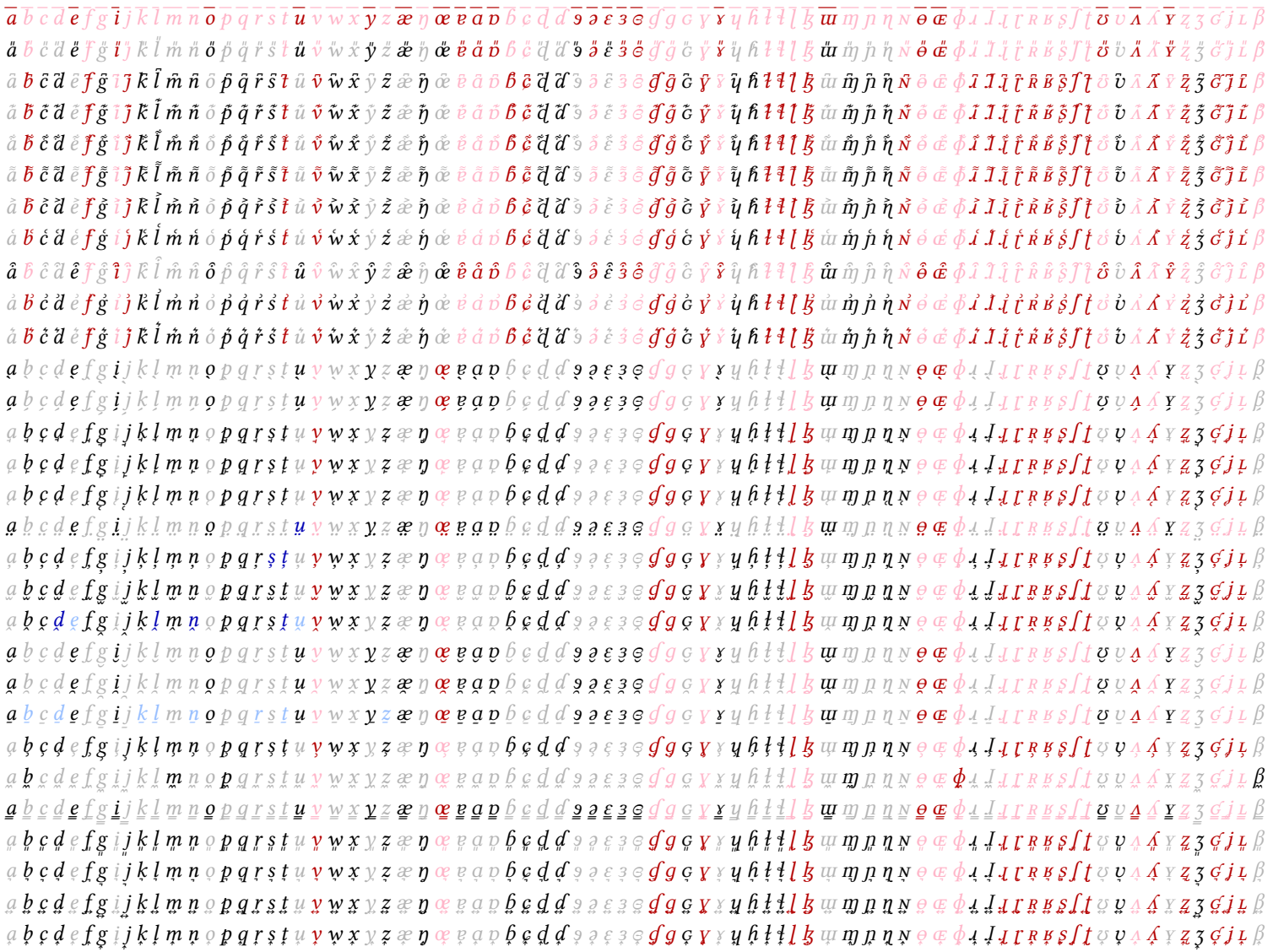
Regular, Bases requiring anchors for rare marks, Rare combining marks



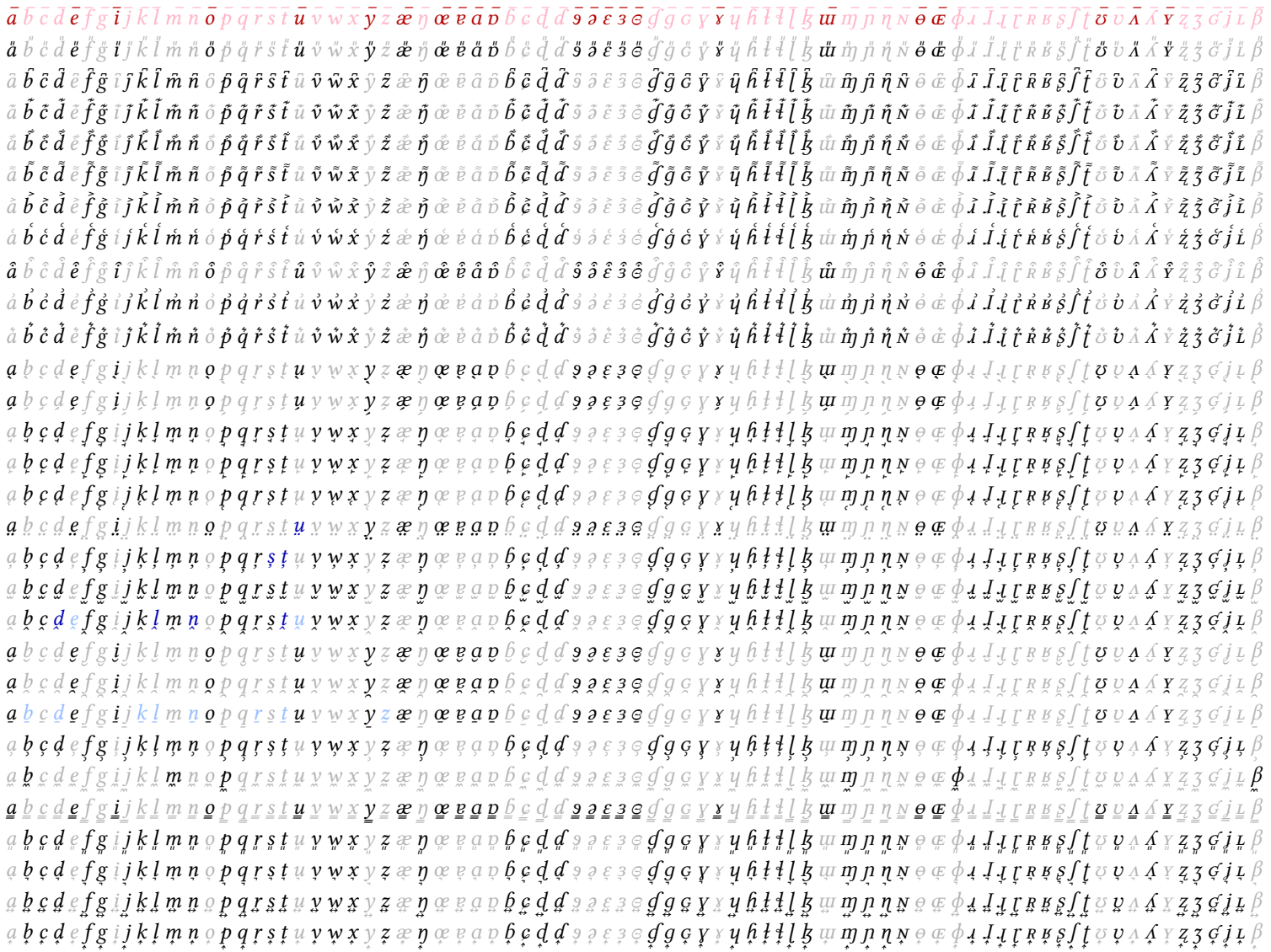
Regular patched, Bases requiring anchors for rare marks, Rare combining marks



Italic, Bases requiring anchors for rare marks, Rare combining marks



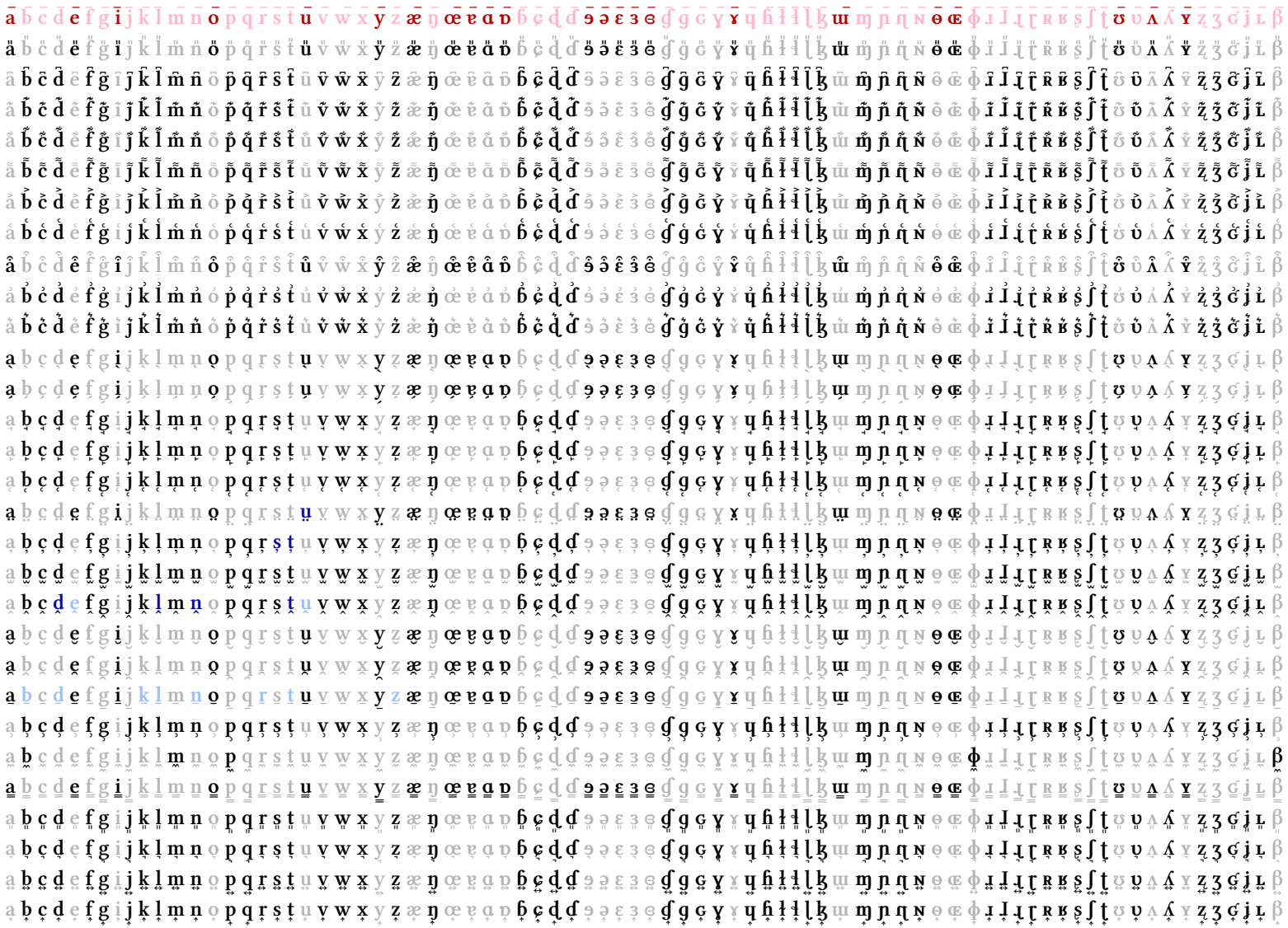
Italic patched, Bases requiring anchors for rare marks, Rare combining marks



Semibold, Bases requiring anchors for rare marks, Rare combining marks



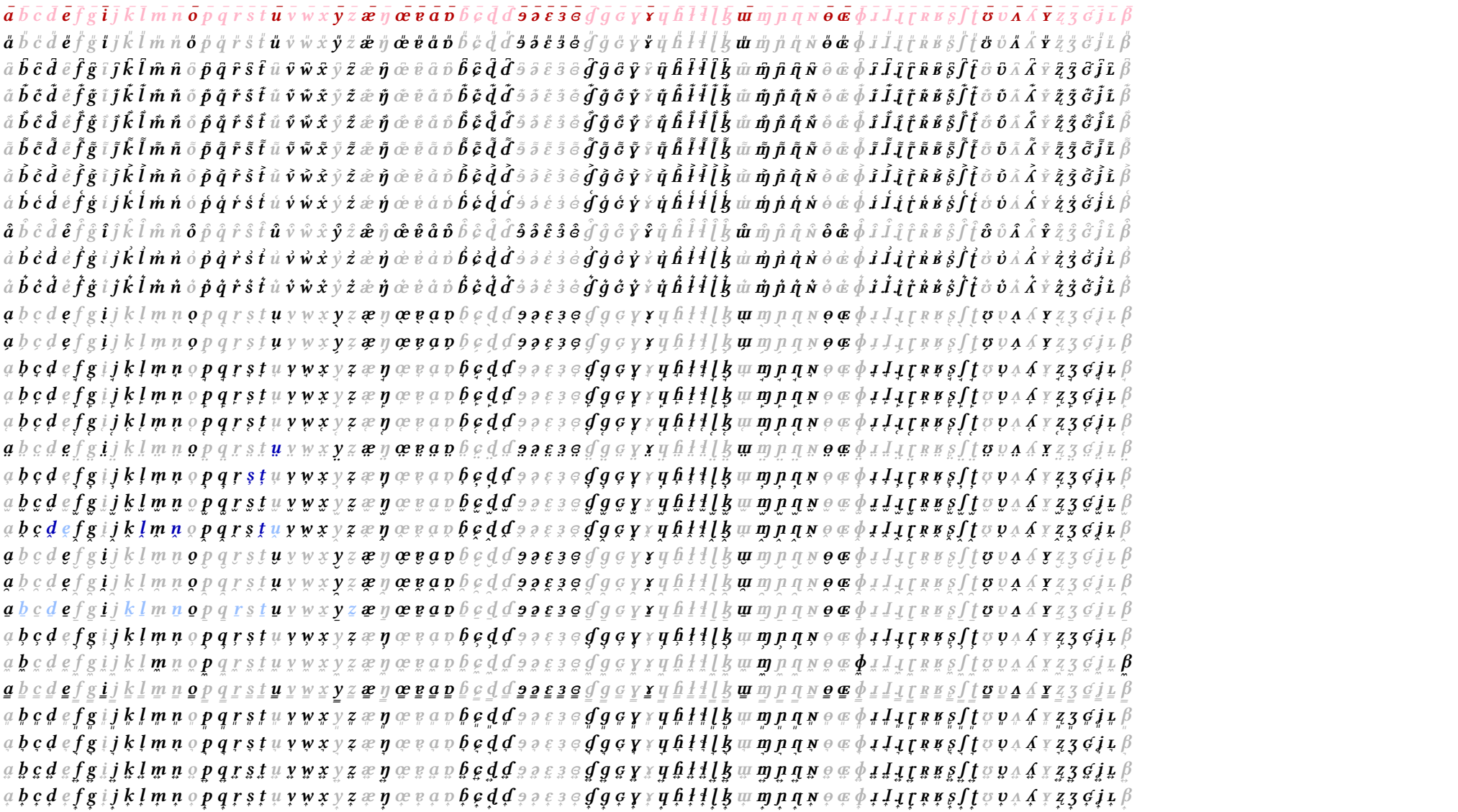
Semibold patched, Bases requiring anchors for rare marks, Rare combining marks



Semibold italic, Bases requiring anchors for rare marks, Rare combining marks



Semibold italic patched, Bases requiring anchors for rare marks, Rare combining marks



Base and mark shaping – Complete

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic. The grids use more complete sets of Latin and IPA bases and above and below marks.

Blue indicates that XeLaTeX uses a precomposed Unicode character glyph.

Black indicates that the mark is positioned using GPOS anchors.

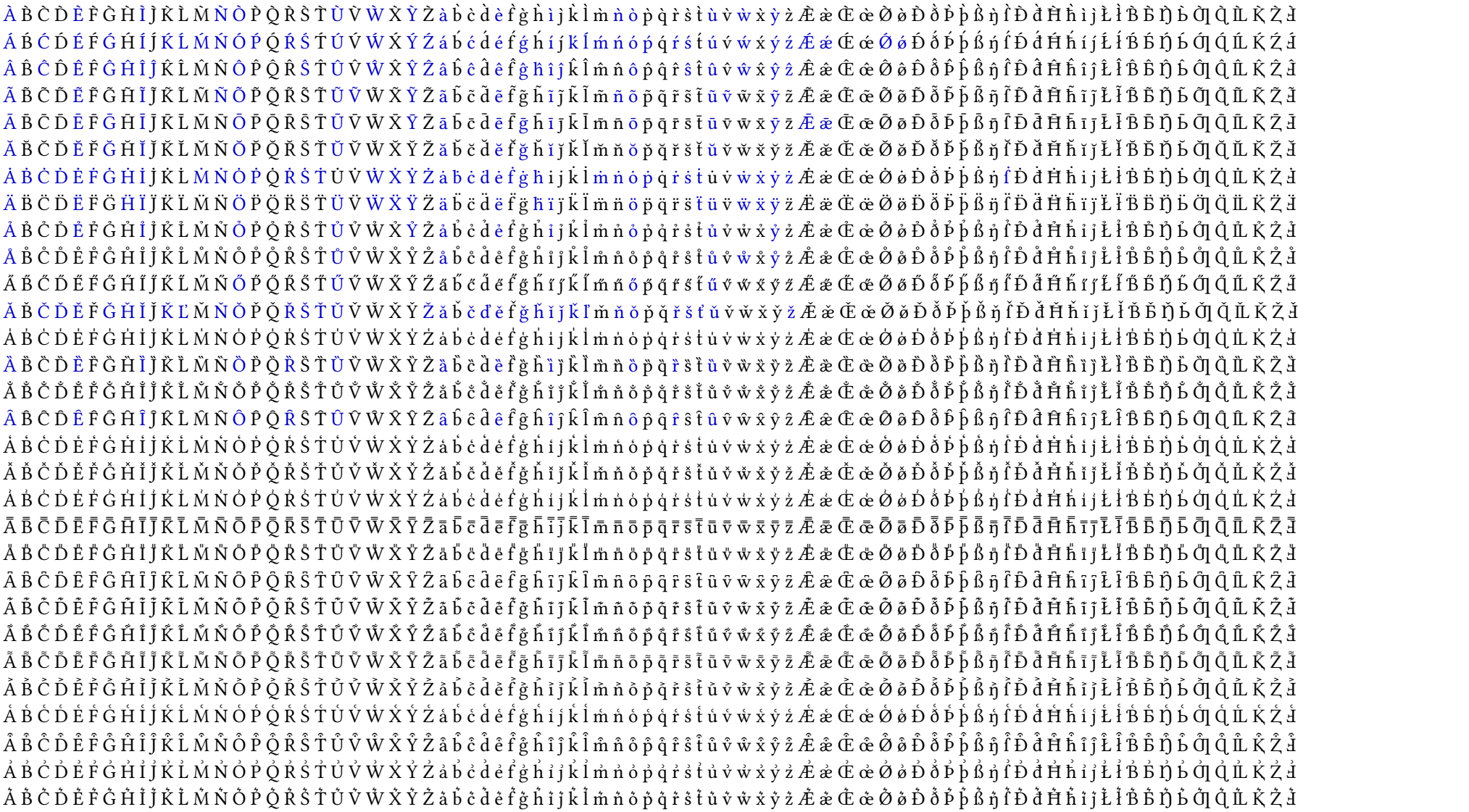
Red indicates HarfBuzz fallback shaping: no precomposed glyph and no GPOS anchors are used.

Light gray indicates that either the base or the mark glyph is missing from the font. In this case only the base glyph is shown.

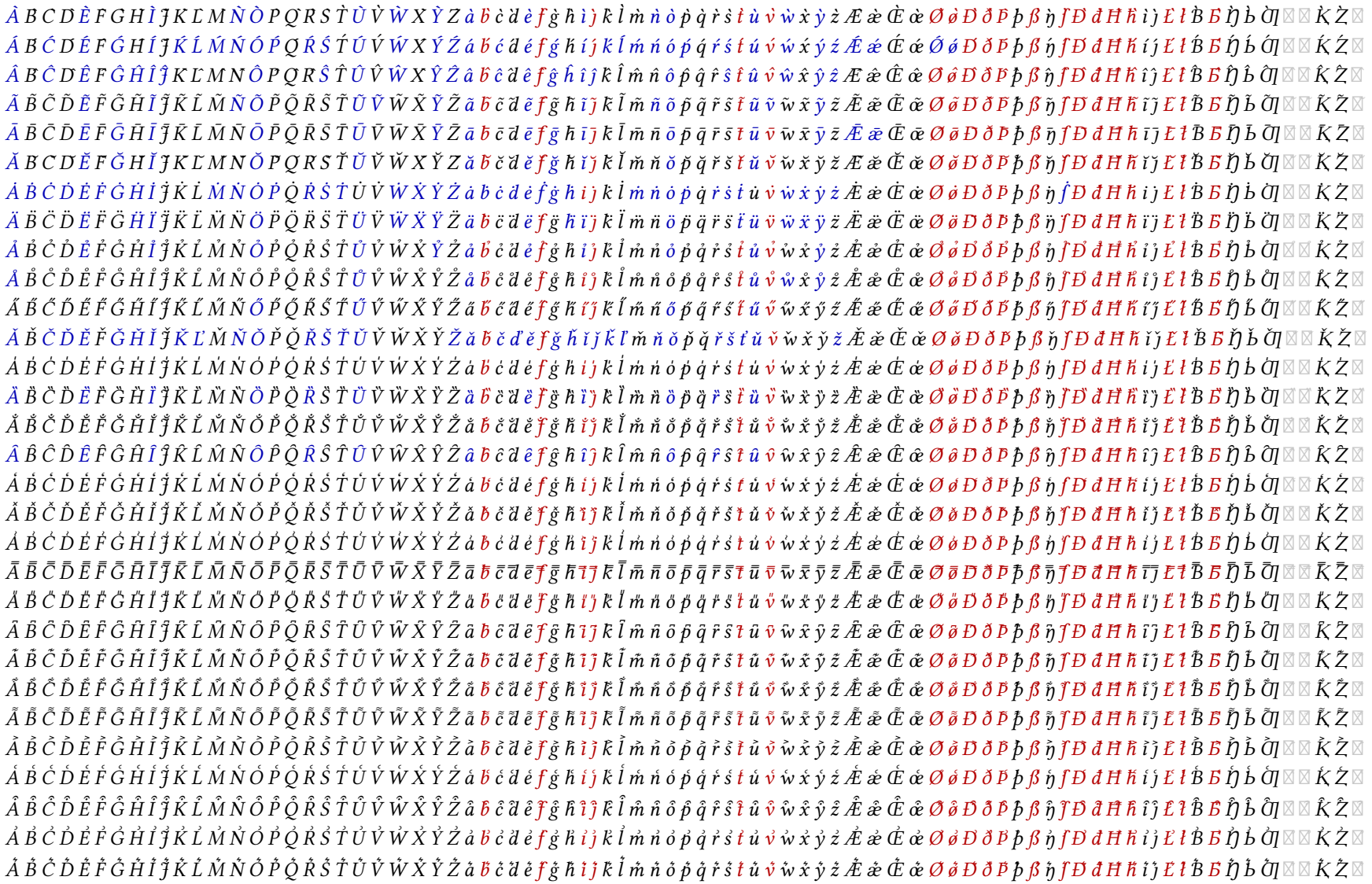
Regular, Latin, Above



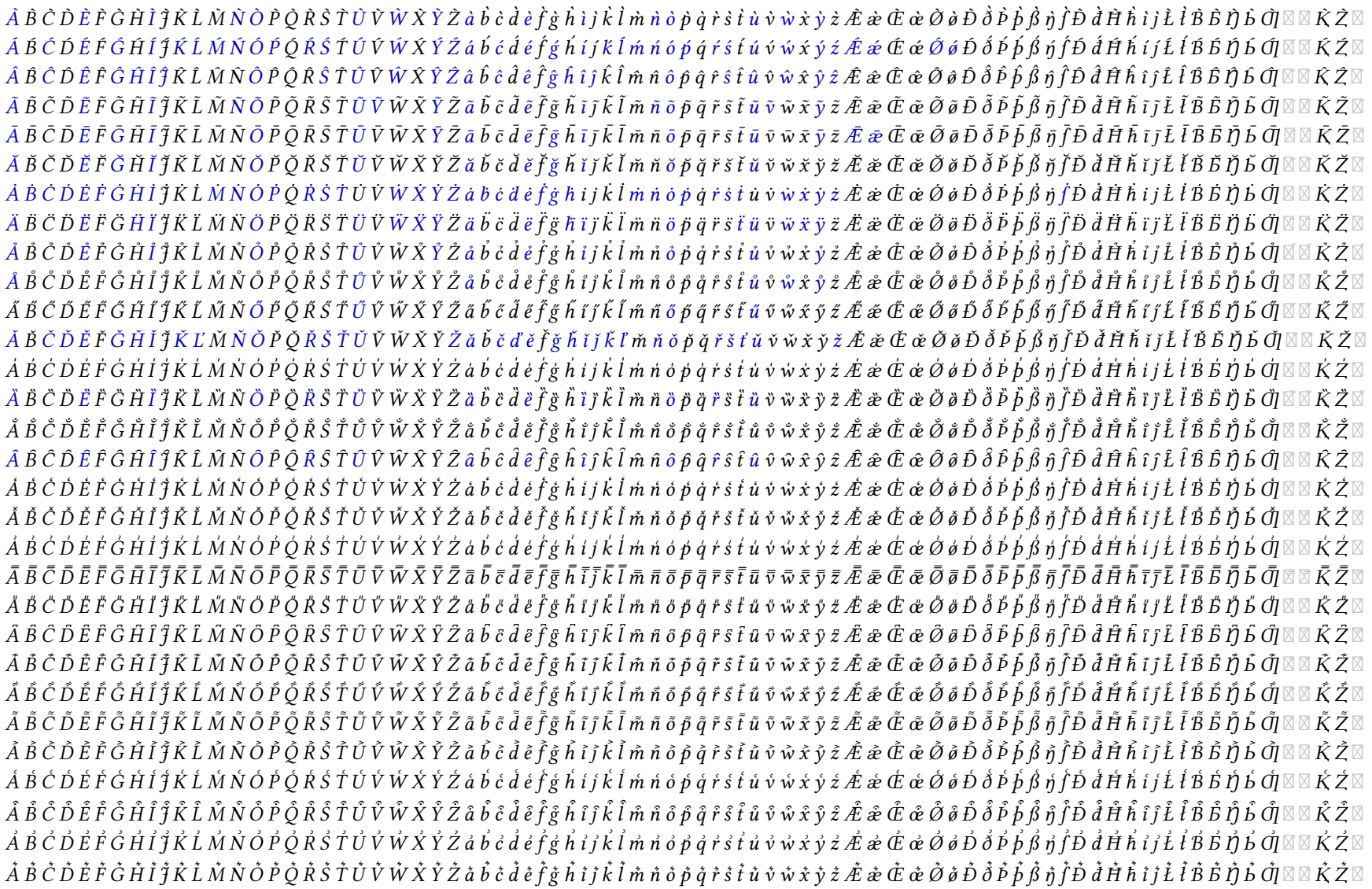
Regular patched, Latin, Above



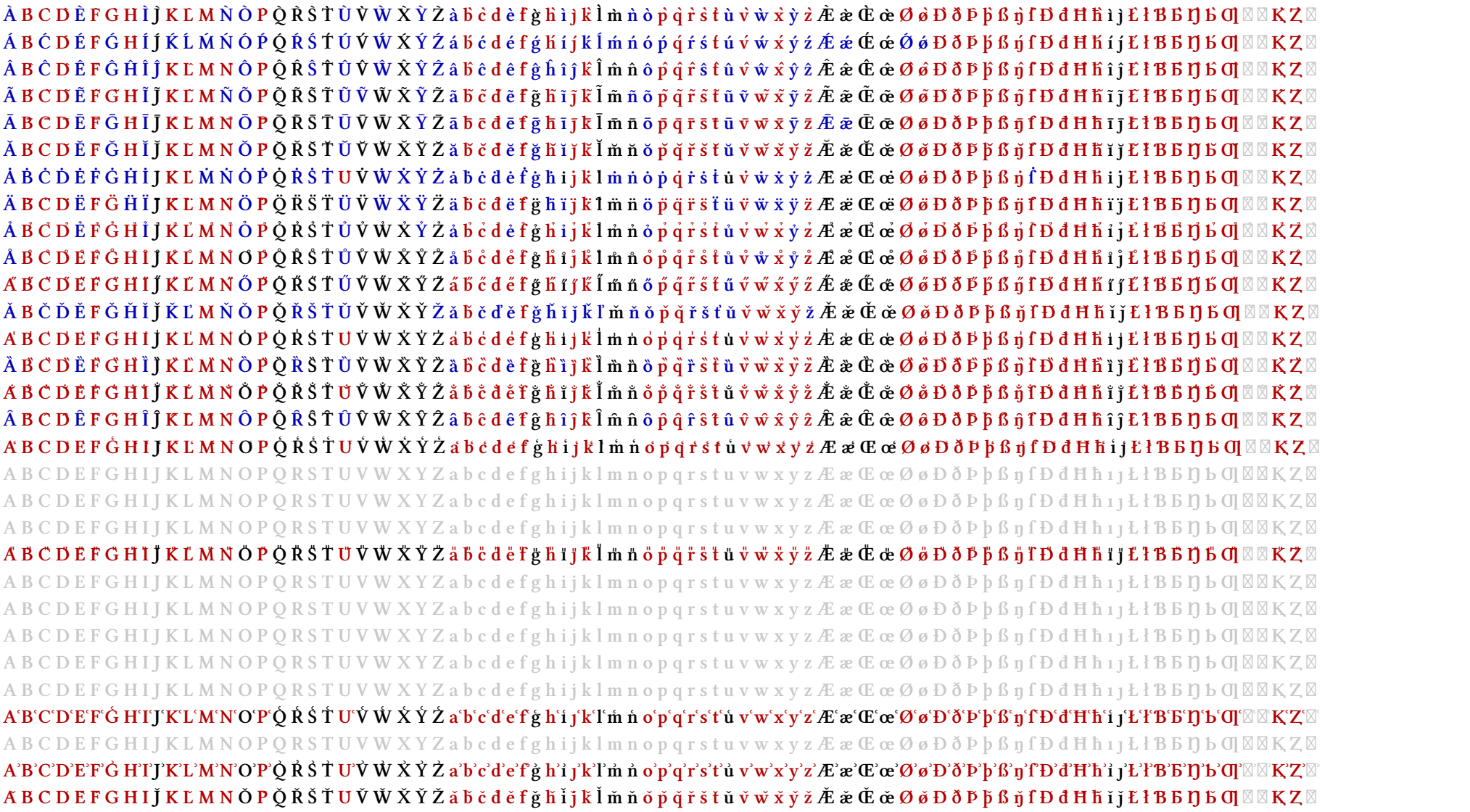
Italic, Latin, Above



Italic patched, Latin, Above



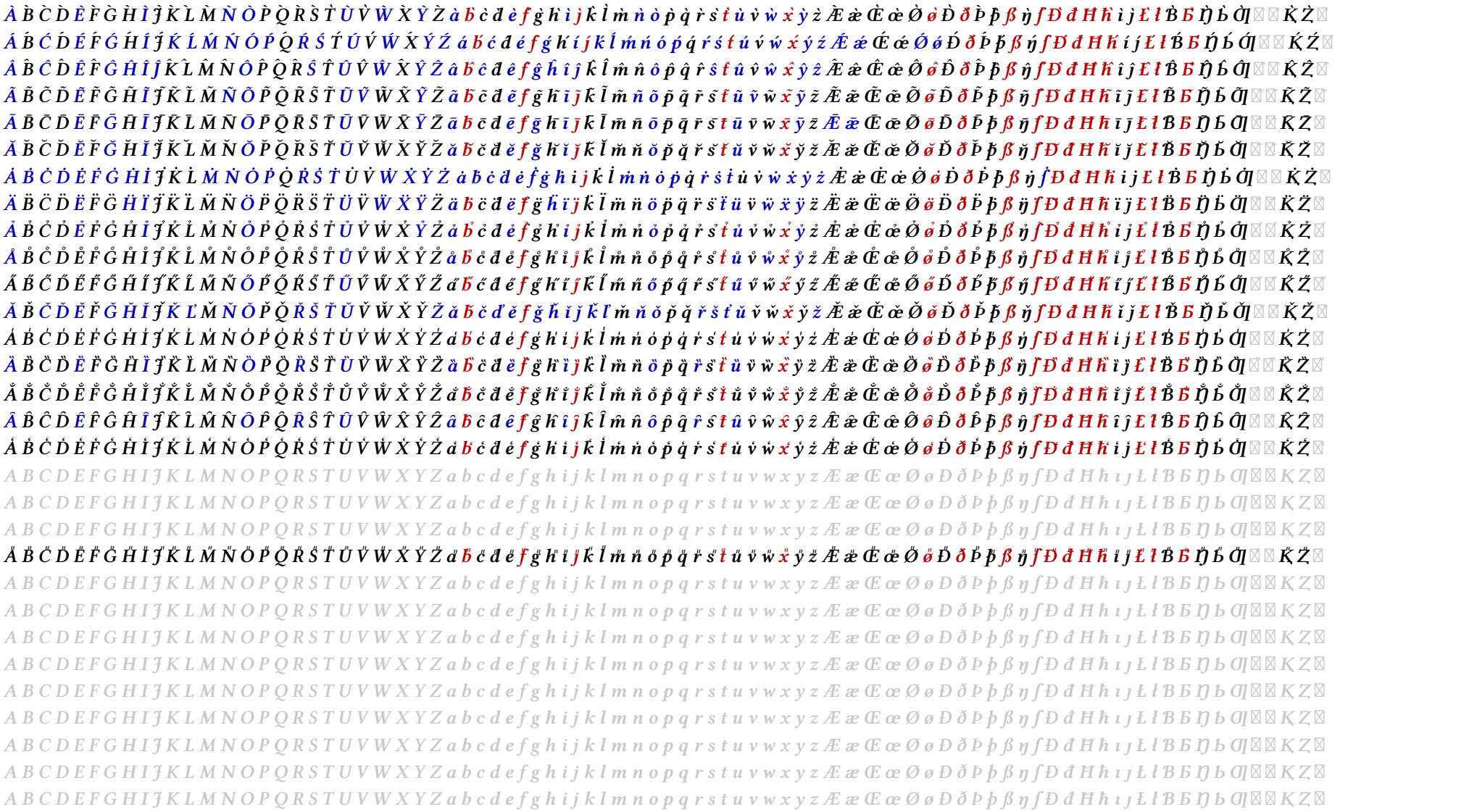
Semibold, Latin, Above



Semibold patched, Latin, Above

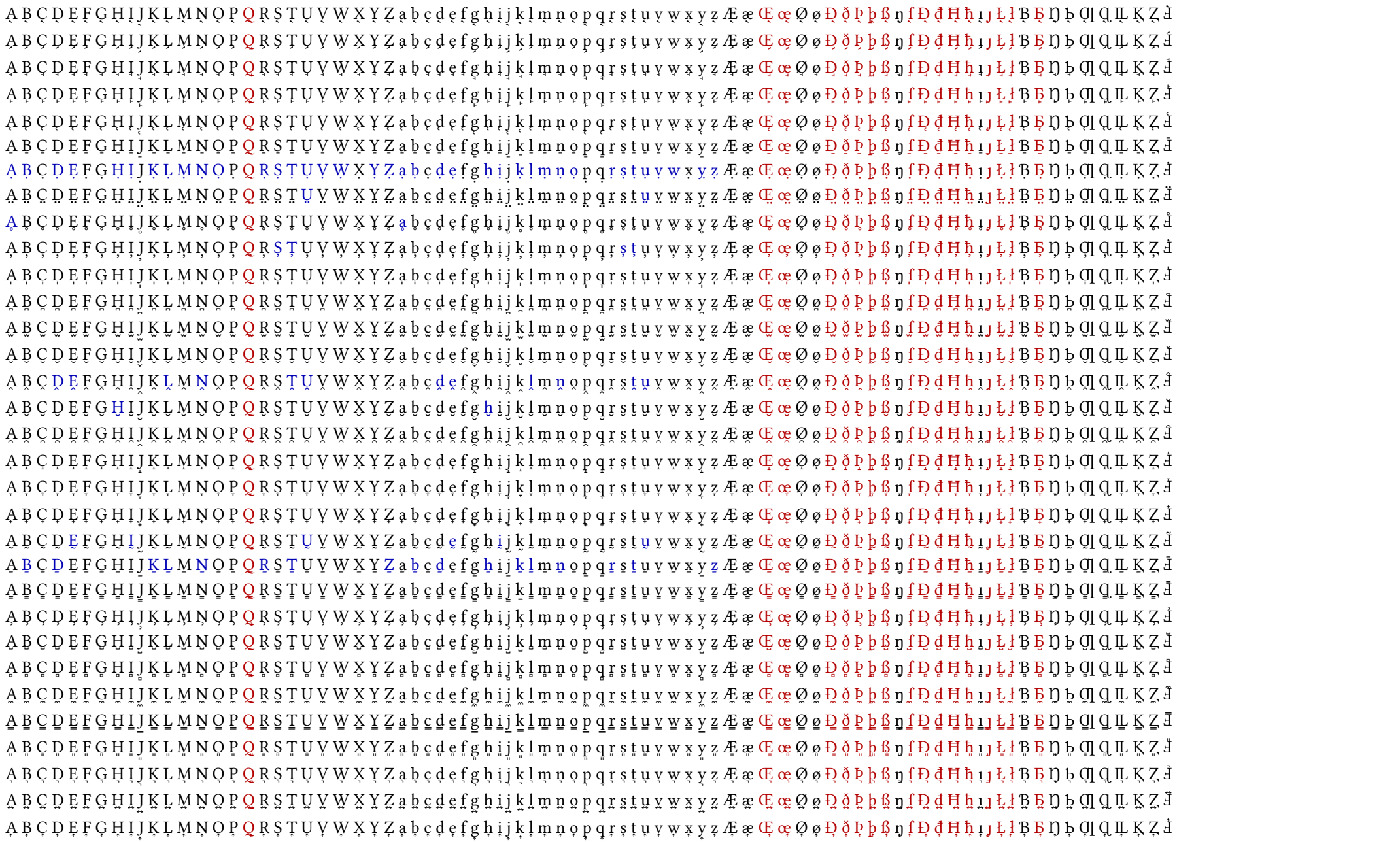


Semibold italic, Latin, Above

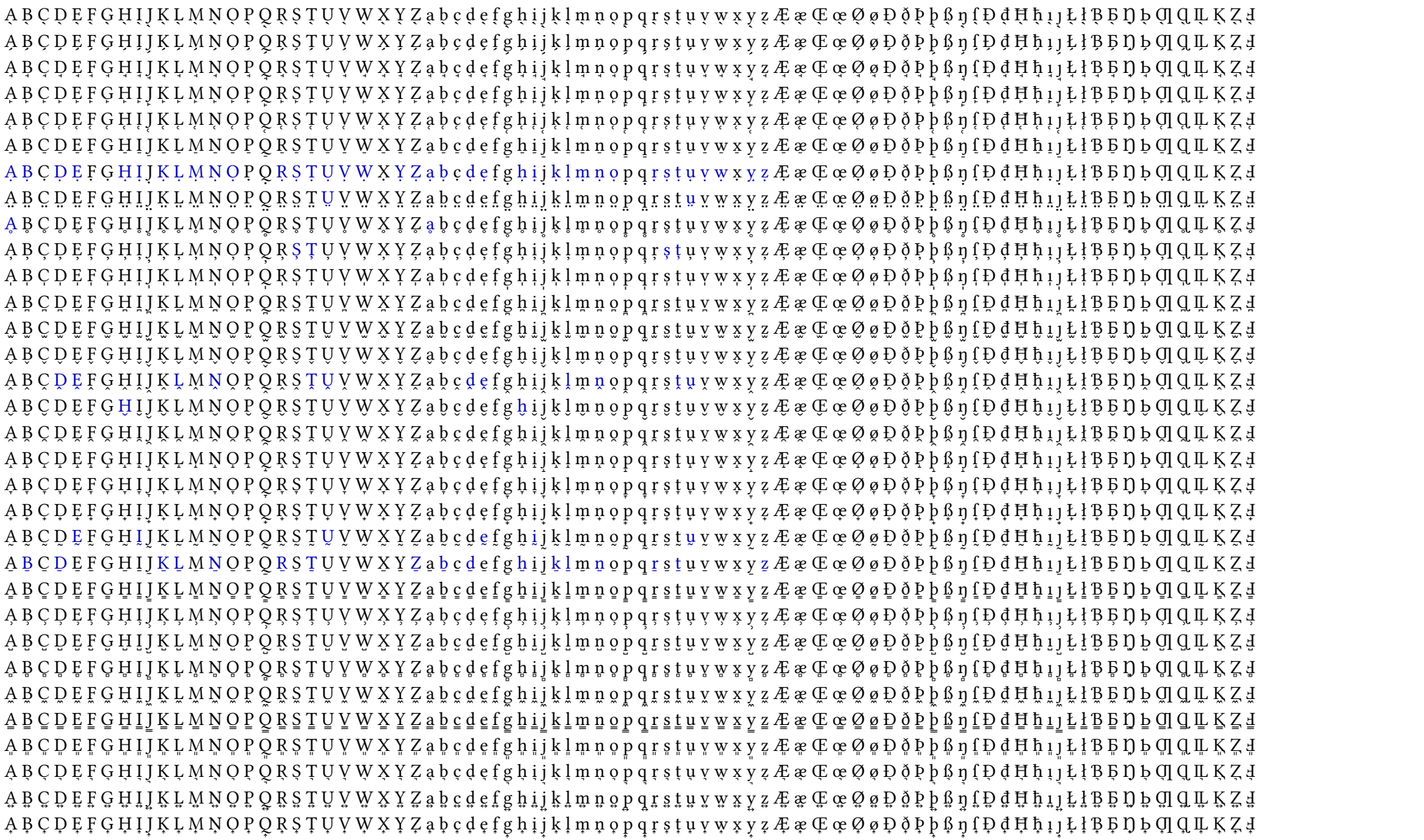


Semibold italic patched, Latin, Above

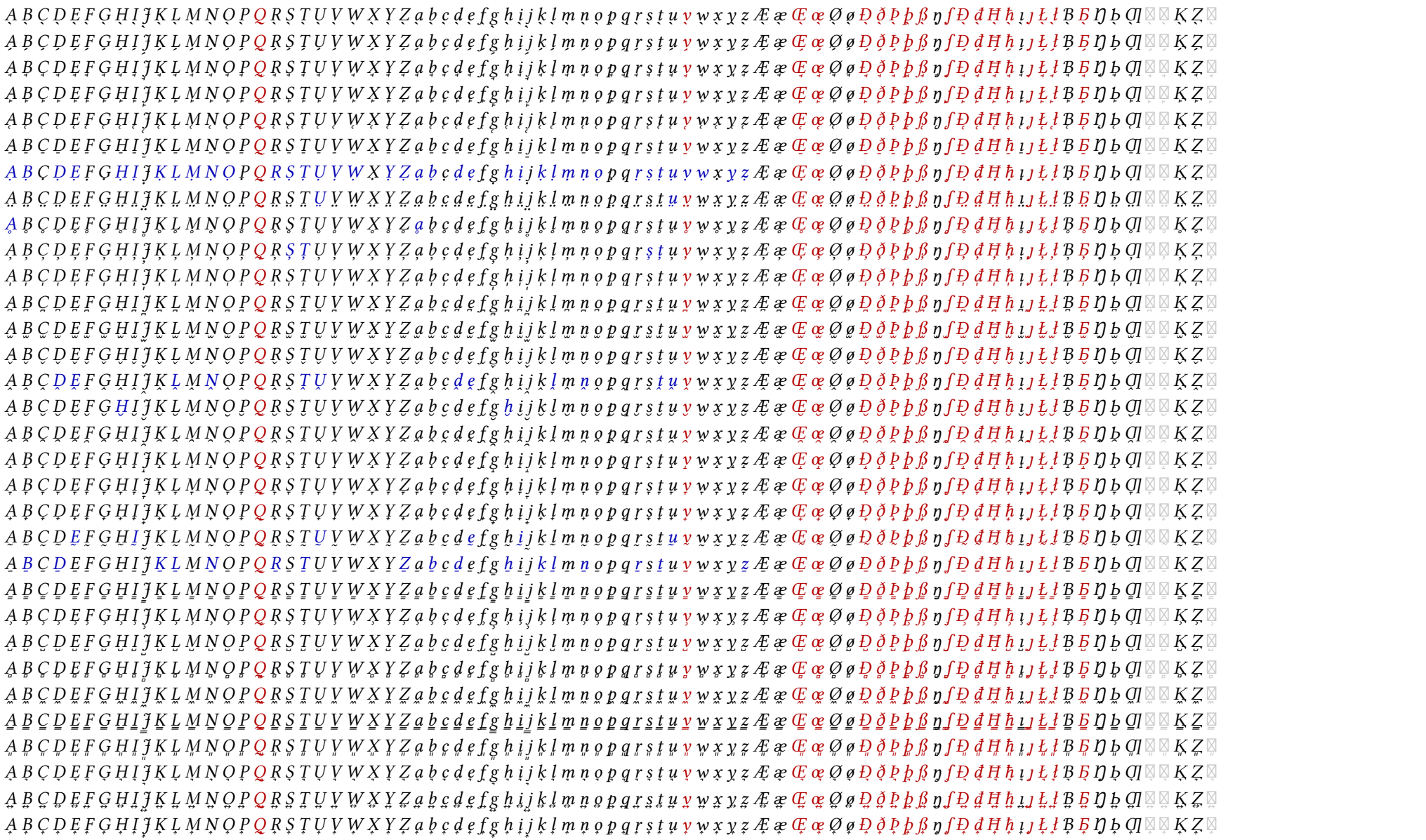
Regular, Latin, Below



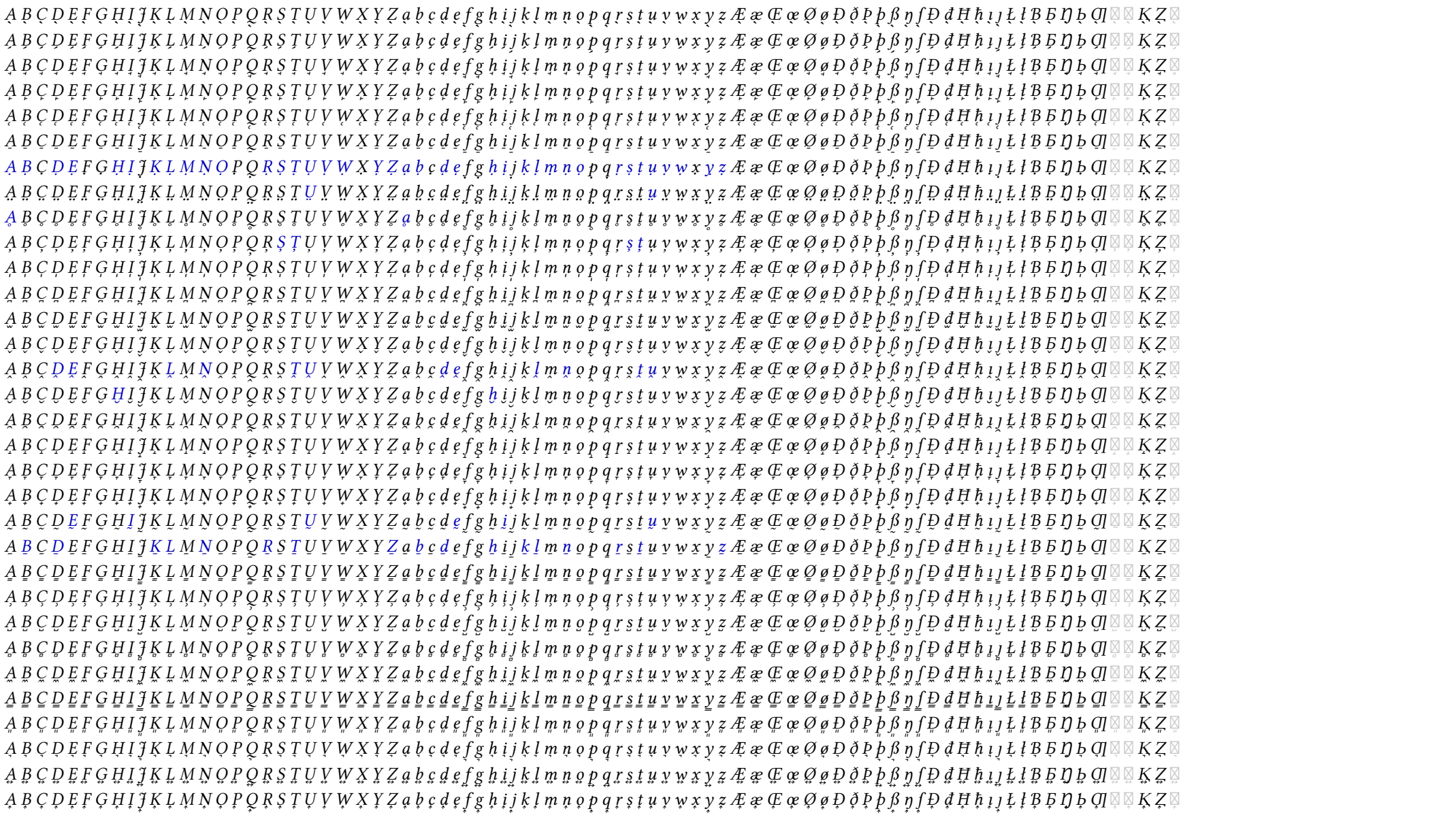
Regular patched, Latin, Below



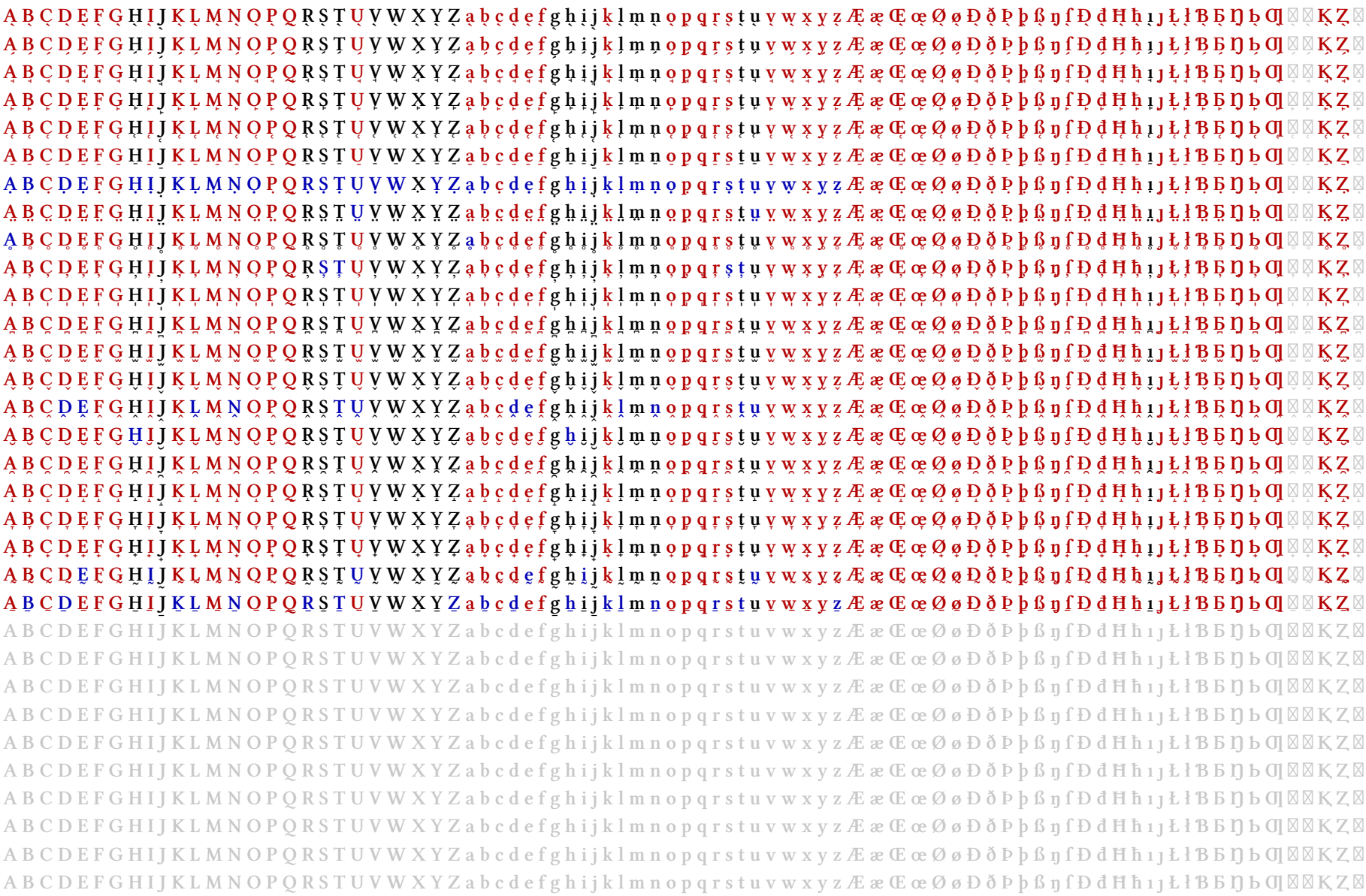
Italic, Latin, Below



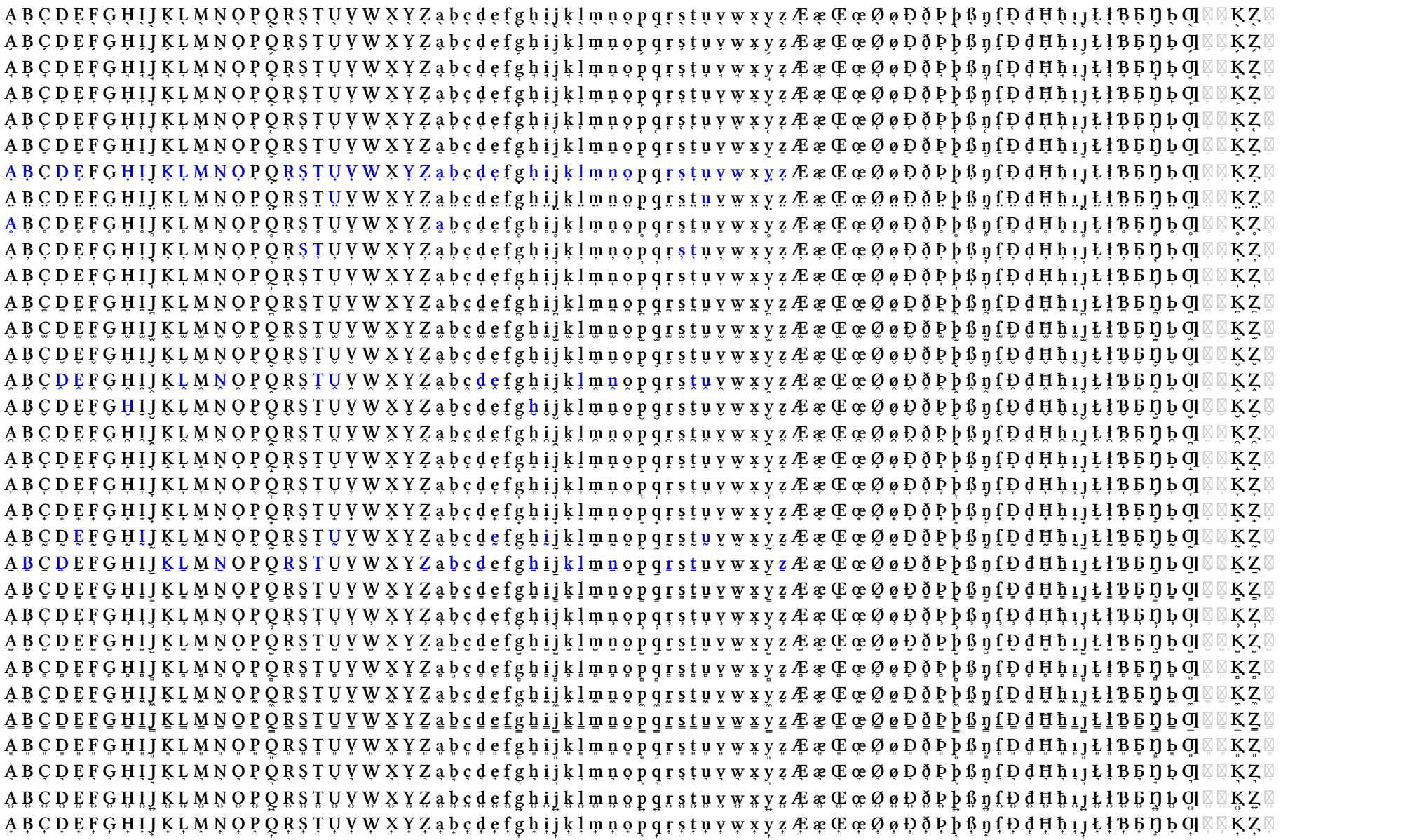
Italic patched, Latin, Below



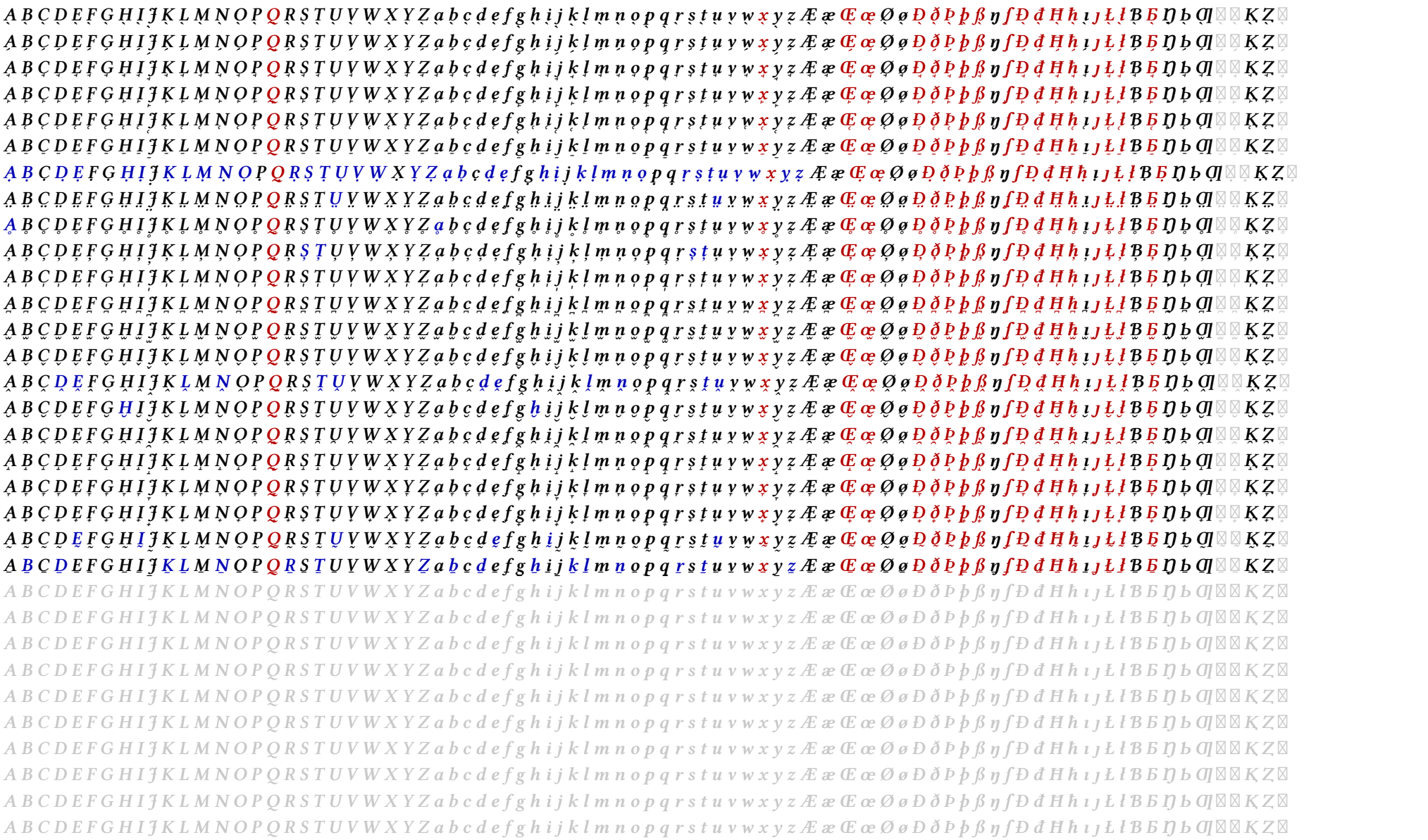
Semibold, Latin, Below



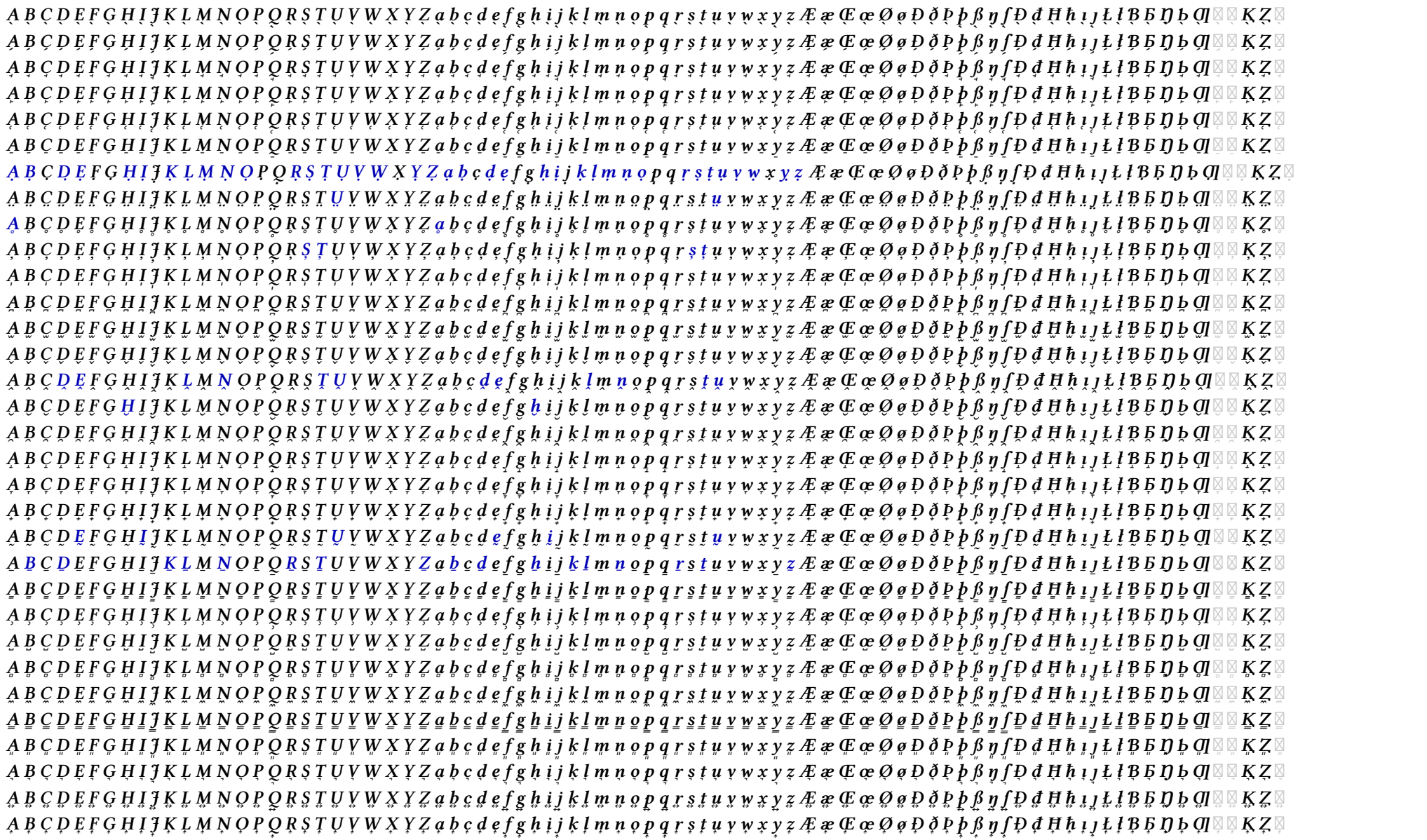
Semibold patched, Latin, Below



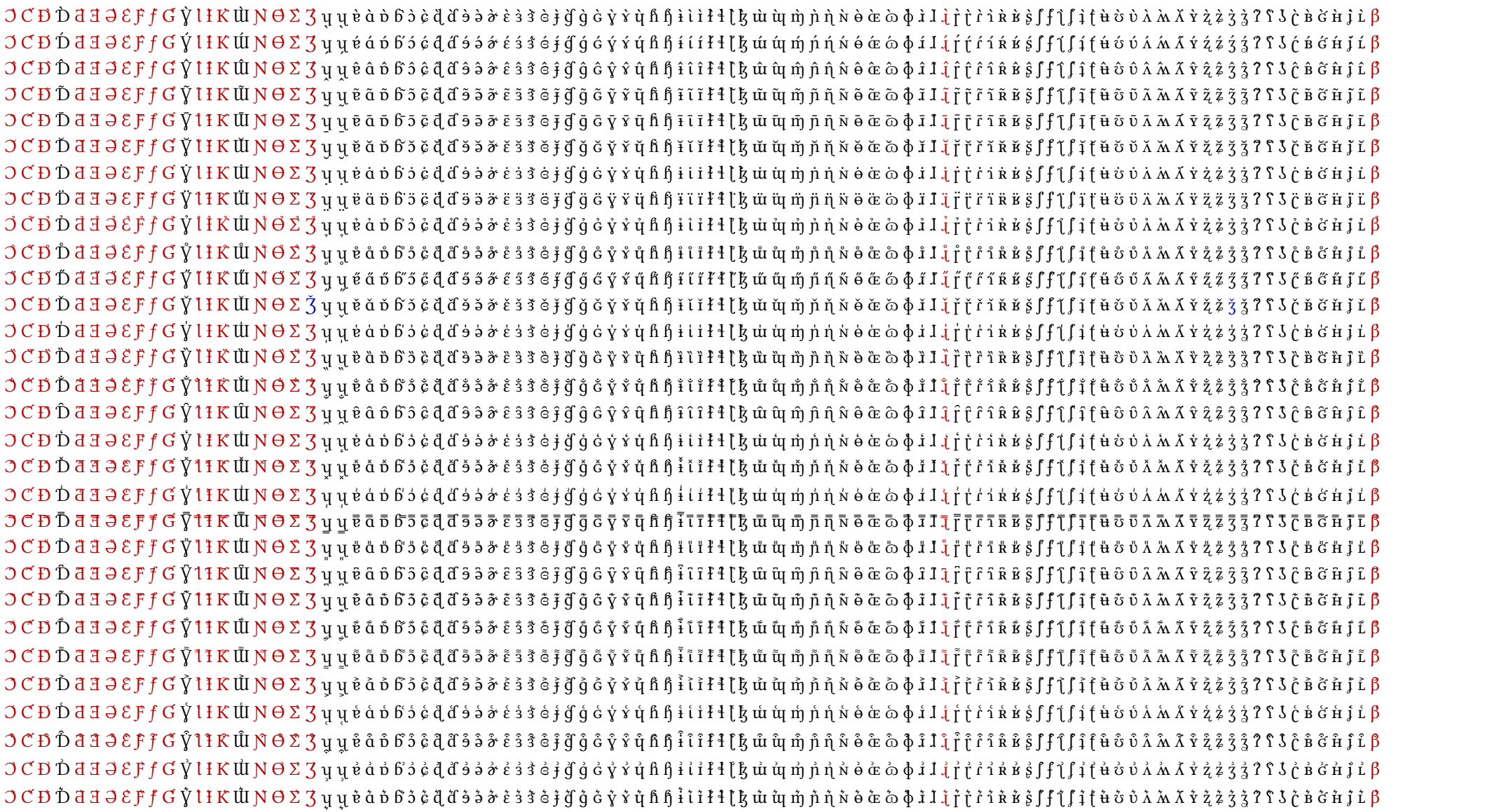
Semibold italic, Latin, Below



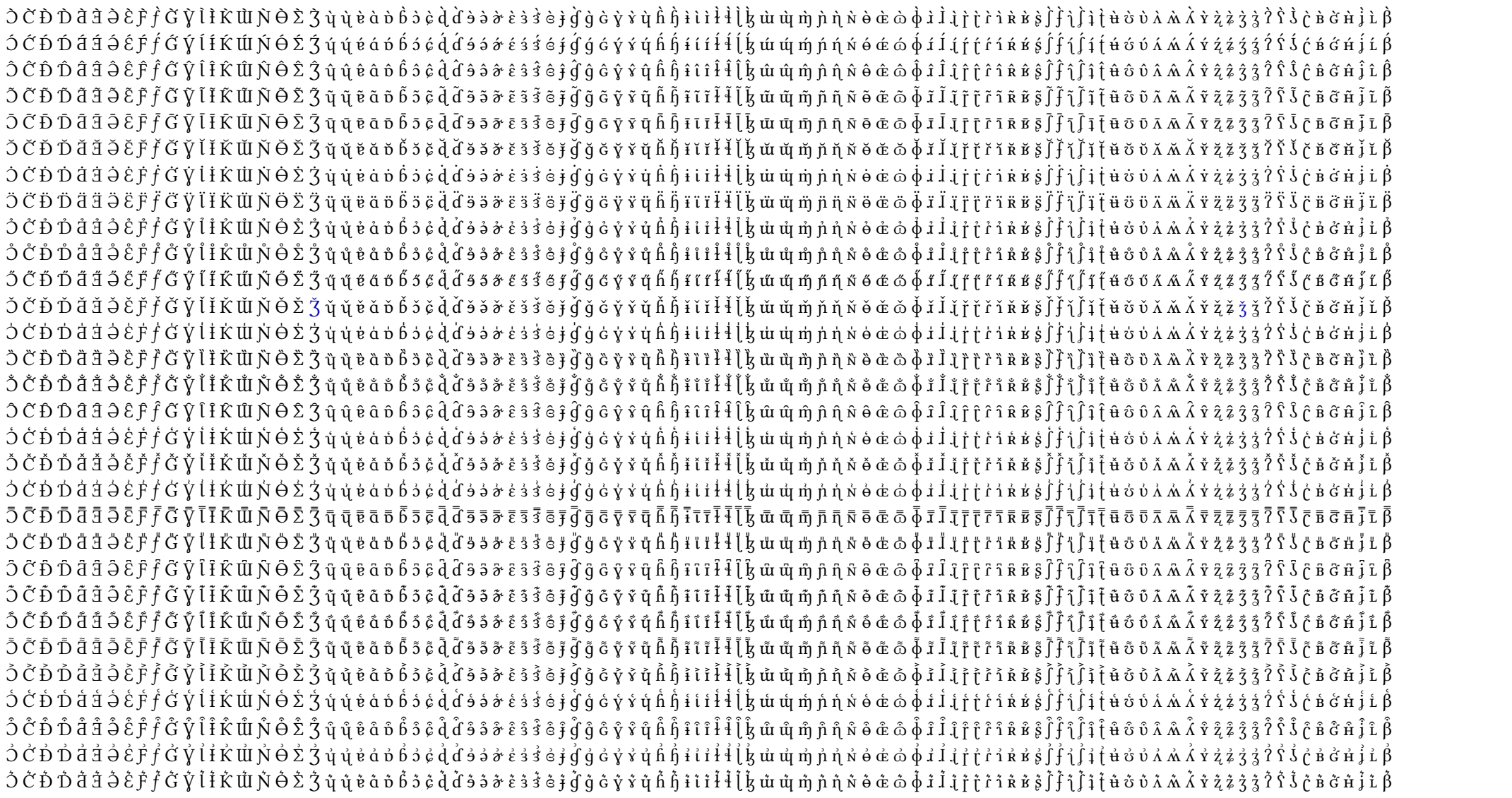
Semibold italic patched, Latin, Below



Regular, IPA, Above



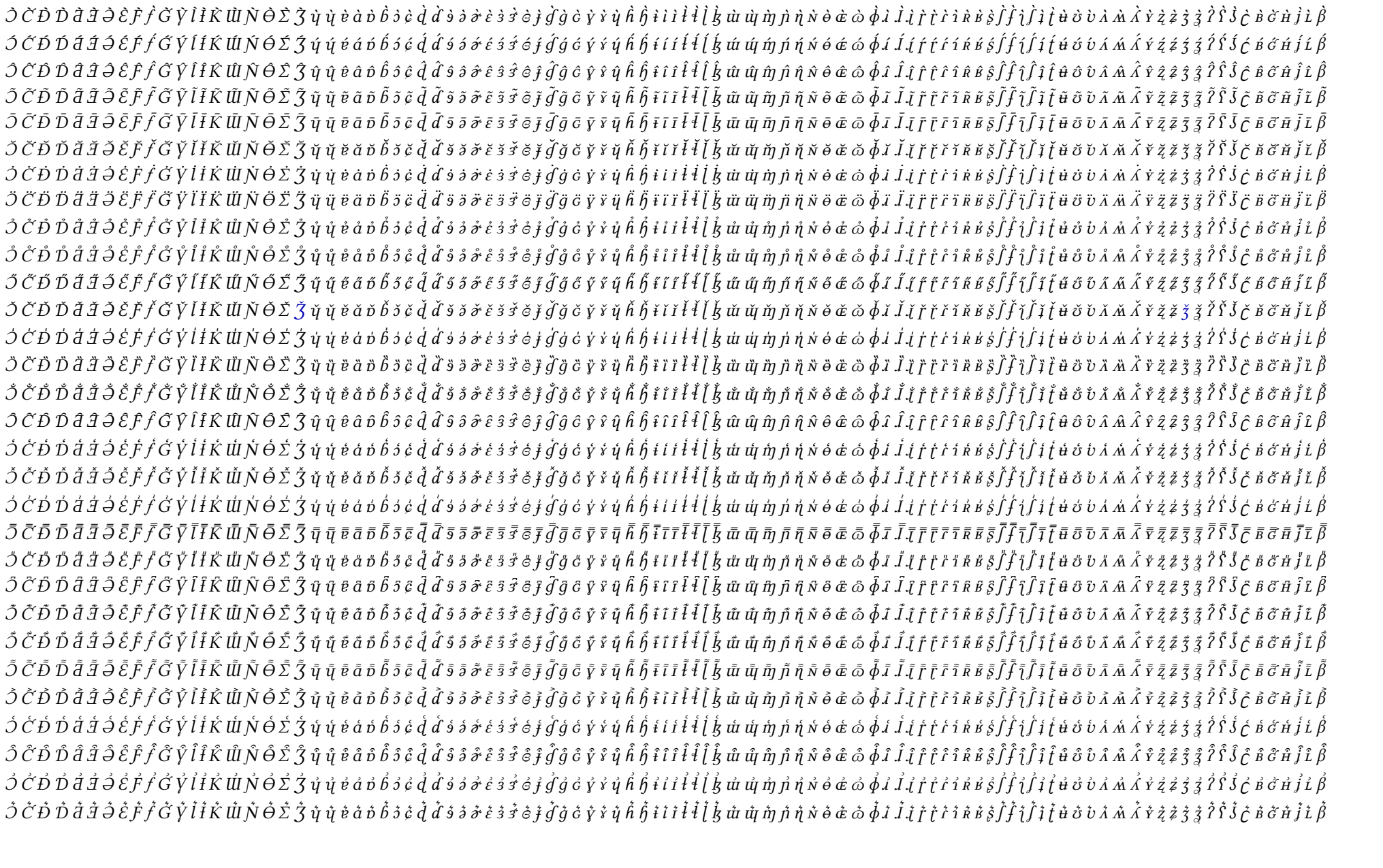
Regular patched, IPA, Above



Italic, IPA, Above



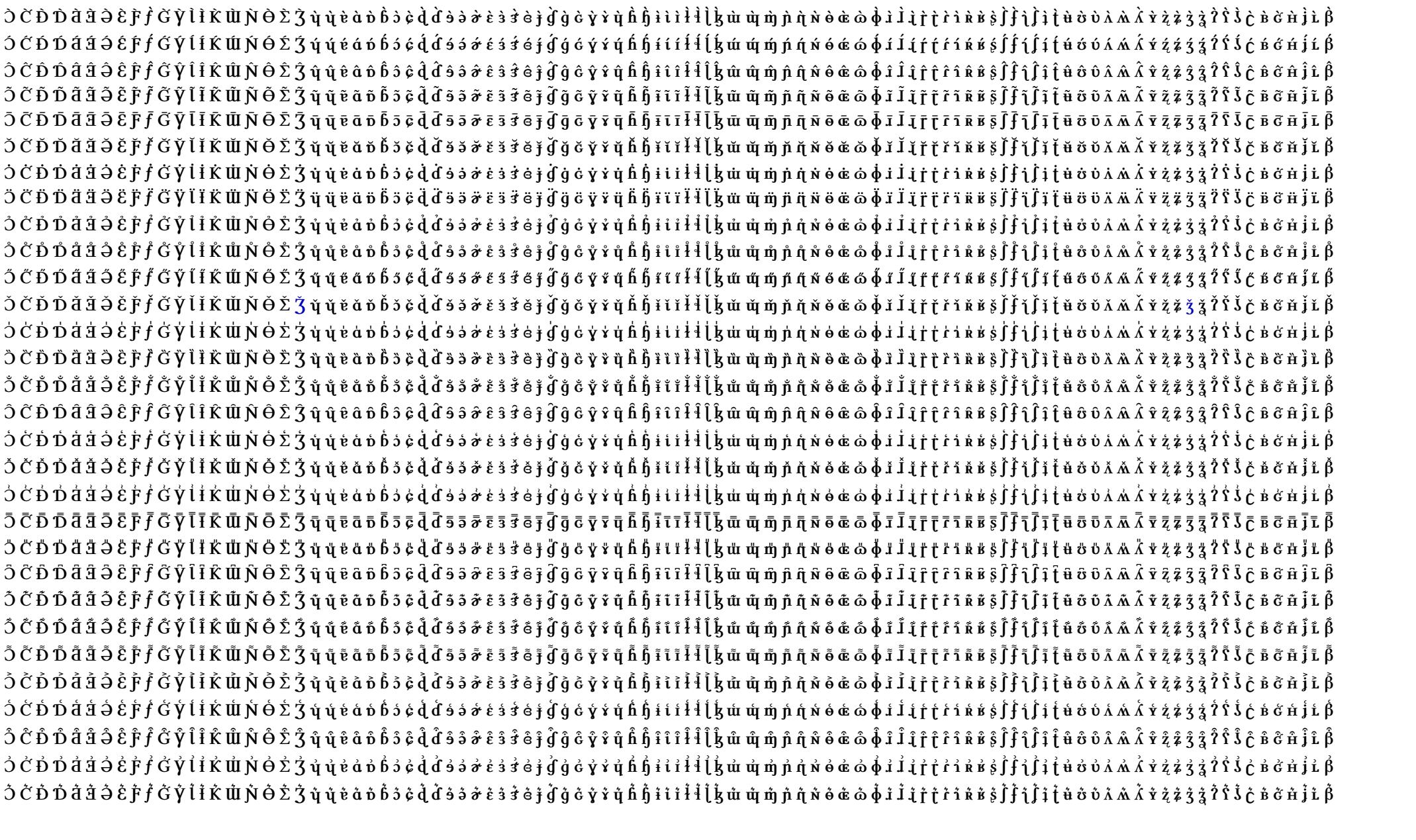
Italic patched, IPA, Above



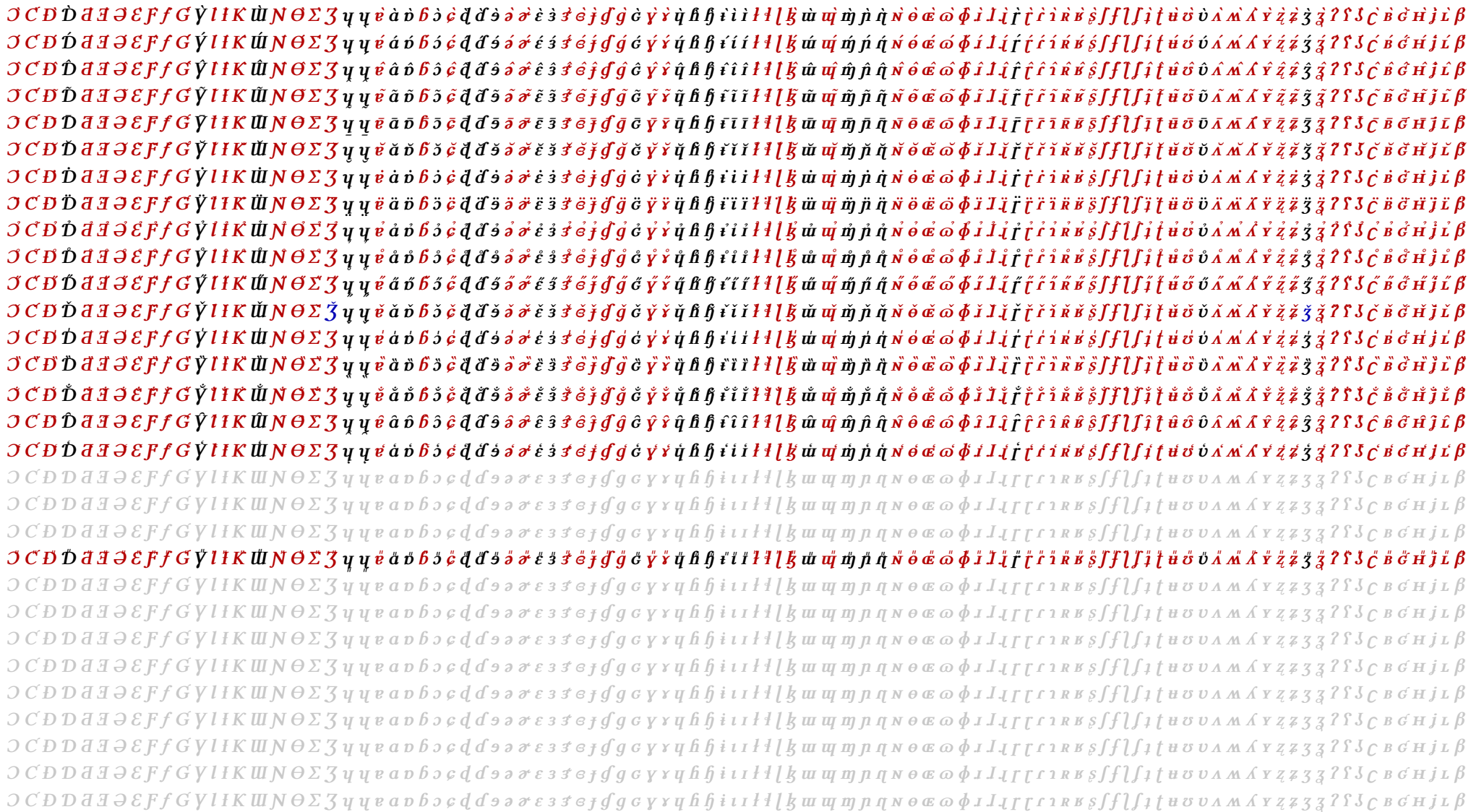
Semibold, IPA, Above



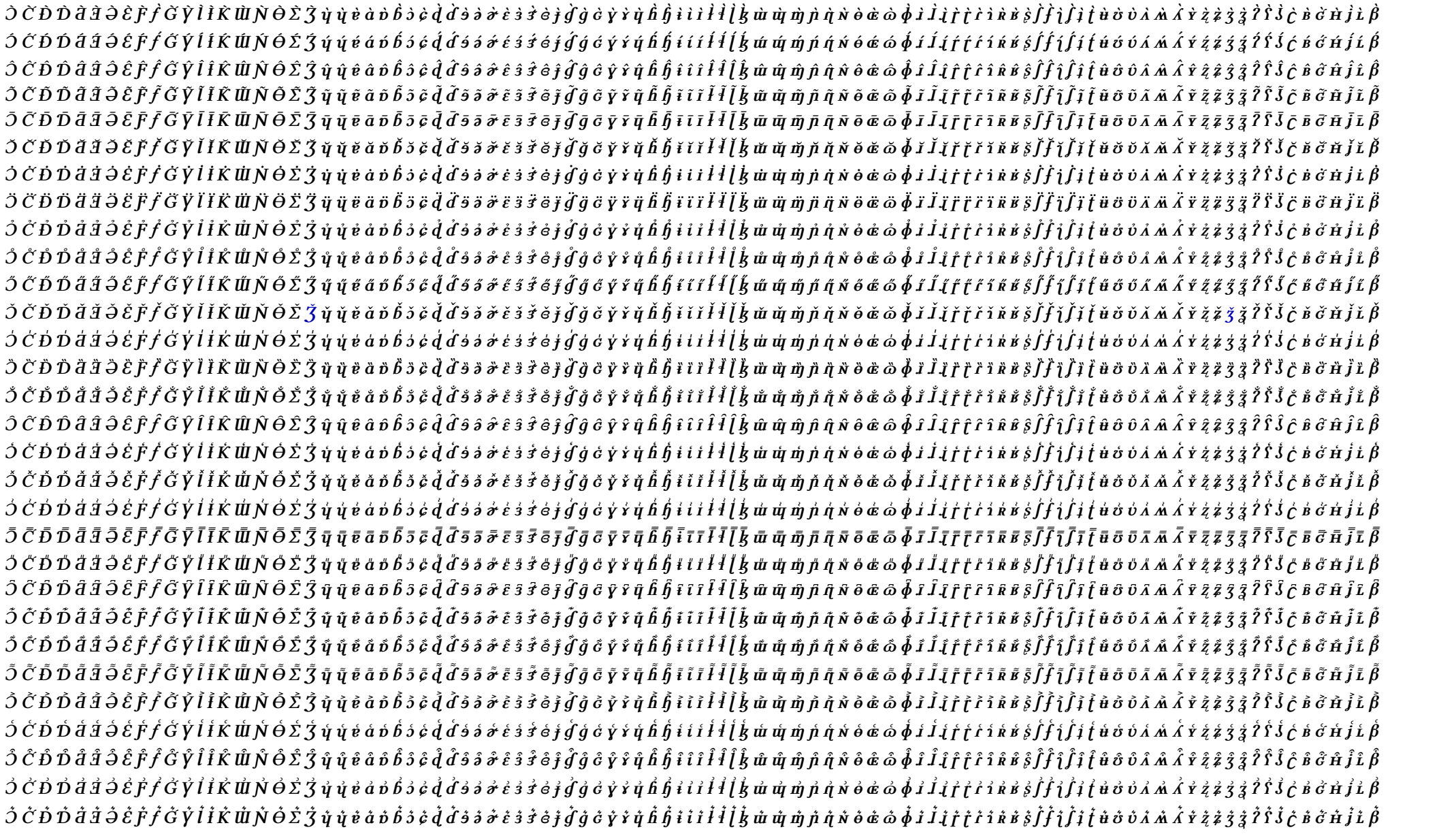
Semibold patched, IPA, Above



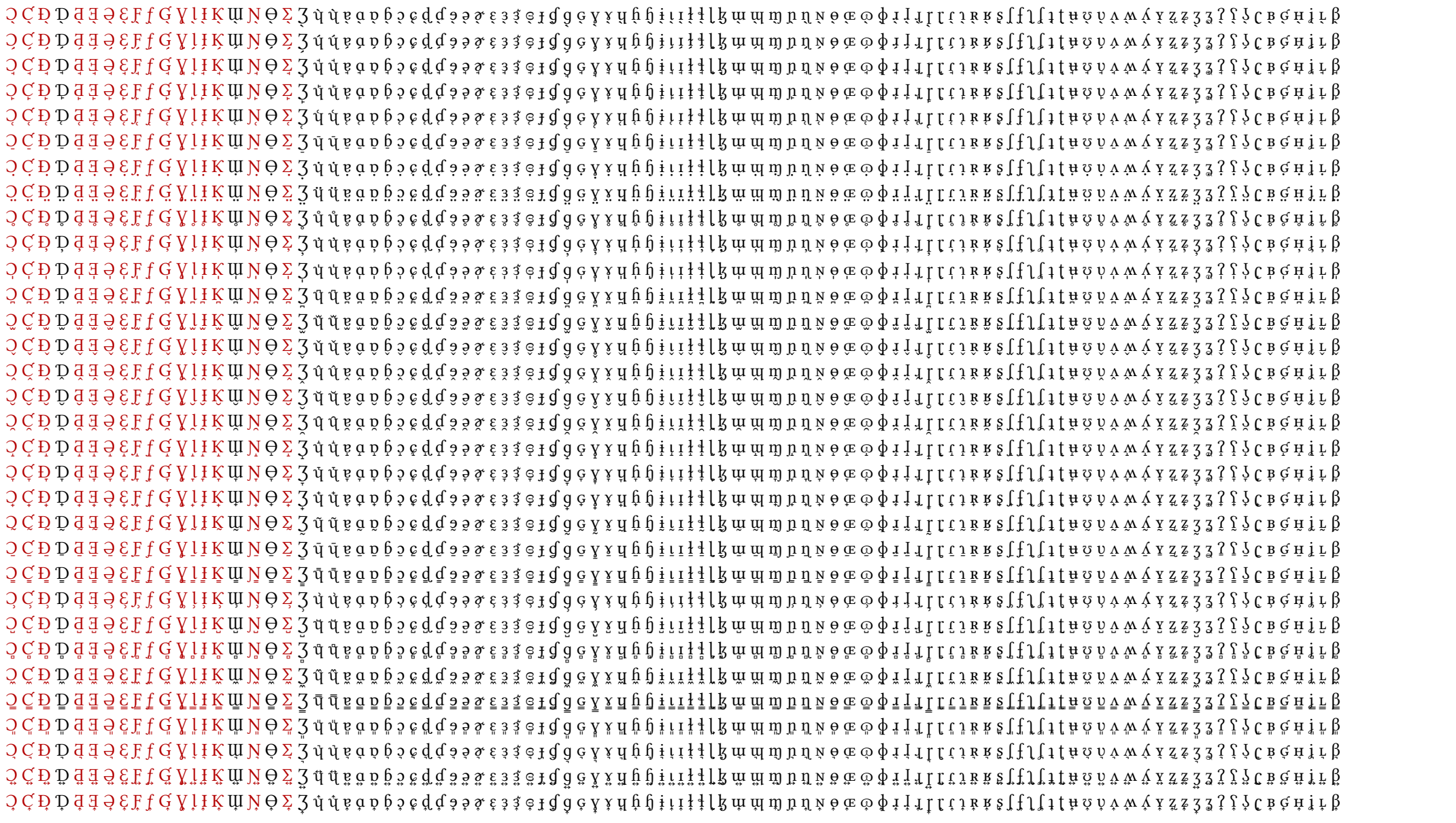
Semibold italic, IPA, Above



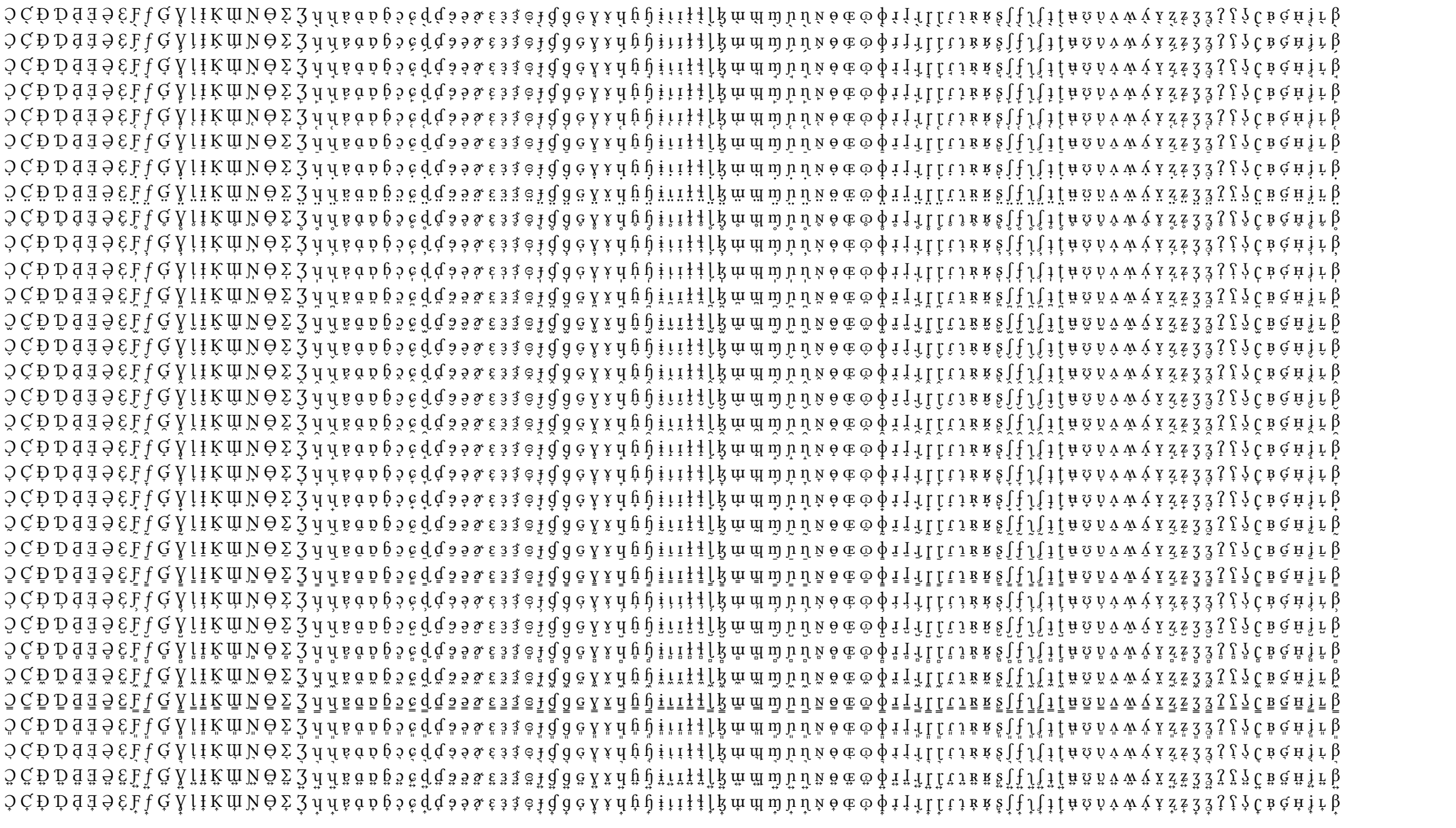
Semibold italic patched, IPA, Above



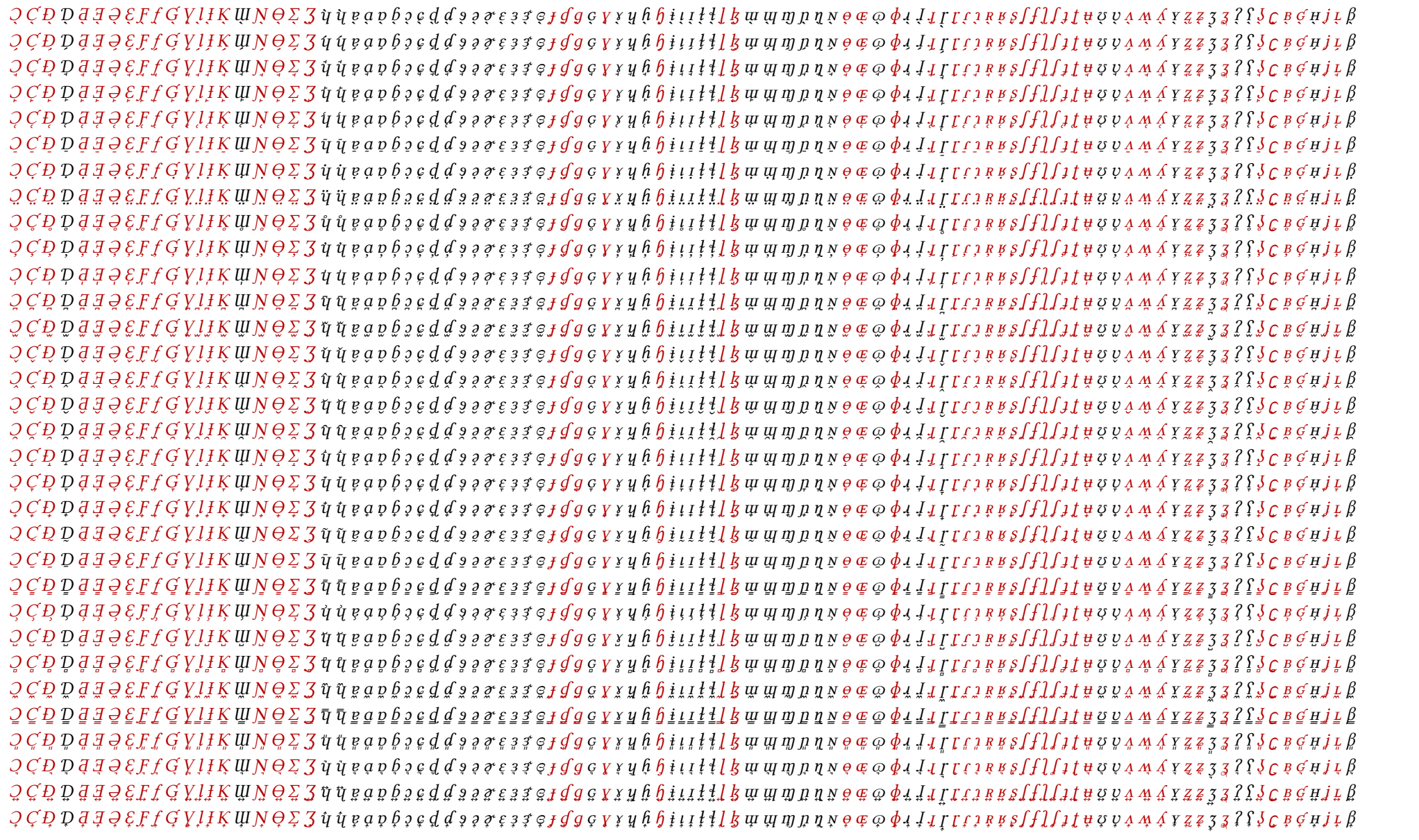
Regular, IPA, Below



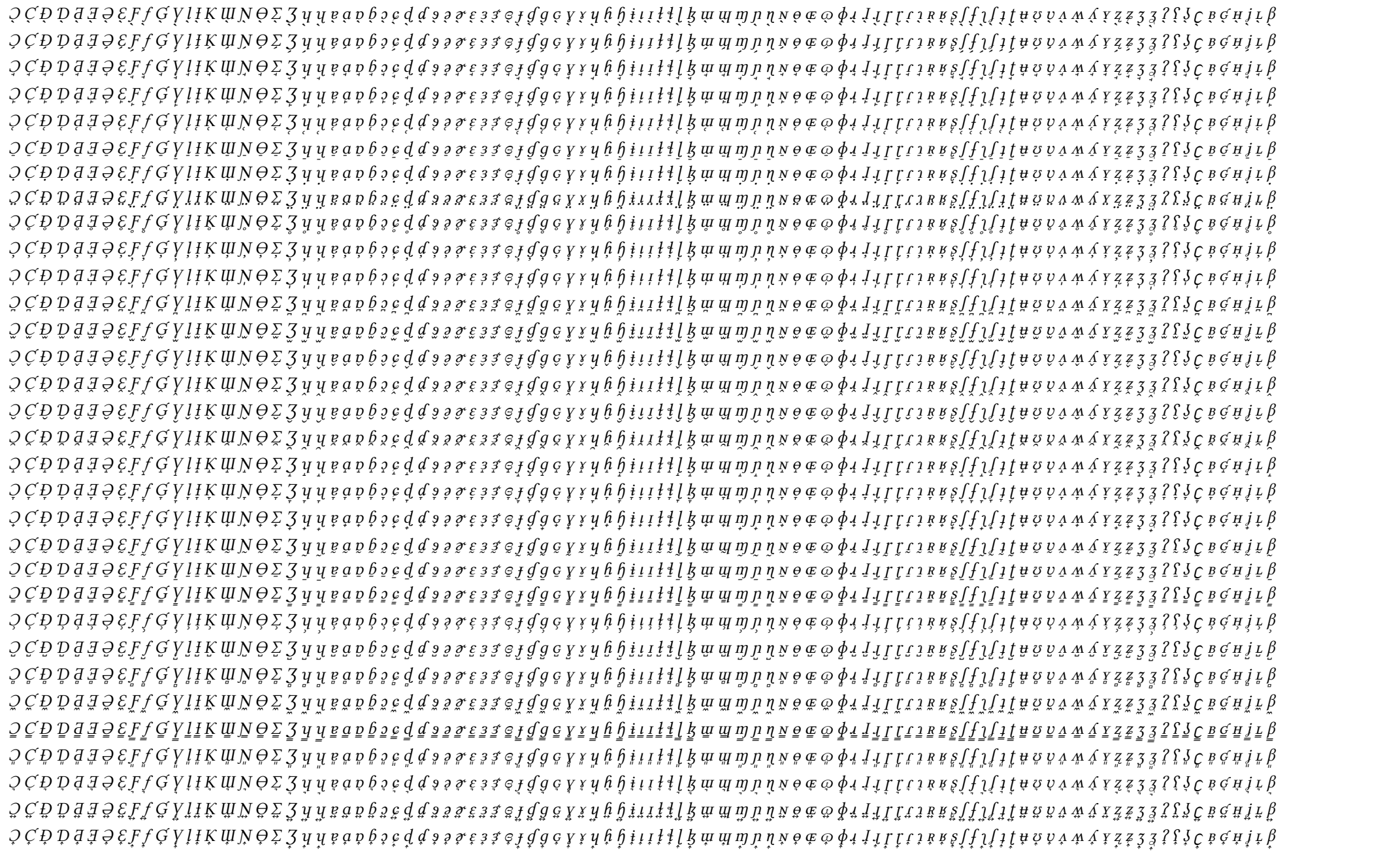
Regular patched, IPA, Below



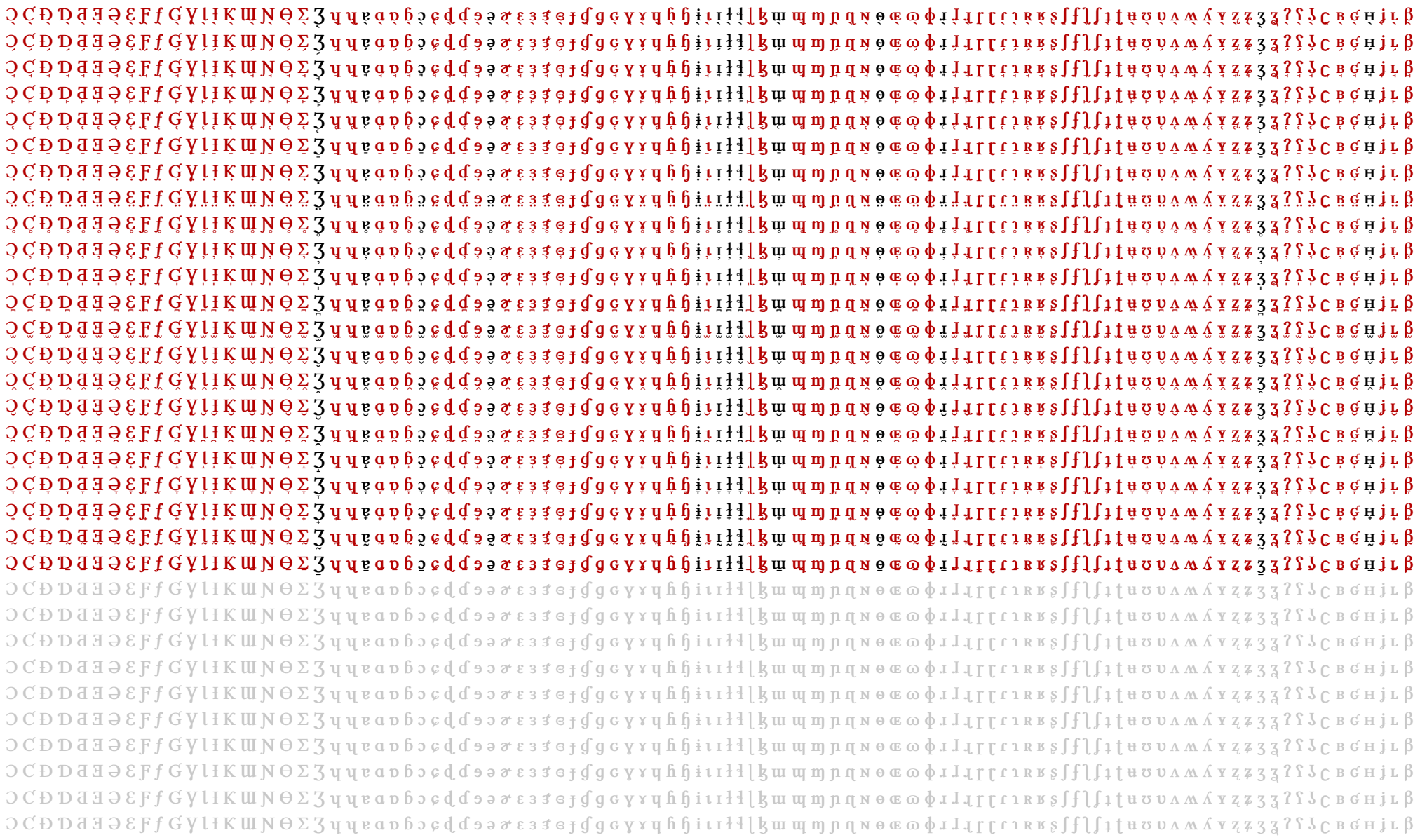
Italic, IPA, Below



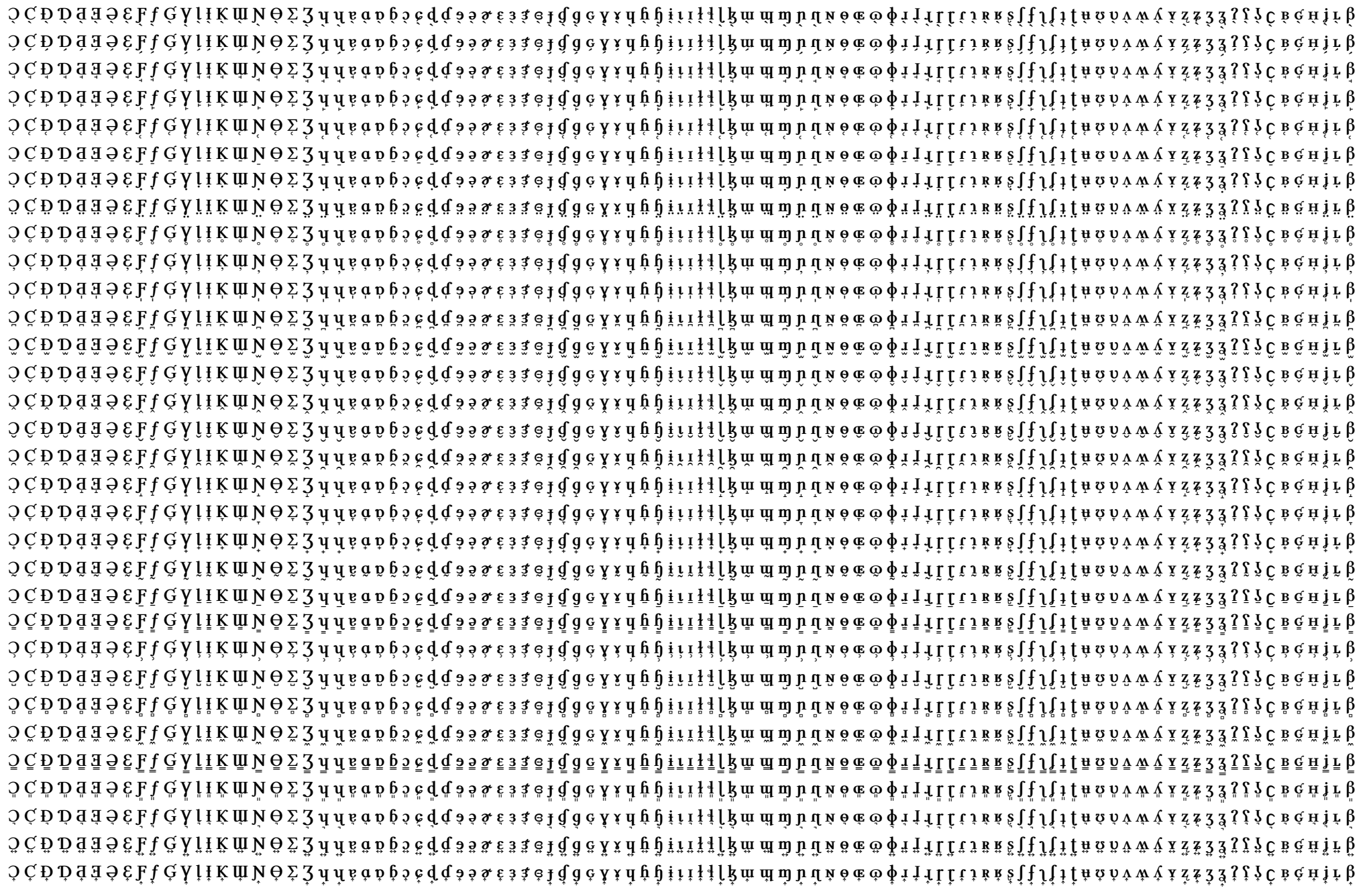
Italic patched, IPA, Below



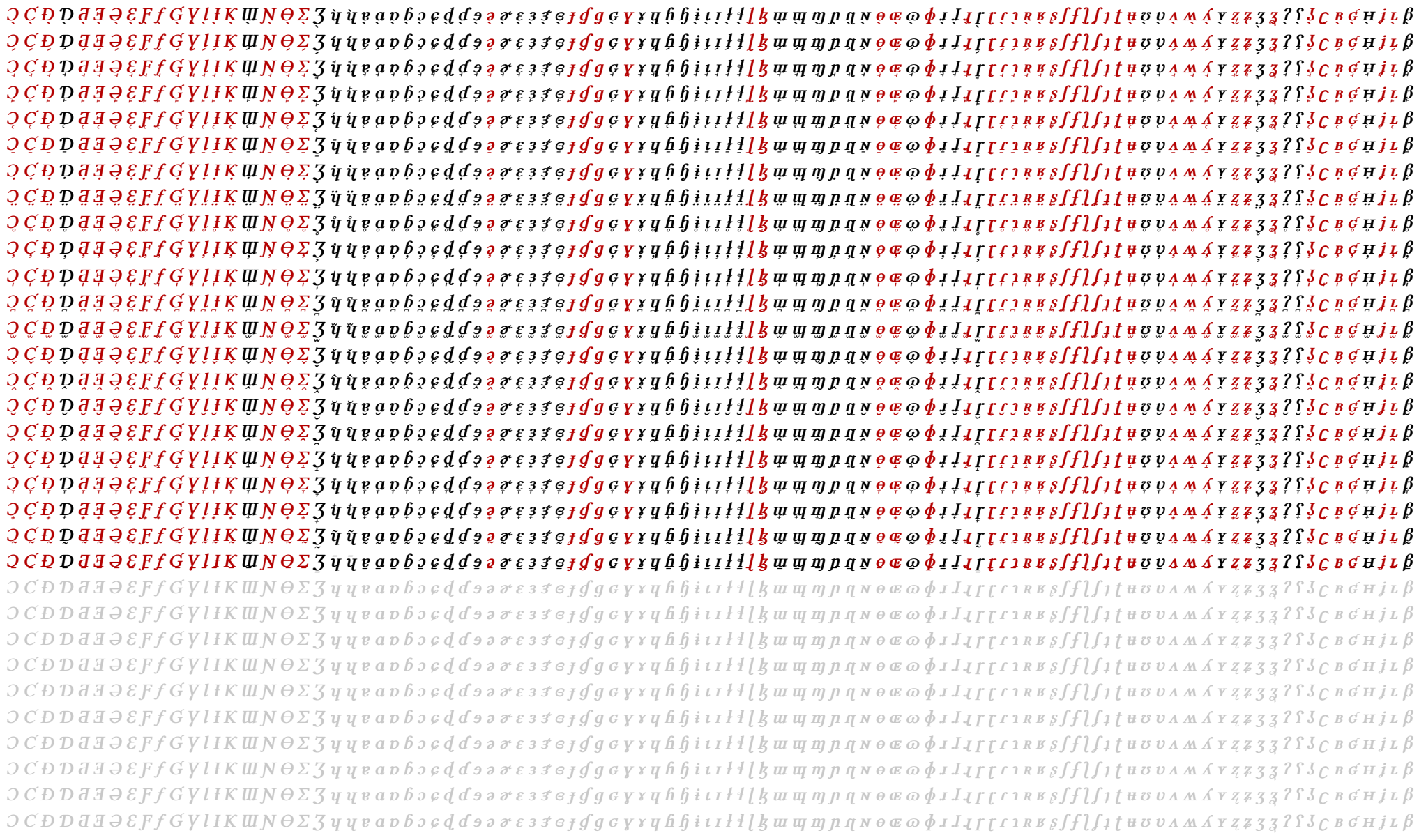
Semibold, IPA, Below



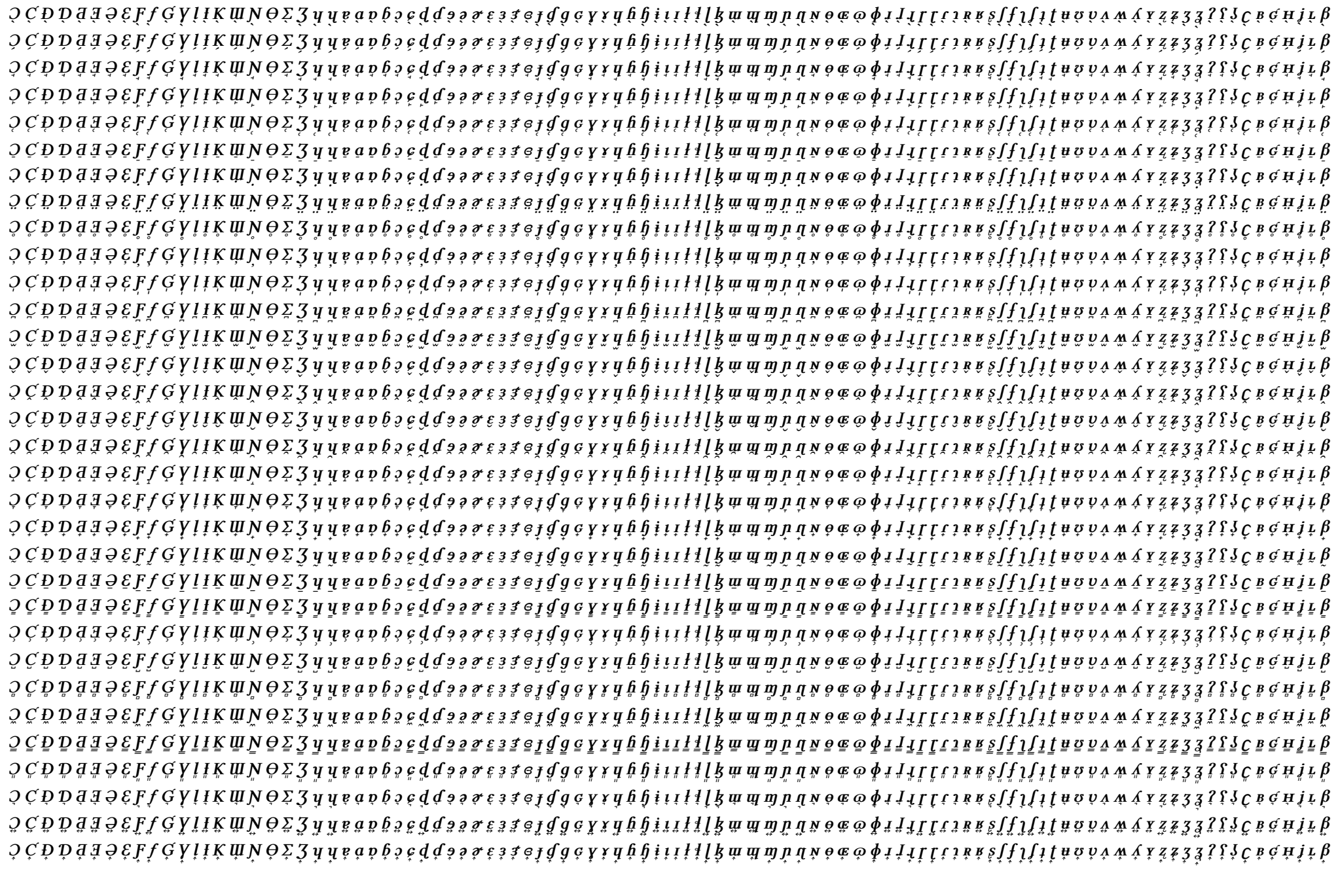
Semibold patched, IPA, Below



Semibold italic, IPA, Below



Semibold italic patched, IPA, Below



Capital alternates

A capital alternate (`grave.cap`, `acute.cap`, ...) is a form of an above-mark that sits lower over a capital, as the accents of the designer's precomposed capitals do. In the original fonts, `ccmp` substitutes only four of them, for the grave, acute, circumflex and breve, although the designer also drew the caron, double acute, double grave, inverted breve, dieresis and hook above (and, in regular and semibold, the dot above). The patched fonts substitute twelve, after every capital listed in `@capitals` of `data/fea/{style}/ccmp.fea` (stage 10): grave, acute, circumflex, tilde, breve, dot above, dieresis, hook above, double acute, caron, double grave and inverted breve. Stage 3 adds the two forms that were missing: `dotaccent.cap` in italic and semibold italic, the dot of the precomposed İ, and `tildecomb.cap` in every style (2026-11-08), the tilde of the precomposed tilded capitals (`tildecomb` itself, except in semibold italic, where the precomposed capitals use a taller tilde, that of Ã). The other above-marks have no capital form and attach with their own anchors. The ring above has none by decision: the precomposed Å sets the ring on the apex, and the combining ring keeps its clearance; nor do the vertical lines above U+030D and U+033E.

In stage 4 of the patched fonts (`font_patching_capmarks.py`), the mark anchor of each capital alternate is set from the designer's precomposed capitals A–Z. The mark's outline is found in each precomposed glyph that is a capital plus that mark, which gives how far the designer moved it there. The Y coordinate is the capital anchor reference (850) less the median height by which the mark is raised in the precomposed glyphs. The X coordinate of a symmetric mark is the centre of its outline on its own axis, $X = C + Y\tan\theta - K$. For the asymmetric grave, acute, double acute and double grave, it is the median position of the mark in the precomposed glyphs relative to the candidate above-anchor of the capital, $W/2 + 850\tan\theta - K$. K is 150 in semibold italic and 0 elsewhere. The precomposed placement is reproduced exactly where a capital's above-anchor is on its candidate; where it is not, the composed mark is displaced by the same amount. Human-curated anchors (stage 7) overrule stage 4.

The table compares each composed capital (base, capital alternate, anchors) with the designer's precomposed glyph, over the capitals that have one. Vertically every capital alternate matches: the median difference is 0 in every style, and no single capital differs by more than 12, except Ỏ, whose precomposed hook above is 24 units higher than the composed one in all styles but semibold italic. The numbers are the median horizontal distance; n is the number of precomposed capitals, and * marks a curated anchor.

mark	n	regular	italic	semibold	semibold italic
grave	8	17	30^*	4	10
acute	17	9	14^*	16	28
circumflex	12–13	4	9^*	14	16
tilde	7–8	5	10	10	25
breve	6	2	6^*	10	10
dot above	20	9	13	10	15
dieresis	7	2	6	8	8
hook above	6	4	12	12	16
double acute	2	0	2^*	18	6
caron	12–15	7	6^*	16	11
double grave	6	12	4^*	15	15
inverted breve	6	2	10^*	12	9

Notes by style:

- Regular. All scripted (2026-11-08). The curated grave $(-175, 723)$ was removed in favour of stage 4's $(-162, 724)$, as in semibold italic: the median distance from the precomposed graves goes from 18 to 17 units, and the median offset from 18 units right to 6, since the graves vary letter by letter. The other curated anchors are reproduced by the script and are commented out in `fontanchors_human`.
- Italic. Eight curated anchors (2026-11-07) overrule stage 4: grave -6 (stage 4: 17), acute -50 (-37), circumflex 15 (24), breve -14 (-1), caron -32 (-29), double acute -109 (-103), double grave -40 (-34), inverted breve -12 (-2). Each is fitted so that the accent lands, as a median over the capitals, where the designer drew it in the precomposed glyphs, with the capitals' actual above-anchors (so the signed median is 0). The dieresis, dot above, hook above and tilde are scripted.
- Semibold. All scripted. The designer's capital above-anchors were near 805, with the capital alternates drawn to compensate; stage 5 moves the capitals to 850, the reference stage 4 computes from.

- Semibold italic. All scripted (2026-11-08). The acute, grave, double acute and double grave had been curated on 2026-11-06 at the mark-axis position, $X = C + Y \tan\theta + d - K$ with d from regular (-236, -194, -282, -214, against stage 4's -195, -168, -243, -167). Those reproduced the precomposed capitals only on the designer's capital anchors, which are left of the candidates (median k 173 against $K = 150$), because the designer drew the precomposed accents on those anchors. Stage 4 is preferred because the precomposed glyphs are used wherever they exist, so a composed capital without one should match them; and since H I J K L N O U V Y Æ were moved on 2026-11-07, it is also closer to them. The larger distances of the acute (28) and tilde (25) come from capitals whose above-anchors are off their candidates (A, B, D, Q, R, W, X; V, Y and E for the tilde), not from the mark anchors.

Notes by mark:

- The grave, acute and their doubles are placed by eye, letter by letter, in the precomposed glyphs, so the implied X varies by up to 190 units over the bases (most for the acute, because of L), and one anchor is a compromise.
- Tilde. Unicode has no precomposed Ḡ, which Guarani writes, so G + U+0303 is composed. With the combining tilde it sat 25 to 47 units higher than the tilde of the precomposed Ñ, Ã, Õ beside it; `tildecomb.cap` matches their height exactly. Gentium Plus does the same with its `tildecomb.LP`.
- Dot above. `dotaccent.cap` is new in italic and semibold italic; in regular, stage 4 raises its anchor by 50 units to match the precomposed dotted capitals.
- Not used as evidence, because the mark outline differs in the precomposed glyph: Ÿ and Ẅ (dieresis), Ľ (comma-style caron), Ũ in semibold italic (tilde), and Ĵ in semibold italic.

The grids show these twelve marks on every Latin and IPA capital of `@capitals`: A–Z, Æ, Œ, Đ, Ø, Ɖ, Ħ, Ł, the IPA and African capitals, and the 59 precomposed capitals with a cedilla, ogonek, horn or below-mark that stage 9 gives an above-anchor (Ç Ą Ę Į Ų Ư Ḍ Ẹ Ọ …; added to `@capitals` 2026-11-08, so that, e.g., Lithuanian Ą́ and Yoruba Ẹ́ take the capital acute like Á and É). The Greek and Cyrillic capitals of `@capitals` are not shown: their combinations with these marks are all precomposed. Blue cells are precomposed glyphs, which the shaper substitutes for base + mark wherever the font has one, so they do not use the capital alternates. Black cells are composed: in the patched fonts, the base with the capital alternate; in the original fonts, the base with the alternate only for the grave, acute, circumflex and breve, and otherwise with the ordinary mark.

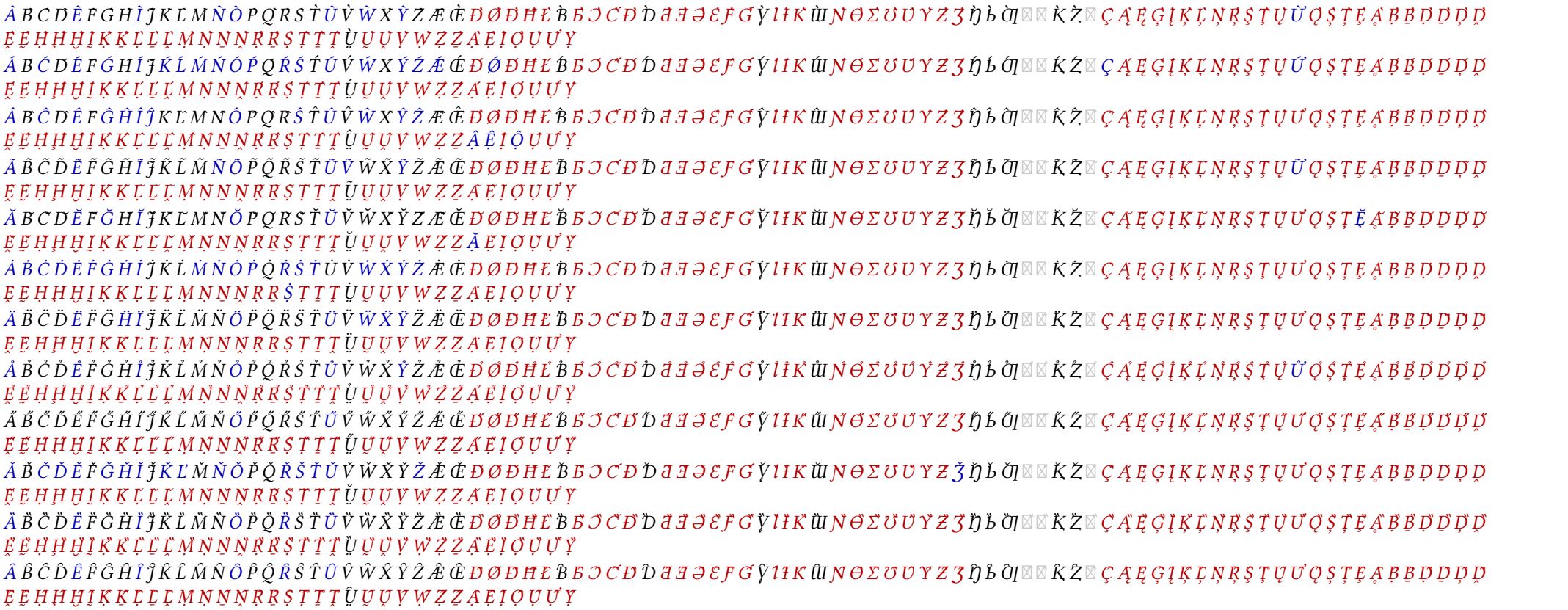
Regular, Capitals, Marks with capital alternates



Regular patched, Capitals, Marks with capital alternates

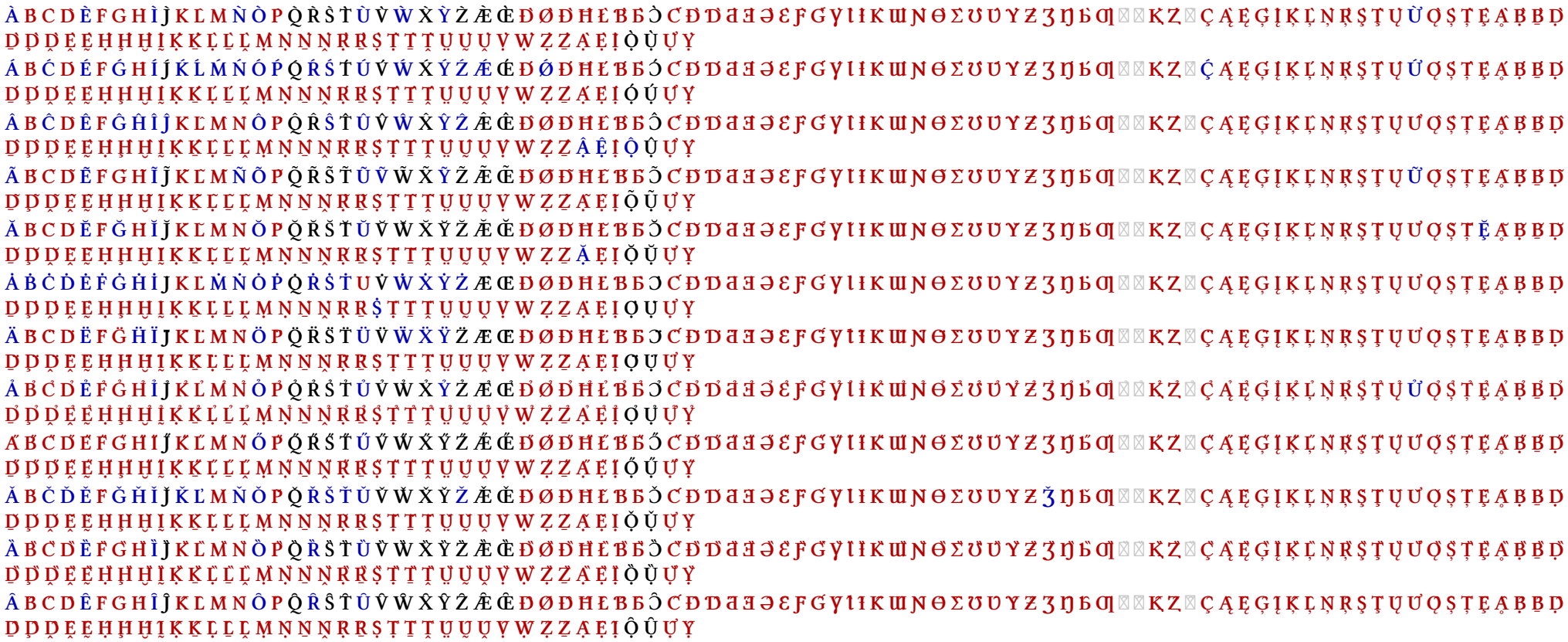


Italic, Capitals, Marks with capital alternates



Italic patched, Capitals, Marks with capital alternates

Semibold, Capitals, Marks with capital alternates



Semibold patched, Capitals, Marks with capital alternates

À B̀ C̀ D̀ È F̀ G̀ H̀ Ì J̀ K̀ L̀ M̀ Ǹ Ò P̀ Q̀ R̀ S̀ T̀ Ù V̀ Ẁ X̀ Ỳ Z̀ Æ Œ Ð Ø Đ Ħ Ł Ƀ Ƃ Ɔ̀ Ć Đ Ð ȁ ȃ ȅ ȇ Ƒ Ǵ Ỳ ȉ ȋ Ќ Ш Ǹ Ө Ѐ Ѝ Ў Ỳ Ż Ʒ Ŋ Ъ Ԍ ̀ ́ Ќ Ż ̀ Ç Ą Ę Ǫ Į Ķ Ļ Ņ Ŗ Ş Ţ Ų Ű Ǫ Ş Ţ Ę Ą̇ Ḅ Ḇ Ḍ
Ḏ Ḑ Ḓ Ḝ Ḗ Ḥ Ḫ Ḯ Ḱ Ḵ Ḷ Ḽ Ṃ Ṇ Ṉ Ṋ Ṛ Ṝ Ṣ Ṭ Ṯ Ṱ Ṳ Ṷ Ṻ Ṽ Ẁ Ẓ Ẕ Ạ Ẹ Ị Ọ Ụ Ự Ỵ
Á B́ Ć D́ É F́ Ǵ H́ Í J́ Ḱ Ĺ Ḿ Ń Ó Ṕ Q́ Ŕ Ś T́ Ú V́ Ẃ X́ Ý Ź Æ Œ Ð Ø Đ Ħ Ł Ƀ Ƃ Ɔ́ Ć Đ Ð ά ȃ ȅ ȇ Ƒ Ǵ Ý ί ȋ Ќ Ш Ń Ө Έ Ѝ Ў Ý Ź Ʒ Ŋ Ъ Ԍ ́ ́ Ќ Ź ́ Ç Ą Ę Ǫ Į Ķ Ļ Ņ Ŗ Ş Ţ Ų Ű Ǫ Ş Ţ Ę Ą̇ Ḅ Ḇ Ḍ
Ḏ Ḑ Ḓ Ḝ Ḗ Ḥ Ḫ Ḯ Ḱ Ḵ Ḷ Ḽ Ṃ Ṇ Ṉ Ṋ Ṛ Ṝ Ṣ Ṭ Ṯ Ṱ Ṳ Ṷ Ṻ Ṽ Ẃ Ẓ Ẕ Ạ Ẹ Ị Ọ Ụ Ự Ỵ
Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ Ǩ Ľ M̌ Ň Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Æ Œ Ð Ø Đ Ħ Ł Ƀ Ƃ Ɔ̂ Ć Đ Ð â ȃ ȇ ȇ Ƒ Ĝ Ŷ Î ȋ Ǩ Ш Ň Ө Ŝ Ŭ Û Ŷ Ź Ʒ Ŋ Ъ Ԍ ̂ ̂ Ǩ Ẑ ̂ Ç Ą Ę Ĝ Į Ǩ Ľ Ň Ř Ŝ Ţ Ų Ŭ Ǫ Ŝ Ţ Ę Ą̂ Ḅ Ḇ Ḍ
Ḏ Ḑ Ḓ Ḝ Ḗ Ḥ Ḫ Ḯ Ḱ Ḵ Ḷ Ḽ Ṃ Ṇ Ṉ Ṋ Ṛ Ṝ Ṣ Ṭ Ṯ Ṱ Ṳ Ṷ Ṻ Ṽ Ẃ Ẓ Ẕ Ậ Ệ Ị Ộ Ụ Ự Ỵ
Ã B̃ C̃ D̃ Ẽ F̃ G̃ H̃ Ĩ J̃ K̃ L̃ M̃ Ñ Õ P̃ Q̃ R̃ S̃ T̃ Ũ Ṽ W̃ X̃ Ỹ Z̃ Æ Œ Ð Ø Đ Ħ Ł Ƀ Ƃ Ɔ̃ Ć Đ Ð ã ȃ ẽ ȇ Ƒ G̃ Ỹ Ĩ ȋ Ḳ Ш Ñ Θ Š ŭ Ũ Ỹ Ž Ʒ Ŋ Ъ Ԍ ̃ ̃ Ḳ Ž ̃ Ç Ą Ę Ǧ Į Ḳ Ľ Ñ Ř Ş Ţ Ų Ű Ǫ Ş Ţ Ę Ą̃ Ḅ Ḇ Ḍ
Ḏ Ḑ Ḓ Ḝ Ḗ Ḥ Ḫ Ḯ Ḱ Ḵ Ḷ Ḽ Ṃ Ṇ Ṉ Ṋ Ṛ Ṝ Ṣ Ṭ Ṯ Ṱ Ṳ Ṷ Ṻ Ṽ Ẃ Ẓ Ẕ Ạ̃ Ệ Ị Ọ Ụ Ự Ỵ
Ă B̆ C̆ D̆ Ĕ F̆ Ğ H̆ Ĭ J̆ Ǩ L̆ M̆ Ŋ̆ Ŏ P̆ Q̆ R̆ S̆ T̆ Ŭ V̆ W̆ X̆ Y̆ Ž Æ Œ Ð Ø Đ Ħ Ł Ƀ Ƃ Ɔ̆ Ć Đ Ð ă ȃ ĕ ȇ Ƒ Ğ Ў ĭ ȋ Ǩ Ш Ŋ̆ Θ Ŏ ŭ Ŭ Ў Ž Ʒ Ŋ Ъ Ԍ ̆ ̆ Ǩ Ž ̆ Ç Ą Ę Ğ Į Ǩ Ľ Ŋ̆ Ř Ş Ţ Ų Ŭ Ǫ Ş Ţ Ę̆ Ą̆ Ḅ Ḇ Ḍ
Ḏ Ḑ Ḓ Ḝ Ḗ Ḥ Ḫ Ḯ Ḱ Ḵ Ḷ Ḽ Ṃ Ŋ̆ Ṋ Ṛ Ṝ Ṣ Ṭ Ṯ Ṱ Ṳ Ṷ Ṻ Ṽ Ẃ Ẓ Ẕ Ạ Ệ Ị Ọ Ụ Ự Ỵ
Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Æ Œ Ð Ø Đ Ħ Ł Ƀ Ƃ Ɔ̈ Ć Đ Ð ä ȃ ë ȇ Ƒ G̈ Ÿ Ï ȋ Ḳ Ш Ñ Θ Š ŭ Ü Ÿ Ž Ʒ Ŋ Ъ Ԍ ̈ ̈ Ḳ Ž ̈ Ç Ą Ę Ǧ Į Ḳ Ľ Ñ Ř Ş Ţ Ų Ű Ǫ Ş Ţ Ę Ä Ḅ Ḇ Ḍ
Ḏ Ḑ Ḓ Ḝ Ḗ Ḥ Ḫ Ḯ Ḱ Ḵ Ḷ Ḽ Ṃ Ṇ Ṉ Ṋ Ṛ Ṝ Ṣ Ṭ Ṯ Ṱ Ṳ Ṷ Ṻ Ṽ Ẅ Ẓ Ẕ Ạ̈ Ệ Ị Ọ Ụ Ự Ỵ
Å B̊ C̊ D̊ Ê F̊ G̊ H̊ Î J̊ K̊ L̊ M̊ N̊ Ů P̊ Q̊ R̊ S̊ T̊ Ů V̊ Ẅ Ẍ Y̊ Ž Æ Œ Ð Ø Đ Ħ Ł Ƀ Ƃ Ɔ̊ Ć Đ Ð å ȃ ê ȇ Ƒ G̊ Y̊ Î ȋ Ǩ Ш Ň Ө Ŝ ů Ů Y̊ Ž Ʒ Ŋ Ъ Ԍ ̊ ̊ Ǩ Ž ̊ Ç Ą Ę Ǫ Į Ǩ Ľ Ň Ř Ş Ţ Ų Ű Ǫ Ş Ţ Ę Ą̊ Ḅ Ḇ Ḍ
Ḏ Ḑ Ḓ Ḝ Ḗ Ḥ Ḫ Ḯ Ḱ Ḵ Ḷ Ḽ Ṃ Ṇ Ṉ Ṋ Ṛ Ṝ Ṣ Ṭ Ṯ Ṱ Ṳ Ṷ Ṻ Ṽ Ẅ Ẓ Ẕ Ạ Ệ Ị Ọ Ụ Ự Ỵ
Ǻ B̊́ Ć D́ É F́ Ǵ H́ Í J́ Ḱ Ĺ Ḿ Ń Ó Ṕ Q́ Ŕ Ś T́ Ű V́ Ẃ X́ Ý Ź Æ Œ Ð Ø Đ Ħ Ł Ƀ Ƃ Ɔ́ Ć Đ Ð ά ȃ é ȇ Ƒ Ǵ Ý ί ȋ Ḱ Ш Ń Ө Ś ű Ű Ý Ź Ʒ Ŋ Ъ Ԍ ́ ́ Ḱ Ź ́ Ç Ą Ę Ǫ Į Ḱ Ĺ Ń Ř Ś Ţ Ų Ű Ǫ Ś Ţ Ę Ą̊ Ḅ Ḇ Ḍ
Ḏ Ḑ Ḓ Ḝ Ḗ Ḥ Ḫ Ḯ Ḱ Ḵ Ḷ Ḽ Ṃ Ṇ Ṉ Ṋ Ṛ Ṝ Ṣ Ṭ Ṯ Ṱ Ṳ Ṷ Ṻ Ṽ Ẃ Ẓ Ẕ Ạ Ệ Ị Ọ Ụ Ự Ỵ
Ă B̆ C̆ D̆ Ĕ F̆ Ğ H̆ Ĭ J̆ Ǩ Ľ M̆ Ň Ŏ P̆ Q̆ Ř Š T̆ Ŭ V̆ W̆ X̆ Y̆ Ž Æ Œ Ð Ø Đ Ħ Ł Ƀ Ƃ Ɔ̆ Ć Đ Ð ă ȃ ĕ ȇ Ƒ Ğ Ў ĭ ȋ Ǩ Ш Ň Ө Ŝ ŭ Ŭ Ў Ž Ʒ Ŋ Ъ Ԍ ̆ ̆ Ǩ Ž ̆ Ç Ą Ę Ğ Į Ǩ Ľ Ň Ř Ş Ţ Ų Ű Ǫ Ş Ţ Ę Ą̆ Ḅ Ḇ Ḍ
Ḏ Ḑ Ḓ Ḝ Ḗ Ḥ Ḫ Ḯ Ḱ Ḵ Ḷ Ḽ Ṃ Ṇ Ṉ Ṋ Ṛ Ṝ Ṣ Ṭ Ṯ Ṱ Ṳ Ṷ Ṻ Ṽ Ẃ Ẓ Ẕ Ạ Ệ Ị Ọ Ụ Ự Ỵ
Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Æ Œ Ð Ø Đ Ħ Ł Ƀ Ƃ Ɔ̈ Ć Đ Ð ä ȃ ë ȇ Ƒ G̈ Ÿ Ï ȋ Ḳ Ш Ñ Θ Š ŭ Ü Ÿ Ž Ʒ Ŋ Ъ Ԍ ̈ ̈ Ḳ Ž ̈ Ç Ą Ę Ǧ Į Ḳ Ľ Ñ Ř Ş Ţ Ų Ű Ǫ Ş Ţ Ę Ä Ḅ Ḇ Ḍ
Ḏ Ḑ Ḓ Ḝ Ḗ Ḥ Ḫ Ḯ Ḱ Ḵ Ḷ Ḽ Ṃ Ṇ Ṉ Ṋ Ṛ Ṝ Ṣ Ṭ Ṯ Ṱ Ṳ Ṷ Ṻ Ṽ Ẅ Ẓ Ẕ Ạ̈ Ệ Ị Ọ Ụ Ự Ỵ
Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ Ǩ Ľ M̂ Ň Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Æ Œ Ð Ø Đ Ħ Ł Ƀ Ƃ Ɔ̂ Ć Đ Ð â ȃ ê ȇ Ƒ Ĝ Ŷ Î ȋ Ǩ Ш Ň Ө Ŝ ŭ Û Ŷ Ẑ Ʒ Ŋ Ъ Ԍ ̂ ̂ Ǩ Ẑ ̂ Ç Ą Ę Ǫ Į Ǩ Ľ Ň Ř Ş Ţ Ų Ű Ǫ Ş Ţ Ę Ą̂ Ḅ Ḇ Ḍ
Ḏ Ḑ Ḓ Ḝ Ḗ Ḥ Ḫ Ḯ Ḱ Ḵ Ḷ Ḽ Ṃ Ṇ Ṉ Ṋ Ṛ Ṝ Ṣ Ṭ Ṯ Ṱ Ṳ Ṷ Ṻ Ṽ Ẅ Ẓ Ẕ Ạ Ệ Ị Ọ Ụ Ự Ỵ

Semibold italic, Capitals, Marks with capital alternates



Semibold italic patched, Capitals, Marks with capital alternates

Precomposed characters

These grids show the IPA above-marks and below-marks on the precomposed Latin letters of stage 9 (`data/precomposed_letters.py`), in the patched fonts only: above-marks on the letters that have no mark above, below-marks on those that have no mark below. Each such letter inherits its base letter's anchor, moved to where the base sits in the precomposed glyph. The excluded letters (ǵ and the horn letters above, and a few letters in one style whose base was redrawn) keep only the anchors they already had.

Every cell is the letter, `\cgj` and the mark (2026-11-08). Without CGJ the shaper decomposes the letter, puts the marks in canonical order and recomposes greedily from the base, so a cell can show a different letter with loose marks instead of the letter's own anchor: Ấ with a dot below becomes Ạ + ring above + acute, Ẹ with a circumflex becomes Ệ. That happened in 674 of the 11 456 below-mark cells. In 130 of them two above-marks were left to stack (Ū–ů, Ā–ǎ, Ấ ắ, Ō–ő, Ē–é, Ố–ớ, Ű–ǘ, Á–ắ, Ế–ễ … with a dot, ring, circumflex, tilde, diaeresis or comma below), and they collide, because Libertinus has almost no mark-to-mark positioning (regular stacks on 11 marks, italic on 4, semibold only on the spacing macron, semibold italic not at all; nothing stacks on the ring). That is what real text without CGJ gets; see the to-do.

Regular patched, Precomposed letters that inherit an above anchor in stage 9, Above

Ç ç Ą ą Ḝ ḝ Ģ Į Ḳ ḳ Ḽ ļ Ṇ ṇ Ṛ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ṣ ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Ḹ Ḻ ḻ Ṃ ṃ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẇ ẇ Ẓ ẓ Ẕ ẕ Ḥ Ą ą Ḝ ḝ Į Ǫ ǫ Ṳ ų Ṻ Ų ų

Ç ç Á á Ḝ ḝ Ǵ Į Ḱ ḱ Ĺ Į Ń ń Ŕ ŗ Ś ś Ţ ţ Ú ú Ṻ Ǫ ǫ Ś ś Ţ ţ Ḗ ḗ Ą ą Ḃ ḃ Ḃ ḃ Ḋ ḋ Ḋ ḋ Ḋ ḋ Ḗ ḗ Ḗ ḗ Ḣ ḣ Ḣ ḣ Ḣ ḣ Į Ḱ ḱ Ḱ ḱ Ĺ Í Ĺ Į Ḿ ḿ Ń ń Ń ń Ń ń Ŕ ŗ Ŕ ŗ Ś ś Ţ ţ Ţ ţ Ṯ ṯ Ṱ ṱ Ú ú Ṻ Ų ų Ṷ ṷ Ẃ ẃ Ź ź Ẑ ẑ Ḥ Á á Ḝ ḗ Í Ǫ ǫ Ú ú Ṻ Ý ý

Ç ç Â â Ḝ ḝ Ĝ Į Ḱ ḱ Ĺ Į Ñ ñ Ŕ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Î Ḻ Į Ṁ ṁ Ṅ ṅ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ṳ ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Â â Ḝ ḝ Î Ǫ ǫ Ṳ ų Ṻ Ý ý

Ç ç Ã ã Ḝ ḝ Ĝ Į Ḱ ḱ Ĺ Į Ñ ñ Ŕ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Ĩ Ḻ Į Ṁ ṁ Ṅ ṅ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ṳ ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Ã ã Ḝ ḝ Ĩ Ǫ ǫ Ṳ ų Ṻ Ý ý

Ç ç Ā ā Ḝ ḝ Ḡ Į Ḳ ḳ Ḻ Į Ṇ ṇ Ṛ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Ī Ḻ Į Ṁ ṁ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ų ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Ā ā Ḝ ḝ Ī Ǫ ǫ Ṳ ų Ṻ Ý ý

Ç ç Ă ă Ḝ ḝ Ğ Į Ḳ ḳ Ḽ Į Ñ ñ Ŕ ŗ Ş ş Ţ ţ Ų ų Ŭ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Ĭ Ḻ Į Ṁ ṁ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ų ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Ă ă Ḝ ḝ Ĭ Ǫ ǫ Ṳ ų Ṻ Ý ý

Ç ç Ą ą Ḝ ḝ Ġ Į Ḳ ḳ Ḷ Į Ṇ ṇ Ṛ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ İ Ḻ Į Ṁ ṁ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ų ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Ą ą Ḝ ḝ İ Ǫ ǫ Ṳ ų Ṻ Ý ý

Ç ç Ä ä Ḝ ḝ Ḡ Į Ḳ ḳ Ḻ Į Ṇ ṇ Ṛ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ä ä Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Ï Ḻ Į Ṁ ṁ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ų ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Ä ä Ḝ ḝ Ï Ǫ ǫ Ṳ ų Ṻ Ý ý

Ç ç Å å Ḝ ḝ Ģ Į Ḳ ḳ Ḻ Į Ṇ ṇ Ṛ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Į Ḻ Į Ṁ ṁ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ų ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Å å Ḝ ḝ Į Ǫ ǫ Ṳ ų Ṻ Ý ý

Ç ç Ą ą Ḝ ḝ Ģ Į Ḳ ḳ Ḻ Į Ṇ ṇ Ṛ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Į Ḻ Į Ṁ ṁ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ų ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Ą ą Ḝ ḝ Į Ǫ ǫ Ṳ ų Ṻ Ý ý

Ç ç Ą ą Ḝ ḝ Ģ Į Ḳ ḳ Ḻ Į Ṇ ṇ Ṛ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Į Ḻ Į Ṁ ṁ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ų ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Ą ą Ḝ ḝ Į Ǫ ǫ Ṳ ų Ṻ Ý ý

Ç ç Ą ą Ḝ ḝ Ģ Į Ḳ ḳ Ḻ Į Ṇ ṇ Ṛ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Į Ḻ Į Ṁ ṁ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ų ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Ą ą Ḝ ḝ Į Ǫ ǫ Ṳ ų Ṻ Ý ý

Ç ç Ą ą Ḝ ḝ Ģ Į Ḳ ḳ Ḻ Į Ṇ ṇ Ṛ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Į Ḻ Į Ṁ ṁ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ų ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Ą ą Ḝ ḝ Į Ǫ ǫ Ṳ ų Ṻ Ý ý

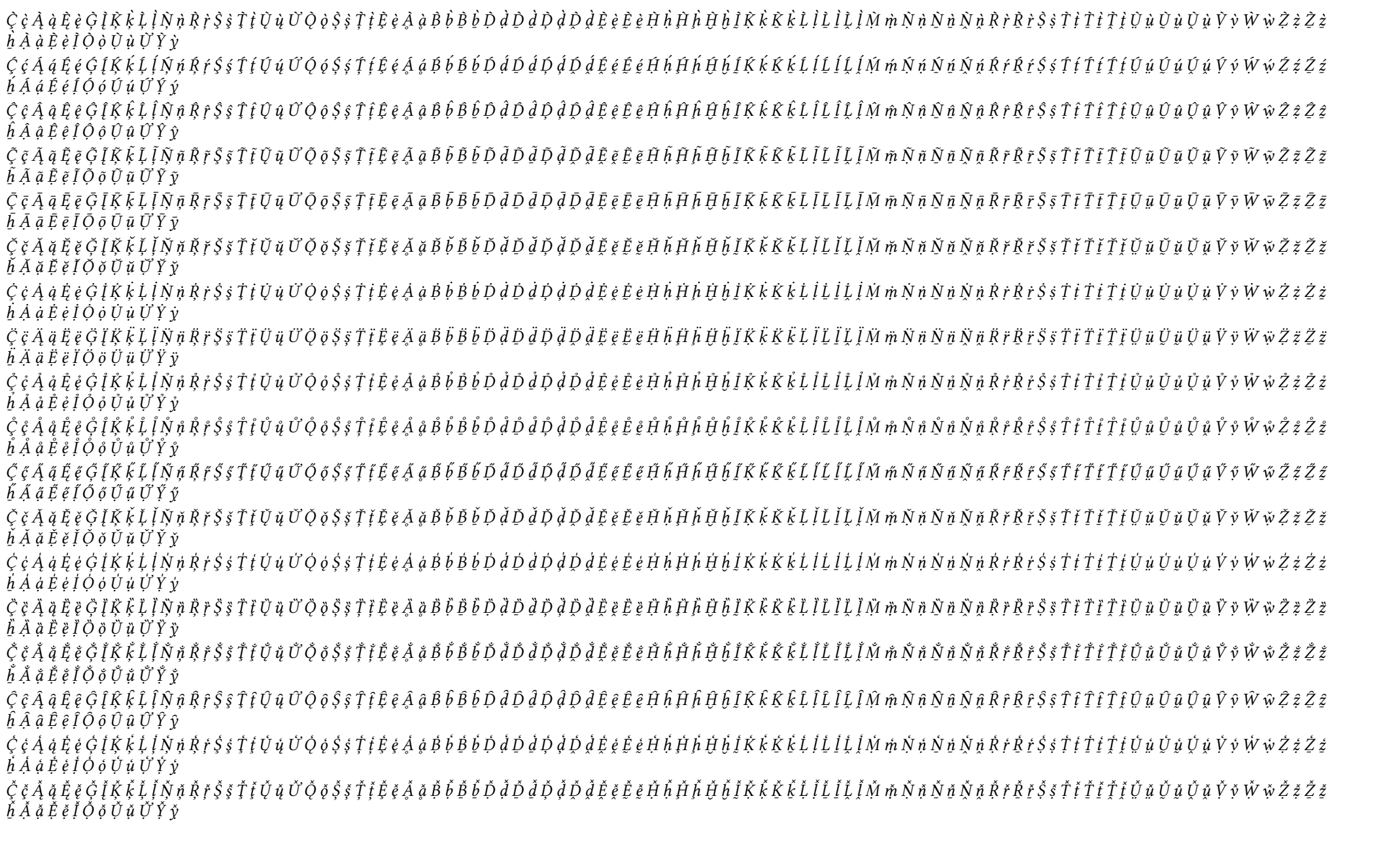
Ç ç Ą ą Ḝ ḝ Ģ Į Ḳ ḳ Ḻ Į Ṇ ṇ Ṛ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Į Ḻ Į Ṁ ṁ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ų ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Ą ą Ḝ ḝ Į Ǫ ǫ Ṳ ų Ṻ Ý ý

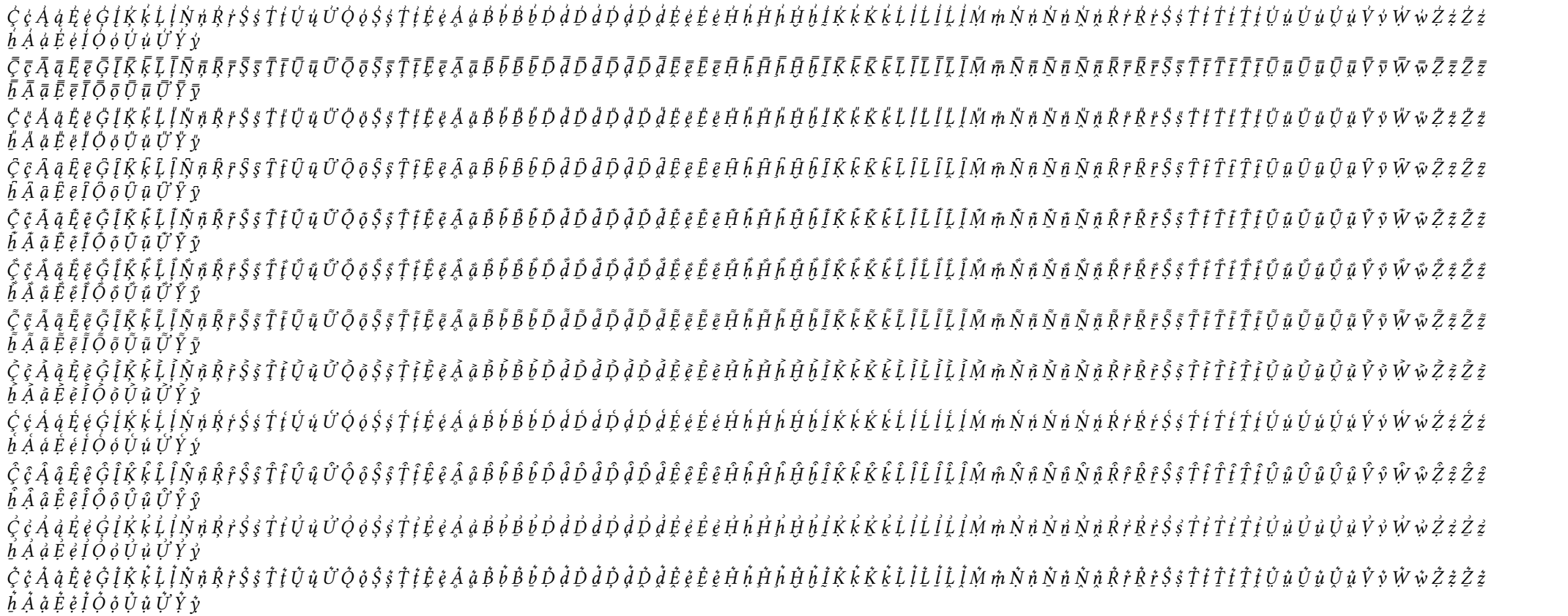
Ç ç Ą ą Ḝ ḝ Ģ Į Ḳ ḳ Ḻ Į Ṇ ṇ Ṛ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Į Ḻ Į Ṁ ṁ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ų ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Ą ą Ḝ ḝ Į Ǫ ǫ Ṳ ų Ṻ Ý ý

Ç ç Ą ą Ḝ ḝ Ģ Į Ḳ ḳ Ḻ Į Ṇ ṇ Ṛ ŗ Ş ş Ţ ţ Ų ų Ṻ Ǫ ǫ Ş ş Ţ ţ Ḙ ḙ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḙ ḙ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Į Ḳ ḳ Ḵ ḵ Ḷ Į Ḻ Į Ṁ ṁ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ţ ţ Ṯ ṯ Ṱ ṱ Ų ų Ṻ Ų ų Ṷ ṷ Ẅ ẅ Ẑ ẑ Ẕ ẕ Ḥ Ą ą Ḝ ḝ Į Ǫ ǫ Ṳ ų Ṻ Ý ý

Ç̄ ç̄ Ā ā Ē ē Ḡ ḡ Ī ī Ḵ ḵ Ḻ ḻ Ṉ ṉ Ō ō Ṟ ṟ Ṣ ṣ Ṯ ṯ Ū ū Ṳ ṳ Ṽ ṽ Ẇ ẇ Ẓ ẓ Ẕ ẕ H̱ ẖ Ā ā Ē ē Ī ī Ō ō Ū ū Ṻ ṻ Ȳ ȳ
Ç̣ ç̣ Ą ą Ę ę Ị ị Ḳ ḳ Ḷ ḷ Ṇ ṇ Ọ ọ Ṛ ṛ Ş ş Ṭ ṭ Ụ ụ Ự ự Ǫ ǫ Ṣ ṣ Ṭ ṭ Ẹ ẹ Ạ ạ Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḥ ḥ Ḫ ḫ Ḯ ḯ Ḵ ḵ Ḽ ḽ Ḹ ḹ Ḿ ḿ Ṋ ṋ Ṉ ṉ Ṓ ṓ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṯ ṯ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẉ ẉ Ẓ ẓ
Ç̌ ç̌ Ǎ ǎ Ě ě Ǧ ǧ Ǐ ǐ Ǩ ǩ Ľ ľ Ň ň Ř ř Š š Ť ť Ǔ ǔ Ǔ ǔ Ǒ ǒ Š š Ť ť Ě ě Ǎ ǎ Ḇ ḇ Ḇ ḇ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḕ ḕ Ḕ ḕ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ǐ ǐ Ǩ ǩ Ǩ ǩ Ľ ľ Ḹ ḹ Ḹ ḹ Ḿ ḿ Ň ň Ň ň Ň ň Ř ř Ř ř Š š Ť ť Ť ť Ť ť Ǔ ǔ Ǔ ǔ Ǔ ǔ Ǔ ǔ Ṽ ṽ Ẇ ẇ Ẓ ẓ
Ç̌ ç̌ Ǎ ǎ Ě ě Ǧ ǧ Ǐ ǐ Ǩ ǩ Ľ ľ Ň ň Ř ř Š š Ť ť Ǔ ǔ Ǔ ǔ Ǒ ǒ Š š Ť ť Ě ě Ǎ ǎ Ḇ ḇ Ḇ ḇ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḕ ḕ Ḕ ḕ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ǐ ǐ Ǩ ǩ Ǩ ǩ Ľ ľ Ḹ ḹ Ḹ ḹ Ḿ ḿ Ň ň Ň ň Ň ň Ř ř Ř ř Š š Ť ť Ť ť Ť ť Ǔ ǔ Ǔ ǔ Ǔ ǔ Ǔ ǔ Ṽ ṽ Ẇ ẇ Ẓ ẓ
Ç̌ ç̌ Ǎ ǎ Ě ě Ǧ ǧ Ǐ ǐ Ǩ ǩ Ľ ľ Ň ň Ř ř Š š Ť ť Ǔ ǔ Ǔ ǔ Ǒ ǒ Š š Ť ť Ě ě Ǎ ǎ Ḇ ḇ Ḇ ḇ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḕ ḕ Ḕ ḕ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ǐ ǐ Ǩ ǩ Ǩ ǩ Ľ ľ Ḹ ḹ Ḹ ḹ Ḿ ḿ Ň ň Ň ň Ň ň Ř ř Ř ř Š š Ť ť Ť ť Ť ť Ǔ ǔ Ǔ ǔ Ǔ ǔ Ǔ ǔ Ṽ ṽ Ẇ ẇ Ẓ ẓ
Ç̌ ç̌ Ǎ ǎ Ě ě Ǧ ǧ Ǐ ǐ Ǩ ǩ Ľ ľ Ň ň Ř ř Š š Ť ť Ǔ ǔ Ǔ ǔ Ǒ ǒ Š š Ť ť Ě ě Ǎ ǎ Ḇ ḇ Ḇ ḇ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḕ ḕ Ḕ ḕ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ǐ ǐ Ǩ ǩ Ǩ ǩ Ľ ľ Ḹ ḹ Ḹ ḹ Ḿ ḿ Ň ň Ň ň Ň ň Ř ř Ř ř Š š Ť ť Ť ť Ť ť Ǔ ǔ Ǔ ǔ Ǔ ǔ Ǔ ǔ Ṽ ṽ Ẇ ẇ Ẓ ẓ
Ç̌ ç̌ Ǎ ǎ Ě ě Ǧ ǧ Ǐ ǐ Ǩ ǩ Ľ ľ Ň ň Ř ř Š š Ť ť Ǔ ǔ Ǔ ǔ Ǒ ǒ Š š Ť ť Ě ě Ǎ ǎ Ḇ ḇ Ḇ ḇ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḕ ḕ Ḕ ḕ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ǐ ǐ Ǩ ǩ Ǩ ǩ Ľ ľ Ḹ ḹ Ḹ ḹ Ḿ ḿ Ň ň Ň ň Ň ň Ř ř Ř ř Š š Ť ť Ť ť Ť ť Ǔ ǔ Ǔ ǔ Ǔ ǔ Ǔ ǔ Ṽ ṽ Ẇ ẇ Ẓ ẓ
Ç̌ ç̌ Ǎ ǎ Ě ě Ǧ ǧ Ǐ ǐ Ǩ ǩ Ľ ľ Ň ň Ř ř Š š Ť ť Ǔ ǔ Ǔ ǔ Ǒ ǒ Š š Ť ť Ě ě Ǎ ǎ Ḇ ḇ Ḇ ḇ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḕ ḕ Ḕ ḕ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ǐ ǐ Ǩ ǩ Ǩ ǩ Ľ ľ Ḹ ḹ Ḹ ḹ Ḿ ḿ Ň ň Ň ň Ň ň Ř ř Ř ř Š š Ť ť Ť ť Ť ť Ǔ ǔ Ǔ ǔ Ǔ ǔ Ǔ ǔ Ṽ ṽ Ẇ ẇ Ẓ ẓ
Ḉ ḉ Á á Ę́ ę́ Ǵ ǵ Ĺ ĺ Ń ń Ọ́ ọ́ Ś ś Ṫ ṫ Ų́ ų́ Ứ ứ Ǫ́ ǫ́ Ś ś Ṫ ṫ Ę́ ę́ Ǻ ǻ Ḃ ḃ Ḃ ḃ Ḋ ḋ Ḋ ḋ Ḋ ḋ Ḋ ḋ Ḗ ḗ Ḗ ḗ Ḯ ḯ Ḱ ḱ Ḱ ḱ Ĺ ĺ Ĺ ĺ Ĺ ĺ Ḿ ḿ Ń ń Ń ń Ń ń Ŕ ŕ Ŕ ŕ Ś ś Ṫ ṫ Ṫ ṫ Ṫ ṫ Ų́ ų́ Ų́ ų́ Ų́ ų́ Ṿ ṿ Ẃ ẃ Ź ź
Ç̂ ç̂ Â â Ê ê Ĝ ĝ Î î Ǩ ǩ Ľ ľ Ñ ñ Ř ř Ŝ ŝ Ť ť Ǔ ǔ Ǔ ǔ Ô ô Ŝ ŝ Ť ť Ê ê Â â Ḃ ḃ Ḃ ḃ Ḋ ḋ Ḋ ḋ Ḋ ḋ Ḋ ḋ Ê ê Ê ê Ḣ ḣ Ḣ ḣ Ḣ ḣ Ḣ ḣ Ǐ ǐ Ǩ ǩ Ǩ ǩ Ľ ľ Ḹ ḹ Ḹ ḹ Ṁ ṁ Ñ ñ Ñ ñ Ñ ñ Ř ř Ř ř Ŝ ŝ Ť ť Ť ť Ť ť Ǔ ǔ Ǔ ǔ Ǔ ǔ Ǔ ǔ V̂ v̂ Ŵ ŵ Ẑ ẑ
Ç̇ ç̇ Ȧ ȧ Ė ė İ ȯ ṗ u̇ Ẇ ẏ Ż ż Ḣ ḣ Ȧ ȧ Ė ė İ Ȯ ȯ Ṗ u̇ Ẇ ẏ
Ç̌ ç̌ Ǎ ǎ Ě ě Ǧ ǧ Ǐ ǐ Ǩ ǩ Ľ ľ Ň ň Ř ř Š š Ť ť Ǔ ǔ Ǔ ǔ Ǒ ǒ Š š Ť ť Ě ě Ǎ ǎ Ḇ ḇ Ḇ ḇ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḕ ḕ Ḕ ḕ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ǐ ǐ Ǩ ǩ Ǩ ǩ Ľ ľ Ḹ ḹ Ḹ ḹ Ḿ ḿ Ň ň Ň ň Ň ň Ř ř Ř ř Š š Ť ť Ť ť Ť ť Ǔ ǔ Ǔ ǔ Ǔ ǔ Ǔ ǔ Ṽ ṽ Ẇ ẇ Ẓ ẓ
Ç̌ ç̌ Ǎ ǎ Ě ě Ǧ ǧ Ǐ ǐ Ǩ ǩ Ľ ľ Ň ň Ř ř Š š Ť ť Ǔ ǔ Ǔ ǔ Ǒ ǒ Š š Ť ť Ě ě Ǎ ǎ Ḇ ḇ Ḇ ḇ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḕ ḕ Ḕ ḕ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ǐ ǐ Ǩ ǩ Ǩ ǩ Ľ ľ Ḹ ḹ Ḹ ḹ Ḿ ḿ Ň ň Ň ň Ň ň Ř ř Ř ř Š š Ť ť Ť ť Ť ť Ǔ ǔ Ǔ ǔ Ǔ ǔ Ǔ ǔ Ṽ ṽ Ẇ ẇ Ẓ ẓ
Ç̌ ç̌ Ǎ ǎ Ě ě Ǧ ǧ Ǐ ǐ Ǩ ǩ Ľ ľ Ň ň Ř ř Š š Ť ť Ǔ ǔ Ǔ ǔ Ǒ ǒ Š š Ť ť Ě ě Ǎ ǎ Ḇ ḇ Ḇ ḇ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḏ ḏ Ḕ ḕ Ḕ ḕ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ḫ ḫ Ǐ ǐ Ǩ ǩ Ǩ ǩ Ľ ľ Ḹ ḹ Ḹ ḹ Ḿ ḿ Ň ň Ň ň Ň ň Ř ř Ř ř Š š Ť ť Ť ť Ť ť Ǔ ǔ Ǔ ǔ Ǔ ǔ Ǔ ǔ Ṽ ṽ Ẇ ẇ Ẓ ẓ

Italic patched, Precomposed letters that inherit an above anchor in stage 9, Above





Semibold patched, Precomposed letters that inherit an above anchor in stage 9, Above

Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ș ș Ṭ ṭ Ẹ ẹ Ạ ạ Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḓ ḓ Ḍ ḍ Ḓ ḓ Ḕ ḕ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Ḯ Ḳ ḳ Ḵ ḵ Ḻ Ḷ Ḽ Ḻ Ṃ ṃ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṟ ṟ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṷ ṷ Ṽ ṽ Ẇ ẇ Ẓ ẓ Ẕ ẕ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ử Ỵ ỵ

Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ş ş Ţ ţ Ę ę Ą ą Ḇ ḇ B̲ b̲ Ḓ ḓ D̲ d̲ Ḏ ḏ D̲ d̲ Ḓ ḓ Ḕ ḕ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḩ ḩ Ḭ Ḵ ḵ K̲ k̲ Ḷ ḷ L̲ l̲ Ḽ ḽ Ḿ ḿ Ṋ ṋ N̲ n̲ Ṓ ṓ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ṯ ṯ T̲ t̲ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẇ ẇ Ẓ ẓ Ẕ ẕ H̱ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ủ Ỷ ỷ

Semibold italic patched, Precomposed letters that inherit an above anchor in stage 9, Above



Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Į Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ư Ǫ ǫ Ş ş Ţ ţ Ę ę Ą ą Ḃ ḃ Ḃ ḃ Ḍ ḍ Ḍ ḍ Ḍ ḍ Ḍ ḍ Ḛ ḛ Ḛ ḛ Ḥ ḥ Ḫ ḫ Ḥ ḥ Ḯ Ḳ ḳ Ḳ ḳ Ḷ ḷ Ḷ ḷ Ḷ ḷ Ṁ ṁ Ṇ ṇ Ṅ ṅ Ṇ ṇ Ṛ ṛ Ṛ ṛ Ṣ ṣ Ṭ ṭ Ṫ ṫ Ṭ ṭ Ụ ụ Ủ ụ Ủ ụ Ṽ ṽ
Ẇ ẇ Ẓ ẓ Ẑ ẑ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ủ ụ Ử Ỵ ỵ

Ç̄ ç̄ Ą̄ ą̄ Ę̄ ę̄ Ģ̄ Į̄ Ķ̄ ķ̄ Ļ̄ Į̄ Ņ̄ ņ̄ Ŗ̄ ŗ̄ Ş̄ ş̄ Ţ̄ ţ̄ Ę̄ ę̄ Ą̄ ą̄ Ḃ̄ ḃ̄ Ḃ̄ ḃ̄ Ḍ̄ ḍ̄ Ḍ̄ ḍ̄ Ḍ̄ ḍ̄ Ḍ̄ ḍ̄ Ḛ̄ ḛ̄ Ḛ̄ ḛ̄ Ḥ̄ ḥ̄ Ḫ̄ ḫ̄ Ḥ̄ ḥ̄ Ḯ̄ Ḳ̄ ḳ̄ Ḳ̄ ḳ̄ Ḹ ḹ Ḹ ḹ Ḹ ḹ Ṁ̄ ṁ̄ Ṇ̄ ṇ̄ Ṅ̄ ṅ̄ Ṇ̄ ṇ̄ Ṝ ṝ Ṝ ṝ Ṣ̄ ṣ̄ Ṭ̄ ṭ̄ Ṫ̄ ṫ̄ Ṭ̄ ṭ̄ Ṷ̄ ṷ̄ Ṷ̄ ṷ̄ Ṷ̄ ṷ̄ Ṽ̄ ṽ̄
Ẅ̄ ẅ̄ Ẓ̄ ẓ̄ Ẑ̄ ẑ̄ ẖ̄ Ạ̄ ạ̄ Ẹ̄ ẹ̄ Ị̄ Ọ̄ ọ̄ Ṷ̄ ṷ̄ Ử̄ Ỵ̄ ỵ̄

Ç̈ ç̈ Ą̈ ą̈ Ë ë Ģ̈ Į̈ Ķ̈ ķ̈ Ļ̈ Į̈ Ņ̈ ņ̈ Ŗ̈ ŗ̈ Ş̈ ş̈ Ţ̈ ţ̈ Ų̈ ų̈ Ư̈ Ǫ̈ ǫ̈ Ş̈ ş̈ Ţ̈ ţ̈ Ë ë Ą̈ ą̈ Ḃ̈ ḃ̈ Ḃ̈ ḃ̈ Ḍ̈ ḍ̈ Ḍ̈ ḍ̈ Ḍ̈ ḍ̈ Ḍ̈ ḍ̈ Ḛ̈ ḛ̈ Ḛ̈ ḛ̈ Ḥ̈ ḥ̈ Ḫ̈ ḫ̈ Ḥ̈ ḥ̈ Ḯ̈ Ḳ̈ ḳ̈ Ḳ̈ ḳ̈ Ḷ̈ ḷ̈ Ḷ̈ ḷ̈ Ḷ̈ ḷ̈ Ṁ̈ ṁ̈ Ṇ̈ ṇ̈ Ṅ̈ ṅ̈ Ṇ̈ ṇ̈ Ṛ̈ ṛ̈ Ṛ̈ ṛ̈ Ṣ̈ ṣ̈ Ṭ̈ ṭ̈ Ṫ̈ ṫ̈ Ṭ̈ ṭ̈ Ṷ̈ ṷ̈ Ṷ̈ ṷ̈ Ṷ̈ ṷ̈ Ṽ̈ ṽ̈
Ẅ̈ ẅ̈ Ẓ̈ ẓ̈ Ẑ̈ ẑ̈ ẖ̈ Ạ̈ ạ̈ Ẹ̈ ẹ̈ Ị̈ Ọ̈ ọ̈ Ṷ̈ ṷ̈ Ử̈ Ỵ̈ ỵ̈

Ç̃ ç̃ Ą̃ ą̃ Ẽ ẽ Ģ̃ Į̃ Ķ̃ ķ̃ Ļ̃ Į̃ Ņ̃ ņ̃ Ŗ̃ ŗ̃ Ş̃ ş̃ Ţ̃ ţ̃ Ų̃ ų̃ Ữ Ǫ̃ ǫ̃ Ş̃ ş̃ Ţ̃ ţ̃ Ẽ ẽ Ą̃ ą̃ Ḃ̃ ḃ̃ Ḃ̃ ḃ̃ Ḍ̃ ḍ̃ Ḍ̃ ḍ̃ Ḍ̃ ḍ̃ Ḍ̃ ḍ̃ Ḛ̃ ḛ̃ Ḛ̃ ḛ̃ Ḥ̃ ḥ̃ Ḫ̃ ḫ̃ Ḥ̃ ḥ̃ Ḯ̃ Ḳ̃ ḳ̃ Ḳ̃ ḳ̃ Ḷ̃ ḷ̃ Ḷ̃ ḷ̃ Ḷ̃ ḷ̃ Ṁ̃ ṁ̃ Ṇ̃ ṇ̃ Ṅ̃ ṅ̃ Ṇ̃ ṇ̃ Ṛ̃ ṛ̃ Ṛ̃ ṛ̃ Ṣ̃ ṣ̃ Ṭ̃ ṭ̃ Ṫ̃ ṫ̃ Ṭ̃ ṭ̃ Ṷ̃ ṷ̃ Ṷ̃ ṷ̃ Ṷ̃ ṷ̃ Ṽ̃ ṽ̃
Ẅ̃ ẅ̃ Ẓ̃ ẓ̃ Ẑ̃ ẑ̃ ẖ̃ Ạ̃ ạ̃ Ẹ̃ ẹ̃ Ị̃ Ọ̃ ọ̃ Ṷ̃ ṷ̃ Ử̃ Ỵ̃ ỵ̃

Ç̆ ç̆ Ą̆ ą̆ Ę̆ ę̆ Ģ̆ Į̆ Ķ̆ ķ̆ Ļ̆ Į̆ Ņ̆ ņ̆ Ŗ̆ ŗ̆ Ş̆ ş̆ Ţ̆ ţ̆ Ų̆ ų̆ Ư̆ Ǫ̆ ǫ̆ Ş̆ ş̆ Ţ̆ ţ̆ Ę̆ ę̆ Ą̆ ą̆ Ḃ̆ ḃ̆ Ḃ̆ ḃ̆ Ḍ̆ ḍ̆ Ḍ̆ ḍ̆ Ḍ̆ ḍ̆ Ḍ̆ ḍ̆ Ḛ̆ ḛ̆ Ḛ̆ ḛ̆ Ḥ̆ ḥ̆ Ḫ̆ ḫ̆ Ḥ̆ ḥ̆ Ḯ̆ Ḳ̆ ḳ̆ Ḳ̆ ḳ̆ Ḷ̆ ḷ̆ Ḷ̆ ḷ̆ Ḷ̆ ḷ̆ Ṁ̆ ṁ̆ Ṇ̆ ṇ̆ Ṅ̆ ṅ̆ Ṇ̆ ṇ̆ Ṛ̆ ṛ̆ Ṛ̆ ṛ̆ Ṣ̆ ṣ̆ Ṭ̆ ṭ̆ Ṫ̆ ṫ̆ Ṭ̆ ṭ̆ Ṷ̆ ṷ̆ Ṷ̆ ṷ̆ Ṷ̆ ṷ̆ Ṽ̆ ṽ̆
Ẅ̆ ẅ̆ Ẓ̆ ẓ̆ Ẑ̆ ẑ̆ ẖ̆ Ặ ặ Ẹ̆ ẹ̆ Ị̆ Ọ̆ ọ̆ Ṷ̆ ṷ̆ Ử̆ Ỵ̆ ỵ̆

Ç̊ ç̊ Ą̊ ą̊ Ę̊ ę̊ Ģ̊ Į̊ Ķ̊ ķ̊ Ļ̊ Į̊ Ņ̊ ņ̊ Ŗ̊ ŗ̊ Ş̊ ş̊ Ţ̊ ţ̊ Ų̊ ų̊ Ư̊ Ǫ̊ ǫ̊ Ş̊ ş̊ Ţ̊ ţ̊ Ę̊ ę̊ Ą̊ ą̊ Ḃ̊ ḃ̊ Ḃ̊ ḃ̊ Ḍ̊ ḍ̊ Ḍ̊ ḍ̊ Ḍ̊ ḍ̊ Ḍ̊ ḍ̊ Ḛ̊ ḛ̊ Ḛ̊ ḛ̊ Ḥ̊ ḥ̊ Ḫ̊ ḫ̊ Ḥ̊ ḥ̊ Ḯ̊ Ḳ̊ ḳ̊ Ḳ̊ ḳ̊ Ḷ̊ ḷ̊ Ḷ̊ ḷ̊ Ḷ̊ ḷ̊ Ṁ̊ ṁ̊ Ṇ̊ ṇ̊ Ṅ̊ ṅ̊ Ṇ̊ ṇ̊ Ṛ̊ ṛ̊ Ṛ̊ ṛ̊ Ṣ̊ ṣ̊ Ṭ̊ ṭ̊ Ṫ̊ ṫ̊ Ṭ̊ ṭ̊ Ṷ̊ ṷ̊ Ṷ̊ ṷ̊ Ṷ̊ ṷ̊ Ṽ̊ ṽ̊
Ẅ̊ ẅ̊ Ẓ̊ ẓ̊ Ẑ̊ ẑ̊ ẖ̊ Ạ̊ ạ̊ Ẹ̊ ẹ̊ Ị̊ Ọ̊ ọ̊ Ṷ̊ ṷ̊ Ử̊ Ỵ̊ ỵ̊

Ç̋ ç̋ Ą̋ ą̋ Ę̋ ę̋ Ģ̋ Į̋ Ķ̋ ķ̋ Ļ̋ Į̋ Ņ̋ ņ̋ Ŗ̋ ŗ̋ Ş̋ ş̋ Ţ̋ ţ̋ Ų̋ ų̋ Ư̋ Ǫ̋ ǫ̋ Ş̋ ş̋ Ţ̋ ţ̋ Ę̋ ę̋ Ą̋ ą̋ Ḃ̋ ḃ̋ Ḃ̋ ḃ̋ Ḍ̋ ḍ̋ Ḍ̋ ḍ̋ Ḍ̋ ḍ̋ Ḍ̋ ḍ̋ Ḛ̋ ḛ̋ Ḛ̋ ḛ̋ Ḥ̋ ḥ̋ Ḫ̋ ḫ̋ Ḥ̋ ḥ̋ Ḯ̋ Ḳ̋ ḳ̋ Ḳ̋ ḳ̋ Ḷ̋ ḷ̋ Ḷ̋ ḷ̋ Ḷ̋ ḷ̋ Ṁ̋ ṁ̋ Ṇ̋ ṇ̋ Ṅ̋ ṅ̋ Ṇ̋ ṇ̋ Ṛ̋ ṛ̋ Ṛ̋ ṛ̋ Ṣ̋ ṣ̋ Ṭ̋ ṭ̋ Ṫ̋ ṫ̋ Ṭ̋ ṭ̋ Ṷ̋ ṷ̋ Ṷ̋ ṷ̋ Ṷ̋ ṷ̋ Ṽ̋ ṽ̋
Ẅ̋ ẅ̋ Ẓ̋ ẓ̋ Ẑ̋ ẑ̋ ẖ̋ Ạ̋ ạ̋ Ẹ̋ ẹ̋ Ị̋ Ọ̋ ọ̋ Ṷ̋ ṷ̋ Ử̋ Ỵ̋ ỵ̋

Ç̌ ç̌ Ą̌ ą̌ Ę̌ ę̌ Ǧ Į̌ Ǩ ǩ Ľ Į̌ Ň ň Ř ř Š š Ť ť Ų̌ ų̌ Ǔ Ǫ̌ ǫ̌ Š š Ť ť Ę̌ ę̌ Ą̌ ą̌ Ḃ̌ ḃ̌ Ḃ̌ ḃ̌ Ḍ̌ ḍ̌ Ḍ̌ ḍ̌ Ḍ̌ ḍ̌ Ḍ̌ ḍ̌ Ḛ̌ ḛ̌ Ḛ̌ ḛ̌ Ḥ̌ ḥ̌ Ḫ̌ ḫ̌ Ḥ̌ ḥ̌ Ḯ̌ Ḳ̌ ḳ̌ Ḳ̌ ḳ̌ Ḷ̌ ḷ̌ Ḷ̌ ḷ̌ Ḷ̌ ḷ̌ Ṁ̌ ṁ̌ Ň ň Ṅ̌ ṅ̌ Ň ň Ř ř Ř ř Š š Ť ť Ť ť Ť ť Ų̌ ų̌ Ǔ ų̌ Ǔ ų̌ Ṽ̌ ṽ̌
Ẅ̌ ẅ̌ Ž̌ ž̌ Ẑ̌ ẑ̌ ẖ̌ Ǎ ǎ Ě ě Ǐ Ǒ ǒ Ǔ ų̌ Ǔ Ǐ ǰ̌

Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Į Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ư Ǫ ǫ Ş ş Ţ ţ Ę ę Ą ą Ḃ ḃ Ḃ ḃ Ḍ ḍ Ḍ ḍ Ḍ ḍ Ḍ ḍ Ḛ ḛ Ḛ ḛ Ḥ ḥ Ḫ ḫ Ḥ ḥ Ḯ Ḳ ḳ Ḳ ḳ Ḷ ḷ Ḷ ḷ Ḷ ḷ Ṁ ṁ Ṇ ṇ Ṅ ṅ Ṇ ṇ Ṛ ṛ Ṛ ṛ Ṣ ṣ Ṭ ṭ Ṫ ṫ Ṭ ṭ Ụ ụ Ủ ụ Ủ ụ Ṽ ṽ
Ẇ ẇ Ẓ ẓ Ẑ ẑ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ủ ụ Ử Ỵ ỵ

Ç̇ ç̇ Ą̇ ą̇ Ė ė Ġ Į̇ Ķ̇ ķ̇ Ļ̇ Į̇ Ṅ ṅ Ṙ ṙ Ṡ ṡ Ṫ ṫ Ė ė Ą̇ ą̇ Ḃ̇ ḃ̇ Ḃ̇ ḃ̇ Ḋ ḋ Ḋ ḋ Ḋ ḋ Ḋ ḋ Ḗ ḗ Ḗ ḗ Ḣ ḣ Ḣ ḣ Ḣ ḣ Ḯ Ḱ ḱ Ḱ ḱ Ḷ̇ ḷ̇ Ḷ̇ ḷ̇ Ḷ̇ ḷ̇ Ṁ ṁ Ṅ ṅ Ṅ ṅ Ṅ ṅ Ṙ ṙ Ṙ ṙ Ṡ ṡ Ṫ ṫ Ṫ ṫ Ṫ ṫ Ụ̇ ụ̇ Ủ̇ ụ̇ Ủ̇ ụ̇ Ṿ ṿ
Ẇ ẇ Ż ż Ż ż ẖ Ạ̇ ạ̇ Ẹ̇ ẹ̇ Ị̇ Ọ̇ ọ̇ Ủ̇ ụ̇ Ử̇ Ẏ ẏ

Ç̂ ç̂ Ą̂ ą̂ Ê ê Ĝ Î Ķ̂ ķ̂ Ļ̂ Î Ñ̂ ñ̂ Ř̂ ř̂ Ŝ ŝ Ť̂ ť̂ Û ų̂ Ư̂ Ô ô Û ų̂ Ư̂ Ê ê Â â Ḃ̂ ḃ̂ Ḃ̂ ḃ̂ Ḋ̂ ḋ̂ Ḋ̂ ḋ̂ Ḋ̂ ḋ̂ Ḋ̂ ḋ̂ Ê̂ ê̂ Ê̂ ê̂ Ĥ ĥ Ĥ ĥ Ĥ ĥ Î Ǩ̂ ǩ̂ Ǩ̂ ǩ̂ Ĵ̌ ĵ̌ Ĵ̌ ĵ̌ Ĵ̌ ĵ̌ Ṁ̂ ṁ̂ Ṅ̂ ṅ̂ Ñ̂ ñ̂ Ṙ̂ ṙ̂ Ṙ̂ ṙ̂ Ŝ̌ ŝ̌ Ť̂ ť̂ Ť̂ ť̂ Ť̂ ť̂ Û̌ ų̂ Û̌ ų̂ Û̌ ų̂ Ṽ̂ ṽ̂
Ẅ̂ ẅ̂ Ẑ̂ ẑ̂ Ẑ̂ ẑ̂ ẖ̂ Ậ ậ Ê ê Î Ô ô Û ų̂ Ư̂ Ŷ ŷ

Ç̉ ç̉ Ả ả Ẻ ẻ Ỉ Ǩ ǩ Ľ Ỉ Ň ň Ỏ ỏ Ř ř Ş ş Ť ť Ủ ụ Ử Ǫ̉ ǫ̉ Ş ş Ť ť Ẻ ẻ Ả ả Ḃ̉ ḃ̉ Ḃ̉ ḃ̉ Ḍ̉ ḍ̉ Ḍ̉ ḍ̉ Ḍ̉ ḍ̉ Ḍ̉ ḍ̉ Ḛ̉ ḛ̉ Ḛ̉ ḛ̉ Ḥ̉ ḥ̉ Ḫ̉ ḫ̉ Ḥ̉ ḥ̉ Ỉ Ǩ̉ ǩ̉ Ǩ̉ ǩ̉ Ľ̉ ľ̉ Ľ̉ ľ̉ Ľ̉ ľ̉ Ṁ̉ ṁ̉ Ṇ̉ ṇ̉ Ṅ̉ ṅ̉ Ṇ̉ ṇ̉ Ṛ̉ ṛ̉ Ṛ̉ ṛ̉ Ṣ̉ ṣ̉ Ť̉ ť̉ Ť̉ ť̉ Ť̉ ť̉ Ủ̉ ụ̉ Ủ̉ ụ̉ Ủ̉ ụ̉ Ṽ̉ ṽ̉
Ẅ̉ ẅ̉ Ẓ̉ ẓ̉ Ẑ̉ ẑ̉ ẖ̉ Ạ̉ ạ̉ Ẻ ẻ Ỉ Ỏ ỏ Ủ ụ Ử Ỷ ỷ

Ç̀ ç̀ Ą̀ ą̀ Ȅ ȅ Ǵ Ị̀ Ḱ ḱ Ĺ Ị̀ Ń ń Ȑ ȑ Ś ś Ṫ ṫ Ȕ ų̀ Ừ Ǫ̀ ǫ̀ Ś ś Ṫ ṫ Ȅ ȅ Ą̀ ą̀ Ḃ̀ ḃ̀ Ḃ̀ ḃ̀ Ḍ̀ ḍ̀ Ḍ̀ ḍ̀ Ḍ̀ ḍ̀ Ḍ̀ ḍ̀ Ḕ ḕ Ḕ ḕ Ḥ̀ ḥ̀ Ḫ̀ ḫ̀ Ḥ̀ ḥ̀ Ị̀ Ḱ̀ ḱ̀ Ḱ̀ ḱ̀ Ĺ̀ ĺ̀ Ĺ̀ ĺ̀ Ĺ̀ ĺ̀ Ṁ̀ ṁ̀ Ń̀ ń̀ Ṅ̀ ṅ̀ Ń̀ ń̀ Ȑ̀ ȑ̀ Ȑ̀ ȑ̀ Ś̀ ś̀ Ṫ̀ ṫ̀ Ṫ̀ ṫ̀ Ṫ̀ ṫ̀ Ȕ̀ ų̀ Ừ̀ ų̀ Ừ̀ ų̀ Ṽ̀ ṽ̀
Ẅ̀ ẅ̀ Ẓ̀ ẓ̀ Ẑ̀ ẑ̀ ẖ̀ Ạ̀ ạ̀ Ȅ̀ ȅ̀ Ị̀ Ǫ̀ ǫ̀ Ȕ̀ ų̀ Ừ̀ Ỳ̀ ỳ̀

Ḉ ḉ Ą́ ą́ Ȩ́ ȩ́ Ǵ Į́ Ḱ ḱ Ĺ Į́ Ń ń Ŕ ŕ Ś ś Ť́ ť́ Ų́ ų́ Ứ Ǫ́ ǫ́ Ś ś Ť́ ť́ Ȩ́ ȩ́ Ą́ ą́ Ḃ́ ḃ́ Ḃ́ ḃ́ Ḍ́ ḍ́ Ḍ́ ḍ́ Ḍ́ ḍ́ Ḍ́ ḍ́ Ḗ ḗ Ḗ ḗ Ḥ́ ḥ́ Ḫ́ ḫ́ Ḥ́ ḥ́ Ḯ Ḱ ḱ Ḱ ḱ Ĺ́ ĺ́ Ĺ́ ĺ́ Ĺ́ ĺ́ Ṁ́ ṁ́ Ń ń Ṅ́ ṅ́ Ń ń Ŕ́ ŕ́ Ŕ́ ŕ́ Ś́ ś́ Ť́ ť́ Ť́ ť́ Ť́ ť́ Ų́ ų́ Ứ ų́ Ứ ų́ Ṽ́ ṽ́
Ẃ ẃ Ẓ́ ẓ́ Ẑ́ ẑ́ ẖ́ Ạ́ ạ́ Ȩ́ ȩ́ Ị́ Ǫ́ ǫ́ Ų́ ų́ Ứ Ý ý

Ç̂ ç̂ Ą̂ ą̂ Ê ê Ĝ Î Ķ̂ ķ̂ Ļ̂ Î Ñ̂ ñ̂ Ř̂ ř̂ Ŝ ŝ Ť̂ ť̂ Û ų̂ Ư̂ Ô ô Ŝ ŝ Ť̂ ť̂ Ê ê Â â Ḃ̂ ḃ̂ Ḃ̂ ḃ̂ Ḋ̂ ḋ̂ Ḋ̂ ḋ̂ Ḋ̂ ḋ̂ Ḋ̂ ḋ̂ Ê̂ ê̂ Ê̂ ê̂ Ĥ ĥ Ĥ ĥ Ĥ ĥ Î Ǩ̂ ǩ̂ Ǩ̂ ǩ̂ Ĵ̌ ĵ̌ Ĵ̌ ĵ̌ Ĵ̌ ĵ̌ Ṁ̂ ṁ̂ Ṅ̂ ṅ̂ Ñ̂ ñ̂ Ṙ̂ ṙ̂ Ṙ̂ ṙ̂ Ŝ̌ ŝ̌ Ť̂ ť̂ Ť̂ ť̂ Ť̂ ť̂ Û̌ ų̂ Û̌ ų̂ Û̌ ų̂ Ṽ̂ ṽ̂
Ẅ̂ ẅ̂ Ẑ̂ ẑ̂ Ẑ̂ ẑ̂ ẖ̂ Ậ ậ Ê ê Î Ô ô Û ų̂ Ư̂ Ŷ ŷ

Ç̉ ç̉ Ả ả Ẻ ẻ Ỉ Ǩ ǩ Ľ Ỉ Ň ň Ỏ ỏ Ř ř Ş ş Ť ť Ủ ụ Ử Ǫ̉ ǫ̉ Ş ş Ť ť Ẻ ẻ Ả ả Ḃ̉ ḃ̉ Ḃ̉ ḃ̉ Ḍ̉ ḍ̉ Ḍ̉ ḍ̉ Ḍ̉ ḍ̉ Ḍ̉ ḍ̉ Ḛ̉ ḛ̉ Ḛ̉ ḛ̉ Ḥ̉ ḥ̉ Ḫ̉ ḫ̉ Ḥ̉ ḥ̉ Ỉ Ǩ̉ ǩ̉ Ǩ̉ ǩ̉ Ľ̉ ľ̉ Ľ̉ ľ̉ Ľ̉ ľ̉ Ṁ̉ ṁ̉ Ṇ̉ ṇ̉ Ṅ̉ ṅ̉ Ṇ̉ ṇ̉ Ṛ̉ ṛ̉ Ṛ̉ ṛ̉ Ṣ̉ ṣ̉ Ť̉ ť̉ Ť̉ ť̉ Ť̉ ť̉ Ủ̉ ụ̉ Ủ̉ ụ̉ Ủ̉ ụ̉ Ṽ̉ ṽ̉
Ẅ̉ ẅ̉ Ẓ̉ ẓ̉ Ẑ̉ ẑ̉ ẖ̉ Ạ̉ ạ̉ Ẻ ẻ Ỉ Ỏ ỏ Ủ ụ Ử Ỷ ỷ

Regular patched, Precomposed letters that inherit a below anchor in stage 9, Below



ŗŖŗŪŭŪŭŬĦĦȦȧŌŏŎŏȮȯŎŏŶŷḂḃḊḋḖḗḘḙḞḟḠḡḢḣḦḧḪḫĮįḰḳḾḿṀṁṄṅŐőŒœȪȫȰȱṔṕṖṗŖŗṠṡṢṣṨṩṪṫŰűŲųṺṻṾṿẀẁẂẃẄẅẆẇẊẋẌẍỲỳỴỵỸỹ
ẘẙỶỷỺẠạẢảẤấẦầẪẫẨẩẮắẰằẴẵẲẳẶặẸẹẼẽẾếỀềỄễỂểỆệỈỉỊịỌọỎỏỐốỒồỖỗỔổỘộỚớỜờỠỡỞởỢợỤụỦủỨứỪừỮữỬửỰựỲỳỶỷỸỹ
ẠẤẬẮẰẶẸẾỀỆỈỊỌỐỒỘỚỜỢỤỨỪỮỬỰỲỴỶỸẚáâãäåèéêëìíîïñòóôõöùúûüýÿĀāĂăĆćĈĉĊċČčĎďĒēĔĕĖėĚěĜĝĞğĠġĦħĨĩĪīĬĭĮįĴĵĹĺĽľŃńŇňŌōŎ
ŏŐőŔŕŘřŞşŜŝŠšŢţŪūŬŭŮůŰűŴŵŶŷŸŹźŻżŽžƠơƯưǍǎǏǐǑǒǓǔǕǖǗǘǙǚǛǜǞǟǠǡǢǣǦǧǨǩǪǫǬǭǮǯȀȁȂȃȆȇȊȋȎȏȒȓ

ŏŐőŔŕŘřŞşŜŝŠšŢţŪūŬŭŮůŰűŴŵŶŷŸŹźŻżŽžƠơƯưǍǎǏǐǑǒǓǔǕǖǗǘǙǚǛǜǞǟǠǡǢǣǤǥǦǧǨǩǪǫǬǭǮǯǰǴǵǸǹǺǻǼǽǾǿȀȁȂȃȄȅȆȇȈȉȊȋȌȍȎȏȐȑȒȓȔȕȖȗȘșȚțȞȟȦȧȨȩȪȫȬȭȮȯȰȱȲȳ
ẀẁẂẃẄẅẆẇẈẉẊẋẌẍẎẏẐẑẔẕẖẗẘẙẚẛẜẝẞẟẠạẢảẤấẦầẨẩẪẫẬậẮắẰằẲẳẴẵẶặẸẹẺẻẼẽẾếỀềỂểỄễỆệỈỉỊịỌọỎỏỐốỒồỔổỖỗỘộỚớỜờỞởỠỡỢợỤụỦủỨứỪừỬửỮữỰựỲỳỴỵỶỷỸỹ
ỺỻỼỽỾỿ

Ṏ Ő ő Ř ŕ Ř ř Ś ś Ŝ ŝ Š š Ť ť Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ẁ ẁ Ŷ ŷ Ÿ Ź ź Ż ż Ž ž Ǫ ǫ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ Ǵ ǵ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ Ȁ ȁ Ȃ ȃ Ȅ ȅ Ȇ ȇ Ȉ ȉ Ȋ ȋ Ȍ Ȓ
ȓ Ȑ ȑ Ț ț Ȟ ȟ Ȧ ȧ Ȫ ȫ Ȯ ȯ Ȱ ȱ Ȳ ȳ Ḃ ḃ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṫ ṫ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẗ
ẘ ẙ ẛ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ
À Á Â Ã Ä Å È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö Ù Ú Û Ü Ý à á â ã ä å è é ê ë ì í î ï ñ ò ó ô õ ö ù ú û ü ý ÿ Ā ā Ă ă Ć ć Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ę ę Ĝ ĝ Ğ ğ Ġ ġ Ģ ģ Ĥ ĥ Ĩ ĩ Ī ī Ĭ ĭ Į į Ĵ ĵ Ĺ ĺ Ļ ļ Ľ ľ Ń ń Ň ň Ō ō Ŏ

Italic patched, Precomposed letters that inherit a below anchor in stage 9, Below

Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỵ Ỷ

Ụ ụ Ḧ ḧ Ạ ạ Ọ̈ ọ̈ Ọ̄ ọ̄ Ọ ọ Ọ̈ ọ̈ Ȳ ȳ Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṍ ṍ Ṏ ṏ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṡ ṡ Ṣ ṣ Ṩ ṩ Ṫ ṫ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ỳ ỳ Ẓ ẓ Ỻ ỻ W̊ ẘ Y̊ ẙ Ỷ ỷ Ỹ ỹ
Ậ ậ Ặ ặ Ẳ ẳ Ằ ằ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ấ ấ Ặ ặ Ậ ậ Ẳ ẳ Ằ ằ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ấ ấ Ệ ệ Ễ ễ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ớ ớ Ợ ợ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ
Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỵ Ỷ

Ụ ụ Ȟ ȟ Ạ ạ Ọ ọ Ỏ ỏ Ọ ọ Ỳ ỳ Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṍ ṍ Ṏ ṏ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṡ ṡ Ṣ ṣ Ṩ ṩ Ṫ ṫ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ T̈ ẘ Y̊ ẙ Ṡ ẛ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỵ Ỷ Ỹ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỵ Ỷ Ỹ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỵ Ỷ Ỹ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỵ Ỷ Ỹ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỵ Ỷ Ỹ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỵ Ỷ Ỹ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỵ Ỷ Ỹ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỵ Ỷ Ỹ

Semibold patched, Precomposed letters that inherit a below anchor in stage 9, Below



Ẋ Ẏ ẏ Ẑ ẑ ẗ ẘ ẙ ẚ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỷ Ỹ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Semibold italic patched, Precomposed letters that inherit a below anchor in stage 9, Below

Ǐ ǐ Î î Ǒ ǒ Ô ô Ř ř Ȓ ȓ Ǔ ǔ Û û Ȟ ȟ Ạ ạ Ǫ ǫ Ȫ ȫ Ọ ọ Ȍ ȍ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṭ ṭ Ṵ ṵ Ṳ ṳ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ỳ ỳ Ẓ ẓ Ỉ ỉ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỷ ỷ Ỹ ỹ
Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ
À Á Â Ã Ä Å Ḕ Ḗ Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö Ù Ú Û Ü Ý à á â ã ä å ḕ ḗ ê ë ì í î ï ñ ò ó ô õ ö ù ú û ü ý ÿ Ā ā Ă ă Ć ć Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ę ę Ğ ğ Ġ ġ Ģ ģ Ĥ ĥ Ĩ ĩ Ī ī Ĭ ĭ İ ĵ J ĺ Ļ ļ Ľ ľ Ń ń Ň
ň Ō ō Ŏ ŏ Ő ő Ř ř Ȑ ȑ Ș ș Ş ş Š š Ţ ţ Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ẁ ẁ Ŷ ŷ Ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ā ā Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ ǰ Ǵ ǵ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ À à Â â Ȅ ȅ Ȇ ȇ
Ḯ ḯ Î î Ǒ ǒ Ô ô Ř ř Ȓ ȓ Ǔ ǔ Û û Ȟ ȟ Ạ ạ Ǫ ǫ Ȫ ȫ Ọ ọ Ȍ ȍ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṭ ṭ Ṵ ṵ Ṳ ṳ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ỳ ỳ Ẓ ẓ Ỉ ỉ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ
À Á Â Ã Ä Å Ḕ Ḗ Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö Ù Ú Û Ü Ý à á â ã ä å ḕ ḗ ê ë ì í î ï ñ ò ó ô õ ö ù ú û ü ý ÿ Ā ā Ă ă Ć ć Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ę ę Ğ ğ Ġ ġ Ģ ģ Ĥ ĥ Ĩ ĩ Ī ī Ĭ ĭ İ ĵ J ĺ Ļ ļ Ľ ľ Ń ń Ň
ň Ō ō Ŏ ŏ Ő ő Ř ř Ȓ ȓ

ŋ̃ Ō ō Ŏ ŏ Ő ő Ŕ ŕ Ř ř Ś ś Ŝ ŝ Ş ş Š š Ţ ţ Ť ť Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ẁ ẁ Ŷ ŷ Ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ À à Â â Ề ề Ệ ệ
Ỉ ỉ Ị ị Ọ ọ Ộ ộ Ờ ờ Ợ ợ Ọ̃ ọ̃ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ḃ ḃ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṗ ṗ Ṛ ṛ Ṡ ṡ Ṩ ṩ Ṫ ṫ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ
ẇ Ẋ ẋ Ẍ ẍ Ỳ ỳ Ẓ ẓ ỻ ỽ ỿ Ả ả Ấ ấ Ậ ậ Ẩ ẩ Ẫ ẫ Ắ ắ Ặ ặ Ẳ ẳ Ẵ ẵ Ẻ ẻ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỹ ỹ
À Á Â Ã Ä Å Ḕ Ḗ Ê Ì Í Î Ï Ñ Ò Ó Ô Õ Ö Ù Ú Û Ü Ý à á â ã ä å ḕ ḗ ê ì í î ï ñ ò ó ô õ ö ù ú û ü ý ÿ Ā ā Ă ă Ç ç Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ę ę Ě ě Ĝ ĝ Ğ ğ Ġ ġ Ḥ ḥ Ĩ ĩ Ī ī Ĭ ĭ İ ĵ Ĵ ĵ Ĺ ĺ Ļ ļ Ľ ľ Ń ń Ň

À Á Â Ã Ä Å È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö Ù Ú Û Ü Ý à á â ã ä å è é ê ë ì í î ï ñ ò ó ô õ ö ù ú û ü ý ÿ Ā ā Ă ă Ć ć Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ę ę Ě ě Ĝ ĝ Ğ ğ Ġ ġ Ģ ģ Ĥ ĥ Ĩ ĩ Ī ī Ĭ ĭ Į į İ Ĵ ĵ Ĺ ĺ Ļ ļ Ľ ľ Ń ń Ņ ņ Ň ň Ő ő Ŏ ŏ Ő ő Ŕ ŕ Ř ř Ś ś Ŝ ŝ Ş ş Š š Ţ ţ Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ŵ ŵ Ŷ ŷ Ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ Ǵ ǵ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ À ȁ Â ȃ Ȅ ȅ Ȇ ȇ Ȉ ȉ Ȋ ȋ Ȍ ȍ Ȏ ȏ Ȑ ȑ Ȓ ȓ Ȕ ȕ Ȗ ȗ Ș ș Ȟ ȟ Ƞ ȡ Ȣ ȣ Ȥ ȥ Ȧ ȧ Ȩ ȩ Ȫ ȫ Ȭ ȭ Ȯ ȯ Ȱ ȱ Ȳ ȳ Ⱥ ⱥ Ȼ ȼ Ƚ ⱦ Ɂ ɂ Ƀ ɀ Ʉ ʉ Ɇ ɇ Ɉ ɉ Ɍ ɍ Ɏ ɏ ɐ ɑ ɒ ɓ ɔ ɕ ɖ ɗ ɘ ə ɚ ɛ ɜ ɝ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɧ ɨ ɩ ɪ ɫ ɬ ɭ ɮ ɯ ɰ ɱ ɲ ɳ ɴ ɵ ɶ ɷ ɸ ɹ ɺ ɻ ɼ ɽ ɾ ɿ ʀ ʁ ʂ ʃ ʄ ʅ ʆ ʇ ʈ ʉ ʊ ʋ ʌ ʍ ʎ ʏ ʐ ʑ ʒ ʓ ʔ ʕ ʖ ʗ ʘ ʙ ʚ ʛ ʜ ʝ ʞ ʟ ʠ ʡ ʢ ʣ ʤ ʥ ʦ ʧ ʨ ʩ ʪ ʫ ʬ ʭ ʮ ʯ